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PROPOSED SYLLABUS 1 to 6 (SESSION 2024-2025)

SYLLABUS	PHYSICS	CHEMISTRY	BIOLOGY
SYLLABUS 1	**» Physics And Measurement** **» Kinematics** **» Laws Of Motion** **» Work, Energy, And Power** ***» Experimental Skills*** * Vernier Calipers-Its Use To Measure The Internal And External Diameter And Depth Of A Vessel. * Screw Gauge-Its Use To Determine Thickness/ Diameter Of Thin Sheet/Wire.	**» Some Basic Concepts of Chemistry** **» Equilibrium,** **» Chemical Thermodynamics** **» Redox reactions**	**» The Living World** **» Biological Classification** **» Plant Kingdom** **» Structural Organisation In Animals (Animal Tissue),** **» Animal Kingdom** **» Cockroach, Frog**
SYLLABUS 2	**» Center Of Mass And Collision** **» Circular Motion** **» Rotational Motion** **» Gravitation** **» Properties Of Solids And Liquids** **» Experimental Skills** * Young'S Modulus Of Elasticity Of The Material Of A Metallic Wire * Surface Tension Of Water By Capillary Rise And Effect Of Detergents, * Co-Efficient Of Viscosity Of A Given Viscous Liquid By Measuring Terminal Velocity Of A Given Spherical Body.	**» Atomic Structure** **» Classification of Elements and Periodicity in Properties** **» Chemical Bonding & Molecular Structure**	**» Cell : The Unit Of Life** **» Cell Cycle And Cell Division** **» Biomolecules, Enzyme** **» Excretory Products And Their Elimination (Excretory System)** **» Breathing And Exchange Of Gases (Respiratory System)** **» Body Fluids And Circulation (Circulatory System)**
SYLLABUS 3	**» Thermodynamics** **» Kinetic Theory Of Gases** **» Oscillations And Waves** **» Experimental Skills** * Simple Pendulum-Dissipation Of Energy By Plotting A Graph Between The Square Of Amplitude And Time. * Metre Scale - The Mass Of A Given Object By The Principle Of Moments. * Speed Of Sound In Air At Room Temperature Using A Resonance Tube, * Specific Hea: Capacity Of A Given (I) Solid And (Ii) Liquid By Method Of Mixtures.	**» Some Basic Principles of Organic Chemistry** **» Hydrocarbons** **» Purification and Characterisation of Organic Compounds.**	**» Photosynthesis in Higher Plants** **» Plant Growth and Development** **» Respiration in Plants** **» Neural Control And Co-ordination (Nervous System)** **» Chemical Co-ordination And Integration (Endocrine System)** **» Locomotion And Movement (Muscles, Skeletal System)**
SYLLABUS 4	**» Electrostatics & Capacitor** **» Current Electricity** **» Experimental Skills** * The Resistivity Of The Material Of A Given Wire Using A Metre Bridge. * The Resistance Of A Given Wire Using Ohm'S Law. * Resistance And Figure Of Merit Of A Galvanometer By Half Deflection Method.	**» Chemical Kinetics** **» Solutions** **» Electrochemistry** **» Principles Related to Pratical Chemistry :-** The chemistry involved in the titrimetric exercises-Acids, Bases and the use of indicators, oxalic acid vs $KMnO_4$, Mohr's salt vs $KMnO_4$. Chemical principles involved in the following experiments. 1. Enthalpy of solution of $CuSO_4$.	**» Sexual Reproduction in Flowering Plants** **» Morphology Of Flowering Plants** **» Anatomy Of Flowering Plants** **» Human Reproduction** **» Reproductive Health**
SYLLABUS 5	**» Magnetic Effects Of Current And Magnetism** **» Electromagnetic Induction And Alternating Currents** **» Electromagnetic Waves**	**» p-block elements** **» d & f-block elements** **» Coordination Compounds** **» Principles Related to Pratical Chemistry :-** 1. The chemistry involved in the preparation of Inorganic compound; Mohr's salt, potash alum. 2. Chemical principles involved in the qualitative salt analysis. 3. Cation – Pb2+ Cu+2 Al3+ Fe3+, Zn+2, Ni2+ Ca2+, Ba2+, Mg2+, NH4+ 4. Anions – CO32- , S2- , SO42- , NO2–, Cl–, Br –, I – (Insoluble Salt Excluded)	**» Principles Of Inheritance And Variation** **» Molecular Basis Of Inheritance, Microbes In Human Welfare** **» Biotechnology : Principles And Processes** **» Biotechnology And Its Applications**
SYLLABUS 6	**» Optics** **» Dual Nature Of Matter And Radiation** **» Atoms And Nuclei** **» Electronic Devices** ***» Experimental Skills*** * The Focal Length Of; (I) Convex Mirror (Ii) Concave Mirror (Iii) Convex Lens, Using The Parallax Method. * The Plot Of The Angle Of Deviation Vs Angle Of Incidence For A Triangular Prism. * Refractive Index Of A Glass Slab Using A Travelling Microscope. * Characteristic Curves Of A P-N Junction Diode In Forward And Reverse Bias. * Characteristic Curves Of A Zener Diode And Finding Reverse Break Down Voltage. * Identification Of Diode. Led,. Resistor. A Capacitor From A Mixed Collection Of Such Items.	**» Organic Compounds Containing Halogens,** **» Organic Compounds Containing Oxygen,** **» Organic Compounds Containing Nitrogen, Biomolecules** **» Principles Related to Pratical Chemistry :-** 1. Detection of extra elements (Nitrogen, Sulphur, halogens) in organic compounds. 2. Detection of the hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketones) , carboxyl and amino functional groups in organic compounds. 3. The chemistry involved in the preparation of Acetanilide , p-nitro acetanilide , aniline yellow and iodoform.	**» Organisms and Populations** **» Ecosystem** **» Biodiversity and Conservation** **» Demography** **» Biology In Human Welfare : Human Health and Disease** **» Origin And Evolution**

PROPOSED MAJOR TEST PLAN FOR LEADER AND ACHIEVER (2024-25)

DATE	DAY	SYLLABUS
13-03-2025	THURSDAY	SYLLABUS -1
17-03-2025	MONDAY	SYLLABUS -2
21-03-2025	FRIDAY	SYLLABUS -3
25-03-2025	TUESDAY	SYLLABUS -4
29-03-2025	SATURDAY	SYLLABUS -5
02-04-2025	WEDNESDAY	SYLLABUS -6
06-04-2025	SUNDAY	FULL SYLLABUS (AIOT)
08-04-2025	TUESDAY	SYLLABUS - 1+2+3
11-04-2025	FRIDAY	SYLLABUS - 4+5+6
14-04-2025	MONDAY	FULL SYLLABUS
17-04-2025	THURSDAY	FULL SYLLABUS
20-04-2025	SUNDAY	FULL SYLLABUS (AIOT)
22-04-2025	TUESDAY	FULL SYLLABUS
25-04-2025	FRIDAY	FULL SYLLABUS
28-04-2025	MONDAY	FULL SYLLABUS
01-05-2025	THURSDAY	FULL SYLLABUS

PHYSICS

1) Which of the following pair represent the same dimensions :-

(1) Watt and Newton-metre
(2) Pascal and Joule
(3) Newton and Joule
(4) Joule and Newton-metre

2) If x is length, $v = \frac{dx}{dt}$ is velocity and $a = \frac{dv}{dt}$ is acceleration, then which option is not possible :- (t = time)

(1) $\left(\frac{dv}{dt} + \frac{d^2x}{dt^2}\right)$
(2) $\left(\frac{dv}{dt} - \frac{d^2x}{dt^2}\right)$
(3) $\left(\frac{d^2x}{dt^2} + x\frac{dv}{dt}\right)$
(4) $\left(v\frac{dv}{dx} + 5\frac{d^2x}{dt^2}\right)$

3) In a relation $y = A\sin(e^{\alpha x} + \beta)$ here y and x represent length then dimensions of $\frac{A}{\alpha\beta}$ will be :-

(1) $[ML^{-1}T^0]$
(2) $[M^0L^2T^0]$
(3) $[M^0L^{-2}T^0]$
(4) $[M^0L^2T]$

4) If force, $F = \rho^a A^b v^c$, here ρ is density, A is area and v is velocity then value of a, b, c will be :-

(1) a = 1, b = 1, c = 2
(2) a = 2, b = 1, c = 1
(3) a = 1, b = 2, c = 1
(4) a = 1, b = 2, c = 2

5) If acceleration $a_t = m^a v^b t^c$ where m is mass, v is velocity and t is time then value of $(a^2 + bc)$ will be :-

(1) Zero

(2) –1

(3) 2

(4) –2

6) Two resistance $R_1 = 1000 \pm 5\%\ \Omega$ and $R_2 = 200 \pm 5\%\ \Omega$ are connected in series. The percentage error in equivalent resistance will be :-

(1) 10%

(2) 25%

(3) 15%

(4) 5%

7) A physical quantity represented as $x = \frac{A^2B^{1/3}}{\sqrt{CD}}$, if percentage error in A is 2%, B is 3%, C is 4% and D is 6%, then percentage error in x will be :-

(1) 5%

(2) 15%

(3) 10%

(4) 12%

8) The time period of simple pendulum is given by $T = 2\pi\sqrt{\frac{\ell}{g}}$. Measured value of ℓ is 40.0 cm and time for 200 oscillations is found to be 400 s by using of resolution 1s. Maximum percentage error in g will be if it is measured from above relation for time period :-

(1) 1.25%

(2) 0.75%

(3) 2.50%

(4) 0.5%

9) The radius of sphere is measured to be (5.00 ± 0.05) cm. If percentage error in its volume and surface area are x% and y% then |x| + |y| will be :-

(1) 2

(2) 5

(3) 7

(4) 3

10) If 30 litres of water per minute is lifted from a well through 10 m depth and delivered with a velocity of $2\sqrt{15}$ m/s then the power of the motor is :- (take $g = 10\ m/s^2$ and $\rho_{water} = 10^3\ kg/m^3$)

(1) 50 watt

(2) 80 watt

(3) 65 watt

(4) 72 watt

11) A block of mass 3 kg is initially at rest on a horizontal frictionless surface. A horizontal force $\vec{F} = (x^2 + x)\hat{i}N$ acts on it, when the block is at x = 0. The maximum kinetic energy of the block between x = 0 to x = 1m is :-

(1) 1.2 J

(2) 0.83 J

(3) 0.72 J

(4) 0.98 J

12) Consider the following two statements :-
(P) Linear momentum of a system of particles is zero.
(Q) Kinetic energy of a system of particles is zero.
Then

(1) P does not imply Q and Q does not imply P.

(2) P implies Q but Q does not imply P.

(3) P does not imply Q but Q implies P.

(4) P implies Q and Q implies P.

13)

Match the following column-

	Column-1		**Column-2**
(a)	Erg	(p)	1.6×10^{-19} J
(b)	Electron volt (eV)	(q)	3.6×10^{6} J
(c)	Calorie (cal)	(r)	10^{-7} J
(d)	Kilowatt hour (kWh)	(s)	4.186 J

(1) (a–r), (b–p), (c–s), (d–q)

(2) (a–p), (b–r), (c–q), (d–s)

(3) (a–p), (b–q), (c–r), (d–s)

(4) (a–s), (b–r), (c–q), (d–p)

14) A uniform chain has a mass m and length □. It is placed on frictionless table with a length □/4 hanging over the edge. The chain begins to slide down. What is the velocity v with which the end slides away from the edge?

(1) $\sqrt{g\ell}$

(2) $\sqrt{\frac{15}{32}g\ell}$

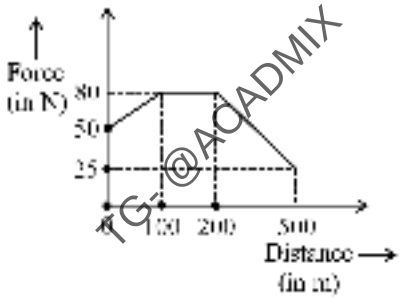
(3) $\sqrt{\frac{8}{16}g\ell}$

(4) $\sqrt{\frac{15}{16}g\ell}$

15) **Assertion (A) :** A spring does negative work only when it is compressed.
Reason (R) : The spring force always act in the opposite direction of deformation in the spring.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

16) A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle becomes



$(\alpha \times 10^3)$J after it has travelled first 200 m distance, then value of α is-

(1) 1.45
(2) 145
(3) 14.5
(4) 14500

17) A ball of mass 800 g is thrown vertically upwards with initial velocity 20 m/s. If due to air resistance and gravity ball reached maximum height of 16 m. The work done by air resistance is- (g = 10 m/s^2)

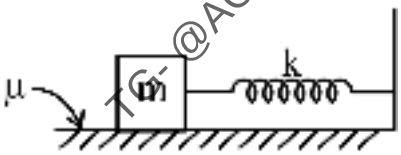
(1) 32 J
(2) 160 J
(3) – 32 J
(4) –160 J

18) A particle in a certain conservative force field has potential energy given by $U = 2\,xyz^{-1}$. The force exerted on it at position (1, 0, –1) is-

(1) $-\hat{i} + 2\hat{j}$
(2) $2\hat{j}$
(3) $-2\hat{i}$
(4) $-2\hat{k} + 2\hat{j}$

19) Find the maximum energy stored in the spring shown in the figure, for which the block remains

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stationary on a rough horizontal surface.

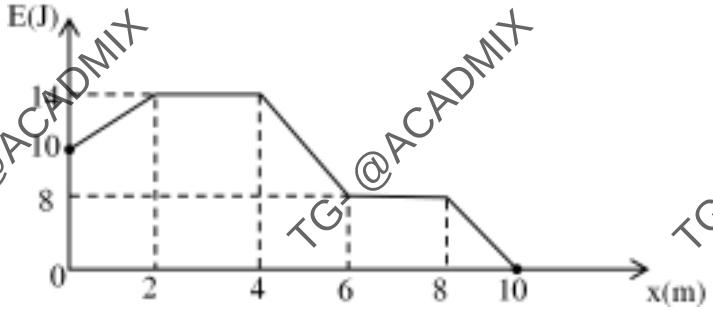
(1) $\frac{\mu mg}{k}$

(2) $\frac{(\mu mg)^2}{k}$

(3) $\frac{\mu (mg)^2}{k}$

(4) $\frac{(\mu mg)^2}{2k}$

20) The kinetic energy (E) of a particle moving along x-axis varies with position (x) as shown in figure. The magnitude of force acting on the particle at x = 5m.



(1) 3 N

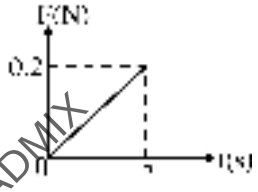
(2) 5 N

(3) 12 N

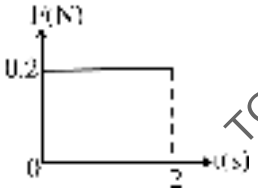
(4) 8 N

21) If kinetic energy of a particle of mass 10g moving along a straight line under the influence of a force varies with time as $K = (2t^2)$ J. The force acting on the particle varies with time as-

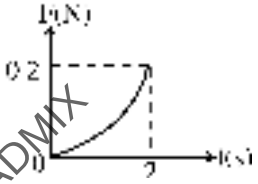
(1)

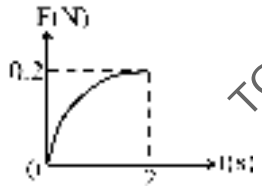


(2)



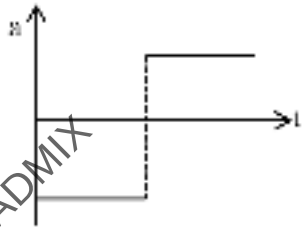
(3)

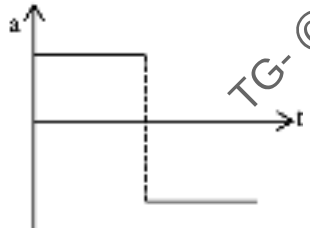


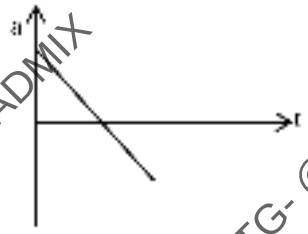
(4) 

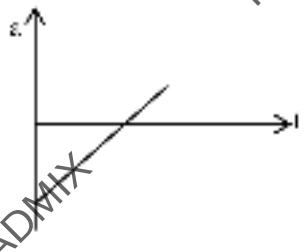
22) Choose correct graph of a-t for given v-t graph :

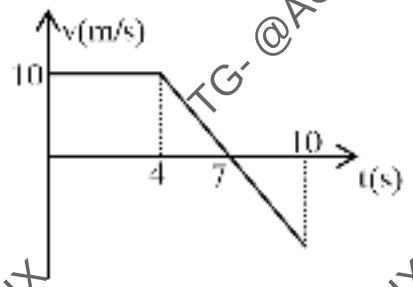


(1) 

(2) 

(3) 

(4) 



23) Find out distance covered by particle in 10s in given graph -

(1) 40 m

(2) 110 m

(3) 70 m

(4) 90 m

24) Speed of boat in still water is 5 km/h and river is flowing with speed 3 km/h. Find out time taken by boat to move 8 km in down stream and come back.

(1) 1 hr

(2) 4 hr

(3) 5 hr

(4) 2 hr

25) Car A is 100 m towards North of car B. Car A starts moving towards east with speed 10 m/s simultaneously car B starts to move towards north with same speed find out time when seperation between the cars become minimum.

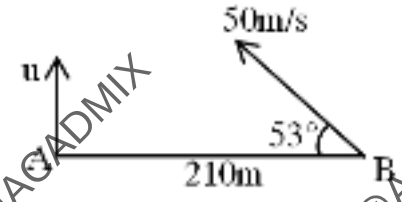
(1) 5 s

(2) $\frac{5}{\sqrt{2}}$s

(3) $5\sqrt{2}$s

(4) $\frac{5}{2}$s

26) Find out projection speed of particle A such that A can collide with particle B in given figure.



(1) 40 m/s

(2) 30 m/s

(3) 50 m/s

(4) 20 m/s

27) A partical is projected with speed 50 m/s at an angle 53° from vertical. Choose the correct option -

(A)	Max. height (m)	(P)	6
(B)	Range (m)	(Q)	120
(C)	Time of flight (s)	(R)	240
		(S)	45

(1) A-Q, B-R, C-S

(2) A-S, B-Q, C-R

(3) A-S, B-R, C-P

(4) A-R, B-S, C-Q

28) If equation of trajectory of projectile is $y = x - \frac{x^2}{80}$ than find out maximum height of this projectile.

(1) 20 m
(2) 40 m
(3) 60 m
(4) 80 m

29) If position of a particle with time is $x = 2t - t^2$ than find out distance covered by it in first 2 s.

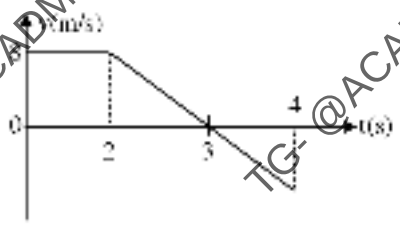
(1) Zero
(2) 1 m
(3) 2 m
(4) 4 m

30) A particle is moving on a circular path of radius 2m. Find out its displacement when it covers angle $\frac{2\pi}{3}$ radian.

(1) $2\sqrt{3}$m
(2) $\frac{2}{\sqrt{3}}$m
(3) 2m
(4) $\sqrt{3}$m

31) A particle has initial velocity $\left(8\hat{i} + 6\hat{j}\right)$ and has acceleration $\left(0.3\hat{i} + 0.4\hat{j}\right)$. It's speed after 10s is :-

(1) $7\sqrt{2}$ m/s
(2) $11\sqrt{2}$ m/s
(3) 15 m/s
(4) $\sqrt{221}$ m/s



32) Find out average speed of particle in 4s in given graph

(1) 6 m/s
(2) 7 m/s
(3) 8 m/s
(4) 10 m/s

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33) Car A is moving towards North with speed 5 m/s. It's driver observes that car B is moving towards East. If car B is moving towards 30° N of E. Find out velocity of car B with respect to car A.

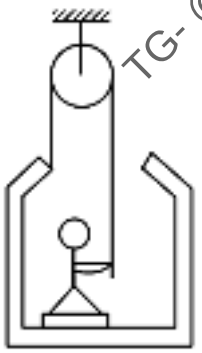
(1) 5 m/s

(2) $5\sqrt{3}$ m/s

(3) $\frac{5}{\sqrt{3}}$ m/s

(4) 10 m/s

34) A man of 60 kg is standing on a weighing machine in a cage of 30 kg in the given figure. If the



system is in equilibrium than find out reading of weighing machine :

(1) 100 N

(2) 150 N

(3) 300 N

(4) 600 N

35) Find out acceleration of block A if block B is moving downward with acceleration 'a' in given

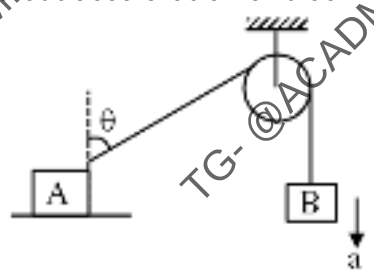


figure :

(1) a cosθ

(2) a sinθ

(3) a secθ

(4) a cosecθ

36) While walking on ice one should take small steps to avoid slipping. This is because small steps ensure –

(1) Large friction

(2) Smaller friction

(3) Larger normal force

(4) Smaller normal force

37) N bullets each of mass 'm' kg are fired with a velocity v ms^{-1} at the rate of 'n' bullets per second

upon a wall. Bullets get embedded into the wall. The reaction offered by the wall to the bullets is given by :

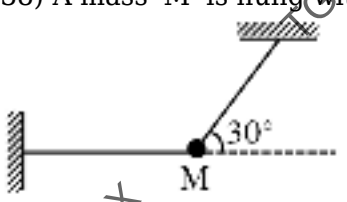
(1) nmv

(2) $\frac{Nmv}{n}$

(3) $n\frac{Nm}{v}$

(4) $n\frac{Nv}{m}$

38) A mass 'M' is hung with a light inextensible string. Tension in horizontal part of string is:



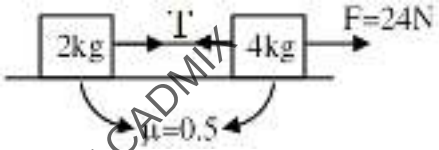
(1) $\sqrt{3}Mg$

(2) $\sqrt{2}Mg$

(3) $\frac{Mg}{\sqrt{3}}$

(4) $\frac{Mg}{2}$

39) Find out tension in the string in given figure :



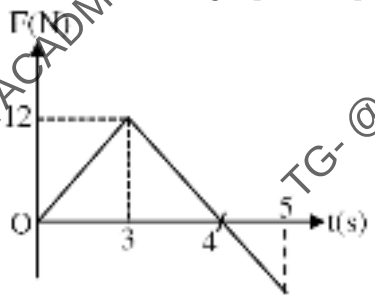
(1) 4 N

(2) 10 N

(3) 24 N

(4) 30 N

40) Force-time graph of a particle is given. Find out impulse in 5 seconds.



(1) 18 N-s

(2) 30 N-s

(3) 9 N-s

(4) 6 N-s

41) A balloon has 2 gm of air. A small hole is projected into it. The air comes out with a velocity of 4 m/s. If the balloon shrinks completely in 2.5 s, the average force acting on the balloon is :

(1) 0.008 N

(2) 0.0032 N

(3) 8 N

(4) 3.2 N

42) Find out tension in the string in given figure :



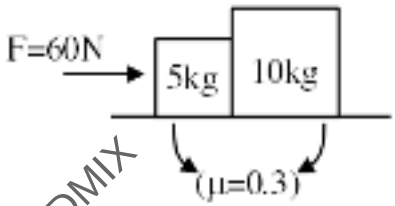
(1) Zero

(2) 80 N

(3) 50 N

(4) 30 N

43) Find out normal contact force between 5 kg and 10 kg in given figure :



(1) 20 N

(2) 30 N

(3) 40 N

(4) 60 N

44) A block of mass 2kg is placed on the floor. The coefficient of static friction is 0.4. If a force of 2.8 N is applied on the block parallel to the floor, the force of friction between the block and the floor (taking g = 10 m/s^2) is :

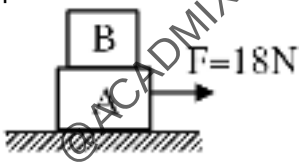
(1) 2.8 N

(2) 8 N

(3) 2 N

(4) Zero

45) A block B of mass 2 kg is placed on another block A of mass 4 kg which rests on a smooth horizontal surface. If μ = 0.5 between A and B and a force of 18N is applied on A, the friction between the blocks :



(1) 10 N

(2) 8 N

(3) 6 N

(4) 4 N

CHEMISTRY

1) What will be the concentration of acetate ion in a 1.0 M solution of acetic acid containing 0.1 M HCl [$K_a = 2 \times 10^{-5}$] :-

(1) 2×10^{-3} M

(2) 2×10^{-4} M

(3) 2×10^{-2} M

(4) 2×10^{-6} M

2) For which reaction will $K_p = K_c$?

(1) $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$

(2) $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

(3) $C(s) + 2H_2O(g) \rightleftharpoons 2H_2(g) + CO_2(g)$

(4) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

3) On doubling the volume of the vessel, which of the following equilibrium will be shifted to right ?

(1) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(2) $H_{2(g)} + Cl_{2(g)} \rightleftharpoons 2HCl_{(g)}$

(3) $2CO_{(g)} + O_{2(g)} \rightleftharpoons 2CO_{2(g)}$

(4) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

4) On adding inert gas to the equilibrium $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ at constant pressure, the degree of dissociation will :-

(1) Not change

(2) Decrease

(3) Increase

(4) None of these

5) The formula used to evaluate the pH of monobasic weak acid is :-

(1) $pH = C\alpha$

(2) $pH = 1/2\ pK_a - 1/2 \log C$

(3) $pH = \sqrt{K_a C}$

(4) $pH = pK_a + \log C$

6) What is the pH of a weak acid HA having 0.004 M concentration and $pK_a = 5$?

(1) 4.70
(2) 3.70
(3) 7.40
(4) 5

7) If pOH ≥ 5, then possible nature of aqueous Solution is :-

(1) acidic
(2) basic
(3) neutral
(4) any one is possible

8) Which of the following will make acidic buffer ?

(1) 50 mL of 0.1M NaOH + 25 mL of 0.1M CH_3COOH
(2) 100 mL of 0.1M HCl + 100 mL of 0.1M NaOH
(3) 100 mL of 0.1M HCl + 200 mL of 0.1M NH_4OH
(4) 25 mL of 0.1M NaOH + 50 mL of 0.1M CH_3COOH

9) 0.1M formic acid solution is titrated against 0.1M NaOH solution. What would be the difference in pH between 1/5 and 4/5 stages of neutralization of acid?

(1) 2log(3/4)
(2) 2log(1/5)
(3) log(1/3)
(4) 2log4

10) What is the pH at which 0.01 M Co^{+2} ions in solution begin to precipitate as $Co(OH)_2$? K_{sp} of $Co(OH)_2 = 25 \times 10^{-16}$:-

(1) 6.3
(2) 7.3
(3) 10.8
(4) 7.7

11) Sodium nitrate, heated in the presence of an excess of H_2, forms H_2O according to two step process:
$2NaNO_3 \rightarrow 2NaNO_2 + O_2$
$2H_2 + O_2 \rightarrow 2H_2O$
How many grams of $NaNO_3$ are required to form 9 g of H_2O ?

(1) 21.3
(2) 42.5
(3) 69.0

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(4) 85.0

12) When mixture of 6 L of methane and propane were burnt, 7 L CO_2 was obtained. Then volume of methane and propane in mixture will be :-

(1) 5.5 L and 0.5 L
(2) 6 L and 1 L
(3) 3 L and 4 L
(4) 5 L and 2 L

13) A 1000 gm sample of NaOH contains 3 moles of oxygen atoms. What is the % purity of NaOH sample :-

(1) 14%
(2) 100%
(3) 12%
(4) 24%

14) If 4 mole of Mn^{2+} will loose 16 mole of electrons, to form a single product then it may form:

(1) MnO_2
(2) MnO_4^-
(3) MnO_4^{2-}
(4) Mn_2O_5

15) Which substance is serving as a reducing agent in the following reaction ?
$14H^+ + Cr_2O_7^{2-} + 3Ni \rightarrow 2Cr^{3+} + 7H_2O + 3Ni^{2+}$

(1) H_2O
(2) Ni
(3) H^+
(4) $Cr_2O_7^{2-}$

16) How many mole of electrons are needed to convert one mole of nitrate ion to hydrazine (N_2H_4) ?

(1) 8
(2) 7
(3) 5
(4) 3

17) Reaction (A) $S^{-2} + 4\ H_2O_2 \rightarrow SO_4^{2-} + 4\ H_2O$
(B) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
The true statement regarding the above reactions is :

(1) H_2O_2 acts as reductant in both the reactions.
(2) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).

(3) H_2O_2 acts as an oxidant in both the reactions.

(4) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction(B)

18) In which of the following, entropy decreases ?

(1) Dissolving sucrose in water

(2) Sublimation of camphor

(3) Melting of ice

(4) Stretching of rubber band

19) The enthalpy change for transition of liquid water to steam is 3.73 kJ mol^{-1} at 373 K. Find ΔS in J/mol K :-

(1) 100 J/mol K

(2) 10 J/mol K

(3) 1000 J/mol K

(4) None of these

20) A gas is allowed to expand from a volume of 1.0 L to 10.0 L against a constant external pressure of 0.50 atm. If the gas absorbs 250 J of heat from the surroundings, what are the values of q, w and ΔE ?

	q	w	ΔE
(1)	250 J	–455 J	–205 J
(2)	–250 J	–455 J	–710 J
(3)	250 J	455 J	710 J
(4)	–250 J	455 J	205 J

(1) 1

(2) 2

(3) 3

(4) 4

21) In a cyclic process, the amount of heat given to a system is equal to

(1) Net work done by the system

(2) Net increase in internal energy

(3) Net decrease in internal energy

(4) Net change in specific heat

22) Enthalpies of formation of CO(g), CO_2(g), N_2O(g) and N_2O_4(g) are –110, –393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of ΔH for the reaction :
$N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$

(1) +778 kJ mol^{-1}

(2) –211.7 kJ mol^{-1}

(3) +211.7 kJ mol^{-1}

(4) –778 kJ mol^{-1}

23) For the hypothetical reaction

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

$\Delta_rG°$ and $\Delta_rS°$ are 20 kJ/mol and – 20 JK^{-1} mol^{-1} respectively at 200 K. If Δ_rC_p is 20 $JK^{-1}mol^{-1}$ then $\Delta_rH°$ at 400 K is :-

(1) 20 kJ/mol

(2) 7.98 kJ/mol

(3) 28 kJ/mol

(4) None of these

24) In order to decompose 9 gm water into H_2 and O_2, 142.5 kJ heat is required. Hence the enthalpy of formation of water is :-

(1) +285 kJ

(2) –285 kJ

(3) +142.5 kJ

(4) –142.5 kJ

25) The heat of combustion of ethylene $C_2H_4(g)$ is –1420 kJ/mole. The number of litres of C_2H_4 at NTP that would evolve 355 kJ on combustion is

(1) 2.8

(2) 8.4

(3) 5.6

(4) 11.2

26) 10 mol ammonium carbamate is heated in a closed container so that at equilibrium, 12 atm total pressure was obtained, what will be K_p for the following reaction ? $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

(1) 512

(2) 256

(3) 144

(4) 1

27) Which is correct statement regarding CrO_5

A : Oxidation number of Cr is +6.

B : Four oxygen-atoms of molecule are involved in peroxide linkage.

C : Only one oxygen-atom has O.N. = –2

(1) All are correct

(2) B, C are correct

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(3) A, C are correct
(4) Only B is correct

28) In the reaction ;
$As_2S_3 + HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$,
the element oxidised is/are :-

(1) As only
(2) S only
(3) N only
(4) As and S both

29) If M_4O_5 compound is written as $M_x(+1)\ M_y(+3)O_5$ then calculate the value of x & y:-

(1) 1, 5
(2) 2, 2
(3) 3, 1
(4) 1, 3

30) How many moles of hydrazine (N_2H_4) can be oxidized into N_2 using 2/3 mole of bromate ions ($BrO_3^- \rightarrow Br^-$) :-

(1) $\frac{1}{3}$
(2) 1
(3) 1.5
(4) $\frac{2}{3}$

31) Calculate the work done when 1 mol of an ideal gas is compressed reversibly from 1 bar to 4 bar at a constant temperature of 300 K -

(1) 3.45 kJ
(2) –8.02 kJ
(3) 18.02 kJ
(4) –14.01 kJ

32) The bond dissociation energies for Cl_2, I_2 and ICl are 242.3, 151 and 211.3 kJ/mol respectively. The enthalpy of sublimation of iodine is 62.8 kJ/mol. What is the standard enthalpy of formation of ICl(g) ?

(1) –211.3 kJ/mol
(2) –14.6 kJ/mol
(3) 16.8 kJ/mol
(4) 33.5 kJ/mol

33) The sublimation enthalpy of I_2(s) is 57.3 kJ/mol and the enthalpy of fusion is 15.5 kJ/mol. The

enthalpy of vaporisation of I_2 is-

(1) 41.8 kJ/mol
(2) –41.8 kJ/mol
(3) 72.8 kJ/mol
(4) –72.8 kJ/mol

34) What is the pH of a solution containing 25.0 mL each of 0.1 M NaOH and 0.08 M HCl and final solution is diluted upto 500 mL ?

(1) 3
(2) 11
(3) 12
(4) 13

35) $xA_{(s)} \rightleftharpoons yB_{(g)} + zC_{(g)}$

If $\frac{K_c}{K_p} = (RT)^{-2}$, then which is correct -

(1) y + z – x = –2
(2) y + z – x = 2
(3) y + z = –2
(4) y + z = 2

36) The number of oxygen atoms in 0.2 mole of sodium carbonate decahydrate is :-

(1) 1.56×10^{24}
(2) 1.56×10^{23}
(3) 1.56×10^{25}
(4) 3.12×10^{24}

37) How many moles of P_4 can be produced by reaction of 0.10 moles $Ca_5(PO_4)_3F$, 0.36 moles SiO_2 and 0.90 moles C according to the following reaction ?
$4Ca_5(PO_4)_3F + 18SiO_2 + 30C \rightarrow 3P_4 + 2CaF_2 + 18CaSiO_3 + 30CO$

(1) 0.060
(2) 0.030
(3) 0.045
(4) 0.075

38) Calculate $\Delta G°$ (kJ/mol) at 127°C for a reaction with $K_{equilibrium} = 10^5$

(1) –38.294
(2) –16.628
(3) –9.16
(4) None of these

39) When CO_2 dissolves in water, the following equilibrium is established
$CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$
for which the equilibrium constant is 3.8×10^{-7} and pH = 6.0. The ratio of $[HCO_3^-]$ to $[CO_2]$ would be :-

(1) 3.8×10^{-13}
(2) 3.8×10^{-1}
(3) 6.0
(4) 13.4

40) The bond energies of C – C, C = C, H – H and C – H bonds are 350, 600, 400 and 410 kJ per mole respectively. The heat of hydrogenation of ethylene (C_2H_4) is :-

(1) –170 kJ mol^{-1}
(2) –260 kJ mol^{-1}
(3) –400 kJ mol^{-1}
(4) –450 kJ mol^{-1}

41) 0.01 M HA(aq.) is 2% ionized, $[OH^-]$ of solution is :-

(1) 2×10^{-4}
(2) 10^{-8}
(3) 5×10^{-11}
(4) 5×10^{-12}

42) If 1 mole of H_3PO_x is completely neutralized by 80 gm of NaOH, select the correct statement :-

(1) x = 2 and acid is monobasic
(2) x = 3 and acid is dibasic
(3) x = 4 and acid is tribasic
(4) All are correct

43) What is the pH of a solution prepared by mixing 20 ml. of 0.250 M sodium formate with 30 mL of 0.100 M HCl
(pK_a for HCOOH = 3.75) :-

(1) 3.57
(2) 4.18
(3) 5.22
(4) None

44) Consider the following data for series of hydrogen halide Bronsted acids :-

Acid	HF	HCl	HBr	HI
K_a	7.2×10^{-4}	1×10^{6}	1×10^{9}	3×10^{11}

Which of these Bronsted acids would have the weakest conjugate base ?

(1) HF

(2) HCl

(3) HBr

(4) HI

45) Heat of combustion of carbon in diamond and amorphous form are –94.3 and –97.6 kcal/mole. The heat required to convert 18 gm carbon from diamond to amorphous form is:-

(1) –4.95 kcal

(2) + 4.95 kcal

(3) –3.3 kcal

(4) +3.3 kcal

BIOLOGY

1) Two organisms belong to same class but belong to different families, they may be kept in same :-

(1) Genus

(2) Species

(3) Order

(4) Variety

2) Which of the following is not a correct statement?

(1) The number of species that are known and described range between 1.7 to 1.8 billion.

(2) The number and types of organisms present on earth referred as biodiversity.

(3) As we explore new areas, and even old ones, new organisms are continuously being identified.

(4) Each different kind of plant, animal or organism that you see, represent a species.

3) **Statement-I :** All living organisms can be classified into different taxa.
Statement-II : The word systematics is derived from the Latin word 'Systema'.

(1) Both statement-I and statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II are incorrect.

4) **Statement-I** : Each scientific name has two components.
Statement-II : The first word in a biological name represents the specific epithet while second word denotes the genus.

(1) Both statement I and II are correct

(2) Both statement I and II are incorrect

(3) Statement I is correct but statement II is incorrect.

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(4) Statement I is incorrect but statement II is correct

5) **Assertion (A) :** In natural classification systems, more weightage will be given to reproductive/sexual characters.
Reason (R) : Vegetative characters are more easily affected by environment as compared to sexual characters.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is correct but **Reason** is incorrect.
(4) **Assertion** is incorrect but **Reason** is correct.

6) Which of the following statements are true?
(a) Classification given by Bentham and Hooker was of non-flowering plants, algae and bryophytes.
(b) Numbers and codes are assigned to floral characters only, in numerical taxonomy.
(c) Phylogenetic classification systems are based on evolutionary relationships between various organisms.
(d) The earliest systems of scientific classification used only gross superficial morphological characters.

(1) a and b only
(2) c and d only
(3) a, c and d only
(4) a, b and c only

7) **Assertion (A)** - The problem of classification becomes more complex at higher taxonomic categories.
Reason (R) - As we go from species to Kingdom, the number of common characteristics goes on decreasing.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)
(3) (A) is correct but (R) is incorrect
(4) (A) is incorrect but (R) is correct

8) Given organisms belongs to how many families ?
Wheat, Mango, Brinjal, Potato, Datura, Lion, Cat, Dog

(1) Seven
(2) Six
(3) Five
(4) Four

9) Find out the family and order of *Triticum aestivum* :-

	Family	**Order**

1	Anacardiaceae	Poales
2	Poaceae	Sapindales
3	Anacardiaceae	Sapindales
4	Poaceae	Poales

(1) 1
(2) 2
(3) 3
(4) 4

10) Monkey, Gorilla, Tiger and Cat belongs to different order but based on common features they all belongs to the same :

(1) Species
(2) Genus
(3) Family
(4) Class

11) Study the following statements :
I. Growth and Reproduction are not the defining character of living organism while metabolism, cellular organisation and consciousness are defining characters.
II. In multicellular organisms, reproduction refer to the production of progeny possessing features more or less similar to those of parents.
III. All higher animals have self consciousness.
IV. All organisms from the prokaryotes to the most complex eukaryotes can sense and respond to environmental cues.
Choose the correct option :-

(1) Only I, III, IV are correct
(2) Only I, II, IV are correct
(3) All are correct
(4) Only II and IV are correct

12) Read the following statements :
A. He was the earliest to attempt a more scientific basis for classification.
B. He use simple morphological characters to classify plants into trees, shrubs and herbs.
C. He also divided animals into two groups, those which had red blood and those that did not.
Above statements are related with -

(1) Aristotle
(2) Linnaeus
(3) Theophrastus
(4) R.H. Whittaker

13) Choose the incorrect statement with respect to three-domain system ?

(1) Carl Woese placed prokaryotes in two domains

(2) Carl Woese placed prokaryotes in two kingdoms
(3) Carl Woese placed eukaryotes in one domain
(4) Carl Woese placed eukaryotes in two kingdom

14) Which one was not the criteria used by R.H. Whittaker in five kingdom classification ?

(1) Thallus organisation
(2) Cell structure
(3) Growth habit
(4) Mode of nutrition

15) Who placed *Chlamydomonas* and *Chlorella* in kingdom Protista ?

(1) Aristotle
(2) Whittaker
(3) Linnaeus
(4) Bentham and Hooker

16) Sole members of kingdom Monera are -

(1) Bacteria
(2) Diatoms
(3) Desmids
(4) Lichens

17) Which one of the following is not the character of bacterial cell ?

(1) Mesosomes
(2) Genomic DNA
(3) 70 s ribosomes
(4) Nuclear membrane

18) A special membranous structure formed by extension or infoldings or invagination of plasma membrane in to the cell is called as -

(1) Mesosomes
(2) Ribosome
(3) Storage granules
(4) Nucleoid

19) Which of the following structure help in DNA replication in bacterial cell ?

(1) Lipopolysaccharide wall
(2) Peptidoglycan wall
(3) Mesosome
(4) Ribosome

20) Chromatophores are present in :-

(1) *Albugo*
(2) Slime moulds
(3) *Trypanosoma*
(4) *Nostoc*

21) Match the column-I and II and select correct option :

Column-I		**Column-II**	
(I)	Halophiles	(p)	Hot springs
(II)	Thermoacidophiles	(q)	Marshy areas
(III)	Methanogens	(r)	Salty areas

(1) I-p, II-q, III-r
(2) I-q, II-p, III-r
(3) I-r, II-p, III-q
(4) I-r, II-q, III-p

22) Select the correct statement from the following regarding protists :-

(1) Members of protista are primarily terrestrial.
(2) All are multicellular eukaryotes
(3) Have well-defined nucleus
(4) No zygote formation.

23) Which of the following statements is correct for flagella of dinoflagellates ?

(1) Only single flagellum lies in the transverse groove between the wall plates
(2) Only single flagellum lies in the longitudinal groove between the wall plates
(3) Two flagella, one lies longitudinally and the other transversely in a furrow between the wall plates.
(4) Flagella are absent

24) 'Diatomaceous earth' is used in :-

(1) Polishing
(2) Filtration of oils
(3) Filtration of syrups
(4) All of the above

25) Which protista are photosynthetic in presence of sunlight but behave like heterotroph when deprived of sunlight by predating on other smaller organisms :-

(1) Chrysophytes
(2) Dinoflagellates

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(3) Euglenoids
(4) Slime moulds

26) Slime moulds are-

(1) Photosynthetic protists
(2) Saprophytic protists
(3) Parasite protists
(4) Both (1) and (2)

27) Vegetative reproduction in fungi can take place by which process :-

(1) Fragmentation
(2) Fission
(3) Budding
(4) All of the above

28) (A) Members of phycomycetes are found in aquatic habitat.
(B) Members of phycomycetes are obligate parasite on plants.
(C) The mycelium of phycomycetes is septate.
(D) In phycomycetes zoospores are produced exogenously.
(E) In phycomycetes, Asexual reproduction take place by non motile aplanospore.
Select the set of correct statements :

(1) A, B and C
(2) B, C and D
(3) C, D and E
(4) A, B and E

29)

Which of the following ascomycetes is the source of antibiotics ?

(1) *Neurospora*
(2) *Penicillium*
(3) *Claviceps*
(4) *Puccinia*

30) Match the following and choose correct option.

(A) Zoospore	(i) Endogenous
(B) Conidia	(ii) Exogenous
(C) Aplanospore	
(D) Ascospore	

(1) (A)-(i), (B)-(ii), (c)-(i), (D)-(i)

(2) (A)-(i), (B)-(ii), (c)-(i), (D)-(ii)
(3) (A)-(i), (B)-(i), (c)-(ii), (D)-(i)
(4) (A)-(ii), (B)-(i), (c)-(ii), (D)-(ii)

31) Which of the following disease is caused by viruses ?

(1) Small pox
(2) Potato spindle tuber disease
(3) Mad cow disease in cattle
(4) Cr-Jakob disease

32) Chlorophyll-bearing, simple, thalloid autotrophic and eukaryotic organisms are belong to:-

(1) Fungi
(2) Algae
(3) Bacteria
(4) Gymnosperm

33) How many of the followings are the members of chlorophyceae :-
Volvox, Fucus, Ectocarpus, Chara, Sargassum, Ulothrix, Gracilaria, Spirogyra
Choose the correct option :-

(1) Five
(2) Three
(3) Seven
(4) Four

34) Match column-I with column-II and select the correct option

Column-I		**Column-II**	
(A)	*Fucus*	I.	Homosporous pteridophyta
(B)	*Pteris*	II.	Brown algae
(C)	*Ginkgo*	III.	Heterosporous pteridophyta
(D)	*Salvinia*	IV.	Naked seed

(1) (A) IV, (B) III, (C) II, (D) I
(2) (A) II, (B) III, (C) I, (D) IV
(3) (A) II, (B) IV, (C) I, (D) III
(4) (A) II, (B) I, (C) IV, (D) III

35) How many of the followings are the members of Rhodophyceae :-
Polysiphonia, *Porphyra*, *Fucus*, *Gracilaria*, *Gelidium*, *Laminaria*, *Dictyota*.
Choose the correct option:-

(1) Three

(2) Two

(3) Six

(4) Four

36) Match the following and select correct option.

(A) Buds of protonema	(i) Haploid
(B) Antheridia	(ii) Diploid
(C) Archegonia	
(D) Spore mother cell	

(1) (A)-(i), (B)-(i), (C)-(i), (D)-(ii)

(2) (A)-(i), (B)-(ii), (C)-(ii), (D)-(ii)

(3) (A)-(ii), (B)-(i), (C)-(i), (D)-(i)

(4) (A)-(ii), (B)-(ii), (C)-(ii), (D)-(i)

37)

Match List-I with List-II

	List-I		**List-II**
A	Gemma cup	i	*Sphagnum*
B	Peat Moss	ii	*Funaria* and *Polytrichum*
C	*Mosses*	iii	*Marchantia*
D	Capsule	iv	Part of sporophyte in bryophytes

Choose the correct answer from the option given below :

	A	**B**	**C**	**D**
1	iii	i	ii	iv
2	ii	iii	iv	i
3	ii	iii	i	iv
4	iii	iv	ii	i

(1) 1

(2) 2

(3) 3

(4) 4

38) Members of Pteridophyta -

A. They produce spores by mitosis in spore mother cells.

B. They form spores which germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

C. They perform fusion of male gamete with the megaspore present in the archegonium result in the formation of zygote.
Choose the correct option :-

(1) Only A is incorrect.
(2) Only B is incorrect.
(3) Only C is incorrect.
(4) Both A and C are incorrect.

39) Which of the following statement is incorrect ?

(1) In Sphenopsida class, *equisetum* is included
(2) Pteridophytes are used for medicinal purposes and as soil binders
(3) Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous
(4) *Salvinia* belongs to class lycopsida in pteridophytes

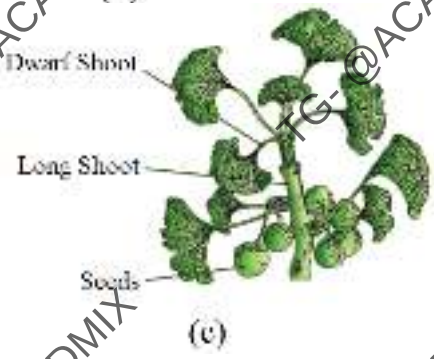
40) Identify the diagrams (a), (b) and (c) :-



(a)



(b)



(c)

(1) (a) *Cycas*, (b) *Pinus*, (c) *Ginkgo*
(2) (a) *Pinus*, (b) *Cycas*, (c) *Ginkgo*

(3) (a) *Ginkgo*, (b) *Cycas*, (c) *Pinus*

(4) (a) *Cycas*, (b) *Ginkgo*, (c) *Pinus*

41) If a root cell of *Cycas* plant contain 24 chromosome then what will be the number of chromosome in female gametophyte cell ?

(1) 24

(2) 36

(3) 12

(4) 48

42) Higher gymnosperms have following character except:

(1) Heterosporous nature

(2) Wind-pollination

(3) Nucellus is protected by envelopes

(4) Gametophyte is monoecious

43) Sporophytic plant body of *Pinus* have :-

(1) Male and female cone found on same plant body.

(2) Unbranched stem.

(3) Coralloid roots

(4) All of the above

44) Match list-I and list-II :-

	List-I		**List-II**
(A)	Pteropsida	(i)	*Lycopodium*
(B)	Lycopsida	(ii)	*Psilotum*
(C)	Sphenopsida	(iii)	*Pteris*
(D)	Psilopsida	(iv)	*Equisetum*

Choose the correct answer from the option given below :-

(1) (A)-(iii), (B)-(i), (C)-(iv), (D)-(ii)

(2) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(3) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)

(4) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

45) Gymnosperms are:

(1) Covered seeded vascular embryophytes

(2) Naked seeded non-vascular embryophytes

(3) Naked seeded vascular non-embryophytes

(4) Naked seeded vascular embryophytes

46)

Tube within tube body plan is absent in :

(1) *Taenia*
(2) *Ascaris*
(3) Earthworms
(4) Cockroach

47) Which one is correct about Porifera :-

(1) Unisexual
(2) Internal fertilisation with indirect development
(3) External fertilisation with direct development
(4) External fertilisation with indirect development

48) Which of the following is exclusively marine :-

(1) Sea walnut
(2) *Amphioxus*
(3) *Ascidia*
(4) All above

49) A vertebrate which doesn't possess jaws :

(1) *Balanoglossus*
(2) *Petromyzon*
(3) *Salamandra*
(4) *Scoliodon*

50) Select the correct option :

	Animal	**Common name**
(a)	*Gorgonia*	(1) Bath sponge
(b)	*Ctenoplana*	(2) Sea fan
(c)	*Taenia*	(3) Comb jelly
(d)	*Euspongia*	(4) Tapeworm

(1) a-3, b-2, c-4, d-1
(2) a-1, b-3, c-4, d-2
(3) a-2, b-3, c-1, d-4
(4) a-2, b-3, c-4, d-1

51) Metagenesis means :-

(1) Polyp produce medusae asexually and medusae produce polyp sexually
(2) Polyp produce medusae sexually and medusae produce polyp sexually
(3) Polyp produce medusae asexually and medusae produce polyp asexually.
(4) Both (1) & (2)

52) The animal group where the adults are less developed but larvae are more developed is :-

(1) Agnatha
(2) Tunicate
(3) Amphibian
(4) Cephalochordates

53) Animals are classified on the basis of :-

(1) Arrangement of cells
(2) Body symmetry
(3) Nature of coelom
(4) All of these

54) In the given examples, how many animals are viviparous ? *Corvus, Columba, Macropus, Pteropus, Calotes, Hemidactylus, Psittacula, Struthio* and *Macaca*

(1) Three
(2) Four
(3) Five
(4) Six

55) Which of the following are incorrect for platyhelminthes ?

(1) All are endoparasite
(2) All are bilateral & Triploblastic
(3) Hooks and suckers are present in the parasitic forms
(4) Both 1 & 2

56) Cyclostomes do not have :-

(1) Scales
(2) Caudal fins
(3) Unpaired fins
(4) Notochord

57) Read the given statements and select the correct option.
Statement-1 : All triploblastic animals are eucoelomates.
Statement-2 : Diploblastic animals have a false coelom.

(1) Both statements 1 and 2 are correct.

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(2) Statements 1 is not correct but statement 2 is correct.

(3) Statement 1 is correct and statement 2 is not correct.

(4) Both statements 1 and 2 are incorrect.

58)

How many of the following animals shows organ system level of body organisation along with bilateral symmetry ?

Pleurobrachia, *Euspongia*, *Ancylostoma*, *Hirudinaria*, *Anopheles*, *Asterias*, *Balanoglossus*

(1) Four

(2) Five

(3) Two

(4) Three

59) Ciliated epithelial cells are known to occur in :-

(1) Eustachian tube and stomach

(2) Fallopian tube and bronchioles

(3) Fallopian tube and urethra

(4) Trachea and oesophagus.

60) Cartilage is formed by :-

(1) Osteoblasts

(2) Fibroblasts

(3) Chondrocytes

(4) Osteocytes

61) How many statements are **true** in the following?

(i) Simple epithelium forms lining for body cavities, ducts and tubes.

(ii) In epithelial tissue, cells are loosely packed with little intercellular matrix.

(iii) Simple epithelium provide protection against chemical and mechanical stresses.

(iv) Gap junction perform cementing to keep neighbouring cells together.

(1) Two

(2) Three

(3) Four

(4) One

62) Which of the following connective tissue connects skeletal muscles to bones in form of tendon :-

(1) Areolar connective tissue

(2) Adipose tissue

(3) Dense regular connective tissue

(4) Dense irregular connective tissue

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63) PCT of nephron in kidney has :-

(1) Simple squamous epithelium with microvilli
(2) Simple cuboidal epithelium with microvilli
(3) Simple columnar epithelium with microvilli
(4) Simple epithelium with cilia and microvilli

64) What is the correct function of "tight junction" ?

(1) Facilitate the cells to communicate with each other.
(2) To stop substances from leaking across a tissue.
(3) To perform cementing to keep neighbouring cells together.
(4) To allows transport of substances in unidirection.

65) 

The given figure represent :-

(1) Adipose tissue
(2) Areolar tissue
(3) Cartilage
(4) Ligament

66) Which type of cells of epithelium get specialised for secretion and are called glandular epithelium :-
(a) Squamous (b) Cuboidal
(c) Columnar

(1) Only a
(2) Only b
(3) Only c
(4) b and c

67) Function of compound epithelium is :-

(1) Mainly Absorption
(2) Only Secretion
(3) Only Excretion
(4) Mainly Covering & Protection

68) Highest regeneration power is present in :

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(1) Epithelial tissue

(2) Connective tissue

(3) Muscular tissue

(4) Nervous tissue

69) For female cockroaches, which of the following statements are **correct**?

(a) The 7th sternum is boat shaped

(b) The 7th sternum together with the 8th and 9th sterna forms a brood or genital pouch

(c) The anterior part of genital pouch contains female gonopore, spermathecal pores and collateral glands

(d) Only right collateral gland form ootheca

(1) a and b

(2) b and d

(3) a, b, c

(4) a and d

70) Another name of Labrum is ?

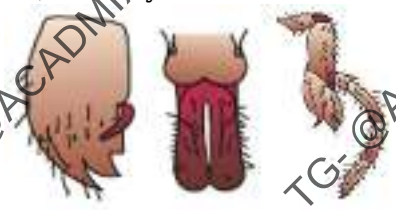
(1) Upper lip

(2) Lower lip

(3) Upper jaw

(4) Lower jaw

71) Identify the structures from figures A, B, C which is not correctly matched with their function.



(1) A – Mandible → Help in grinding.

(2) B – Labrum → Upper lip of cockroach.

(3) C – Maxilla → Form lateral wall of pre oral cavity.

(4) B – Hypopharynx → Tongue

72) Which of the following is not a part of cockroach body associated with excretion ?

(1) Malpighian tubules

(2) Nephrocytes

(3) Uricose gland of male reproductive system

(4) Collateral glands of female reproductive system

73) Fore wings of cockroach are :-

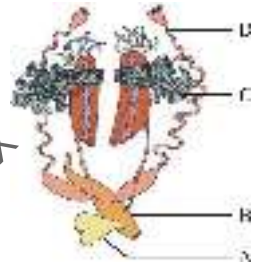
(1) meta thoracic wings known as Tegmina

(2) Pro thoracic wings known as Tegmina
(3) Meso thoracic wings knows as Tergites
(4) Meso thoracic wings known as Tegmina

74) In male cockroaches, sperms are stored in which part of the reproductive system ?

(1) Testes
(2) Vas deferens
(3) Seminal vesicles
(4) Mushroom gland

75) Given below is a diagramatic representation of female reproductive system of frog labelled A, B, C and D :



Find out incorrect one :-

(1) There is functional connection between C and kidney.
(2) B is small, median chamber that is used to pass faecal matter, urine and ova to the exterior.
(3) D is one pair arising from the ovaries opens into the cloaca separately.
(4) A is thin walled urinary bladder is present ventral to rectum which also opens in cloaea.

76) In frogs, testes are attached to kidneys by a double fold of peritoneum known as :-

(1) Urinary bladder
(2) Mesovarium
(3) Urinogenital duct
(4) Mesorchium

77) Sinus venosus in the heart of frog joins the

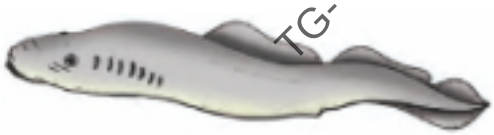
(1) Left atrium
(2) Ventricle
(3) Truncus arteriosus
(4) Right atrium

78) Which of the following is **incorrect** w.r.t Amphibians ?

(1) Skull is dicondylic
(2) They can respire through skin only
(3) Some amphibians are found in sea
(4) Both (2) and (3)

79)

Which one is correct feature for given animal specimen :



(1) 4 pair of gill slits
(2) Sucking and circular mount with Jaw
(3) Marine and show migration
(4) All of these

80) Limbless amphibian is :

(1) Hyla
(2) Ichthyophis
(3) Salamandra
(4) Bufo

81) Select the incorrect statement for chordates :

(1) They posses a post anal tail and a closed circulatory system.
(2) Protochordata subphylum include class urochordata and cephalochordata.
(3) Heart is ventral.
(4) Single dorsal hollow nerve cord present.

82) Match the column I & II and select the correct option :

Column-I		**Column-II**	
(i)	Myxine	(a)	Mammalia
(ii)	Aptenodyte	(b)	Cyclostomata
(iii)	Felis	(c)	Urochordata
(iv)	Salpa	(d)	Aves

(1) (i)-a, (ii)-b, (iii)-c, (iv)-d
(2) (i)-b, (ii)-d, (iii)-a, (iv)-c
(3) (i)-d, (ii)-c, (iii)-a, (iv)-b
(4) (i)-d, (ii)-c, (iii)-b, (iv)-a

83) **Assertion (A) :** Main functions of simple cuboidal epithelium are secretion and absorption.
Reason (R) : The function of ciliated epithelium is to move particles in a specific direction.

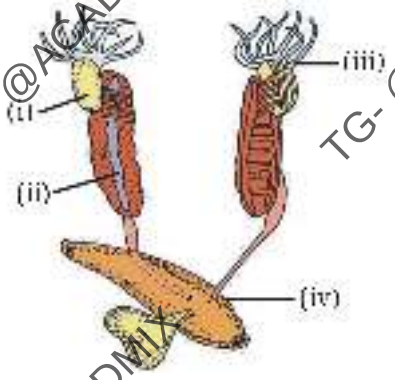
(1) Both (A) and (R) is correct and (R) is correct explanation of (A).
(2) Both (A) and (R) is correct but (R) is not correct explanation of (A).
(3) Both (A) and (R) incorrect
(4) (A) is correct but (R) is incorrect.

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84) Most of the exchange surfaces of multicellular animals are lined with :

(1) Epithelial tissue
(2) Connective tissue
(3) Neural tissue
(4) Smooth muscle cells

85) Given below is the diagram of reproductive system of an animal labelled by (i), (ii), (iii) and (iv),



identify the animal with correct labelling of (i) to (iv)

(1) Male frog-(i) Testes (ii) Adrenal gland (iii) Fat bodies (iv) Cloaca
(2) Female frog-(i) Kidney (ii) Adrenal gland (iii) Ovary (iv) Cloaca
(3) Male frog-(i) Testes (ii) Bidder canal (iii) Adrenal gland (iv) Cloaca
(4) Female frog-(i) Ovary (ii) Oviduct (iii) Adrenal gland (iv) Cloaca

86) Vasa-efferentia in male frog open into :

(1) Vas deferens
(2) Rete testis
(3) Cloaca
(4) Bidder's canal

87) Frog can protect it self from enemies by ?

(1) Teeth
(2) Camouflage
(3) Clawed digit
(4) Both (1) and (2)

88) In frog ureter act as urinogenital duct in :-

(1) Only in male
(2) Only in female
(3) Both male & female
(4) It is absent in frog

89) The number of ganglia in thorax and abdomen of cockroach are respectively.

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(1) Three and Six
(2) Six and Three
(3) Two and Eight
(4) Three and Seven

90) Arthrodial membrane (Articular membrane) connects the :

(1) Legs of cockroach
(2) Antennal segments of cockroach
(3) Sclerites of cockroach
(4) Head to thorax of cockroach

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	2	1	2	4	3	2	2	3	2	3	1	4	4	3	3	2	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	3	3	1	1	3	1	3	1	4	1	2	2	4	2	1	1	1	1
Q.	41	42	43	44	45															
A.	2	3	3	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	1	3	2	2	4	4	4	4	2	1	3	3	2	2	2	4	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	1	2	3	2	1	4	4	2	1	3	1	2	4	1	1	1	2	1
Q.	86	87	88	89	90															
A.	3	2	1	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	1	3	1	2	1	3	4	4	2	1	4	3	2	1	4	1	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	3	4	3	2	4	4	2	1	1	2	4	4	4	1	1	4	4	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	1	1	4	1	2	4	2	4	1	2	4	1	1	1	4	1	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	2	2	4	4	4	1	3	1	2	4	4	3	1	4	4	4	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	2	1	1	4	2	1	1	3										

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SOLUTIONS

PHYSICS

2) For sum & difference, dimensions of physical quantity must be same.

$\frac{d^2x}{dt^2}$ has dimensions $[LT^{-2}]$

$x\frac{dv}{dt}$ has dimensions $[L^2T^{-2}]$

3) $y = A \sin(e^{\alpha x} + \beta)$

Here dimensional formula for $A = [y] = [L]$

For $[\alpha x] = [M^0L^0T^0] \Rightarrow [\alpha] = [L^{-1}]$

$[\beta] = [M^0L^0T^0]$

$$\frac{A}{\alpha\beta} = \frac{[L]}{[L^{-1}][M^0L^0T^0]}$$

⇒ Dimensional formula for $= [M^0L^2T^0]$

4) $F = \rho^a A^b v^c$

$[MLT^{-2}] = [ML^{-3}]^a [L^2]^b [LT^{-1}]^c$

$[MLT^{-2}] = [M^a L^{-3a + 2b + c} T^{-c}]$

⇒ $a = 1$, $c = 2$, $-3a + 2b + c = 1$

⇒ $b = 1$

a, b, c = (1, 1, 2)

5) $a_t = m^a v^b t^c$

⇒ $[LT^{-2}] = [M]^a [LT^{-1}]^b [T]^c$

$[LT^{-2}] = [M^a L^b T^{-b + c}]$

⇒ $a = 0$, $b = 1$, $-b + c = -2$

$c = -1$

⇒ $a^2 + bc = -1$

6) $R_{eq} = R = R_1 + R_2 = 1200\ \Omega$

$$\frac{\Delta R_1}{R_1} \times 100 = 5\%$$

$$\Rightarrow \Delta R_1 = \frac{5 \times 1000}{100} = 50\ \Omega$$

$$\frac{\Delta R_2}{R_2} \times 100 = 5\%$$

$$\Rightarrow \Delta R_2 = \frac{5 \times 200}{100} = 10\ \Omega$$

$\Delta R = \Delta R_1 + \Delta R_2 = 50 + 10 = 60\Omega$

$$\Rightarrow \frac{\Delta R}{R} \times 100 = \frac{60}{1200} \times 100 = 5\%$$

7) $x = \frac{A^2B^{1/3}}{\sqrt{CD}}$

$\Rightarrow$ % error in x = $2(2) + \frac{1}{3}(3) + \frac{1}{2}(4) + \frac{1}{2}(6)$

= 10%

8) $g = \frac{4\pi^2\ell}{T^2}$

$\Rightarrow \frac{\Delta g}{g} \times 100$

$= \left(\frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}\right)100$

$= \left(\frac{0.1}{40.0} + 2\left(\frac{1}{400}\right)\right) \times 100$

$= \left(\frac{1}{4} + \frac{1}{2}\right)$ % = 0.75 %

9) Here → r = (5.00 ± 0.05) cm

$\Rightarrow \frac{\Delta r}{r} \times 100 = \frac{0.05}{5.00} \times 100 = 1\%$

Surface area = S = $4\pi r^2$ and volume = $\frac{4}{3}\pi r^3$

$\Rightarrow \frac{\Delta s}{s} \times 100 = 2\left(\frac{\Delta r}{r} \times 100\right) = 2\% = y$

$\Rightarrow \frac{\Delta V}{V} \times 100 = 3\left(\frac{\Delta r}{r} \times 100\right) = 3\% = x$

$\Rightarrow |x| + |y| = 5$

10) Power of motor = Total energy drawn per sec from motor

$$P = \frac{mgh}{t} + \frac{\frac{1}{2}mv^2}{t}$$

$$P = \frac{(10^3 \times 30 \times 10^{-3})(10)(10)}{60} + \frac{1}{2}\frac{(10^3 \times 30 \times 10^{-3})}{60} \times (2\sqrt{15})^2$$

P = 65 watt

11) By WET :-

Workdone by force = Change in KE

$$\int_{x_1}^{x_2} F dx = KE_{final} - KE_{initial}$$

$$\int_0^1 (x^2 + x)dx = KE_F - 0$$

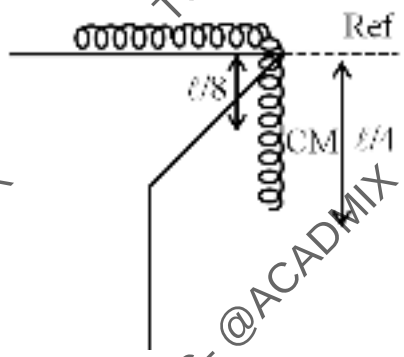
$$\Rightarrow \left(\frac{x^3}{3} + \frac{x^2}{2}\right)_0^1 = KE_{Final}$$

$\therefore \quad KE_{Final} = \left(\frac{1^3}{3} + \frac{1^2}{2}\right) - (0)$

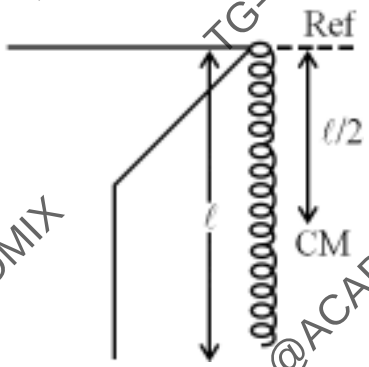
$\therefore \quad KE_{max} \Rightarrow \frac{5}{6} J$

= 0.83 J

12) Linear momentum is a vector quantity while kinetic energy is a scalar quantity. This means that even if the linear momentum of a system is zero, its KE may not be zero. How ever if the KE of a system is zero, then its linear momentum must also be zero. This is because for the total KE to be zero, the velocity of each particle must be zero.



14)



Gain in KE = Loss in PE

$\frac{1}{2} mv^2 - 0 = \frac{-mg}{4}\left(\frac{\ell}{8}\right) - \left(-mg\left(\frac{\ell}{2}\right)\right)$

$\frac{1}{2}mv^2 \Rightarrow mg\left(\frac{\ell}{2} - \frac{\ell}{32}\right)$

$\frac{1}{2}mv^2 = \frac{15}{32}mg\ell$

$v = \sqrt{\frac{15}{16}g\ell}$

15) Spring force always act in the opposite direction of deformation, so work done by spring force is always negative whenever it is elongated or compressed from its natural length.

16)

$w = \Delta KE = \frac{1}{2}mv^2 - 0$

$\frac{1}{2}(50 + 80) \times 100 + (80 \times 100) = KE$

$\therefore$ KE = 14500 J
$\Rightarrow 14.5 \times 10^3$ J

17) $w_g + w_{air} = KE_f - KE_i$

$-mgh + w_{air} = 0 - \frac{1}{2}mv_1^2$

$-0.8 \times 10 \times 16 + w_{air} = -\frac{1}{2} \times 0.8 \times (20)^2$

$w_{air} = -160 + 128 = -32J$

18) $\vec{F} = \frac{-\partial U}{dx}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$

$\frac{\partial U}{\partial x} = \frac{2y}{z}, \frac{\partial U}{\partial y} = \frac{2x}{z}, \frac{\partial U}{\partial z} = \frac{-2xy}{z^2}$

$\vec{F} \Rightarrow -\frac{2y}{z}\hat{i} - \frac{2x}{z}\hat{j} + \frac{2xy}{z^2}\hat{k}$

at (1, 0, –1)

$\vec{F} = 2\hat{j}$

19) $f_{max} = \mu mg$

$\therefore kx_{max} = \mu mg$

$x_{max} = \frac{\mu mg}{k}$

$\therefore$ Maximum potential energy stored in the spring is :-

$U_{max} = \frac{1}{2}kx_{max}^2$

$= \frac{1}{2}k\left(\frac{\mu mg}{k}\right)^2$

$= \frac{(\mu mg)^2}{2k}$

20)

Slope of kinetic energy (E) and position (x) curve gives force.

$Fdx = dK$

$F = \frac{dk}{dx}$

at x = 5m $\rightarrow$ Magnitude of slope of curve = $\frac{14-8}{6-4} = \frac{6}{2} = 3$

21) m = 10 g = 10^{-2} kg

Linear Momentum (p) = $\sqrt{2mK} = \sqrt{2(10^{-2})(2t^2)} = 0.2t$

$\therefore$ force f = $\frac{dp}{dt}$ = 0.2 (constant)

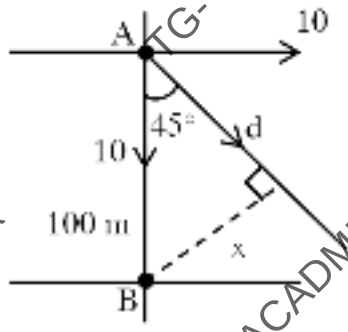
23)

Distance = Area under speed - time graph
= 40 + 15 + 15
= 70 m

24) $\rightarrow V_b = 5 + 3 = 8$ km/h $\quad t_1 = \frac{8}{8} = 1hr$

$\leftarrow V_b = 5 - 3 = 2$ km/h $\quad t_2 = \frac{8}{2} = 4hr$

$t = t_1 + t_2 = 5hr$



25)

$\frac{d}{100} = \cos 45°$ $\quad t = \frac{x}{V_{AB}}$

$x = 50\sqrt{2}$ m $\quad t = \frac{50\sqrt{2}}{10\sqrt{2}}$

t = 5s

26) For collision
$(u_y)_A = (u_y)_B$
$u = 50 \sin 53°$
$u = 40$

27) $u_y = 50 \cos 53° = 30$
$u_x = 50 \sin 53° = 40$

$H = \frac{u_y^2}{2g} = 45\,m$

$R = \frac{2u_x \times u_y}{g} = 240m$

$T = \frac{2u_y}{g} = 6\,s$

28) $y = x\tan\theta\left[1 - \frac{x}{R}\right]$

$y = x\left[1 - \frac{x}{80}\right]$

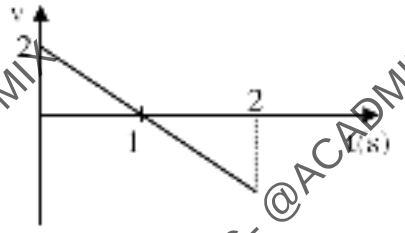
$\tan\theta = 1$, $R = 80$

$\therefore \tan\theta = \frac{4H}{R}$

$H = \frac{80}{4} = 20m$

29) $v = 2 - 2t$

at $t = 1s \rightarrow v = 0$



distance = Area under speed-time graph

distance = 2 m

30) $S = 2r \sin\frac{\theta}{2}$

$= 2(2) \sin\left(\frac{\pi}{3}\right)$

$= 2\sqrt{3}m$

31) $\vec{v} = \vec{u} + \vec{a}t$

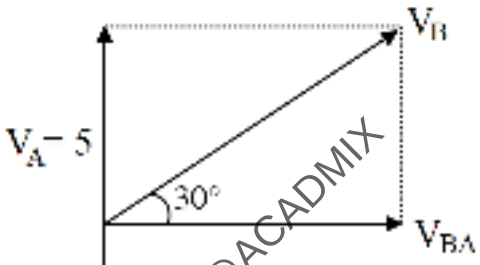
$\vec{v} = (8\hat{i} + 6\hat{j}) + (0.3\hat{i} + 0.4\hat{j}) \times 10$

$\vec{v} = 11\hat{i} + 10\hat{j}$

$|\vec{v}| = \sqrt{221}$ m/s

32) $\langle v \rangle = \frac{\text{Distance}}{\text{time}} = \frac{\int |\vec{v}|dt}{\Delta t}$

$\langle v \rangle = \frac{(16 + 4 + 4)}{4} = 6m/s$



33)

$\tan 30^\circ = \frac{5}{v_{BA}}$

$v_{BA} = 5\sqrt{3}m/s$



34)

T + N = 600 .....(i) 2T = 900 ⇒ T = 450 .....(ii)

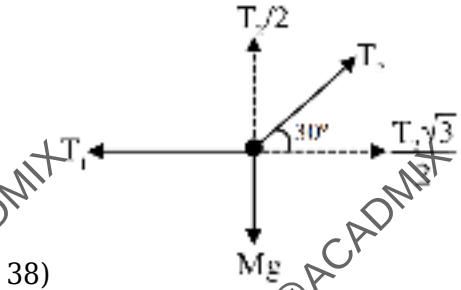
From eq.(i) and (ii)
N = 150 newton

35) $a_0.\sin\theta = a$

$a^0 = \frac{a}{\sin\theta} = a\ \text{cosec}\theta$

37) $F = \frac{dP}{dt}$

$F = n(mv)$



38)

$\frac{T_2}{2} = Mg$ ...(1)

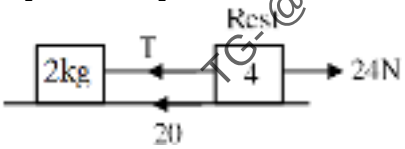
$\frac{T_2\sqrt{3}}{2} = T_1$ ...(2)

from (1) & (2)

$T_1 = \sqrt{3}Mg$

39) $N_1 = 20$ $f_1 = 10$

$N_2 = 40$ $f_2 = 20$



$T + 20 = 24$

$\boxed{T = 4N}$

40) $\vec{I} = \int \vec{F}\,dt$

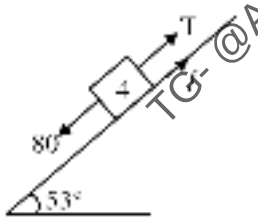
$= \frac{1}{2} \times 3 \times 12$

$= 18$ N-s

41) $F = v\frac{dm}{dt}$

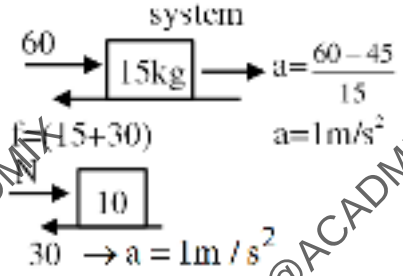
$F = \frac{4(2 \times 10^{-3})}{2.5} = \frac{16}{5} \times 10^{-3}N$

$F = 3.2 \times 10^{-3}$ N $= 0.0032$ N



42)
N = 60, f = μN = 30
T + f = 80
T = 80 – 30
= 50 N

43) $N_1 = 50 \rightarrow f_1 = 15$
$N_2 = 100 \rightarrow f_2 = 30$



N – 30 = 10(1)
N = 40

44) $f_{\square} = \mu N$
= 0.4 (20)
= 8 N
f_s = 2.8 N

45) $a = \frac{18}{6} = 3\ m/s^2$
f = 2 × 3 = 6 N (< $f_{\square}$)

CHEMISTRY

47)

$K_p = K_c (RT)^{\Delta ng}$
If Δng = 0 $K_p = K_c$

64)

$H_2O(\square) \rightarrow H_2O(g)$
$\Delta S = \frac{\Delta H}{T}$

67)

$\Delta H = \Sigma\Delta H_{f(P)} - \Sigma\Delta H_{f(R)}$
or $\Delta H = [81 + 3(-393)] - [9.7 + 3(-110)]$
or $\Delta H = -1098 + 320.3$
or $\Delta H \simeq -778$ kJ mol^{-1}

68)

$\Delta G = \Delta H - T\Delta S$
$\Delta H = \Delta G + T\Delta S$
$= 20000 + 200 \times (-20)$
$= 16000$ J/mol
Using Kirchoff's equation,
$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$
$\Delta H_2 = \Delta H_1 + \Delta C_p (T_2 - T_1)$
$= 16000 + 20 (400 - 200)$
$= 20000$ J/mol
$= 20$ kJ/mol.

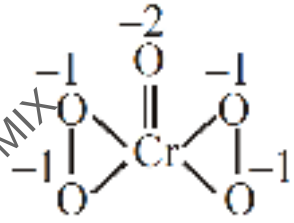
70)

$C_2H_{4(g)} + 3\ O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(\ell)}$ $\Delta H = -1420$ KJ
1420KJ → 1 mole
355 KJ → 0.25 mole
Volume at STP = $0.25 \times 22.4 = 5.6$L
OR
∴ 1420 kJ heat evolved from 1 mole C_2H_4

∴ 355 kJ heat evolved from $\frac{1}{1420} \times 355$ mole C_2H_4

$\frac{355}{1420} \times 22.4 = 5.6$ litre

72) The structure of CrO_5 is



73) $As_2S_3 \rightarrow H_3AsO_4 + H_2SO_4$
(+3) (–2) (+5) (+6)

74) M_4O_5
$M_x(+1)\ M_y(+3)\ O_5$
$x + y = 4$

$$\begin{array}{r} x + 3y = 10 \\ \hline -2y = -6 \end{array}$$

y = 3 ; x = 1

76) $w = 2.303nRT\log_{10}\left(\frac{P_2}{P_1}\right)$

77) $\frac{1}{2}Cl_2(g) + \frac{1}{2}I_2(g) \rightarrow ICl_{(g)}\ \Delta H = ?$

$\Delta H = \frac{62.8 + 151 + 242.3}{2} - 211.3$

= 16.75 kJ/mol

78) $\Delta H_{Sub} = (\Delta H)_{Fusion} + (\Delta H)_{Vap}$

79) Milli-equivalents of HCl = 25 × 0.08 = 2
Milli-equivalents of NaOH = 25 × 0.1 = 2.5
remaining $[OH^-] = \frac{2.5 - 2}{500} = 10^{-3}$, pOH = 3
or pH = 11

80) $K_p = K_c \times (RT)^{\Delta n_g}$

Given :- $\frac{K_c}{K_p} = (RT)^{-2} \Rightarrow \frac{K_p}{K_c} = (RT)^2$

$\therefore \Delta n_g = 2$
y + z = 2

81) 1 mole of $Na_2CO_3.10H_2O$ has O atoms = 13 × N_A
0.2 mol of $Na_2CO_3.10H_2O$ has O atoms
= 0.2 × 13 × $6.02 \times 10^{23} = 1.56 \times 10^{24}$

82) SiO_2 = Limiting reactant

$18SiO_2 \xrightarrow{\text{give}} 3P_4$

0.36 mol $SiO_2 \rightarrow \frac{3}{18} \times 0.36 = 0.06$ mol P_4

83) $\Delta G° = -2.303\ RT \log K_{eq}$

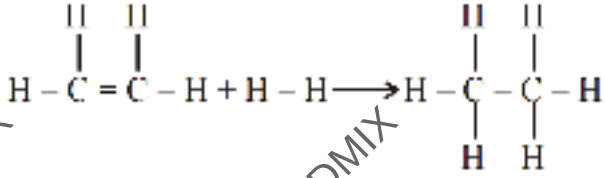
$= -2.303 \times \left(\frac{8.314}{1000}kJ\right) \times (400) \times \log(10^5)$

= –38.29 kJ

84) $Ka = \frac{[H_3O^+][HCO_3^-]}{[CO_2]}$

$\frac{[HCO_3^-]}{[CO_2]} = \frac{Ka}{[H_3O^+]}$

$= \frac{3.8 \times 10^{-7}}{10^{-6}} = 3.8 \times 10^{-1}$



85)
ΔH reaction = ΣB.E. (Rectant) – ΣB.E. (product)
= [B.E.(4C – H) + B.E. (C = C) + B.E. (C – C)]
– [B.E.6(C – H) + B.E.(C – C)]
= B.E. (C = C) + B.E. (H – H) – 2 B.E.(C – H) – B.E. (C – C)
= 600 + 400 – 2 × 410 – 350 kJ
= – 170 kJ

86) $[H^+]_{\text{weak acid}} = C\alpha = 0.01 \times \left(\frac{2}{100}\right)$
$= 2 \times 10^{-4}$
$[H^+] \times [OH^-] = K_w = 10^{-14}$
$\therefore [OH^-] = \frac{10^{-14}}{2 \times 10^{-4}} = 5 \times 10^{-11}$

87) Eqv. of H_3PO_x = Eqv. of NaOH
$1 \text{ mole} \times \text{Basicity} = \frac{80\,gm}{(E_{NaOH} = 40)}$
∴ Basicity = 2
∴ ($H_3PO_x = H_3PO_3$)

88) HCOONa + HCl → HCOOH + NaCl

t=0	0.25 × 20ml =5 milli-mole	0.1 × 30ml =3 milli-mole	0	0
End	2 milli-mole	0	3 milli-mole	3milli-mole

$pH = pKa + \log \frac{[HCOONa]}{[HCOOH]}$
$= 3.75 + \log \frac{(2/50ml)}{(3/50ml)}$
= 3.57

89) (K_a = Highest) ⇒ Strongest acid
⇓
weakest conjugate base

90) $C_{(D)} + O_2 \rightarrow CO_2$; ΔH = –94.3
$C_{(A)} + O_2 \rightarrow CO_2$; ΔH = –97.6

$C_{(s)} \rightarrow C_{(g)}$; $\Delta H = -94.3 - 197.6 = +3.3$ kcal

∴ 1 mol = 12 gm → 3.3 kcal

18 gm → $\frac{3.3}{12} \times 18 = 4.95$ kcal

BIOLOGY

91) NCERT (XI) Pg # 07

92) NCERT XI Pg. 3, 4

93) NCERT XI pg.- 5

94)

XI NCERT Pg. No. # 4

95) NCERT-XI Pg.# 23

96) NCERT-XI Pg.# 23,24

97)

XI NCERT Pg. No. # 7, 8

98) NCERT-XI, Pg. No. # 6, 7
Wheat : Poaceae,
Mango : Anacardiaceae
Brinjal, Potato, Datura, : Solanaceae
Cat, Lion : Felidae
Dog : Canidae

99) NCERT-XI Pg.# 8

100) XI NCERT Pg. No. # 7

101) XI OLD NCERT Pg. No. # 04, 05

102) XI NCERT Pg. No. # 10

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103) NCERT XI Page No. # 11

104) NCERT Pg. # 11

105) NCERT Pg. # 10 - 12

106) NCERT Pg. # 12

107) NCERT Pg. # 90

108) NCERT Pg. # 91

109) NCERT - XI, Pg. # 91

110)

NCERT-XI, Pg. # 91

111) NCERT-XI, Pg. # 13

112) NCERT-XI, Pg. # 14

113)

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114) NCERT XI Pg.# 14

115) NCERT Pg.# 15

116) NCERT (XI) Pg # 15

117) NCERT, Pg. # 16

118) NCERT Pg. # 17

119) NCERT XI, Pg. # 17
Penicillium

120) NCERT Pg. No. # 17, 18

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121)

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122)

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123) NCERT-XI Pg.# 26
Volvox, Chara, Ulothrix, Spirogyra

124) NCERT-XI, Pg. 26, 27

125) NCERT-XI Pg # 27

126) NCERT Pg. No. # 29, 30 (Diagram)

127) XI NCERT Pg. No. # 29, 30

128) NCERT Pg. # 30, 32

129) NCERT (XI) Pg # 32

130)

NCERT XI - P. No. - 33

131) NCERT-XI, Pg. # 33
Root cell = 2n
Female gametophyte = n

132) NCERT-XII Page No. 32-33

133) NCERT-XII Page No. 32-33

134) NCERT-XI, Pg. # 32

135) NCERT Page No. 32

136) Pg. No. 42 NCERT 2022 - 2023 Edition

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137)

NCERT (XI) Pg. # 49

138)

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139) Pg. No. 56 NCERT 2022 - 2023 Edition

140) NCERT (XIth) Pg. # 50, 51

141)

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142)

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144)

NCERT Pg. # 48,49,50

145) NCERT (XI) Pg. # 51

146) NCERT (XI) Pg. # 56

147)

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148) Pg. No. 42, 45 NCERT 2022 - 2023 Edition

149) NCERT (XIth) Pg. # 101

150) NCERT (XIth) Pg. # 104

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151) NCERT XI Pg # 101, 102

152) NCERT–XI, Pg. # 102

153)

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154) NCERT XI (E), Page No. 102

155) NCERT (XI) Pg. # 103

156) Module

157)

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159)

XI-NCERT

160) NCERT Pg.#112

161) NCERT-XI Pg. # 112

162) NCERT XII Pg # 114

163) NCERT Pg. # 112

164) NCERT-XI, Pg. # 114

165) Pg. No. 119 NCERT 2022 - 2023 Edition

166) Pg. No. 120 NCERT 2022 - 2023 Edition

167) NCERT XI Pg. 118

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168) NCERT Pg. # 57

169)

NCERT XI, Pg. No. 47

170)

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171)

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172)

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173)

Old XI, NCERT, Pg. # 101

174) OLD NCERT Biology XI Page no. 101 Para 7.1.2

175) NCERT-XI, Pg # 119

176) NCERT, Pg # 83

177) NCERT, Pg # 8

178)

NCERT Pg.# 82

179) NCERT XI, Page # 114

180) NCERT Pg.#111 (E), 112 (H)

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CHEMISTRY

1) In chromium atom number of electrons which have n = 2 and $\ell = 1$ is :-

(1) 6
(2) 4
(3) 2
(4) 1

2) Among the orbitals $3d_{xy}$, $2p_x$, 3s and 2s, which one has maximum number of nodal planes and radial nodes respectively :-

(1) 3s and 2s
(2) $3d_{xy}$ and 2s
(3) $2p_x$ and 3s
(4) $3d_{xy}$ and 3s

3) In the energy level sequence for a multi-electron atom, 4f-orbital lies between :-

(1) 6s and 5d
(2) 5p and 5d
(3) 4p and 5p
(4) 5d and 6d

4) The first emission line of Balmer series for H spectrum has the wave no. equal to ?

(1) $\frac{9R}{40}$
(2) $\frac{7R}{144}$
(3) $\frac{36}{5R}$
(4) $\frac{5R}{36}$

5) On the basis of Bohr's model, the radius of 3rd orbit is :-

(1) Equal to the radius of first orbit
(2) Three times the radius of first orbit
(3) Nine times the radius of first orbit
(4) Five times the radius of first orbit

6) The ratio of the radii of first orbits of H, He^+ and Li^{2+} is :-

(1) 1 : 2 : 3
(2) 6 : 3 : 2
(3) 1 : 4 : 9
(4) 9 : 4 : 1

7) Match the following :

	Column-I		Column-II
i	Aufbau principle	a	$\lambda = h/(mv)$
ii	de broglie	b	$\lambda = hc/\Delta E$
iii	Hund's rule	c	Max. unpaired e^-
iv	Planck's law	d	Electronic configuration

(1) (i) d (ii) b (iii) c (iv) a
(2) (i) c (ii) a (iii) d (iv) b
(3) (i) d (ii) a (iii) c (iv) b
(4) (i) c (ii) b (iii) d (v) a

8) The e/m ratio is maximum for :-

(1) D^+
(2) He^+
(3) H^+
(4) He^{2+}

9) The de-broglie wavelength associated with particle of mass 10^{-6} kg with a velocity of 10 ms^{-1} is :-

(1) 6.63×10^{-22} m
(2) 6.63×10^{-29} m
(3) 6.63×10^{-31} m
(4) 6.63×10^{-34} m

10) Distance between 2nd and 3rd shell of H-atom is :-

(1) 1.2 Å
(2) 2.65 Å
(3) 4.30 Å
(4) 1.65 Å

11) Which of the following set of quantum number is not possible :-

(1) n = 3, □ = 2, m = +1, s = –1/2
(2) n = 4, □ = 1, m = +2, s = +1/2

(3) n = 4, ℓ = 3, m = 0, s = +1/2

(4) n = 3, ℓ = 1, m = 0, s = –1/2

12) The total number of atomic orbitals in the third energy level of an atom is

(1) 9

(2) 8

(3) 18

(4) 32

13) Select the **incorrect** statement about 8th period ?

(1) The maximum no of electrons in this period sub shell = 50

(2) The maximum no. of elements in this period = 50

(3) The types of subshell in this period = 5

(4) The maximum no of orbitals in new sub shell in this period = 10

14) The compound of vanadium with chlorine has magnetic moment 1.73 B. M. The vanadium chloride has the formula :-

(1) VCl_2

(2) VCl_3

(3) VCl_4

(4) VCl_5

15) $[Rn]6d^2, 7s^2$ is the configuration of :-

(1) Ac, d-block

(2) Th, f-block

(3) Pa, f-block

(4) Hf, d-block

16) On moving in a transition series from left to right increase in the Z_{eff} is very less due to

(1) Some Ionic potential

(2) Abnormal behaviour of d-block series

(3) poor shielding effect of d-orbitals

(4) Shielding effect of (n – 1) shell is more.

17) Match **List-I** with **List-II** :

	List-I Elements		**List-II Atomic radii (pm)**
(a)	O	(i)	88
(b)	C	(ii)	74

(c)	B	(iii)	66
(d)	N	(iv)	77

Choose the **correct answer** from the options given below :

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

18) Which is the correct order of ionic sizes $_{63}Eu$, $_{71}Lu$, $_{70}Yb$ & $_{64}Gd$

(1) Gd > Eu > Yb > Lu

(2) Eu > Gd > Yb > Lu

(3) Eu > Gd > Lu > Yb

(4) Lu > Yb > Gd > Eu

19) Which of the following process requires maximum amount of energy ?

(1) $M^-_{(g)} \rightarrow M_{(g)}$

(2) $M_{(g)} \rightarrow M^+_{(g)}$

(3) $M^+_{(g)} \rightarrow M^{+2}_{(g)}$

(4) $M^{2+}_{(g)} \rightarrow M^{+3}_{(g)}$

20) If the configuration of Y^+ & X^- is $2s^2\ 2p^6$ then

(1) IP of X = IP of Y

(2) EA of X > EA of Y

(3) IP of X^- = IP of Y^+

(4) IP of X^- > IP of Y^+

21) Match the column :

Element (outer most shell e^- configuration)		**Successive value of I.E. (eV/atom)**	
(A)	$3s^1$	(P)	22, 25, 36, 42, 250
(B)	$3s^2$	(Q)	18, 26, 32, 240, 260
(C)	$3s^23p^1$	(R)	20, 24, 210, 230
(D)	$3s^23p^2$	(S)	15, 180, 200, 220

Match the type of element, with the corresponding successive value of I.E :

(1) A-S, B-P, C-Q, D-R

(2) A-R, B-S, C-P, D-Q
(3) A-S, B-R, C-Q, D-P
(4) A-S, B-R, C-P, D-Q

22) Which of the following E.A. order is **not correct**?

(1) N < O < S
(2) Cl > O > N > C
(3) O < S < F < Cl
(4) I < Br < F < Cl

23) Which of the following process is non spontaneous?

(1) $F^{+}_{(g)} + Cl_{(g)} \rightarrow F_{(g)} + Cl^{+}_{(g)}$
(2) $O^{-}_{(g)} + S_{(g)} \rightarrow O_{(g)} + S^{-}_{(g)}$
(3) $Be^{-}_{(g)} + B_{(g)} \rightarrow Be_{(g)} + B^{-}_{(g)}$
(4) $N_{(g)} + O^{+}_{(g)} \rightarrow N^{+}_{(g)} + O_{(g)}$

24) **Statement-1 :** Electron gain enthalpy of N is +ve while that of P is –ve.
Statement-2 : Smaller atomic size of N in which there is a considerable electron-electron repulsion and hence the additional electron is not accepted easily.

(1) Both statements are true and the statement-2 is the correct explanation of statement-1.
(2) Both statements are true but statement-2 is not the correct explanation of statement-1.
(3) Statement-1 is true but the statement-2 is false.
(4) Statement-1 is false but the statement-2 is true

25) Identify the incorrect statement :-

(1) He has the highest IE_1 in periodic table
(2) Hg & Br_2 are liquid at room temperature
(3) F has highest electron affinity
(4) Zn, Cd, Hg are most volatile metals of periodic table.

26) Select correct order of given property :-

(1) HOF < HOCl < HOBr < HOI Acidic nature
(2) HF > HCl > HBr > HI Acidic nature
(3) $N_2O < N_2O_3 < N_2O_4 < N_2O_5$ Acidic nature
(4) $H_2O > H_2S > H_2Se > H_2Te$ Acidic nature

27) Identify incorrect order :-

(1) $Mg^{+2} < K^{+} < S^{-2} < Se^{-2}$; Ionic radius
(2) $F^{-}(aq) > Cl^{-}(aq) > Br^{-}(aq) > I^{-}(aq)$; Electrical conductance

(3) Cl > F > Br > I ; Electron Affinity

(4) $Li^{+} < Mg^{+2} < Al^{+3}$; Hydration Energy

28) Which is wrong statement?

(1) The decreasing order of bond angle is $H_2O > H_2S > H_2Se > H_2Te$

(2) The decreasing order of bond angle is $NH_3 > PH_3 > AsH_3 > SbH_3$

(3) The decreasing order of bond dissociation energy is $Cl_2 > F_2 > Br_2 > I_2$

(4) The decreasing order of bond angle is $CH_4 > NH_3 > H_2O$

29) Which of the following combination of orbitals does not form covalent bond (x-axis is internuclear axis):-

(1) $s + p_y$

(2) $p_y + p_y$

(3) $d_{yz} + d_{yz}$

(4) $d_{xy} + d_{xy}$

30) **Assertion :** At room temperature O_2 and N_2 are stable molecule but S_2 and P_2 are less stable.
Reason : $2p_\pi - 2p_\pi$ collateral overlapping is effective but $3p_\pi - 3p_\pi$ overlapping is weak.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not correct explanation of assertion.

(3) Assertion is correct, reason is incorrect

(4) Both assertion and reason are incorrect.

31)

Consider the following table :-

Steric no. (σ bp + □.p.)	**□.p. on central atom**	**Shape of molecule**
5	a	Linear
b	1	See-Saw
6	c	Square Planar
d	2	T-Shaped

The value of a + b + c + d is :-

(1) 10

(2) 9

(3) 15

(4) 8

32) Which of the following do not have coordinate bond ?

(1) PH_3

(2) $P_2H_6^{+2}$

(3) $P_2H_5^{\oplus}$

(4) PH_4^+

33) Which of the order is correct ?

(1) $ClO^{\ominus} < ClO_2^{\ominus} < ClO_3^{\ominus}$ (Bond order)

(2) $CO_3^{-2} > CO > CO_2$ (Bond strength)

(3) $PO_4^{-3} > SO_4^{-2}$ (no. of coordinate bonds)

(4) $C_3O_2 > C(CN)_4$ (no. of π bond)

34) **Statement-1 (S_1) :** Dipole moment of CH_3Cl is more than CH_3F.
Statement-2 (S_2) : Dipole moment depends upon difference in electronegativity of C – X bond only

(1) Both S_1 and S_2 are correct

(2) S_1 is correct but S_2 is incorrect

(3) Both S_1 and S_2 are incorrect

(4) S_1 is incorrect but S_2 is correct.

35) Which of the following is the correct order of dipole moment ?

(1) $NH_3 < BF_3 < NF_3 < H_2O$

(2) $BF_3 < NF_3 < NH_3 < H_2O$

(3) $BF_3 < NH_3 < NF_3 < H_2O$

(4) $H_2O < NF_3 < NH_3 < BF_3$

36) Correct order of boiling point is :

(1) $H_2O < CCl_4 < CS_2 < CO_2$

(2) $CO_2 < CS_2 < CCl_4 < H_2O$

(3) $CS_2 < H_2O < CO_2 < CCl_4$

(4) $CCl_4 < H_2O < CO_2 < CS_2$

37) Which of the following statement is incorrect ?

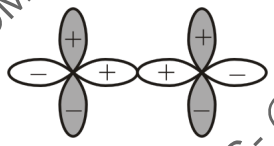
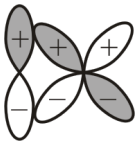
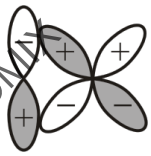
(1) The magnitude of the H-bonding depends on the physical state of the compound

(2) ……H – F…..H – F…..H – F…..
x y
x & y are bond lengths then x > y

(3) H-bond strength - $HF > H_2O > NH_3$

(4) Hydrogen bonds have strong influence on the structure and properties of the compounds.

38) Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists.

	List-I		List-II
(P)		(1)	p – d (π antibonding)
(Q)		(2)	d – d (σ bonding)
(R)		(3)	p – d (π bonding)
(S)		(4)	s – s (σ antibonding)

Code :

	P	Q	R	S
(1)	2	1	3	4
(2)	4	3	1	2
(3)	2	3	1	4
(4)	4	1	3	2

(1) 1

(2) 2

(3) 3

(4) 4

39) Which of the following order is incorrect :-

(1) $O_2^+ > O_2^-$ Number of electrons in BMO

(2) $B_2 > C_2$ Paramagnetism

(3) $B_2 < C_2$ Bond order

(4) $NO_2^+ > NO_2^-$ Bond angle

40) Order of covalent character is not correct for :-

(1) $NaF < Na_2O < Na_3N$

(2) $PbO < PbS$

(3) $CdCl_2 < CaCl_2$

(4) AgCl > KCl

41) Dipole-dipole (Keesom) attraction is present in :-

(1) $KCl + H_2O$

(2) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3-C\equiv N$

(3) $NO_3^- + Cl_2$

(4) $HCl + Cl_2$

42) Pick number of species which can exist ?

PH_5, $XeF_5^{\ominus}$, SiF_6^{-2}, PI_6^-, SBr_6, F_3^-, ClF_7

(1) 1

(2) 2

(3) 3

(4) 4

43) The radius and charge of each of six ions are shown in the tables.

ion	K^+	L^+	M^+	X^-	Y^-	Z^{-2}
radius/nm	0.14	0.18	0.15	0.14	0.18	0.15

The ionic solids KX, LY and M_2Z are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first :-

(1) $KX > LY > M_2Z$

(2) $KX > M_2Z > LY$

(3) $LY > M_2Z > KX$

(4) $M_2Z > KX > LY$

44) Which of the following relation is not true :-

(1) $\mu = q \times d$

(2) F.C. = VE + $\frac{\ell p(e^-)}{2}$

(3) B.O. = $\frac{N_b - N_a}{2}$

(4) All

45) Match the following :-

	Bonds		BE (kJ mole^{-1})
(i)	O–O	(p)	464
(ii)	O–H	(q)	138
(iii)	S–S	(r)	339
(iv)	S–H	(s)	213

The correct match is -

(1) i-q, ii-p, iii-s, iv-r

(2) i-q, ii-p, iii-r, iv-s

(3) i-p, ii-q, iii-s, iv-r

(4) i-r, ii-q, iii-p, iv-s

BIOLOGY

1) DNA is not present in –

(1) Chloroplast

(2) Ribosomes

(3) Nucleus

(4) Mitochondria

2) Match the column-I with column-II and select correct option :-

Column-I		Column-II	
a	Cristae	I	Primary constriction in chromosome
b	Thylakoids	II	Disc shaped sacs in Golgi apparatus
c	Centromere	III	Infoldings in mitochondria
d	Cisternae	IV	Flattened membranous sacs in stroma of plastids

(1) a – III, b – IV, c – I, d – II

(2) a – II, b – III, c – IV, d – I

(3) a – IV, b – III, c – II, d – I

(4) a – I, b – IV, c – III, d – II

3) Outermost covering is cell membrane in case of-

(1) Plant

(2) Fungi

(3) Algae

(4) Animal

4) Read the following statements and choose the correct option.

Statement - I : The centriole forms the axoneme of cilia and flagella and spindle fibres that give rise to spindle apparatus during cell division in plant cells.
Statement - II : The cytoskeleton in a cell is involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.

(1) Both Statements are correct.
(2) Both Statements are incorrect.
(3) Statement-I is correct but statement-II is incorrect.
(4) Statement-I is incorrect but statement-II is correct.

5) Select the incorrect statements :
(a) Golgi apparatus is the important site for synthesis of lipid.
(b) The cell envelope in a bacterial cell consist of cell wall and plasma membrane only.
(c) Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel.
(d) The hydrolytic enzymes of lysosomes are optimally active at basic pH.

(1) a, b and c
(2) b, c and d
(3) a, b and d
(4) a, c and d

6) Centriole consist of 9 evenly spaced peripheral ____fibrils of tubulin :

(1) Doublet
(2) Singlet
(3) Triplet
(4) Singlet and doublet

7) Which of the following statement is incorrect ?

(1) Nucleus as a cell organelle was first described by Robert brown.
(2) Material of nucleus is stained by acidic dye.
(3) Chromatin term was given by flemming.
(4) Normally, there is only one nucleus per cell.

8) Algal cell wall is made up of-

(1) Cellulose, Galactans, Hemicellulose, Pectins
(2) Cellulose, Hemicellulose, Pectins, Protein
(3) Cellulose, Galactans, Mannans & Minerals likes $CaCO_3$
(4) Cellulose, Hemicellulose, Galactans, Mannans

9) Which among the following holds or glues the different neighbouring plant cells together ?

(1) Primary wall
(2) Secondary wall

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(3) Tertiary wall
(4) Middle lamella

10) Which type of plastid is involved in storage of starch?

(1) Chloroplast
(2) Elaioplast
(3) Chromoplast
(4) Amyloplast

11) No organelles like the eukaryotes are found in prokaryotes except -

(1) Nucleus
(2) Lysosome
(3) Golgibody
(4) Ribosome

12) Chlorophyll pigments are present in which part of the chloroplast ?

(1) Stroma
(2) Thylakoid
(3) Outer chloroplastidial membrane
(4) Both (1) and (2)

13) Microbodies are :

(1) Membrane bound minute vesicles
(2) Contain various enzymes
(3) Present in both plant and animal cells
(4) All of these

14) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Anaphase-II begins with the simultaneous splitting of the centromere of each chromosome allowing them to move toward opposite poles of the cell.
Reason (R) : The shortening of microtubules attached to kinetochore takes place during anaphase-II.
Choose the correct answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the correct explaination of (A).
(2) (A) is true but (R) is false.
(3) (A) is false but (R) is true.
(4) Both (A) and (R) are true and (R) is correct explaination of (A).

15) Which of the following is true for telophase-I ?

(1) Nuclear membrane reappears

(2) Nucleolus disappears

(3) Chromatin condensed into chromosome

(4) Each chromosome has one chromatid

16) Match the following-

	Column-I		Column-II
(a)	Prophase-I	(i)	Bivalent chromosomes align on equatorial plate
(b)	Metaphase-I	(ii)	Homologous chromosome get separated
(c)	Anaphase-I	(iii)	Synaptonemal complex formation
(d)	Telophase-I	(iv)	Nucleolus reappear

(1) (a–i), (b–ii), (c–iii), (d-iv)

(2) (a–iii), (b–ii), (c–iv), (d-i)

(3) (a–iii), (b–i), (c–ii), (d-iv)

(4) (a–iv), (b–iii), (c–ii), (d-i)

17) A very significant contribution of mitosis is-

(1) Cell repair

(2) Reduction in number of chromosomes

(3) Increase genetic variability

(4) Recombination

18) A diploid cell contain 8 chromosomes then what will be number of chromosome in each daughter nuclei after meiosis-I ?

(1) 8

(2) 4

(3) 6

(4) 16

19) **Assertion :** Quiescent stage (G_0) of cell cycle is called inactive stage.
Reason : Cells in G_0 stage remain metabolically active but no longer proliferate.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

20) Which process involves division of cells resulting in formation of two identical daughter cells ?

(1) Meiosis

(2) Mitosis
(3) Both 1 and 2
(4) Meiosis-I

21) Which of the following statement is correct for mitotic metaphase ?

(1) Nucleolus, Golgi complex and ER reform
(2) Chromosome cluster at opposite spindle poles and their identity is lost as discrete elements
(3) Chromosomal material start to condense to initiate formation of distinct chromosome.
(4) Chromosome are moved to spindle equator and get aligned along metaphase plate through spindle fibers to both poles.

22) The complete disintegration of __A__ marks the start of __B__ of mitosis, hence the chromosomes are spread through the cytoplasm of the cell.
Identify A and B.

(1) A–Cell membrane, B–Prophase
(2) A–Nuclear membrane, B–Prophase
(3) A–Nuclear membrane, B-Metaphase
(4) A–Cell membrane, B-Metaphase

23) Which stage of mitosis follows S and G_2 phase of interphase?

(1) Prophase
(2) Metaphase
(3) Anaphase
(4) Telophase

24) Find out the incorrect statement-

(1) In zygotene, formation of synaptonemal complex takes place
(2) In pachytene, formation of recombination nodules takes place
(3) In diakinesis, dissolution of synaptonemal complex takes place
(4) Crossing over takes place during pachytene

25) In which phase of cell cycle, synthesis of histone protein takes place ?

(1) G_1 phase
(2) S phase
(3) G_2 phase
(4) G_0 phase

26) Fill up the blanks in the following statements with the appropriate options.
(I) The complex formed by a pair of synapsed homologous chromosomes is called as __A__ .
(II) Enzyme involved in crossing over is __B__ .
(III) The X-shaped structure formed during diplotene is __C__ .

(IV) Dyad of cells are formed at the end of __D__ .

	A	B	C	D
(1)	Synapsis	Permease	Synaptonemal complex	Telophase-I
(2)	Bivalent	Recombinase	Chiasmata	Meiosis-I
(3)	Tetrad	Lyase	Bivalent	Meiosis-I
(4)	Nodules	Endonucleas	Chiasmata	Anaphase-I

(1) 1

(2) 2

(3) 3

(4) 4

27) In a diploid cell after S-phase quantity of DNA is 20 pico gram then after meiosis-II each daughter nucleus will be :-

(1) Diploid with 10 pico gram DNA

(2) Haploid with 10 pico gram DNA

(3) Haploid with 5 pico gram DNA

(4) Diploid with 5 pico gram DNA

28) Yeast can progress through the cell cycle in only about -

(1) 60 minutes

(2) 30 minutes

(3) 120 minutes

(4) 90 minutes

29) Given Below are two statements :-
Statement - I : A protein is imagined as a line, the left end represented by first amino acid (N - terminal) and right end represented by last amino acid (C - terminal).
Statement - II : Adult human haemoglobein, consists of four subunits (two sub units of α-type and two subunits of β-types)
In the light of above statements, choose the correct answer from the options given below.

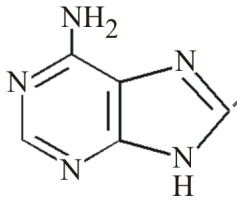
(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

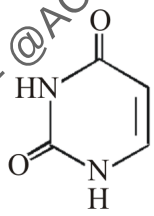
(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

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(A)



(B)

30) (A) and (B) in the above diagram are :-

	(A)	(B)
(1)	Guanine	Cytosine
(2)	Adenine	Uracil
(3)	Adenosine	Uridine
(4)	Guanine	Uracil

(1) 1
(2) 2
(3) 3
(4) 4

31)

Choose the odd one out with respect to homopolymers :

(1) Cellulose
(2) Glycogen
(3) Protein
(4) Chitin

32) Which of the following is considered as the simplest amino acid -

(1) Alanine
(2) Glycine
(3) Arginine
(4) Tryptophane

33) Which of the following is not a protein?

(1) GLUT-4
(2) Collagen
(3) Inulin
(4) Antibody

34) Which of the following pair is incorrect regarding secondary metabolites ?

(1) Pigments – Anthocyanins
(2) Essential oil – Lemon grass oil

(3) Toxins – Curcumin
(4) Drugs – Vinblastin

35) Arachidonic acid and palmitic acid has how many carbon atoms in their structure respectively ?

(1) 19, 15
(2) 15, 19
(3) 20, 16
(4) 16, 20

36) Which functional group is found in glucose ?

(1) Aldehyde
(2) Ketone
(3) Alcohol
(4) Ether

37) Which of the following is linear polymer of fructose ?

(1) Chitin
(2) Cellulose
(3) Pectins
(4) Inulin

38) Codeine is a :-

(1) Toxins
(2) Drug
(3) Alkaloid
(4) Polymeric substance

39) Substrate binds at which site of enzyme ?

(1) Allosteric site
(2) Inhibitory site
(3) Active site
(4) None of these

40) The class which includes all the enzymes catalysing inter-conversion of optical, geometric or positional isomers is called as :

(1) Ligases
(2) Isomerases
(3) Hydrolases
(4) Transferases

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41) Select the correct statements :
(a) Some nucleic acid that behave like enzyme are called ribozyme.
(b) Proteins with catalytic power are named enzymes.
(c) Prosthetic groups are only organic compounds which are loosely bound to apoenzyme.
(d) Catalytic activity of an enzyme is lost when co-factor is removed

(1) a and d
(2) b, c and d
(3) a, b and d
(4) a and d

42) _____ is a metal ion cofactor for the proteolytic enzyme carboxypeptidase.

(1) Mn
(2) Zn
(3) Cl
(4) Mg

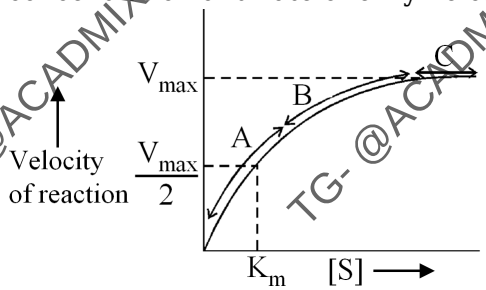
43) Which of the following co-factor is a prosthetic group ?

(1) NAD^+
(2) Zinc
(3) Haem
(4) Niacin

44) Essential chemical component of NAD and NADP is :-

(1) Zinc
(2) Haem
(3) Niacin
(4) Mg^{2+}

45) Which of the following statement is not correct regarding relationship between substrate concentration and rate of enzyme catalysed reaction as depicted in the graph :-



(1) K_m is measure of catalytic activity of an enzyme.
(2) With increase in substrate concentration, the velocity of enzymatic reactions rise in segment 'A' of the curve
(3) Enzymatic reaction velocity remains constant in segment 'B' of the curve with the increase in substrate concentration.

(4) In segment 'C' of the curve, the enzyme molecules are fewer than the substrate molecules and after saturation of these molecules, there are no free enzyme molecules to bind with the additional substrate molecules.

46) Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme _____.

(1) Thrombin
(2) Thromboplastin
(3) Thrombokinase
(4) Prothrombin

47) Read the following statements and select the correct option.
Statement-A : In each cardiac cycle two prominent sounds are produced which can be easily heard through a stethoscope.
Statement-B : First heart sound is (lub) associated with closure of semilunar valves.

(1) Both Statement are correct & Statement-B is correct explanation of Statement-A.
(2) Both Statement are correct but Statement-B is not a correct explanation of Statement-A.
(3) Only Statement-A is correct.
(4) Only Statement-B is correct.

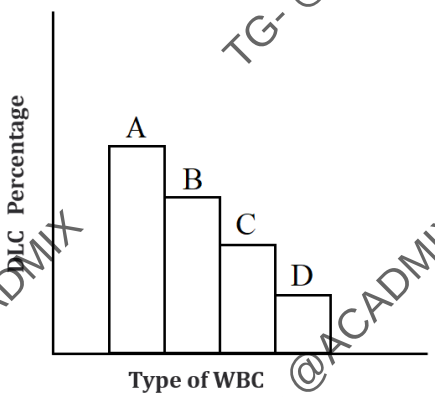
48) Which of the following can increase cardiac output?

(1) Sympathetic nerve
(2) Adrenal medullary hormones
(3) Both (1) & (2)
(4) Thyrocalcitonin

49) Read the following statements & select the correct option.
Statement-A : Lymph is a coloured fluid containing specialised lymphocytes which are responsible for immune response of the body.
Statement-B : Lymph is also important carrier for nutrients, hormones etc.

(1) Both Statement are correct & Statement-B is correct explanation of Statement-A.
(2) Both Statement are correct but Statement-B is not a correct explanation of Statement-A.
(3) Only Statement-A is correct.
(4) Only Statement-B is correct.

50)



Recognise A,B, C,D

(1) A-Neutrophils, B-Lymphocytes, C-Basophils, D-Monocyte

(2) A-Neutrophils, B-Lymphocytes, C-Monocyte, D-Acidophils

(3) A-Lymphocytes, B-Neutrophils, C-Acidophils, D-Basophils

(4) A-Monocytes, B-Neutrophils, C-Lymphocytes, D-Eosinophils

51) Which of the following cells are phagocytic in nature ?

(1) Acidophils

(2) Neutrophils

(3) Monocytes

(4) Both (2) & (3)

52) Read the following statements carefully and choose the correct option :-
Statement-1 :- A reduction in number of thromobocytes can lead to clotting disorders.
Statement-2 :- Megakaryocytes are fragments of thrombocytes which are involved in coagulation.

(1) Both the statement-1 and 2 are correct

(2) Statement-1 is correct but statement-2 is incorrect

(3) Statement-2 is correct but statement-1 is incorrect

(4) Both the statements-1 and 2 are incorrect

53) Incorrect statement about A,B and Rh antigens ?

(1) Present on the surface of RBCs

(2) Can induce immune response

(3) Anti-B antibodies destroys antigen A

(4) Can stimulate the production of antibodies

54) A ___A___ person if exposed to ___B___ blood, will form specific antibodies against Rh antigen :-

(1) A → Rh –ve, B → Rh +ve

(2) A → Rh +ve, B → Rh –ve

(3) A → Rh +ve, B → Rh +ve

(4) A → Rh –ve, B → Rh –ve

55) Which of the following is/are absent in lymph ?

(1) RBCs

(2) WBCs

(3) Larger proteins

(4) Both (1) & (3)

56) Opening of bicuspid and tricuspid valves occurs at :-

(1) Starting of atrial contraction
(2) Starting of ventricular contraction
(3) Starting of joint diastole
(4) Starting of atrial diastole

57) **Assertion (A) :** CO_2 is converted to bicarbonate at tissue level & transported to the alveoli is released out as CO_2.
Reason (R) : Carbonic anhydrase facilitates the trapping of CO_2

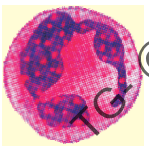
(1) Both Assertion and Reason are false
(2) Assertion and Reason both are true but reason is not correct explanation of assertion
(3) Both assertion & reason are true and reason is correct explanation of assertion
(4) Assertion is true but Reason is false

58) Globulins contained in human blood plasma are primarily involved in :-

(1) Clotting of blood
(2) Osmotic pressure
(3) Defence mechanism
(4) Oxygen transport

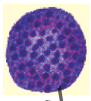
59) Which of the following statements are **correct** ?
(A) A fall in glomerular blood flow can activate the JG cells to release renin.
(B) Angiotensin II is powerful vasodialator.
(C) Angiotensin II also activates the adrenal medulla to release aldosterone.
(D) Aldosterone causes reabsorption of Na^+ and water from the distal parts of the tubule.
(E) Renin converts angiotensinogen in blood to angiotensin I and further to angiotensin II.
Choose the correct answer from the options given below :

(1) B, C and D only
(2) A, B and E only
(3) A, D and E only
(4) A, C and D only

60) Identify the cell shown in the diagram :- 

(1) Neutrophil
(2) Basophil
(3) Eosinophil
(4) Monocytes

61) Find out the correct match from the table given below :

	Cells	Name	Percent	Function
1.		Eosinophils	6–8%	Associated with allergic reactions
2.		Basophils	0.5–1%	Inflammatory reaction
3.		Neutrophils	60–65%	Heparin Secretion
4.		Monocytes	2–3%	Phagocytosis

(1) 1

(2) 2

(3) 3

(4) 4

62) Identify the blood groups shown in the diagram.

Red blood cell type	**(i)**	**(ii)**	**(iii)**	**(iv)**
Antibodies in plasma	Anti-B	Anti-A	None	Anti-A and Anti-B

	(i)	(ii)	(iii)	(iv)
1.	A	O	AB	B
2.	A	B	AB	O
3.	A	B	O	AB
4.	A	AB	B	O

(1) 1

(2) 2

(3) 3

(4) 4

63) In which of the following animals, respiration occurs through lungs ?

(1) Bird

(2) Fish

(3) Cockroach

(4) Earthworm

64)

During swallowing glottis can be covered by _____ to prevent the entry of food into larynx.

(1) Pharynx
(2) Lips
(3) Epiglottis
(4) Alveoli

65) In human respiration takes place in

(1) Cells lining the lungs cavity
(2) Cells found in blood
(3) All living cells of the body
(4) Only RBC

66)

Primary site for exchange of gases in human is-

(1) Nostrils
(2) Nasal passage
(3) Pharynx
(4) Alveoli

67)

PO_2 in deoxygenated blood is-

(1) 40 mmHg
(2) 95 mmHg
(3) 45 mmHg
(4) 159 mmHg

68) When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe?

(1) rising CO_2 concentration
(2) falling CO_2 concentration
(3) rising CO_2 and falling O_2 concentration
(4) falling O_2 concentration

69) Select the correct statement.

(1) Expiration occurs due to contraction of external intercostal muscles
(2) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration.
(3) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.
(4) Expiration is initiated due to contraction of diaphragm.

70) TV + ERV is

TG- @ACADMIX

(1) RV

(2) IC

(3) EC

(4) FRC

71) Read the following statements :-

(a) Respiratory rhythm centre present in pons region of brain

(b) The role of oxygen in the regulation of respiratory rhythm is quite insignificant

(c) Receptors associated with aortic arch and carotid artery can recognise changes in CO_2 and $H^{\oplus}$ Concentration.

(d) Pneumotaxic centre can reduce the duration of inspiration and thereby alter the respiratory rate.

How many statements are correct about regulation of Respiration ?

(1) One

(2) Four

(3) Two

(4) Three

72) Find out **correct** statement :-

(1) Inspiration is a passive process

(2) Inspiration is an active process

(3) Expiration is an active process

(4) Both expiration and inspiration are passive process

73)

Match the following columns.

Column-I		**Column-II**	
A	Earthworms	I	Tracheal-Tubes
B	Aquatic arthropods	II	Lungs
C	Molluscs	III	Gills
D	Reptiles	IV	Moist cuticle
E	Insects		

(1) A= I, B= II, C= III, D= IV, E= III

(2) A= IV, B= III, C= III, D= II, E= I

(3) A= IV, B= II, C= I, D= III, E= III

(4) A= IV, B= II, C= III, D= I, E= III

74) Match the items in column I with column II and choose the correct option :-

	Column-I		**Column-II**
(A)	2500 to 3000 ml	(i)	Tidal volume

(B)	1000 - 1100 ml	(ii)	Inspiratory reserve volume
(C)	500 ml	(iii)	expiratory reserve volume
(D)	5100 - 5800 ml	(iv)	Total lung capacity
(E)	1100 - 1200 ml	(v)	Residual volume

	A	**B**	**C**	**D**	**E**
(1)	(ii)	(iv)	(i)	(iii)	(v)
(2)	(ii)	(iii)	(i)	(v)	(iv)
(3)	(ii)	(iii)	(i)	(iv)	(v)
(4)	(ii)	(v)	(i)	(iv)	(iii)

(1) 1
(2) 2
(3) 3
(4) 4

75) "Chemoreceptors" which recognise the change concentration of CO_2 and H^+ are found in -

(1) Aortic arch
(2) Carotid artery
(3) Medulla
(4) All of these

76)

The vital capacity of human is about :-

(1) 2500 ml
(2) 500 ml
(3) 4100 ml
(4) 6000 ml

77)

Fibrous lungs can be seen in-

(1) Asthma
(2) Emphysema
(3) Occupational respiratory disorders
(4) Flu

78) Which of the following volume is not included in vital capacity ?

(1) ERV
(2) TV
(3) IRV
(4) RV

79) Which of the following statement is not correct about excretion?

(1) Ascending limb of Loop of Henle is impermeable to water
(2) On an average, human excretes 25 to 30 gm urea per day
(3) Glucose and amino acids are reabsorbed passively in renal tubule.
(4) 70-80 percent of electrolytes and water are reabsorbed in PCT

80)

Which of the following statement is/are **true**?
(I) Glomerular filtrate is isotonic to plasma.
(II) When the urine passes into collecting duct, it becomes hypotonic.
(III) Filtrate is isotonic in proximal convoluted tubule.
(1) I and II
(2) II and III
(3) I and III
(4) All

81) The peritubular capillaries of the nephron arise from the :-

(1) Afferent arterioles
(2) Renal artery
(3) Efferent arterioles
(4) Vasa Recta

82) Which of the following part of nephron is found in medulla of kidney?

(1) Malpighian corpuscle
(2) PCT
(3) DCT
(4) Loop of Henle

83) Which of the following structure have stretch receptors for micturition?

(1) Urinary bladder
(2) Ureters
(3) Renal pelvis
(4) Column of Bertini

84)

TG- @ACADMIX

In the given how many are the secretions of sebaceous glands ?

Sterols, Lactic acid, Hydrocarbons, Waxes, Urea

(1) Five

(2) Four

(3) Three

(4) Two

85) Match the abnormal conditions given in Column I with their explanations given in Column II and choose the correct option :-

Column I		Column II	
(A)	Glycosuria	(i)	Accumulation of uric acid in joints
(B)	Renal calculi	(ii)	Inflammation in Glomerulus
(C)	Glomerulo nephritis	(iii)	Mass of crystallised salts within the kidney
(D)	Gout	(iv)	Presence of glucose in urine

(1) A–i, B–iii, C–ii, D–iv

(2) A–iii, B–ii, C–iv, D–i

(3) A–iv, B–iii, C–ii, D–i

(4) A–iv, B–ii, C–iii, D–i

86) **Assertion :-** Analysis of urine helps in clinical diagnosis of malfunctioning of kidney.
Reason :- Glucosuria is indicative of diabetes mellitus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

87) In interstitial fluid of kidney an increasing osmolarity from cortex to the inner medulla. This gradient is mainly caused by :-

(1) Na^+, K

(2) NaCl, H_2O, HCO_3

(3) NaCl and Urea

(4) Urea, HCO_3

88) Which segment does not play a role in the maintenance of pH and ionic balance of blood by selective secretion :-

(1) PCT

(2) Bowman's capsule

(3) DCT

(4) Collecting duct

89)

In human, ammonia produced by metabolism is converted in urea in the-

(1) Liver
(2) Kidney
(3) Blood
(4) Lung

90) Ammonia is the chief excretory substance in-

(1) Camel and whale
(2) Cartilaginous fishes
(3) Whale and tortoise
(4) Bony fishes

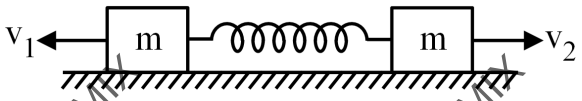
PHYSICS

1) A two particle system of masses 2 kg and 3 kg present at (15, 0) and (0, 20) respectively in a xy plane then its centre of mass lies on the lines :-

(a) $y = 2x$ (b) $\frac{x}{12} + \frac{y}{24} = 1$
(c) $3x - y = 6$ (d) $y = x$

(1) Only (a) is correct
(2) Both (a) and (b) are correct
(3) (a), (b) and (c) are correct
(4) (b), (c) and (d) are correct

2) Two students were given a physics problem for finding maximum extension of spring if blocks are imparted velocities v_1 and v_2 when spring is unstretched.



By student A : $\frac{1}{2}m(v_1 + v_2)^2 = \frac{1}{2}kx^2$

By student B : $\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}kx^2$

(1) Student A is correct, student B is wrong.
(2) Student B is correct, student A is wrong.
(3) Both are correct.
(4) Both are wrong.

3) If velocities of 3 particles of mass 1 kg (each) in a system varies with time $\vec{v}_1 = t^2\hat{i}$, $\vec{v}_2 = (t+1)\hat{j}$

as and $\vec{v}_3 = \left(\frac{3}{2}t\hat{i} + t^2\hat{j}\right)$ respectively, then momentum of centre of mass of system at t = 2 sec is :-

(1) $7\sqrt{2}$ units

(2) $4\sqrt{2}$ units

(3) $3\sqrt{2}$ units

(4) $5\sqrt{2}$ units

4) Mass is distributed uniformly over a thin triangular plate and position of two vertices are given by (–2,3) and (–4,–2). What is the position of third vertex if centre of mass of the plate lies at the origin ?

(1) (6,1)

(2) (1, –6)

(3) (–6, 1)

(4) (6, –1)

5) A bullet of mass 0.01 kg travelling at a speed of 500 m/s strikes a block of mass 2 kg which is suspended by a string of length 5m. The centre of gravity of block is found to rise a vertical distance of 0.2 m. Then find the speed of bullet with which it emerges out from the block ($g = 10\ m/s^2$) :-

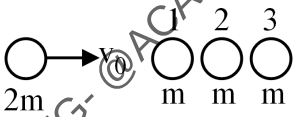
(1) 200 m/s

(2) 140 m/s

(3) 100 m/s

(4) 280 m/s

6) A ball of mass 2m moving with a velocity v_0 collides head-on elastically with a cradle of three identical balls each of mass m. Determine velocity of ball 3 after collision :-



(1) $v_0/3$

(2) $v_0/2$

(3) $4v_0/3$

(4) v_0

7) A ball falls from a height 180 m upon a fixed horizontal plane and rebounds.($e = 0.5$, $g = 10\ m/s^2$).
(a) Velocity of ball after second collision is 25 m/s.

(b) Average speed for whole interval is $\frac{50}{3}$ m/s.
(c) Total distance travelled before rebounding has stopped is 300 m.
Options :-

(1) Only (a) is correct

(2) Only (a) and (b) are correct

(3) Only (b) and (c) are correct

(4) All (a), (b) and (c) are correct

8) A boy of mass 40 kg is standing in a gravity free space at a height 20m above the floor. He throws a ball of mass 0.2 kg downwards with a speed 5 m/s.
Statement-1 : When ball reaches the floor, the distance of the man above the floor will be 20.1 m.
Statement-2 : When ball reaches the floor the displacement of centre of mass of (boy + ball) system is 0.1 m.

(1) Only statement-1 is correct.
(2) Only statement-2 is correct.
(3) Both statement-1 and statement-2 are correct.
(4) Both statement-1 and statement-2 are wrong.

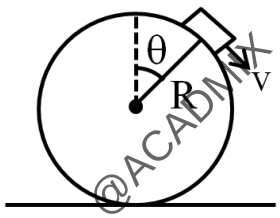
9) A ball is thrown vertically downwards with velocity $\sqrt{gh}$ from a height h above the ground after colliding with the ground, it just reaches at height h/2 above the ground. Coefficient of restitution is :-

(1) $\frac{1}{2}$
(2) 1
(3) $\frac{1}{\sqrt{2}}$
(4) $\frac{1}{\sqrt{3}}$

10) A particle of mass 1 kg describes a circle of radius 2 m. The centripetal acceleration of the particle is 8 m/s^2. What will be the momentum in $\left(\frac{kg-m}{sec}\right)$ of the particle ?

(1) 4
(2) 2
(3) 8
(4) 1

11) A small block of mass 'm' moves on the smooth surface of a fixed sphere as shown. Force applied by block on the sphere at the given instant is-



(1) $mg\cos\theta - \frac{mv^2}{R}$ radially outwards
(2) $mg\cos\theta - \frac{mv^2}{R}$ radially inwards
(3) $mg\sin\theta$
(4) None

12) A cyclist speeding 10 km/h on a level road takes a sharp circular turn of radius 3 m without

reducing the speed. The coefficient of static friction between the road and the tyres is 0.1 then-

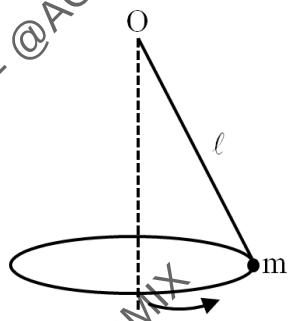
(1) He will slip

(2) He will not slip

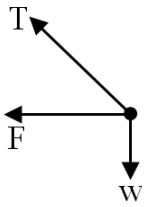
(3) both (1) and (2) can possible

(4) Data insufficient

13) A point mass m is suspended from a light thread of length ℓ, fixed at O, is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are :-



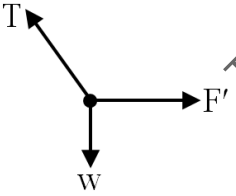
(1)



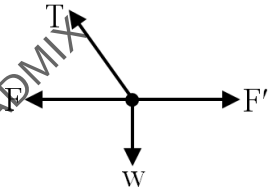
(2)



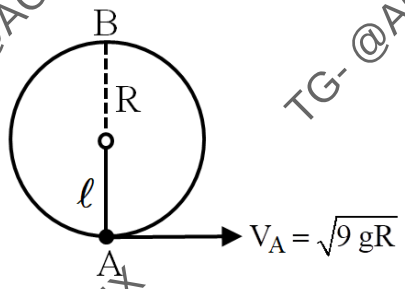
(3)



(4)



14) A stone is attached to one end of a string and is whirled in a vertical circle of radius R as shown in fig. Find the tension in string at highest point 'B' ? (m = mass of stone)



(1) $T_B = 6$ mg

(2) $T_B = 4$ mg

(3) $T_B = 3$ mg

(4) $T_B = 9$ mg

15) Match the following columns and choose the correct option from the codes given below for uniform circular motion.

	Column-I		**Column-II**
(A)	Speed	(1)	Constant
(B)	Velocity	(2)	Variable
(C)	Magnitude of acceleration	(3)	Zero
(D)	Acceleration		

(1) (A)-(1), (B)-(2), (C)-(2), (D)-(1)

(2) (A)-(1), (B)-(2), (C)-(1), (D)-(2)

(3) (A)-(2), (B)-(1), (C)-(2), (D)-(1)

(4) None

16) The speed of a particle moving in a circle of radius r = 2m varies with time t as $v = t^2$. The net acceleration at t = 2 sec is :-

(1) $\sqrt{40}$

(2) $\sqrt{80}$

(3) $\sqrt{90}$

(4) 8

17) **Statement-I :-** Torque is a vector whose direction is along the applied force

Statement-II :- $\vec{\tau} = \vec{r} \times \vec{F}$

(1) **Statement-I** is true, **Statement-II** is true and **Statement-II** is correct explanation of **Statement-I**

(2) **Statement-I** is true, **Statement-II** is true but **Statement-II** is not a correct explanation of **Statement-I**

(3) **Statement-I** is true, but **Statement-II** is false

(4) **Statement-I** is false, but **Statement-II** is true



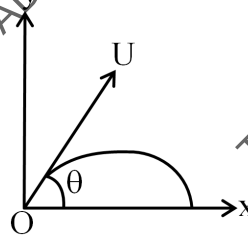
18) Given uniform Rod is hinged about one end and other end is supported by string as shown in fig. If string is cut then angular acceleration of rod when it makes an angle of 37° from Horizontal

(1) $\alpha = \frac{6g}{L}$

(2) $\alpha = \frac{3g}{2L}$

(3) $\alpha = \frac{6g}{5L}$

(4) $\alpha = \frac{2g}{3L}$



19) For the given Projectile of mass m the angular momentum about point of projection when particle is at maximum height is

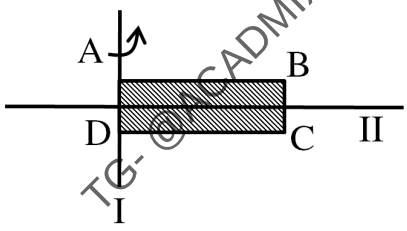
(1) $\frac{mU^3\sin^2\theta\cos\theta}{2g}$

(2) $\frac{mU^3\sin\theta\cos^2\theta}{g}$

(3) $\frac{mU^2\sin\theta^2\cos\theta}{g}$

(4) mUcosθ

20) A rectangular slab ABCD have dimensions □ × 2□ as shown in figure. Then radius of gyration about axis-I is



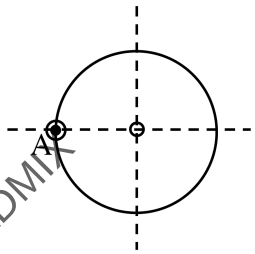
(1) $\sqrt{\frac{2}{3}}$□

(2) $\frac{2\ell}{\sqrt{3}}$

(3) $\frac{\ell}{\sqrt{3}}$

(4) $\sqrt{3}$

21) A thin wire of length and mass m is bent in the form of a circle as shown. Its moment of inertia



about an axis passing through A and perpendicular to plane ?

(1) $\frac{m\ell^2}{\pi^2}$

(2) $m\ell^2$

(3) $\frac{2m\ell^2}{\pi^2}$

(4) $\frac{m\ell^2}{2\pi^2}$

22) Angular momentum of a particle rotating under a central force is constant due to :

(1) Constant torque

(2) Constant force

(3) Constant linear momentum

(4) Zero torque

23) A cylinder rolls on a horizontal plane surface. If the speed of the highest point is 50 m/s then speed of the centre is:

(1) 100 m/s

(2) 25 m/s

(3) 12.5 m/s

(4) $\frac{50}{\sqrt{2}}$m/s

24) **Statement-1 :** For a point (other than points on the axis) on a rotating body, acceleration can't be zero.
Statement-2 : All points (other than points on the axis) on a rotating body moves on circular path.

(1) Both **Statement-1** and **Statement-2** are true and **Statement-2** is the correct explanation of **Statement-1**.

(2) Both **Statement-1** and **Statement-2** are true but **Statement-2** is NOT the correct explanation of **Statement-1**.

(3) **Statement-1** is true but **Statement-2** is false

(4) **Statement-1** is false but **Statement-2** is true.

25) Let I_A and I_B be moments of inertia of a body about two axes A & B respectively. The axis A passes through the centre of mass of the body but B does not then :

(1) $I_A < I_B$ always

(2) If the axes are parallel $I_A < I_B$

(3) If the axes are parallel $I_A = I_B$

(4) If the axes are not parallel then $I_A \geq I_B$

26) The speed of center of mass of a homogenous sphere after rolling down an inclined plane through a vertical height h from rest without sliding is :

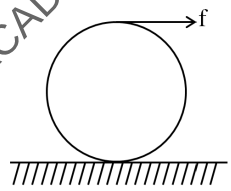
(1) $\sqrt{gh}$

(2) $\sqrt{gh/5}$

(3) $\sqrt{4gh/3}$

(4) $\sqrt{\frac{10gh}{7}}$

27) A disc of radius R and mass M is under pure rolling when a force f is applied at the top most point (as shown in fig.). If there is sufficient friction between the disc and the horizontal surface then



:

(1) $a_{CM} = \frac{f}{M}$

(2) $a_{CM} < \frac{f}{M}$

(3) $a_{CM} > \frac{f}{M}$

(4) None

28) If density of earth is increase by 3% and radius is increase by 1% then percentage change in acceleration due to gravity at the surface of earth is :

(1) 4%

(2) 2%

(3) 3%

(4) No change

29) A point mass 'm' is placed inside a spherical shell of radius R and mass M at a distance of R/2 from centre of the shell. The gravitational force exerted by the shell on the point mass is :

(1) $\frac{GMm}{R^2}$

(2) $-\frac{GMm}{R^2}$

(3) Zero

(4) $\frac{4GMm}{R^2}$

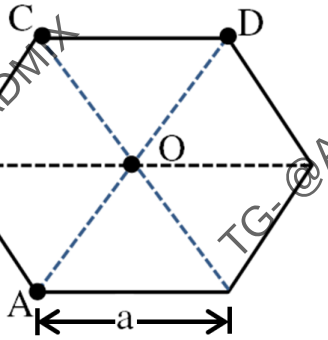
30) A body weighs 36 N on the surface of earth. When it is taken to a height R/2 from surface of earth, where R is radius of earth. Its weight would be _____

(1) 16 N
(2) 28 N
(3) 32 N
(4) 72 N

31) Two particles of equal mass go round a circle of radius R under the action of their mutual gravitation attraction. Speed of each particle is :

(1) $\sqrt{\frac{Gm}{2R}}$
(2) $\sqrt{\frac{Gm}{4R}}$
(3) $\sqrt{\frac{Gm}{R}}$
(4) $\sqrt{\frac{2Gm}{R}}$

32) Four point masses each of mass m are placed at four corner A, B, C and D as shown.



Gravitational field at O is :

(1) $\frac{\sqrt{3}Gm}{a^2}$
(2) $\frac{Gm}{\sqrt{3}a^2}$
(3) $\frac{\sqrt{3}Gm}{2a^2}$
(4) $\frac{2Gm}{a^2}$

33) The escape velocity from the earth is about 11 km/s. The escape velocity from a planet having twice the radius and the same mean density as the earth is :

(1) 22 km/s
(2) 5.5 km/s

(3) 11 km/s

(4) 15.5 km/s

34) Work done by an external agent in bringing three particles of mass m each at the vertices of an equilateral triangle of length '□' is :

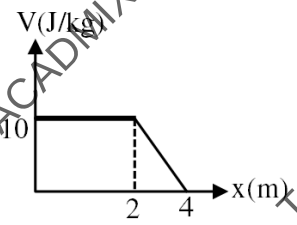
(1) $\frac{3Gm^2}{\ell}$

(2) $\frac{3Gm^2}{2\ell}$

(3) $-\frac{3Gm^2}{\ell}$

(4) $-\frac{3Gm^2}{2\ell}$

35) Gravitational potential varies along x-axis as shown. Gravitational field at x = 3m is :



(1) $\vec{I} = -5\hat{i}$

(2) $\vec{I} = +5\hat{i}$

(3) zero

(4) $\vec{I} = -2.5\hat{i}$

36) In a certain region of space, the potential is given by : $V = k[2x^2 - y^2 + z^2]$. The electric field at the point (1,1,1) has magnitude = :-

(1) $k\sqrt{6}$

(2) $2k\sqrt{6}$

(3) $2k\sqrt{3}$

(4) $4k\sqrt{3}$

37) A cube made of material having a density of 0.9×10^3 kg/m^3 floats between water and a liquid of density 0.8×10^3 kg/m^3, which is immiscible with water. What part of the cube is immersed in water ?

(1) $\frac{1}{2}$

(2) $\frac{2}{3}$

(3) $\frac{3}{4}$

(4) $\frac{3}{7}$

38) Sixty four spherical rain drops of equal size are falling vertically through air with a terminal velocity 1.5 ms^{-1}. If these drops coalesce to form a big spherical drop. then terminal velocity of big drop is:

(1) 8 ms^{-1}

(2) 16 ms^{-1}

(3) 24 ms^{-1}

(4) 32 ms^{-1}

39) A container of height 10 m is filled with water. An orifice is made on a side will of the container such that the water coming out from orifice can achieve maximum range. Then find the value of this maximum range.

(1) 5 m

(2) $10\sqrt{2}$ m

(3) 10 m

(4) 8 m

40) If a capillary tube is dipped into liquid and the levels of the liquid inside and outside are same then the angle of contact is -

(1) 0°

(2) 30°

(3) 90°

(4) 120°

41) Elastic potential energy density stored in a strained wire is U. If wire is replaced by identical wire of twice youngs modulus then elastic potential energy density for the same strain will be -

(1) 4U

(2) 2U

(3) 8U

(4) 16U

42) The work done in increasing the size of a rectangular soap film with dimensions 8 cm × 3.75 cm to 10 cm × 6 cm is 2×10^{-4} J. The surface tension of soap solution will be -

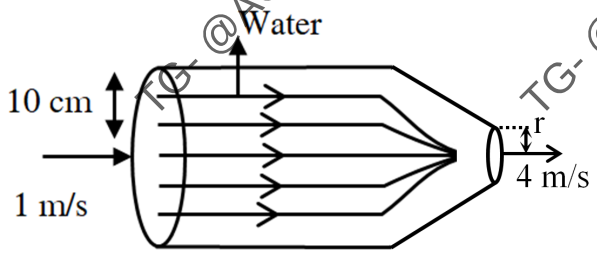
(1) 1.65×10^{-2} N/m

(2) 3.3×10^{-2} N/m

(3) 6.6×10^{-2} N/m

(4) 8.25×10^{-2} N/m

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43) For adjoining fig, the value or r is -

(1) 5 cm
(2) 10 cm
(3) 7 cm
(4) None

44) A man is walking near a railway track then a train passes on the track. As train passes man feels a pull towards the train. Which concept describes this pull.

(1) Newton's Law of viscosity
(2) Stoke's Law
(3) Bernoulli's theorem
(4) Momentum conservation.

45) A fluid is moving through a horizontal pipe. The pressure at point 1 is P_1 = 4000 Pa and the velocity is 4 m/s. At point 2, the velocity is 2 m/s. What is the pressure at point 2? (Density of fluid = 800 kg/m^3)

(1) 4000 Pa
(2) 4400 Pa
(3) 8000 Pa
(4) 8800 Pa

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ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	4	1	4	3	2	3	3	2	2	2	1	4	3	2	4	3	2	4	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	4	1	3	3	2	3	1	1	3	1	1	2	2	2	2	3	1	3
Q.	41	42	43	44	45															
A.	2	2	4	2	1															

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	4	4	3	3	2	3	4	4	4	2	4	4	1	3	1	2	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	3	2	2	3	4	1	2	3	2	3	3	3	1	4	3	3	2
Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
A.	3	2	3	3	3	1	3	3	4	2	4	2	3	1	4	3	3	3	3	1
Q.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125
A.	2	2	1	3	3	4	1	1	2	3	4	2	2	3	4	3	3	4	3	3
Q.	126	127	128	129	130	131	132	133	134	135										
A.	3	4	1	3	3	2	3	2	1	4										

PHYSICS

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
A.	3	4	1	4	3	3	3	1	4	1	2	1	3	2	2	2	4	3	1	2
Q.	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
A.	4	4	2	1	2	4	3	1	3	1	2	1	1	3	2	2	1	3	3	3
Q.	176	177	178	179	180															
A.	2	2	1	3	4															

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CHEMISTRY

10)

$r_n = 0.529 \frac{n^2}{Z}$ Å

13)

5g subshell have 9 orbitals

14)

Magnetic moment = 1.73 BM
The no. of unpaired e^- = 1
$V = [Ar]\ 4s^2\ 3d^3$
$V^{+4} = [Ar]4s^0\ 3d^1$

In VCl_4 Oxidation state of 'V' is '+4'.

16) fact

17)

	List-I Elements		List-II Atomic radii (pm)
(a)	O	(iii)	66 pm
(b)	C	(iv)	77 pm
(c)	B	(i)	88 pm
(d)	N	(ii)	74 pm

B > C > N > O
As we move across the period from left to right atomic radius decreases due to increase in nuclear charge.

18)

Factual theory based.

19) $\boxed{I.P. \propto Z_{eff} \propto \frac{\oplus ve\ charge}{\ominus ve\ charge}}$

20)

Configuration of Y^+ & X^- = $2s^2\ 2p^6$ (noble gas) hence by gaining of an electron. X^- has noble gas configuration so electron affinity (EA) of X very high to become noble gas like. EA of X > EA of Y.

21)

Concept

22) EA : Cl > O > C > N
ΔHeg_1 of N is positive

23) $N_{(g)} + O^+_{(g)} \rightarrow N^+_{(g)} + O_{(g)}$
$N_{IP} > O_{IP}$
$N \rightarrow N^+$ endo

24)

$N(g) + e^- \rightarrow N^-(g)$ $\Delta_{eg}H \rightarrow$ positive value (energy absorbed)
$P(g) + e^- \rightarrow P^-(g)$ $\Delta_{eg}H \rightarrow$ negative value (energy released)

25) F has maximum electronegativity.
Cl has max. electron affinity.

26)

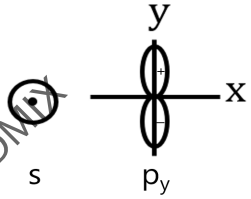
Acidic ch. ∝ oxidation state

27) HE $F^- > Cl^- > Br^- > I^-$
 Electrical conductance
$F^- < Cl^- < Br^- < I^-$

28)

B.E ⇒ $Cl_2 > Br_2 > F_2 > I_2$

29) S + P_y does not form any Bond if I.N.A. is 'x'



30)

O = O $2p_\pi - 2p_\pi$ N ≡ N $2p_\pi - 2p_\pi$

S = S $3p_\pi - 3p_\pi$ P ≡ P $3p_\pi - 3p_\pi$

Strength of overlapping → $2p_\pi - 2p_\pi > 3p_\pi - 3p_\pi$

31)

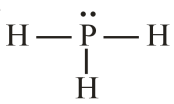
Steric No. = σ + □.P.

5 = 2 +(a = 3)

(b = 5) = 4 + 1

6 = 4 + (c = 2)

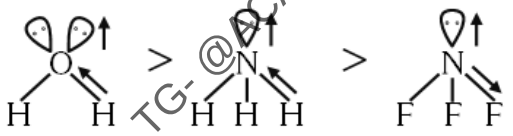
(d = 5) = 3 + 2

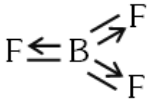
32) 

33) π bond strength ↑↑ distance ↓↓ (B.O.↑↑)

34)

μ = q × d

35) 



(μ = 0)

36) H_2O has intermolecular H-bonding

37) Properties of H-bond. (X is covalent bond and F----H is H-bond)

38) (P) Along the I.N.A. ⇒ σ

(Q) p + d ⇒ π bonding

(R) Opposite phase → π antibonding

(S) Opposite phase → σ bond (s-s)

39) $O_2^+ = \sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma 2p_z^2$

$\left[\pi^2_{2p_x} = \pi^2_{2p_y}\right] . \left[\pi^{*1}_{2p_x} = \pi^{*}_{2p_y}\right]$

$O_2^- = \sigma^2_{1s}\sigma^{*2}_{1s}\sigma^2_{2s}\sigma^{*2}_{2s}\sigma 2p_z^2$

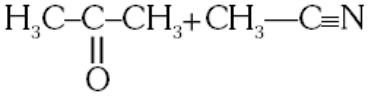
$\left[\pi^2_{2p_x} = \pi^2_{2p_y}\right] . \left[\pi^{*2}_{2p_x} = \pi^{*1}_{2p_y}\right]$

both have same number in BMO.

40) Covalent character ∝ Polarisation.

41)

KCl + H_2O ⇒ ion-dipole attraction



dipole-dipole attraction

HCl + Cl_2 ⇒ dipole induced dipole attraction

42) $XeF_5^{\ominus}$ & SiF_6^{-2} exist

43) Concept

45) BE of → O-O < S – S < S – H < O–H

BIOLOGY

46) NCERT-XI, Pg. # 96, 97, 98, 100

47) NCERT-XI, Pg. # 97, 98, 101

48) NCERT (XI) Pg # 91

49) NCERT XI Pg. # 98, 99, 100

50) NCERT XI Page No. # 90, 96, 99

51) NCERT XI page No. # 100

52) NCERT XI Page. No :- 100

53) NCERT-XI, Pg. 94

54) NCERT-XI, Pg. 94

55) NCERT (XI) Pg # 97

56) NCERT (XI) Pg # 88

57) NCERT-XI Pg # 136

58) NCERT XI Pg # 102

59)

NCERT-XI, Pg. # 127

60) NCERT, Pg. # 127

61) NCERT-XI, Pg # 126, 127

62) NCERT-XI, Pg. # 125

63) NCERT-XI, Pg. # 127

64) NCERT-XI, Pg. # 122

65) NCERT (XI) Pg # 122

66) NCERT-XI Pg. # 123

67) NCERT-XI, Pg. # 123

68) NCERT (XI) Pg # 122

69) NCERT (XI) Pg # 126

70) NCERT (XI) Pg # 121

71) XI-NCERT Page No. # 126

72) NCERT-XI, Pg. # 126 to 128

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73) NCERT XI Pg. # 121

74) NCERT XI Pg : 111-112

75) NCERT-XI, Pg. # 107

76)

NCERT-XI, Pg. # 110

77) NCERT XI Pg. No. 106

78)

NCERT XI, Pg. No. # 109

79) NCERT XI ; Page 108

80) NCERT XI Pg. # 106

81) NCERT Pg. # 107

82) NCERT XI Pg. # 110

83) NCERT XI Pg. # 108

84) NCERT-XI, Pg # 113

85)

NCERT-XI, Pg # 117

86) NCERT XI, Pg. # 112, 118

87) NCERT XI Pg. # 118

88) NCERT XI Pg. # 118

89) NCERT Pg. # 118

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90) NCERT XI, Pg # 116

91) NCERT Pg #281

92) NCERT Pg #285

93) NCERT Pg #287

94) NCERT Pg #282

95) Neutrophils > Lymphocytes > Monocytes > Eosinophils > Basophils
Percentage of : 60 - 65% > 20 - 25% > 6 - 8% > 2 - 3% > 0.5 - 1%

96) NCERT, Pg. # 194

97) NCERT, Pg. # 280

98) NCERT, Pg. # 280

99) NCERT, Pg. # 196

100) NCERT, Pg. # 197

101) NCERT, Pg. # 200

102) NCERT, Pg. # 202

103) NCERT Pg # 279

104) NCERT Page No. # 297

105) NCERT Pg. # 194

106) NCERT Pg. # 194

107) NCERT Pg. # 195

110)

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The correct answer is All living cells of the body.

Here's why:Cellular Respiration: This is the process by which cells break down glucose (or other energy-rich molecules) to release energy in the form of ATP. This process occurs in the mitochondria of all living cells.

113)

A. CO_2 Sensitivity: The body's respiratory system is more sensitive to changes in CO_2 levels than O_2 levels.

B. pH Change: As CO_2 accumulates in the blood, it reacts with water to form carbonic acid, which lowers the blood pH.

C. Chemoreceptor Activation: Specialized sensors called chemoreceptors detect this pH change and send signals to the brain.

D. Respiratory Center Stimulation: The brain's respiratory center receives these signals and initiates the urge to breathe.

While a decrease in O_2 levels can also contribute to the urge to breathe, it's typically a secondary factor. The body's primary mechanism for regulating breathing is through monitoring CO_2 levels.

114)

NCERT Pg. No. 185

115)

SOLUTION-Option 3: EC (Expiratory Capacity)

Explanation:

A. TV (Tidal Volume) is the amount of air inhaled or exhaled during normal, relaxed breathing.

B. ERV (Expiratory Reserve Volume) is the amount of air that can be forcibly exhaled after the normal tidal exhalation.

116)

NCERT PAGE NO. 190

117) NCERT, Page#271

118)

NCERT Page no. 183

119)

NCERT (XI) Pg. # 171

120) NCERT XI, Pg.# 275

121)

NCERT Pg. # 272

123)

NCERT Pg. no. 272

124)

NCERT Pg No#295 - 298

125)

NCERT Pg No#295 – 298
Following statements are true-
(I) Glomerular filtrate is isotonic to plasma.
(III) Filtrate is isotonic in Proximal Convoluted Tubule(PCT).

126)

NCERT Pg. No. 208

127)

NCERT XI Pg.# 293

128) NCERT–XI, Para-6, Pg # 297 (E)

129)

NCERT–XI, Para–4, Pg # 298,

TG- @ACADMIX

Para - (7)(H) Pg. # 298

130)

NCERT–XI, Pg. # 299

131) NCERT XI Pg # 298

132)

NCERT-XI, Page # 211

133)

NCERT Page # 209

135)

NCERT–XI, Pg. # 298

PHYSICS

136) $x_{cm} = \frac{2 \times 15 + 3 \times 0}{5} = 6$

$y_{cm} = \frac{2 \times 0 + 3 \times 20}{5} = 12$

These values of x and y satisfy lines given is a, b and c.

137) By COME

$\frac{1}{2}\mu v_{rel}^2 = \frac{1}{2}kx^2$

$\frac{1}{2}\frac{(m)(m)}{m+m}(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$

$\frac{1}{2}\frac{m}{2}(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$

$\frac{1}{4}m(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$

138) NCERT, Pg. # 99

$\vec{v}_{cm} = \frac{1(t^2\hat{i}) + 1(t+1)\hat{j} + 1\left(\frac{3}{2}t\hat{i} + t^2\hat{j}\right)}{3}$

at t = 2 sec

$\vec{v}_{cm} = \frac{4\hat{i} + 3\hat{j} + 3\hat{i} + 4\hat{j}}{3}$

$\vec{v}_{cm} = \frac{7\hat{i} + 7\hat{j}}{3}$

$\therefore \vec{p}_{cm} = m\vec{v}_{cm}$

$= 3\left(\frac{7\hat{i}+7\hat{j}}{3}\right) = (7\hat{i}+7\hat{j})$

$\therefore p_{cm} = \sqrt{7^2+7^2} = 7\sqrt{2}$ units

139) NCERT, Pg. # 98

 COM will be at centroid

$\therefore \frac{-2-4+x}{3} = 0 \Rightarrow x = 6$

$\frac{3-2+y}{3} = 0 \Rightarrow y = -1$

140) NCERT, Pg. # 100

$v_{block} = \sqrt{2 \times 10 \times 0.2} = 2$ m/s

$(mv)_{bullet} = (mv)_{block} + (mv')_{bullet}$

$5 = 4 + (mv')_{bullet}$

$1 = (0.01)\, v'_{bullet}$

$v'_{bullet} = \frac{1}{0.01} = 100$ m/s

141) Velocity of ball 1 just after collision with mass 2m

$v_1 = \frac{m-(1)(2m)}{m+2m}(0) + \frac{(1+1)(2m)}{m+2m}v_0$

$v_1 = \frac{4}{3}v_0$

Now velocities between 1 and 2 will exchange because masses are equal and head-on elastic collision

$\therefore v_2 = \frac{4}{3}v_0$

Similarly $v_3 = v_2$ after collision and ball (2) comes at rest

$\therefore v_3 = \frac{4}{3}v_0$

142) (a) $v_n = e^n\sqrt{2gh}$

$\therefore v_2 = \left(\frac{1}{2}\right)^2\sqrt{2 \times 10 \times 180}$

$\therefore v_2 = 15$ m/s

(b) $v_{avg} = \sqrt{\frac{gh}{2}}\left(\frac{1+e^2}{(1+e)^2}\right)$

$= \sqrt{\frac{10 \times 180}{2}}\left(\frac{1+1/4}{(3/2)^2}\right)$

$= (30)\left(\frac{5}{9}\right) = \frac{50}{3}$ m/s

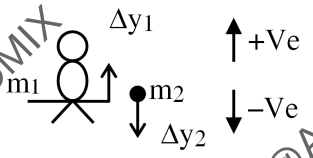
(c) Total distance

$s = h\left(\frac{1+e^2}{1-e^2}\right)$

$$= 180\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right) = 180\left(\frac{5/4}{3/4}\right)$$

= 300 m

143) No external force acting on (boy + ball) system, therefore displacement of CoM is zero.



$\Delta y_{cm} = 0$

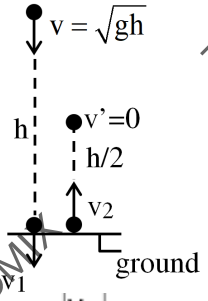
$m_1\Delta y_1 + m_2\Delta y_2 = 0$

(40) $\Delta y_1 = -(0.2)(-20)$

$\Delta y_1 = \frac{4}{40} = 0.1$ m

When ball reaches the floor, the distance of the man above the floor will be (20 + 0.1) = 20.1 m.

144) $v_1 = \sqrt{v^2 + 2gh} = \sqrt{3gh}$



$$e = \frac{|v_2|}{|v_1|}$$

$$e = \frac{\sqrt{2g(h/2)}}{\sqrt{3gh}}$$

$$e = \frac{1}{\sqrt{3}}$$

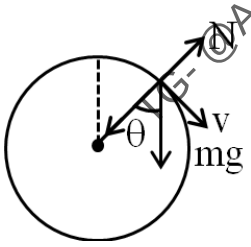
145) NCERT Pg # 41

$$a_c = \frac{v^2}{R}$$

$$8 = \frac{v^2}{2} \Rightarrow v = 4$$

P = mv

= m × 4 = 4M



146)

$mg\cos\theta - N = \frac{mv^2}{R}$

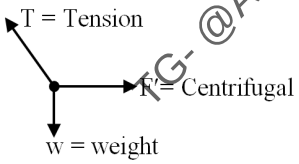
$N = mg\cos\theta - \frac{mv^2}{R}$

147) NCERT Pg # 64

$V_{max} = \sqrt{\mu_s Rg} = \sqrt{\frac{1}{10} \times 3 \times 9.8}$ = 1.71 m/s

$V = 10 \times \frac{5}{18}$ = 2.7 m/s

$V > V_{max}$



148)

149) $T_B = \frac{mV_B^2}{R} - mg = \frac{m\ 5gR}{R} - mg$

= 4 mg

150) NCERT Pg. # 41

In U.C.M. speed is constant and acceleration is changing in direction.

151) NCERT Pg. # 41

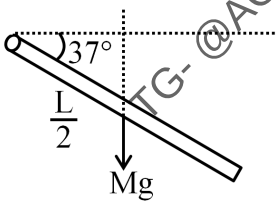
$a_T = \frac{dV}{dt} = 2t = 4$

$a_C = \frac{v^2}{R} = \frac{(t^2)^2}{R} = \frac{(4)^2}{2} = 8$

$a_{net} = \sqrt{a_C^2 + a_T^2} = \sqrt{80}$

152) NCERT Pg.#106

Torque is axial vector

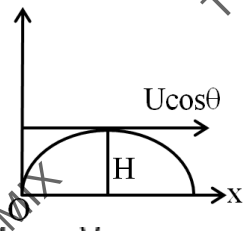


153) $\tau = I\alpha$

$Mg\frac{L}{2}\frac{4}{5} = \frac{ML^2}{3}\alpha$

$\frac{6g}{5L} = \alpha$

154) NCERT Pg. #106



$L = mVr_{\perp ar}$

$L = \frac{mU\cos\theta U^2\sin^2\theta}{2g}$

155)

NCERT Pg.#116

$\frac{1}{3}M(2\ell)^2 = MK^2$

$K = \frac{2\ell}{\sqrt{3}}$

156) $I_{AB} = I_{CM} + Md^2$

$I_A = mR^2 + mR^2$

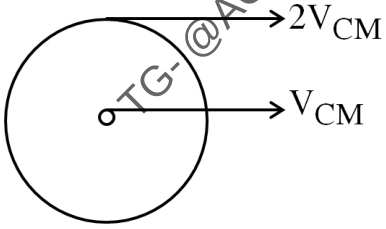
$I_A = 2mR^2$

$2\pi R = \square \Rightarrow R = \frac{\ell}{2\pi}$

$I_A = 2m\left\{\frac{\ell}{\pi}\right\}^2 = \frac{m\ell^2}{2\pi^2}$

157) $\tau = \frac{dL}{dt} = 0$

L = Const.



158)

160) $I_{AB} = I_{CM} + Md^2$

161) $v = \sqrt{\frac{2gh}{1+\frac{k^2}{R^2}}} = \sqrt{\frac{2gh}{1+\frac{2}{5}}}$

$v = \sqrt{\frac{10gh}{7}}$

162) $\frac{3}{2}MR^2\alpha = f(2R)$

$\alpha = \frac{4f}{3MR}$

$\alpha R = \frac{4f}{3M}$

163) $g = \frac{GMe}{Re^2} = \frac{G}{Re^2}\rho\frac{4}{3}\pi Re^3 = \frac{4}{3}G\pi\rho Re$

$\frac{\Delta g}{g} \times 100 = \left(\frac{\Delta\rho}{\rho} + \frac{\Delta Re}{Re}\right)100$

= (3 + 1)%

= 4%

164) $I_{inside} = 0$

F = mI

F = 0

165) $g_h = \frac{g}{\left(1+\frac{h}{R}\right)^2} = \frac{g}{\left(1+\frac{R/2}{R}\right)^2} = \frac{4}{9}g$

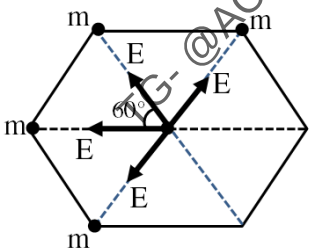
$W' = mgh = m \times \frac{4}{9} \times g = \frac{4}{9}W = \frac{4}{9} \times 36$

W' = 16 N

166) Gravitational force provide centripetal force

$\frac{Gm^2}{4R^2} = \frac{mv^2}{R}$

$\boxed{v = \sqrt{\frac{Gm}{4R}}}$



167) $E_{net} = \frac{\sqrt{3}Gm}{a^2}$

168) $V_e \propto R_e\sqrt{\rho}$

169) $W = U_f - U_i$

$W = -\frac{3Gm^2}{\ell}$

170) $E = -\frac{dv}{dx} = -(\text{slope})$

171) $\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$

$\vec{E} = -k\left[4x\hat{i} - 2y\hat{j} + 2z\hat{k}\right]$

At (1,1,1)

$\left|\vec{E}\right| = k\sqrt{16+4+4}$

$= 2k\sqrt{6}$

172) Let in water immersed volume is v_1 then

$w = (Th)_o + (Th)_w$

$V\rho_B g = (V-V_1)\rho_{og} + V_1\rho_w g$

$V \times 0.9\times10^3 = (V-V_1) \times 0.8 \times10^3 + V_1 \times 10^3$

$V(0.9\times10^3 - 0.8\times10^3) = V_1(10^3 - 0.8\times10^3)$

$\frac{V_1}{V} = \frac{1}{2}$

173)

$R = n^{1/3}r = (64)^{1/3}r$

$R = 4r$

$V_T \propto r^2$

$\frac{(V_T)_{big}}{(V_T)_{small}} = \left(\frac{R}{r}\right)^2 = \left(\frac{4r}{r}\right)^2 = 16$

$(V_T)_{big} = 16 \times 1.5 = 24\ ms^{-1}$

174) For $R_{max} \Rightarrow h = \frac{H}{2}$

$v = \sqrt{2gh} = \sqrt{gH}$

$R_{max} = v\sqrt{\frac{2h}{g}} = \sqrt{gH}\sqrt{\frac{2H}{2g}}$
H = 10 m

175) For θ = 90° then in capillary liquid no rise and no fall.

176) $\frac{E}{V} = \frac{1}{2}Y.Strain^2$

177) W = T × ΔA
= T × [2(SA_f – SA_i)]
$T = \frac{W}{2(SA_f - SA_i)} = \frac{2 \times 10^{-4}}{2(10 \times 6 - 8 \times 3.75) \times 10^{-4}}$
= 3.3 × 10^{-2} N/m

178) Equation of continuity,
$a_1v_1 = a_2v_2$
$\pi r_1^2 \times v_1 = \pi r_2^2 \times v_2$
$r_2 = \sqrt{\frac{r_1^2 v_1}{v_2}} = \sqrt{\frac{10 \times 10 \times 1}{4}} = 5cm.$

179) Pull or attraction by a train is application of bernoullis theorem.

180) Given,
Pressure at point 1 P_1 = 4000 Pa
Pressure Velocity V_1 = 4 m/s
Velocity V_2 = 2 m/s
⇒ ρ = 800 kg/m^3
Using Bernoullis equation -
$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$
$\Rightarrow 4000 + \frac{1}{2}800(4)^2 = P_2 + \frac{1}{2}(800)(2)^2$
$\Rightarrow 4000 + 6400 = P_2 + 1600$
P_2 = 10,400 – 1600
⇒ 8800 Pa

BIOLOGY

1) **Statement-I :-** Photosynthesis is a physico-chemical process.
Statement-II :- Ultimately all living forms on earth depend on sunlight for energy.

(1) Only Statement I is correct and Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I & II are correct.
(4) Both Statement I & II are incorrect.

2) __A__ in 1770 performed a series of experiments that revealed the essential role of air in the growth of green plants.
Identify the A.

(1) Jan Ingenhousz
(2) T.W. Engelmann
(3) Julius Von sachs
(4) Joseph Priestley

3) Identify the correct statements :
A. T.W. Engelmann used prism and green algae, *Cladophora* placed in a suspension of aerobic bacteria.
B. All living forms on earth not depend on sun light for energy.
C. Variegated leaf experiment proves that green parts of plant and sunlight is essential for photosynthesis.
D. Julius von Sachs provided evidence for production of glucose when plants grow. Glucose is usually stored as starch.
Choose the correct answer from the options given below.

(1) A and D only
(2) A, C and D only
(3) B and C only
(4) A, B and C only

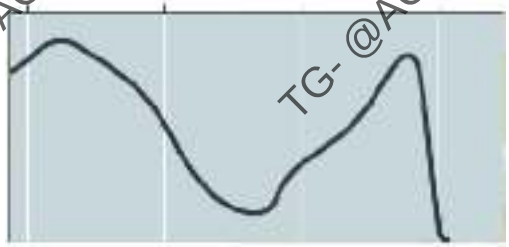
4) Where exactly enzymes of dark reaction are present within chloroplast ?

(1) Thylakoid membrane
(2) Grana
(3) Stromal lamella
(4) Stroma of chloroplast

5) Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :- O_2 evolved by the green plant comes from H_2O not from CO_2.
Reason (R) :- In green plants H_2O is the hydrogen donor.
In the light of above statements, choose the most appropriate answer from the options given below :

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.



6) Identify the below graph :

(1) Action spectrum of photosynthesis
(2) Absorption spectrum of chlorophyll a
(3) Absorption spectrum of chlorophyll b
(4) Action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a

7) **Assertion :-** There is no complete one to one overlapping between absorption spectrum of chlorophyll 'a' and the action spectrum of photosynthesis.
Reason :- Accessory pigments absorb wider range of wavelength and transfer energy to chlorophyll 'a'.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

8) Match the list-I with list-II.

List-I		List-II	
(a)	Light reaction	(i)	Yellow green
(b)	Chlorophyll b	(ii)	ATP + NADPH
(c)	Accessory Pigments	(iii)	Blue and Red region of spectrum
(d)	Maximum photosynthesis	(iv)	Chl b, xanthophyll and carotene

Choose the correct answer from the options given below :
(1) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)
(2) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(3) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(4) (a) – (i), (b) – (iv), (c) – (ii), (d) – (iii)

9) Select the correct statements :

(a) Pigments are substances that have an ability to absorb light at specific wavelength.

(b) Dark reaction occur in darkness only.

(c) We can separate the leaf pigments of any green plant through paper chromatography.

(d) Accessory pigments enable a wider range of wavelength of incoming light to be utilized for photosynthesis.

(1) a, b and c

(2) a, c and d

(3) a and d only

(4) c and d only

10) Match List I with List II and select the correct option.

List- I		**List- II**	
A.	NADP reductase enzyme	(i)	Transmembrane channel
B.	CF_0	(ii)	Inner side of the membrane
C.	CF_1	(iii)	Stroma side of membrane
D.	Splitting of the water molecule	(iv)	Protrudes on the outer surface of the thylakoid membrane

(1) A – ii, B – iv, C – iii, D – i

(2) A – iii, B – i, C – iv, D – ii

(3) A – iv, B – iii, C – ii, D – i

(4) A – iii, B – ii, C – iv, D – i

11) Match the column I with column II

	Column I		**Column II**
i	3-phosphoglyceric acid	a	First stable product in C_3 cycle
ii	Oxalo-acetic acid	b	First stable product in C_4 cycle
iii	RuBP	c	Primary CO_2 acceptor in C_3 cycle
iv	Phosphoenol pyruvate	d	Primary CO_2 acceptor in C_4 cycle

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii, b-i, c-iii, d-iv

(3) a-iii, b-i, c-ii, d-iv

(4) a-iv, b-iii, c-ii, d-i

12) Which of the following characteristic is not shown by C_4 plants ?

(1) Kranz anatomy
(2) Can tolerate higher temperature
(3) Response to high light intensities
(4) Photorespiration.

13) Which of the following condition is favourable for photorespiration ?

(1) High light intensity
(2) Low O_2
(3) High CO_2
(4) Low temperature

14) Given below are two statements :-
Statements–I :- For every CO_2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 of NADPH are required.
Statements–II :- In C_4 plants photorespiration does not occur. This is because they have a mechanism that increases the concentration of CO_2 at the enzyme (RuBisCO) site.
In the light of the above statements, choose the correct answer form the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct
(4) Both **Statement I** and **Statement II** are correct.

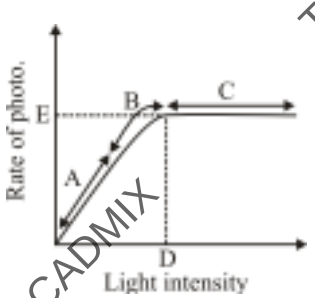
15) Major limiting factor for photosynthesis in terrestrial environment ?

(1) O_2 concentration
(2) CO_2 concentration
(3) H_2O
(4) Temperature

16) The plant or internal factors of photosynthesis depends on -

(1) Growth of plants
(2) Genetic predisposition
(3) Both (1) & (2)
(4) None of these

17) Recognise the figure & find out the correct match :-



	Column-I		Column-II
(i)	B-represents	(a)	Light is no longer limiting
(ii)	C-represents	(b)	Some factor other than light intensity is becoming the limiting factor
(iii)	D-represents	(c)	Maximum rate of photosynthesis
(iv)	E-represents	(d)	Light saturation point

(1) i-a, ii-b, iii-c, iv-d
(2) i-c, ii-d, iii-b, iv-a
(3) i-b, ii-a, iii-d, iv-c
(4) i-a, ii-b, iii-d, iv-c

18) Which one of the following statement is incorrect about cellular respiration in plants ?

(1) Breaking of complex compound through oxidation with in cell, lead to release of considerable amount of energy
(2) During respiration trapped energy is used for synthesis of ATP
(3) During respiration all energy contained in compounds is released in a single step.
(4) During respiration energy is released in a series of slow step wise reactions.

19) Which of the following is not a reason why plant can get along without respiratory organs ?

(1) Each plant part takes care of its own gas-exchange needs.
(2) Plants do not present great demands for gas exchange.
(3) Distance that gases must diffuse even in large, bulky plants is not great.
(4) Lenticels are present at each part of the plant.

20) The process that leads to a complete oxidation of organic substances in the presence of oxygen is :-

(1) Aerobic respiration
(2) Anaerobic respiration
(3) Both (1) and (2)
(4) Fermentation

21) Match the column :-

	Column-I		Column-II
(a)	Final product of glycolysis	(i)	In cytoplasm and mitochondria
(b)	Aerobic respiration	(ii)	In Yeast cell
(c)	Anaeobic respiration	(iii)	Pyruvate

(1) a-(iii), b-(ii), c-(i)

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(2) a-(iii), b-(i), c-(ii)
(3) a-(ii), b-(i), c-(iii)
(4) a-(ii), b-(iii), c-(i)

22) The process of respiration common to both facultative anaerobe as well as obligate anaerobe is :-

(1) Glycolysis
(2) Link reaction
(3) Kreb's cycle
(4) Electron transport system

23) What will be net gain of ATP through substrate level phosphorylation during Glycolysis.

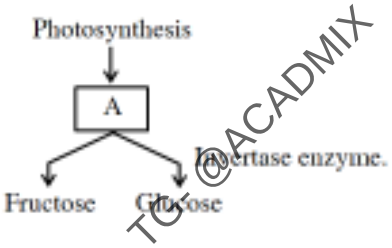
(1) 4 ATP
(2) 2 ATP
(3) 8 ATP
(4) 6 ATP

24) In glycolysis .....A..... undergoes .....B..... oxidation to form two molecules of .....C.....
Find out A, B and C :-

	A	B	C
(1)	Glucose	Complete	Pyruvic acid
(2)	Glucose	Partial	PEP
(3)	Starch	Partial	Pyruvic acid
(4)	Glucose	Partial	Pyruvic acid

(1) 1
(2) 2
(3) 3
(4) 4

25) Study the flow chart and identify A



(1) Galactose
(2) Sucrose
(3) Maltose
(4) Lactose

26) Site of link reaction is –

(1) Cytoplasm
(2) Matrix of mitochondria
(3) Inner mitochondrial membrane
(4) Lumen of thylakoid

27) How many of ATP produced through one turn of Kreb's cycle during oxidation of glucose ?

(1) 30 ATP
(2) 24 ATP
(3) 12 ATP
(4) 38 ATP

28) Electron transport system and ATP synthase are present in which of the following part of mitochondria ?

(1) Outer membrane
(2) Inner membrane
(3) Matrix
(4) Peri mitochondrial space

29) The movement of electrons from NADH + H^+ occurs via complex :-
(a) Complex II - Succinate dehydrogenase
(b) Complex I - NADH dehydrogenase
(c) Complex III - Cytochrome bc_1
(d) Complex IV - Cytochrome C oxidase

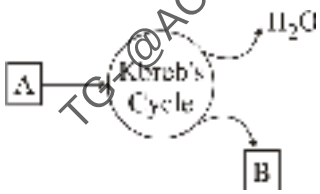
(1) a → c → d
(2) b → c → d
(3) a → b → c → d
(4) a → d → c

30) At how many steps CO_2 is/are released in aerobic respiration ?

(1) One or two
(2) Three
(3) Five
(4) Twelve

31) How many ATP are produced through ETS during oxidation of one molecule of glucose ?

(1) 8
(2) 24
(3) 6
(4) 34



32) Find A & B in the pathway given below :-

(1) A - Pyruvic acid, B - O_2
(2) A - Acetyl CoA, B - O_2
(3) A - Acetyl CoA, B - CO_2
(4) A - Glycerol, B - O_2

33) Which of the following statement is incorrect ?

(1) R.Q for carbohydrates is one
(2) R.Q for fat is less than one
(3) R.Q for organic acids is more than one
(4) R.Q for protein is more than one

34) How many ATP are produced as net gain during anaerobic respiration when pyruvate is converted into lactic acid ?

(1) One
(2) Two
(3) Three
(4) Zero

35) Select the correct statement/s :-
(a) Growth can be defined as an irreverssible permanent increase in size of an organ or its parts.
(b) Growth is accompanied by metabolic process.
(c) The swelling of piece of wood when placed in water involves imbibition.

(1) Only a and b
(2) Only b and c
(3) Only a and c
(4) a, b & c

36) Which statements is/are correct in given below ?
(I) Plant retain the capacity for unlimited growth throughout their life
(II) Vascular cambium and cork cambium appear later in life in all plants.
(III) Vascular cambium and cork cambium are permanent tissues.
(IV) Interfascicular cambium and cork cambium are not capable of dividing.

(1) Only I
(2) II and III
(3) II, III, IV
(4) I and II

37) Cells in watermelon may increase in size by upto how many times?

(1) 17,500 times
(2) 3,50,000 times
(3) 2,10,000 times
(4) 200 times

38) Root elongation at constant rate examplify :

(1) Arithmetic growth
(2) Geometrical growth
(3) Sigmoid growth
(4) Both (2) and (3)

39) Which of the following is an intrinsic factor that affects plant growth ?

(1) Light
(2) Plant growth regulators
(3) Water
(4) Temperature

40) Which statement/statements are correct for plant growth ?
(A) At cellular level growth is increase in amount of the protoplasm.
(B) Cells in watermelon may increase in size by upto 17500 times.
(C) In a dorsiventral leaf, growth is measured by increase in surface area.
Select the correct option :-

(1) A & B
(2) B & C
(3) A & C
(4) A, B & C

41) Decapitation is generally done in tea plantations, and hedge making, what could be the possible reason for this ?

(1) Auxin is generally produced by growing apices which leads to apical dominance.
(2) More lateral branches required in tea plantations and hedge making.
(3) Decapitation helps the plants in growing taller.
(4) Both 1 and 2

42) Spraying juvenile conifers with GA_s hasten the maturity period thus leading to early production of which part of the plant ?

(1) Fruit
(2) Root
(3) Seed

(4) Stem

43) ABA plays a major role in :-

(1) seed development
(2) seed maturation
(3) seed dormancy
(4) All of the above

44) Photochemical phase of photosynthesis includes :

(1) light absorption
(2) water splitting
(3) oxygen release
(4) All of the above

45) **Statement – I :** The bakanae disease of rice seedlings, was caused by a fungal pathogen *Gibberella Fujikurai*.
Statement – II : The appearance of symptoms of the bakane disease is due to active substance cytokinin.

(1) Statement I and II are correct
(2) Statement I and II are incorrect
(3) Statement I is correct and II is incorrect
(4) Statement II is correct and I is incorrect

46) **Statement-I :** The afferent nerve fibres transmit impulses from organs to the CNS.
Statement-II : Visceral nervous system is the part of the peripheral nervous system.

(1) Both statements are incorrect.
(2) Both statements are correct
(3) Only statement-I is incorrect
(4) Only statement-II is incorrect

47) **Statement-I :** Neurons can detect, receive and transmit different kind of stimuli.
Statement-II : In *Hydra* the neural system is very simple and composed of brain and ganglia.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement-I is correct
(4) Only statement-II is correct

48) Read the following statement :
(a) Axonal membrane is comparatively less permeable to K^+ ions in resting stage.
(b) Axonal membrane is highly permeable for Na^+ ions in resting stage.
(c) Ionic gradients across the resting membrane are maintained by the sodium-potassium pump.
(d) Neurons are excitable cells because their membranes are in a polarised state.

Choose the **correct** option :-

(1) a and b
(2) c and d
(3) b and c
(4) a and d

49) Corpora quadrigemina are part of :-

(1) Hind brain
(2) Fore brain
(3) Mid brain
(4) Spinal cord

50) Nerve impulses are initiated in neurons when its membrane become permeable to :-

(1) K^+ ion
(2) Ca^{2+} ion
(3) Na^+ ion
(4) Cl^- ion

51) Brain stem is made up of –

(1) Mid brain + pons + medulla oblongata
(2) Mid brain + Fore brain
(3) Fore brain + Pons
(4) Hind brain only

52) Sodium-potassium pump transport.

(1) $3Na^+$ out and $2K^+$ in
(2) $2Na^+$ out and $3K^+$ in
(3) $3Na^+$ in and $2K^+$ out
(4) $2Na^+$ in and $3K^+$ out

53) Which part of the CNS is responsible for regulating basic vital life functions such as breathing and heart beat ?

(1) Cerebellum
(2) Medulla oblongata
(3) Cerebrum
(4) Thalamus

54) "Amygdala" is a deeply associated structure present in ?

(1) Cerebellar hemisphere

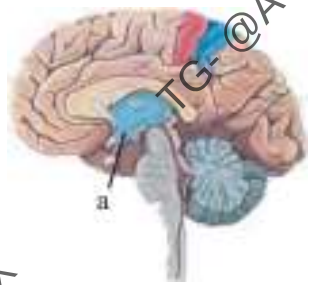
(2) Cerebral hemisphere
(3) Hypothalamus
(4) Medulla oblongata

55) Unmyelinated nerve fibre is :

(1) Not enclosed by a Schwann cell hence no myelin sheath
(2) Enclosed by a Schwann cell that does not form a myelin sheath
(3) Not enclosed by a Schwann cell but enclosed by a myelin sheath formed by Oligodendrocyes.
(4) Only found in the central neural system.

56) Memory & communications is the function of :-

(1) Hypothalamus
(2) Association area
(3) Thalamus
(4) Amygdala



57) Which one is not related with 'a' ?

(1) Regulation of sexual behaviour
(2) Neurosecretory cells
(3) Intersensory associations and balance
(4) Regulation of body temperature

58) Cerebral cortex are greyish in appearance. It is mainly due to :

(1) Bundle of Axon
(2) Corpus callosum
(3) Concentration of neuron cell bodies
(4) Presence of myelin sheath

59) How many of the following function are related with limbic system ?
(a) Motivation
(b) Rage
(c) Pleasure
(d) Excitement

(1) 3
(2) 4
(3) 2

(4) 1

60)

Glucocorticoids stimulates :-
(A) Gluconeogenesis (B) Lipolysis

(C) Proteolysis
(1) Only A and B
(2) Only A
(3) Only B
(4) All A, B and C

61) Hormone responsible for development of alveolar parts of mammary gland :

(1) Estrogen
(2) Progesterone
(3) Prolactin
(4) Oxytocin

62) Which one of the following hormones in general is not involved in sugar metabolism ?

(1) Glucagon
(2) Cortisol
(3) Aldosterone
(4) Insulin

63) Which one of the following pair of hormones, are those that can easily pass through the cell membrane of the target cell ?

(1) Insulin, glucagon
(2) Thyroxine, insulin
(3) Somatostatin, oxytocin
(4) Cortisol & Testosterone

64) Identify A, B and C from table :-

Thymosin	Thymus gland	(A)
(B)	Pancreas	Secretion inhibited by somatostatin
Cortisol	(C)	Suppress immunity during organ transplantation

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	C	B	A
(1)	Adrenal cortex	Thyroxin	Antibody formation by T-lymphocyte
(2)	Testis	Somatostatin	Ca^{+2} level maintain in blood
(3)	Adrenal medulla	Glucagon	Maturation of B lymphocyte
(4)	Adrenal cortex	Insulin	Maturation of T-lymphocyte

(1) 1

(2) 2

(3) 3

(4) 4

65) The hormone responsible for regulation of calcium and phosphorous metabolism is secreted by :–

(1) Pancreas

(2) Pineal

(3) Thymus

(4) Parathyroid

66) Which hormone is not produced in Intestine

(1) CCK

(2) Gastrin

(3) GIP

(4) Secretin

67) **Assertion :** F.S.H and L.H are released from anterior pituitary due to secretions from hypothalamus.

Reason : GnRH is released from hypothalamus and works on anterior pituitary.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

68) Which one is not a secondary messenger ?

(1) cAMP

(2) IP_3

(3) Ca^{+2}

(4) K^{+}

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69) **Statement-I :-** Aldosterone helps in the maintenance of blood pressure
Statement-II :- Aldosterone acts mainly at renal tubule and stimulate reabsorption of K^+ and water.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

70) Match List-I with List-II.

	List-I		**List-II**
(a)	Aldosterone	(i)	RBC production
(b)	Androgenic steroid	(ii)	Osmotic pressure
(c)	Cortisol	(iii)	Pupillary dilation
(d)	Catecholamines	(iv)	Growth of pubic hair

Choose the most appropriate answer from the options given below :-
(1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

71) Acromegaly is a disease caused by :

(1) Over secretion of growth hormone in childhood
(2) Over secretion of growth hormone in adulthood
(3) Under secretion of growth hormone in adulthood
(4) Deficiency of calcium and phosphorous in the diet.

72) Which one of the following hormones though synthesised elsewhere, is stored and released by the pituitary gland ?

(1) Melanocyte stimulating hormone
(2) Antidiuretic hormone
(3) Luteinizing hormone
(4) Prolactin

73) Parathormone (parathyroid hormone) act on bone and stimulate which of the following action ?

(1) Carbohydrate metabolism
(2) Bone resorption
(3) Bone formation
(4) Fat metabolism

74) During muscles contraction ATP bind with :

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(1) Actin
(2) Troponin
(3) Myosin
(4) Tropomyosin

75) Cardiac muscles are :-

(1) Involuntary, Unbranched, Striated
(2) Voluntary, Unbranched, Nonstriated
(3) Involuntary, Branched, Striated
(4) Voluntary, Branched, Nonstriated

76) Match List-I with List-II :-

	List-I		**List-II**
(A)	Cranium	(I)	Gliding joint
(B)	Between adjacent vertebrae	(II)	Pivot joint
(C)	Between carpals	(III)	Fibrous joint
(D)	Between atlas and axis	(IV)	Cartilagenous joint

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

77) Which of following is a wrong statement, regarding muscle contraction ?

(1) I-band gets reduced
(2) A-band retain the same length
(3) H-zone remains unchanged
(4) Sarcomere shortens

78) Incorrect about skeleton muscles is :-

(1) They are syncytium.
(2) They have light and dark bands.
(3) Based on appearance skeleton muscles are unstriated
(4) They are voluntary in nature

79) Which joint is similar to elbow joint ?

(1) Joint between humerus and scapula
(2) Joint between atlas and axis
(3) Joint between Femur, Tibia and patella

(4) Joint between Radius and Ulna

80) Joint between carpal and metacarpals of thumb :

(1) Saddle
(2) Hinge
(3) Pivot
(4) Ball and socket

81) Pelvic girdle consists of :-

(1) 1 coxal bone
(2) 2 coxal bone
(3) 3 coxal bone
(4) 6 coxal bone

82) Acromion process present on ?

(1) Scapula
(2) Clavicle
(3) Humerus
(4) Femur

83) The example of fibrous joint is ?

(1) Vertebral column
(2) Cranium
(3) Knee
(4) Elbow

84) Which of the following is incorrect match.

(1) Ear ossicles- Malleus, Incus and stapes
(2) Floating ribs-8th, 9th and 10th
(3) Pectoral girdle- Scapula and clavicle
(4) Pelvic girdle- Ilium, ischium and pubis

85) Which statement is not correct about osteoporosis ?

(1) In this, bone mass is decreased
(2) It is an age related disorder
(3) Increased levels of estrogen is a common cause of it.
(4) In this, chances of fractures are increased.

86) In voluntary muscle "Fascia" term is used for

TG- @ACADMIX

(1) A collagenous connective tissue by which muscle bundles are held together
(2) A elastic protein which present between I-Bands
(3) A calcium rich structure in sarcoplasm
(4) Central gap between thick filaments.

87) Which structure of muscle is an active ATPase enzyme.

(1) Globular head of meromyosin
(2) Globular head of actin
(3) 'F' - actin helically wound to each other
(4) Complex troponin protein

88) Character of white muscle fibres is -

(1) anaerobic process for energy
(2) amount of Sarcoplamic reticulum is high
(3) Few mitochondria
(4) All of these

89) What is the main reason for tetany?

(1) High level of Ca^{++} ion in bones.
(2) Low level of Ca^{++} ion in bones.
(3) High level of Ca^{++} ion body fluids.
(4) Low level of Ca^{++} ion in body fluids.

90) Which of the following is a genetic disorder?

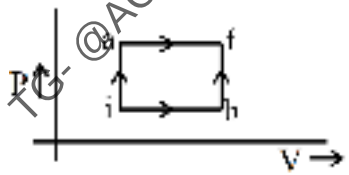
(1) Myasthenia gravis
(2) Muscular dystrophy
(3) Tetany
(4) Gout

PHYSICS

1) Oxygen and hydrogen both are at same temperature T. The ratio of mean kinetic energy of oxygen molecules to that of the hydrogen molecules will be :

(1) 16 : 1
(2) 1 : 1
(3) 4 : 1
(4) 1 : 4

2) When a system is taken from state 'i' to state 'f' along path 'iaf', it is found that Q = 50 cal and W = 20 cal. Along the path 'ibf' Q = 36 cal, then 'W' along this path (Q is heat supplied to gas and W is



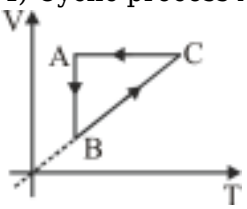
work done by gas) :-

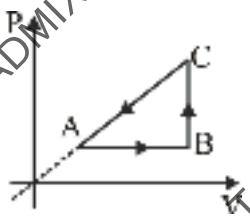
(1) 6 cal
(2) 16 cal
(3) 66 cal
(4) 14 cal

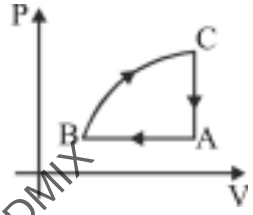
3) A carnot engine operates with source at 127 °C and sink at 27 °C. If source supplies 40 kJ of heat energy, the work done by engine is :-

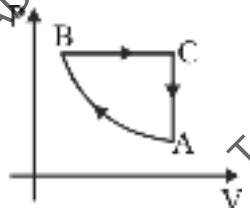
(1) 30 kJ
(2) 10 kJ
(3) 4 kJ
(4) 1 kJ

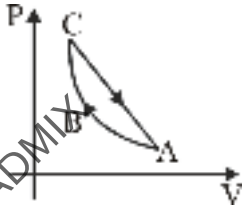
4) Cyclic process ABCA is shown in V–T diagram corresponding process on the P–V diagram is :



(1)


(2)


(3)


(4)


5) A gas is contained in a container at pressure 10 kPa. When gas is expanded isothermally from

volume $9m^3$ to $27m^3$ work done will be.

(1) 27×10^3 (n 3)

(2) 9×10^4 (n 3)

(3) $9 \times 10^3 \left(\ell n \frac{1}{3}\right)$

(4) $\frac{27 \times 10^3}{\left(\ell n \frac{1}{3}\right)}$

6) Two spheres are made of same metal and have same mass. One is solid and other is hollow. When heated to the same temperature, which of the following statements is correct about the percentage increase in their diameter?

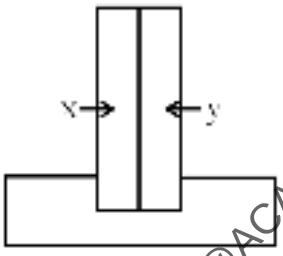
(1) It will be more for hollow sphere

(2) It will be more for solid sphere

(3) It will be same for both spheres

(4) It may be more or less for any sphere depending upon the ratio of the diameters of the two spheres

7) A bimetallic strip consists of metals x and y. It is mounted rigidly at the base as shown in figure. The metal x has a higher coefficient of expansion as compared to that for metal y. When the



bimetallic strip is placed in cold bath?

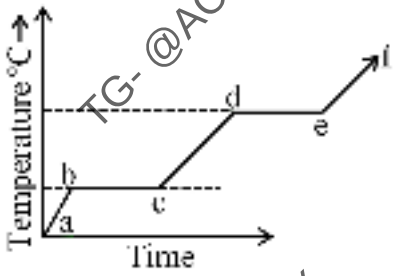
(1) It will bend toward the right

(2) It will bend toward the left

(3) It will not bend but shrink

(4) It will neither bend nor shrink

8) The following figure represents the temperature versus time plot for a given amount of a substance when heat energy is supplied to it at a fixed rate and at a constant pressure. Which part of



the below plot represents a phase change?

(1) a to b and e to f

(2) b to c and c to d

(3) d to e and e to f

(4) b to c and d to e

9) A body cools from 60°C to 50°C in 10 minutes. If the room temperature is 25°C and assuming Newton's Law of cooling to hold good, the temperature of the body at the end of the next 10 minutes will be :

(1) 38.5°C
(2) 40°C
(3) 42.85°C
(4) 45°C

10) Gravitation is necessary for -

(1) For conduction
(2) For natural convection
(3) For radiation
(4) None of the above

11) Total radiation energy incident on body is 400 Joule. If 30% of incident radiation reflected back and 120 J is absorbed by body. Then find out transmittive power in percentage.

(1) 60%
(2) 40%
(3) 50%
(4) None

12)

For a black body λ_{max} is 300 Å at 727°C. Find λ_{max} for same black body at 1227°C.

(1) 500 Å
(2) 1000 Å
(3) 1500 Å
(4) 200 Å

13) A black body at a temperature of 227°C radiates heat at the rate 7 cal/cm^{2}sec. At temperature of 727°C, the rate of heat radiated in the same units will be :-

(1) 50
(2) 112
(3) 80
(4) 60

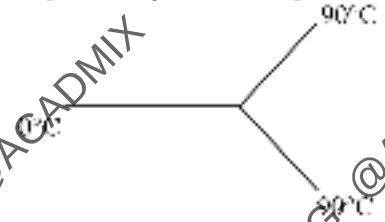
14) Under steady state the temperature of a body :

(1) Increases with time
(2) Decreases with time

(3) Doesn't change with time and is same at all points of the body

(4) Doesn't change with time but is different at different cross - sections of the body.

15) Three rods made of the same material and having the same cross - section have been joined as shown in figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of the junction of the three rods will be :



(1) 45°C

(2) 60°C

(3) 30°C

(4) 20°C

16) A mass m executes simple harmonic motion with frequency n when suspended from a weightless spring. The spring is cut into two equal parts and an another mass 2m is suspended from one of the part. The frequency of oscillation of this mass 2 m is :-

(1) n

(2) 2n

(3) $\frac{n}{\sqrt{2}}$

(4) $n(2)^{2}$

17) Infinite springs with force constant k, 2k, 4k, 8k ..... respectively are connected in series combination. The effective force constant will be :-

(1) 2k

(2) k

(3) k/2

(4) 4k

18) Two simple harmonic motions represented by $y_1 = 4\sin(4\pi t + \frac{\pi}{2})$ and $y_2 = 3\cos(4\pi t)$ superimpose over each other then the resultant amplitude is (all quantities are in SI units) :-

(1) 7 m

(2) 1 m

(3) $(2 + \sqrt{3})$ m

(4) $(2 - \sqrt{3})$ m

19) The angle made by the string of a simple pendulum with vertical depends upon time as $\theta = \frac{\pi}{90}\sin\pi t$. Find the length (in metre) of the pendulum. (If $g = \pi^2 m/s^2$).

(1) 1
(2) 2
(3) 3
(4) 4

20) For S.H.M. of a particle, amplitude is 6cm. At a certain instant its potential energy is half of the total energy then the distance of particle at this instant from its mean position is :-

(1) 3 cm
(2) 4.2 cm
(3) 5.8 cm
(4) 6 cm

21) A simple pendulum whose time period is 2 sec at the surface of earth is now suspended in a satellite orbiting around the earth at a height 3R, where R is the radius of earth. The time period of this pendulum is :-

(1) zero
(2) $2\sqrt{3}\,\text{sec}$
(3) 4 sec
(4) infinite

22) The time period of a simple pendulum is 2 sec. If its length becomes 16 times, then its time period becomes :-

(1) 16 sec
(2) 12 sec
(3) 8 sec
(4) 4 sec

23) The kinetic energy of a particle executing S.H.M is 16 J, when it is in its mean position. If the amplitude of oscillation is 25 cm and the mass of the particle is 5.12 kg, the time period of its oscillation is :-

(1) $\frac{\pi}{5}$ sec
(2) 2π sec.
(3) 20π sec
(4) 5π sec

24) The particle is executing SHM on a line 4 cm long. If its velocity at mean position is 12cm/s, its frequency (in hertz) will be -

(1) $\frac{2\pi}{3}$
(2) $\frac{3}{2\pi}$

(3) $\frac{\pi}{3}$

(4) $\frac{3}{\pi}$

25) Which of the following relationship between acceleration 'a' and displacement 'x' of a particle represent the simple harmonic motion :-

(1) $a = 2x$

(2) $a = -25x$

(3) $a = 4x^2$

(4) $a = -100x^2$

26) The displacement of a particle performing linear simple harmonic motion of amplitude A in one time period is :-

(1) A

(2) 2A

(3) zero

(4) 3A

27) For S.H.M match the following

	Column-I		**Column-II**
A	Displacement and velocity	P	Phase difference is zero
B	Displacement and acceleration	Q	Phase difference is π/2
C	Velocity and acceleration	R	Phase difference is π

(1) A → P, B → Q, C → R

(2) A → Q, B → R, C → Q

(3) A → Q, B → P, C → R

(4) A → R, B → P, C → R

28) **Assertion** :- If a pendulum is suspended in a lift and lift accelerates upwards, then its time period will decrease as compared to that when it is in stationary lift. **Reason** :- Effective value of acceleration due to gravity will be $g_{eff} = g + a$ in this case

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

29) The equation of a stationary wave is $y = 0.8 \cos\left(\frac{\pi x}{20}\right) \sin 200\pi t$, where x and y are in cm and t is in second. The separation between consecutive nodes will be :-

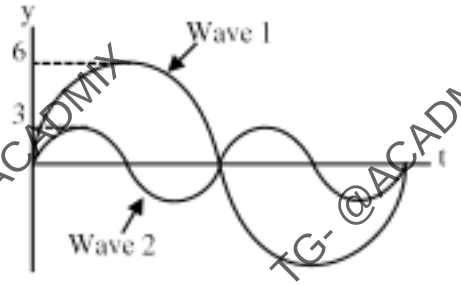
(1) 20 cm

(2) 10 cm
(3) 40 cm
(4) 30 cm

30) The fundamental frequency of a closed organ pipe is 220 Hz. If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is :-

(1) 220 Hz
(2) 440 Hz
(3) 880 Hz
(4) 1760 Hz

31) Two waves propagating in same medium are represented by y-t curves in the figure. The ratio of their average intensities is :-

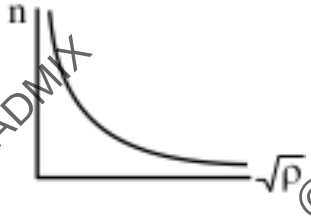


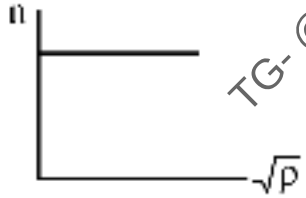
(1) 4
(2) 1/4
(3) 9
(4) 1

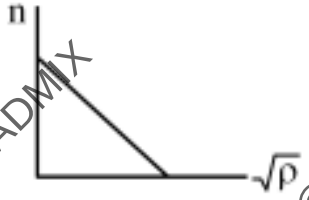
32) When the string of a sonometer of length L between the bridges vibrates in the second overtone, the amplitude of vibration is maximum at :-

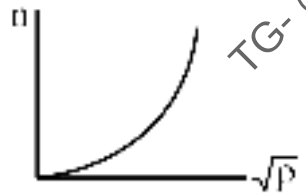
(1) $\frac{L}{2}$
(2) $\frac{L}{4}$ and $\frac{3L}{4}$
(3) $\frac{L}{6}$ and $\frac{3L}{6}$
(4) $\frac{L}{8}$, $\frac{3L}{8}$ and $\frac{5L}{8}$

33) The correct graph between the frequency (n) of wave in a stretched wire and square root of density (ρ) of a wire, keeping its length, radius and tension constant, is :-

(1) 

(2) 

(3) 

(4) 

34) What is the phase difference, at a given instant of time, between two particles 25m apart, when the wave y = 0.03 sin π(2t – 0.01 x) travels in a medium (x, y is in m and t in second) ?

(1) $\frac{\pi}{8}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$

(4) π

35) A string of linear mass density 4g/cm is vibrating according to equation y = A sin (120πt) cos $\left(\frac{2\pi}{5}x\right)$, where x and y are in cm. The tension in the string is :-

(1) 3.6 N

(2) 36 N

(3) 7.2 N

(4) 72 N

36) A tuning fork T_1 produces 4 beats with tuning fork T_2 of frequency 256 Hz. When T_1 is filed, beats are found to occur at shorter intervals. What was the original frequency of tuning fork T_1 ?

(1) 260 Hz

(2) 252 Hz

(3) 254 Hz

(4) 258 Hz

37) An astronaut cannot hear his companion at the surface of the moon because :-

(1) Produced frequencies are above the audible frequencies

(2) There is no medium for sound propagation

(3) Temperature is too low during night and high during day
(4) There are too many craters on the surface of the moon

38) Which of the following properties of a wave does not change with change in medium ?

(1) Frequency
(2) Wavelength
(3) Velocity
(4) Amplitude

39) Two waves of intensities I_1 and I_2 pass through a region at the same time in the same direction superimpose on each other, the sum of the maximum and minimum intensities is :-

(1) $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$
(2) $2(I_1 + I_2)$
(3) $I_1 + I_2$
(4) $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

40) A wave is represented by the relation $y = a_0 \sin 2\pi (nt – x/\lambda)$ if the maximum particle velocity is three times the wave velocity, the wavelength λ of the wave is :-

(1) $\frac{\pi a_0}{3}$
(2) $\frac{2\pi a_0}{3}$
(3) πa_0
(4) $\frac{\pi a_0}{2}$

41) The equation of a travelling wave is
y = 60 cos(1800t – 6x)
where x and y is in metre and t is in sec. The ratio of maximum particle velocity to velocity of wave propagation is :-

(1) 3.6
(2) 36
(3) 360
(4) 0.36

42) 25 tuning forks are arranged in series in the order of decreasing frequency. Any two successive forks produce 3 beats/sec. If the frequency of the first tuning fork is the octave of the last fork, then the frequency of the 21st fork is :-

(1) 72 Hz
(2) 78 Hz
(3) 84 Hz

(4) 87 Hz

43) An open organ pipe 40 cm long and a closed organ pipe 31 cm long, both having same diameter, are producing their first overtone, and these are in unison. The end correction of closed organ pipe is :-

(1) 1 cm
(2) 2 cm
(3) 1.5 cm
(4) 3 cm

44) Which of the following changes with time when damping is considered for a S.H.M. : -
(A) Time period (B) Angular frequency
(C) Amplitude (D) Energy

(1) A & B
(2) C & D
(3) B & C
(4) A, B, C & D

45) If in a resonance tube a oil of density higher than that of water is used then the resonance frequency would be :-

(1) Increased
(2) Decreased
(3) Slightly increased
(4) Remain the same

CHEMISTRY

1) Number of π bonds in ethylene chloride.

(1) 0
(2) 1
(3) 2
(4) 3

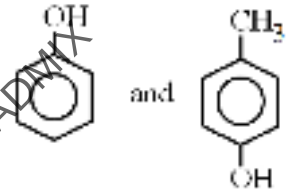
2) The number of ether isomers represented by formula $C_4H_{10}O$ is (only structural) :

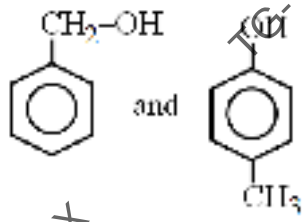
(1) 4
(2) 3
(3) 2
(4) 7

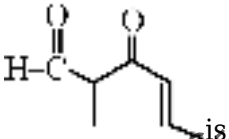
3) Which of the following pair are isomers ?

(1) $CH_3\ CH_2\ OH$ and $CH_3\ \overset{OH}{\underset{|}{C}}H\ CH_3$

(2) $CH_3\ \overset{O}{\overset{\|}{C}}\ CH_3$ and $CH_3–CH_2–O–CH_3$

(3)  and

(4)  and

4) The IUPAC name of the compound  is.

(1) 5-formylhex-2-en-3-one

(2) 5-methyl-4-oxohex-2-en-5-al

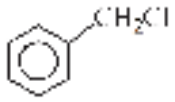
(3) 3-keto-2-methylhex-5-enal

(4) 3-keto-2-methylhex-4-enal

5) Which of the following IUPAC name is correct:

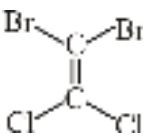
(1) CCl_4 carbon terachloride

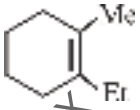
(2) CH_2Cl_2 methylene chloride

(3)  chlorophenylmethane

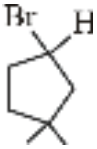
(4) $CH_2 = CH – CH_2 – Br$ Allyl bromide

6) Which of the following molecule can show geometrical isomerism ?

(1) 

(2) 

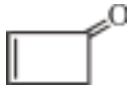
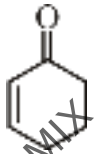
(3) 

(4) 

7) **Assertion :** Staggered conformation of ethane is most stable while eclipsed conformation is least stable.
Reason : Staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain

(1) Assertion is correct, reason is correct, reason is a correct explanation for assertion
(2) Assertion is correct, reason is correct, reasion is not a correct explanation for assertion
(3) Assertion is correct, reason is incorrect.
(4) Assertion is incorrect, reason is correct.

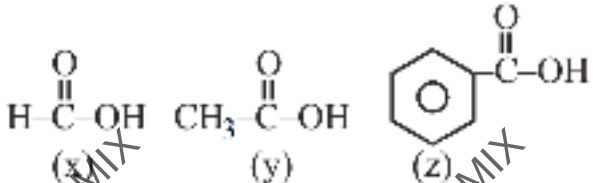
8) Which of the following compound will have most stable enol ?

(1) 
(2) 
(3) 
(4) 

9) In which of the following compound tautomerism takes place : 

(1) Both (I) and (II)
(2) Both (II) and (III)
(3) (III) Only
(4) Both (I) and (III)

10) Arrange the following acids in decreasing order of acidic strength:-



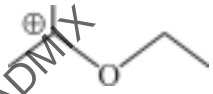
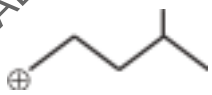
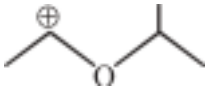
(1) x > z > y

(2) x > y > z
(3) z > x > y
(4) z > y > x

11) Correct order of –I effect is :-

(1) –CN > –COOH > $–NO_2$
(2) –COOH > –CN > $–NO_2$
(3) $–NO_2$ > –COOH > –CN
(4) $–NO_2$ > –CN > –COOH

12) Most stable carbocation is :-

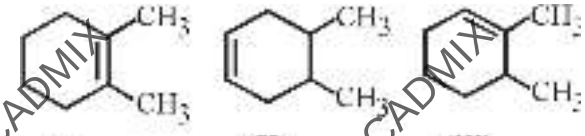
(1) 
(2) 
(3) 
(4) 

13) Consider the following compounds / species.



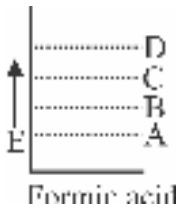
The number of compounds which obey Huckel's rule is :-

(1) 6
(2) 4
(3) 5
(4) 3



14) Which of following orders is correct for heat of hydrogenation of these compounds ?

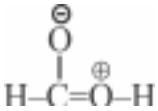
(1) I > II > III
(2) III > II > I
(3) II > III > I
(4) III > I > II

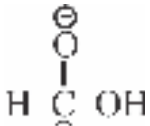


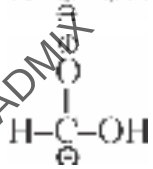
15) Formic acid

A, B, C, D are resonating structures of formic acid. Structure of B is:

(1) 

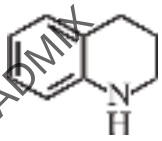
(2) 

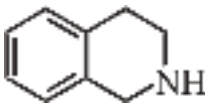
(3) 

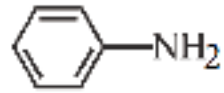
(4) 

16) The compound that is easiest to protonate is ?

(1) 

(2) 

(3) 

(4) 

17) Correct decreasing order of basic strength :-

(A) $(C_2H_5)_2NH$ (B) $C_2H_5NH_2$

(C) $(C_2H_5)_3N$ (D) 

(1) A > B > C > D

(2) D > A > C > B

(3) D > C > B > A

(4) A > C > B > D

18) Given below are two statements

Statement-I : The energy of actual structure of the molecule (the resonance hybrid) is lower than

that of any of the canonical structure.
Statement-II : The difference in energy between the actual structure and the highest energy resonance structure is called the resonance stabilisation energy.

(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I incorrect and statement-II is correct.
(4) Statement-I correct and statement-II is incorrect.

19) Match List - I with List - II.

	List-I (Compounds)		**List-II (pK_a-Value)**
(a)	o-Cresol	(i)	10
(b)	m-Cresol	(ii)	15.9
(c)	Ethanol	(iii)	10.1
(d)	Phenol	(iv)	10.2

(1) a-(iv), b-(iii), c-(ii), d(i)
(2) a-(iv), b-(ii), c-(iii), d(i)
(3) a-(iii), b-(iv), c-(ii), d(i)
(4) a-(ii), b-(iv), c-(i), d(iii)

20) A liquid compound (x) can be purified by steam distillation only if it is

(1) Steam volatile, immiscible with water
(2) Not steam volatile, miscible with water
(3) Steam volatile, miscible with water
(4) Not steam volatile, immiscible with water

21) Match List-I with List-II.

	List-I		**List-II**
(a)	Separation of aniline-water mixture	(i)	Fractional distillation
(b)	Separation of aniline-chloroform mixture	(ii)	Distillation under reduced pressure
(c)	Separation of Glycerol from spent-lye	(iii)	Simple distillation
(d)	Separation of different fractions of crude oil	(iv)	Steam distillation

Choose the correct answer from the options given below :-

(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

22) 0.1 gm of organic compound was analysed by Kjeldahl's method. In analysis produced NH_3 absorbed in 30 ml $\frac{M}{10}$ H_2SO_4. The remaining acid required 20 ml N/10 NaOH for neutralisation. Calculate percentage of nitrogen in organic compound :-

(1) 28 %

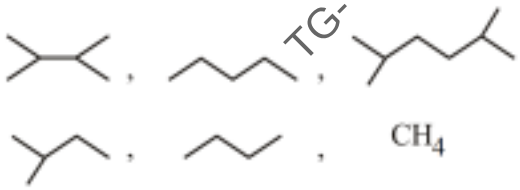
(2) 56 %

(3) 35 %

(4) 70 %

23)

How many compound can be obtained in good yield by wurtz reaction.



(1) 6

(2) 3

(3) 4

(4) 5

24) Match column I with column II

	Column-I (Reaction)		**Column-II (Product)**
I.	$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$	P.	$2CH_3OH$
II.	$(CH_3)_3CH \xrightarrow{KMnO_4}$	Q.	$2CH_3COOH + 2H_2O$
III.	$2CH_4 + O_2 \xrightarrow{Cu/523K/100atm.}$	R.	$HCHO + H_2O$
IV.	$2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$	S.	$(CH_3)_3C–OH$

(1) I–R, II–P, III–Q, IV–S

(2) I–P, II–S, III–R, IV–Q

(3) I–R, II–S, III–P, IV–Q

(4) I–R, II–P, III–S, IV–Q

25) Match the reactions in column-I with their types in column-II

Column-I (Reaction)		Column-II (Types)	
(a)	$CH_3–CH_2–OH \xrightarrow[\Delta]{H^+}$	(p)	Electrophilic addition reaction
(b)	$CH_3–CH=CH_2 \xrightarrow[\text{Peroxide}]{HBr}$	(q)	Free radical addition reaction
(c)		(r)	Elimination reaction
(d)		(s)	Electrophilic substitution reaction

(1) a-r, b-q, c-s, d-p

(2) a-q, b-s, c-r, d-p

(3) a-p, b-r, c-s, d-q

(4) a-s, b-p, c-q, d-r

26) Which of the following alcohol on dehydration with conc. H_2SO_4 at 440 K will give isobutylene.

(I) Isobutyl alcohol

(II) Tert. butyl alcohol

(III) n-butyl alcohol

(IV) Isopropyl alcohol

(1) I, III

(2) II, III

(3) III, IV

(4) I, II

27) **Statement-I:** The presence of weaker π bonds make alkenes less stable than alkanes.
Statement-II: The strength of the double bond is greater than that of carbon-carbon single bond.

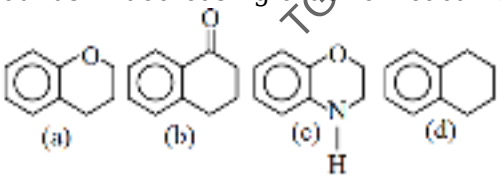
(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

28) Rank the following compounds in decreasing order of reactivity towards electrophilic aromatic substitution reaction (ESR).



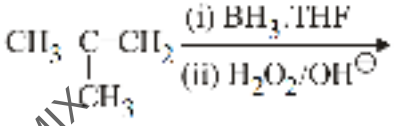
(1) (c) > (a) > (b) > (d)

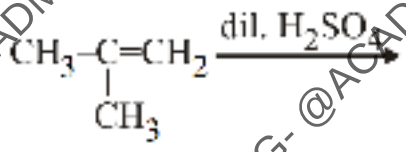
(2) (c) > (a) > (d) > (b)

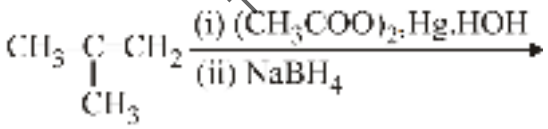
(3) (d) > (c) > (b) > (a)

(4) (a) > (c) > (d) > (b)

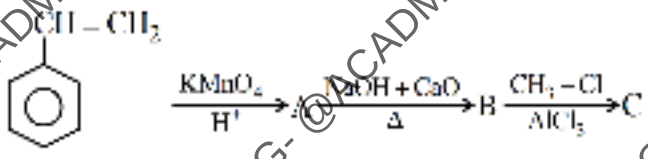
29) Primary alcohol can be formed as major product by :-

(1) 

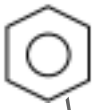
(2) 

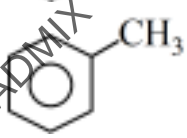
(3) 

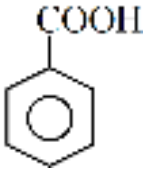
(4) All of these

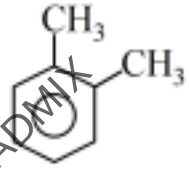
30) 

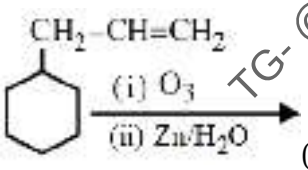
Identify C ?

(1) 

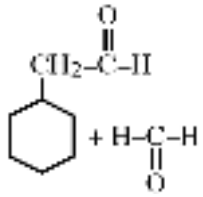
(2) 

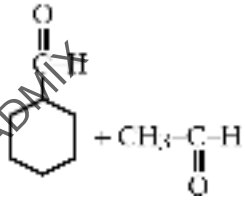
(3) 

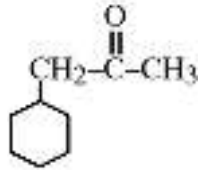
(4) 

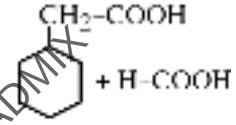
31)  (Major product) :-

Major product will be :-

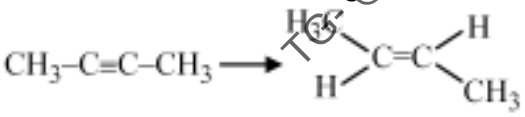
(1) 

(2) 

(3) 

(4) 

32) The most suitable reagent for the following conversion is :-

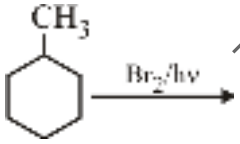


(1) Na/liquid NH_3

(2) H_2, Pd/C, Quinoline

(3) Zn/HCl

(4) Hg^{+2}/H^+, H_2O

33)  Monobromo derivatives.

The number of possible monobromo product is (excluding stereoisomers)

(1) 4

(2) 5

(3) 8

(4) 10

34) Which of the following alkyl bromides will eliminate HBr at fastest rate in elimination reaction.

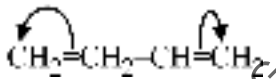
(1) Ethyl bromide

(2) Propyl bromide

(3) Isopropyl bromide

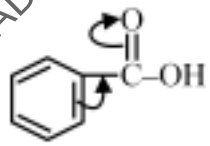
(4) t-Butyl bromide

35) Which of the following is correct representation of delocalisation of electrons during resonance :

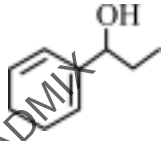
(1) 

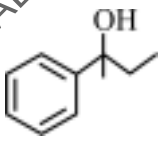
(2) 

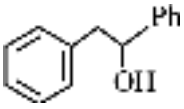
(3) 

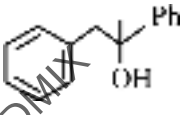
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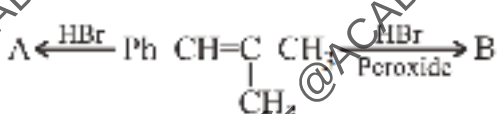
36) Which of the following alcohol is benzylic as well as secondary :-

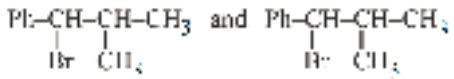
(1) 

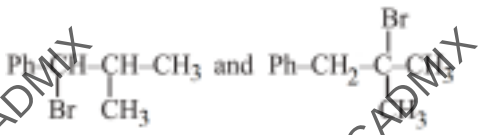
(2) 

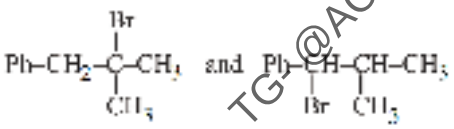
(3) 

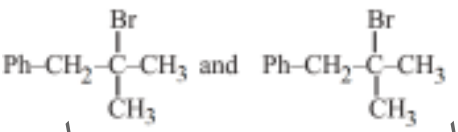
(4) 

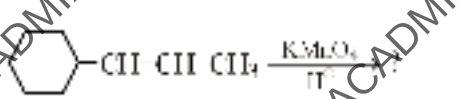
37)  , A and B is :-

(1) 

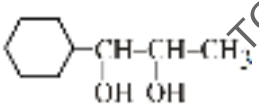
(2) 

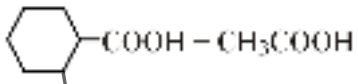
(3) 

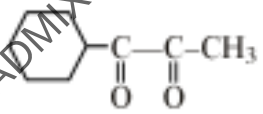
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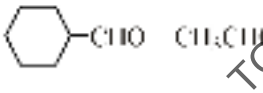
38) 

What are product in above reaction ?

(1) 

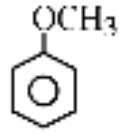
(2) 

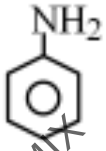
(3) 

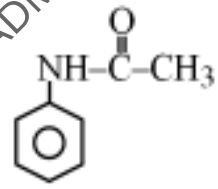
(4) 

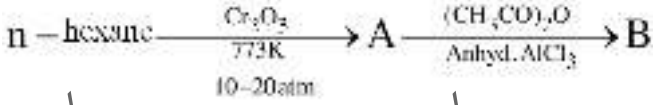
39) Which of the compound will not undergo Friedel craft's reaction -

(1) 

(2) 

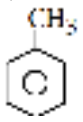
(3) 

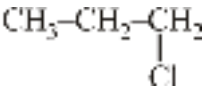
(4) 

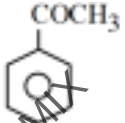


40)

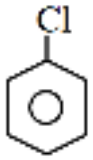
Product B is

(1) 

(2) 

(3) 

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(4) 

41) Which of the following compound will not give a precipitate with Tollen's, reagent :-

(1) Ethyne
(2) 1-Butyne
(3) 2-Pentyne
(4) 1-Pentyne

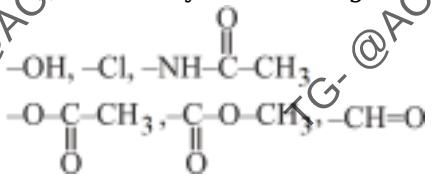
42) Given below are two statements :
Statement-I :- The boiling point of three isomeric pentanes follows the order
n-pentane > isopentane > neopentane
Statement-II :- When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.
In the light of the above statements, choose the most appropriate answer from the option given below :

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

43)  'A' (major product).
Number of σ bonds in 'A' is :-

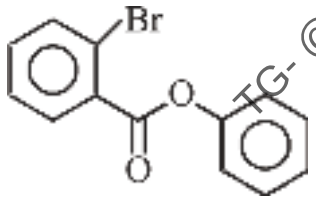
(1) 6
(2) 12
(3) 18
(4) 24

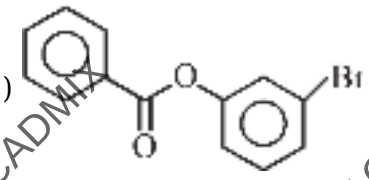
44) How many of following are ortho-para directing groups in electrophilic substitution reaction

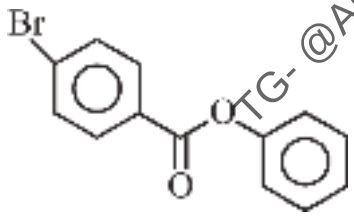


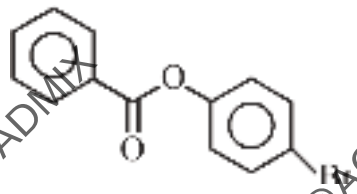
(1) 4
(2) 6
(3) 5
(4) 3

45) The major product formed on monobromination of phenylbenzoate is :-

(1) 

(2) 

(3) 

(4) 

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ANSWER KEYS

BIOLOGY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	2	4	1	1	1	3	2	2	1	4	1	4	2	3	3	3	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	2	4	2	2	3	2	2	2	4	3	4	4	4	1	2	1	2	3
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	3	4	4	3	2	3	2	3	3	1	1	2	2	2	2	3	3	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	2	3	4	4	4	2	1	4	2	1	2	2	2	3	3	2	3	3	3	1
Q.	81	82	83	84	85	86	87	88	89	90										
A.	2	1	2	2	3	1	1	4	4	2										

PHYSICS

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	2	3	2	3	2	4	3	2	2	4	2	4	2	1	3	1	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	3	1	4	2	3	2	1	1	3	4	3	1	2	1	1	2	1	2	2
Q.	131	132	133	134	135															
A.	3	3	2	2	4															

CHEMISTRY

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
A.	1	2	4	4	3	3	1	1	3	1	4	2	2	3	2	3	4	4	1	1
Q.	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
A.	3	2	2	3	1	4	1	2	1	2	1	1	2	4	4	3	2	2	3	3
Q.	176	177	178	179	180															
A.	3	1	3	1	4															

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SOLUTIONS

BIOLOGY

1) NCERT-XI, Pg. # 133

2) NCERT-XI, Pg. # 134

3)

NCERT-XI, Pg. # 134, 135

4) NCERT Pg. No. # 136

5) NCERT-XI, Pg. # 135

6) NCERT-XI, Pg. # 137, Fig. 11.3(b)

7) NCERT-XI, Pg. # 137, 138

8)

NCERT-XI, Pg. # 136, 137, 138

9) NCERT-XI, Pg. # 137, 138

10) NCERT-XI, Pg. # 140, 141

11) NCERT-XI, Pg. # 143, 144, 145, 146

12) NCERT-XI, Pg. # 145, 147

13) NCERT-XI, Pg. # 147, 148

14)

NCERT-XI, Pg. # 145, 147

15) NCERT, Pg # 150

16) NCERT Pg. No. # 149

17) NCERT Pg. # 149

18) NCERT Pg # 154

19) NCERT-XI, Pg. # 154, 155

20) NCERT Pg. # 155

21) NCERT Pg. # 155,156,157,158

22) NCERT Pg. # 155

23) NCERT, Pg. # 156

24) NCERT Pg. # 156

25) NCERT Pg. # 156

26) NCERT, Pg. # 158

27) NCERT-XI, Pg. # 159

28) NCERT-XI, Pg. # 159, 160

29) NCERT-XI, Pg. # 160

30)

NCERT-XI, Pg. # 158, 159

31) NCERT-XI, Pg. # 158, 159, 160

32) NCERT, Pg. # 159

33) NCERT Pg # 163-164

34) NCERT Pg. # 157

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35) NCERT, Pg. # 167

36)

NCERT, Pg. # 168

37) NCERT Pg # 168

38)

NCERT-XI, Pg. # 170

39) NCERT Pg. # 174

40) NCERT Pg. # 168

41) NCERT Pg. # 175

42) NCERT Pg. # 176

43) NCERT, Pg. # 178

44) NCERT Pg. No. # 138

45) NCERT, Pg. # 175

46) NCERT Pg. No. # 231

47) NCERT Pg. No. # 231

48) NCERT Pg. No. # 232

49) NCERT Pg. No. # 236

50) NCERT-XI, Pg. # 317, Para (21.3.1)

51) NCERT, Pg. # 321

53) NCERT Pg. # 236

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54) NCERT Pg. # 321

55)

NCERT-XI Pg.#317

56) NCERT Pg. # 321

57) NCERT Pg. # 320

58) NCERT_Page No.321

60) NCERT Pg. # 335, 22.2.7

61) NCERT XI Page No. 337

62)

NCERT XI, Pg. # 337

63) NCERT - XI, Pg # 338

64) NCERT XI pg.# 334, 336

65) NCERT-XI, Pg. # 334

66) NCERT-XI, Pg. # 338

67) NCERT XI Page No. # 332

68) NCERT-XI, Pg. # 248

69) NCERT Pg. # 244

70) NCERT Pg. # 244, 245

71)

NCERT XI Pg. # 241

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72)

SOLUTION/EXPLANATION/CALCULATION:

The hormone that is synthesized elsewhere but stored and released by the master gland (pituitary gland) is Antidiuretic hormone (ADH).

ADH is synthesized in the hypothalamus but stored in the posterior pituitary gland. It is released when the body needs to conserve water, such as during dehydration or low blood pressure.

73) NCERT Pg.# 335

75) NCERT Pg. # 219

76) NCERT Pg. # 227

77) NCERT XIth Pg # 307

78) NCERT-XI Pg. # 304

79)

NCERT-XI Pg#227

80) NCERT-XI, Pg. # 312, Para (20.4)

81) NCERT Pg. # 311

82)

NCERT Pg. # 311

83) NCERT Pg. # 312

84) NCERT Pg. # 309-311

85)

NCERT (XI) Pg. # 312

86) NCERT Pg. No. # 219

87) NCERT Pg. No. # 221

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88) NCERT Pg. No. # 229

89) NCERT Pg. No. # 227

90) NCERT Pg. No. # 227

PHYSICS

91) $\boxed{KE = \frac{f}{2}nRT}$

92)

From FLOT
$Q = \Delta U + W$

For path 'iaf'
$50 = \Delta U + 20$
$\Delta U = 30$ Cal

For path 'ibf'
$\Delta U = 30$ Cal.
$Q = \Delta U + W$
$W = 36 - 30$
$= 6$ cal

93) $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{400} = \frac{1}{4}$

$\because \eta = \frac{\text{Work done by engine}}{\text{Heat supplied by source}}$

$\frac{1}{4} = \frac{W}{40}$

$W = 10$ kJ

94)

AB temperature constant V ↓ so P ↑
in BC $T \propto V$, P = constant V ↑
in CA volume constant T ↓ P ↓

95) $W = P_1V_1 \ln \frac{V_2}{V_1}$

$W = 10 \times 10^3 \times 9 \times \ln \frac{27}{9}$

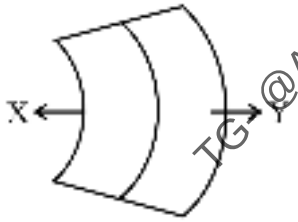
96) For both spheres:

$$\frac{\Delta V}{V} = \gamma \Delta T \Rightarrow \frac{\Delta V}{V} = \frac{3\Delta d}{d} = \gamma \Delta T$$

Same for both spheres

97)

On cooling following happens



98)

During state change heat supplied does not increase temperature. So b to c & d to e are state change process.

99) $\because \frac{60-50}{10} = k(55-25)$ ... (1)

$$\frac{50-T}{10} = k\left(\frac{50+T}{2} - 25\right) \quad \text{... (2)}$$

On solving - T = 42.85°C

100) In convection hot fluid goes upward due to low density and gravitation is necessary for it.

101) $Q = Q_r + Q_a + Q_t$

$400 = 120 + 120 + Q_t$

$Q_t = 160$

$$= \frac{Q_t}{Q} \times 100\%$$

$$\frac{160}{400} \times 100\%$$

= 40%

102) $\lambda_m \propto \frac{1}{T}$

$$\frac{\lambda m_1}{\lambda m_2} = \frac{T_2}{T_1}$$

103) $E = \sigma A T^4$

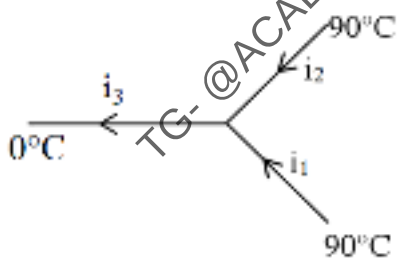
$$\frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4$$

$T_1 = 227 + 273 = 500$ K

$\frac{E_2}{7} = \left(\frac{1000}{500}\right)^4 = 16$ $T_2 = 727 + 273 = 1000$ K

$E_2 = 112 cal/cm^2 sec.$

104) At steady state different parts of conductor are at different temperatures but temperature of each part remains constant with time.



105)

$\because i_3 = i_1 + i_2$

$\frac{T-0}{R} = \frac{90-T}{R} + \frac{90-T}{R}$

$3T = 180°$

$T = 60°C$

106) $n = \frac{1}{T} = \frac{\sqrt{k}}{2\pi\sqrt{m}}$ $n \propto \sqrt{\frac{k}{m}}$

$\frac{n'}{n} = \sqrt{\frac{k'}{m'}} \times \sqrt{\frac{m}{k}} = \sqrt{\frac{2k}{2m}} \times \sqrt{\frac{m}{k}} = 1$

$n' = n$

107) $\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + ...$

$\frac{1}{k_{eq}} = \frac{1}{k}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + ...\right)$

$\frac{1}{k_{eq}} = \frac{1}{k}\left(\frac{1}{1-1/2}\right) = \frac{2}{k}$

$k_{eq} = \frac{k}{2}$

108) $y_1 = 4 \sin(4\pi t + \pi/2) = 4\cos 4\pi t$

$y_2 = 3 \cos 4\pi t$

Phase difference between both is zero, both are along the same line

$A_{res.} = 4 + 3 = 7$ m

109) $T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}}$

$\frac{\ell}{g} = \frac{1}{\pi^2}$

ℓ = 1 m

110) P.E = $\frac{T.E}{2}$

$\frac{1}{2}kx^2 = \frac{\frac{1}{2}ka^2}{2}$

$x = \frac{a}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2} = 4.2cm$

111)

$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$

$g_{eff} = 0$

$T = \infty$

112) $T \propto \sqrt{\ell}$

$\frac{T'}{T} = \sqrt{\frac{16\ell}{\ell}} = 4$

$T' = 4T = 4 \times 2$

$T' = 8$ sec

113) $\frac{1}{2}m\omega^2a^2 = 16$

$\omega^2 = \frac{32}{5.12 \times 625 \times 10^{-4}}$; $\omega = 10$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{10} = \frac{\pi}{5}$

114) $a = 2 \times 10^{-2}$m

$V_{mean} = 12 \times 10^{-2}$m/s

$V_{mean} = a\omega$

$12 \times 10^{-2} = 2 \times 10^{-2} \times 2\pi n$

$n = \frac{3}{\pi}$Hz

115) Acceleration in S.H.M is directly proportional to the displacement and opposite to the displacement.

116) Initial and final position are same.

117) $x = A \sin\omega t$

$V = \omega A \cos\omega t = \omega A \sin(\omega t + \pi/2)$

$a = -\omega^2 A \sin\omega t = \omega^2 A \sin(\omega t + \pi)$

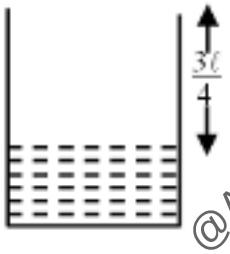
118) $T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$ $g_{eff} = g + a$ (when lift accelerates upwards)

119) Standard equation $y = 2a\cos kx \sin\omega t$

By comparing this equation with given equation

$K = \frac{2\pi}{\lambda} = \frac{\pi}{20} \Rightarrow \lambda = 40$ cm

Distance between two nodes $= \frac{\lambda}{2} = \frac{40}{2} = 20$ cm

120) 

$\frac{V}{4\ell} = 220\,Hz$

$n' = \frac{3}{4}\frac{V}{\frac{3}{4}\ell} = \frac{V}{\ell} = 880\,Hz$

121) $I = 2\pi^2 n^2 a^2 \rho V$

$\frac{I_1}{I_2} = \frac{n_1^2}{n_2^2} \times \frac{a_1^2}{a_2^2} = \frac{1}{4} \times \frac{36}{9} = 1$

122) Second overtone means three loops :

$3\frac{\lambda}{2} = L$ or $\lambda = \frac{2L}{3}$

Anti-nodes will be obtained at $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ or $\frac{L}{6}, \frac{3L}{6}, \frac{5L}{6}$

123) $n = \frac{P}{2\ell}\sqrt{\frac{T}{\pi r^2 \rho}}$ $n \propto \frac{1}{\sqrt{\rho}}$

Graph between n and $\sqrt{\rho}$ will be hyperbola.

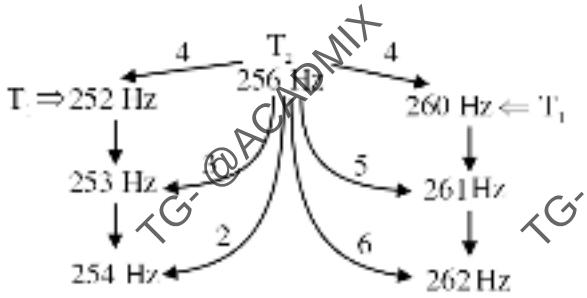
124) Phase difference $\Delta\phi = \frac{2\pi}{\lambda}\Delta x = K\Delta x$

$= \pi \times 0.01 \times 25 = \frac{\pi}{4}$

125) $V = \frac{\omega}{K} = \frac{120\pi}{\frac{2\pi}{5\times10^{-2}}} = 3\,m/s$

$V = \sqrt{\frac{T}{m}} \Rightarrow 3 = \sqrt{\frac{T}{0.4}} \Rightarrow T = 9 \times 0.4$

T = 3.6 N

126) 

The beat frequency should increase, so answer is 260 Hz.

127) Sound waves requires medium for propagation.

128) $V \propto \lambda$

depends on medium

$V = f\lambda$

f not depends on medium, it depends on source.

129) $I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$

$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$

$I_{max} + I_{min} = 2(I_1 + I_2)$

130) $V_P = 3V_w$

$a_0\omega = 3n\lambda$

$a_0 \times 2\pi n = 3n\lambda$

$\lambda = \frac{2\pi a_0}{3}$

131) NCERT Pg. # 284

$\frac{V_P}{V_W} = \frac{\omega A}{\omega/K} = AK$ = 60 × 6 = 360

132) Frequency of first and last tuning fork are 2n & n respectively.

2n – 24 × 3 = n ⇒ n = 72 Hz

Frequency of 21st tuning fork = (2 × 72 – 20 × 3) = 84 Hz

133) $\frac{V}{(40+2e)} = \frac{3V}{4(31+e)}$

3(40 + 2e) = 4(31 + e)

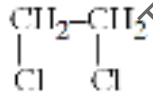
2e = 4

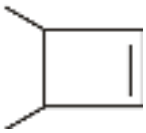
e = 2 cm

134) Amplitude and energy are exponential function of time.

135) Resonance frequency is independent from the density of reflecting medium (denser medium)

CHEMISTRY

136) 
ethylene chloride (no π bonds)

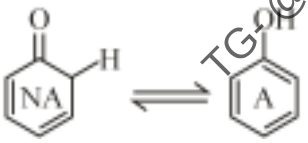
141) 
Can show GI along ring (Restricted rotation along σ bond)

142)

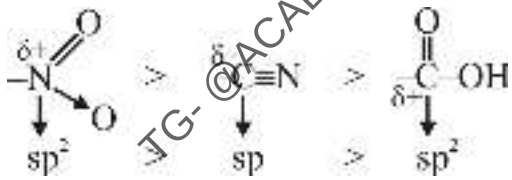
Fact

143)

Enol form is aromatic in nature



145) According to pK_a value

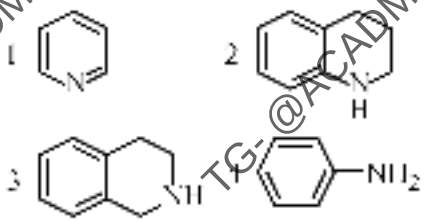
146)  (–I) order
electronegativity

148)  are aromatic

149) Heat of hydrogenation $\propto \frac{1}{\text{stability}}$

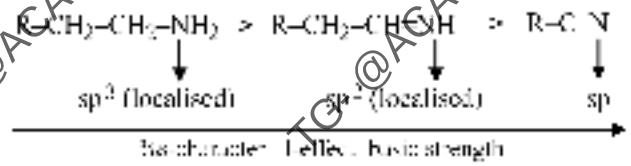
151) **A. Question: The question asks which of the following is most basic**

B. Given Data: Options are:



C. Concept:

A. **Basic strength of nitrogeneous compound depends upon the hybridised state of nitrogen.**



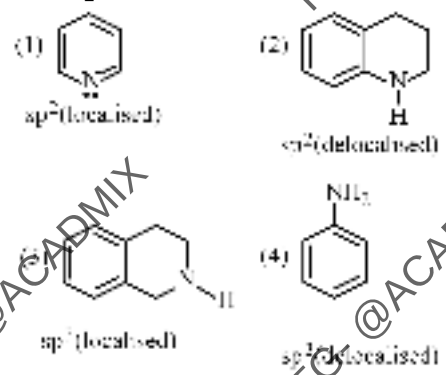
A. **If %s-character is same than Kb = localised lone pair > delocalised lone pair**

$R-CH_2-CH=NH > R-CH=CH-NH_2$ sp^2

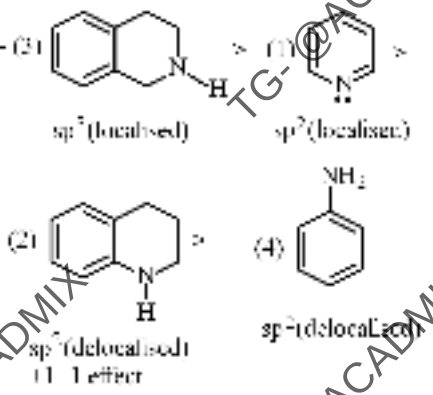
sp2 (localised) sp2 (delocalised)

The –NH, group is attached directly to the conjugated system or benzene ring. It results in the unshared electron pair on nitrogen atom to be in conjugation with the conjugated system or benzene ring and thus making it less available for protonation and decreases basic strength.

D. Explanation:



Kb= sp' > sp2(localised lone pair)> sp2(delocalised lone pair)
Kb= (3) > (1) > (2),(4)



E. Final Answer: 3

154)

Fact based

155) Compounds purified by steam distillation which are immiscible in water but steam volatile.

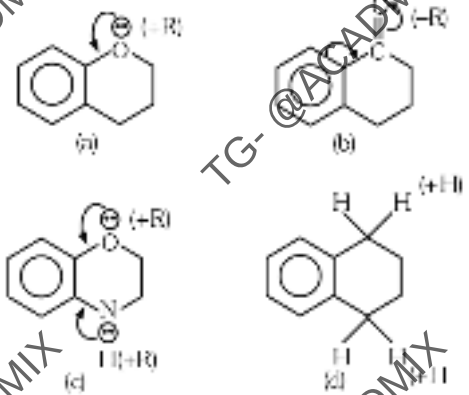
157) $\%N = \frac{1.4}{w} \quad TV = NV = \frac{1.4}{0.1} \times 4 = 56\%$

159) Reaction in column I are oxidation of alkane under different condtion.

161) 

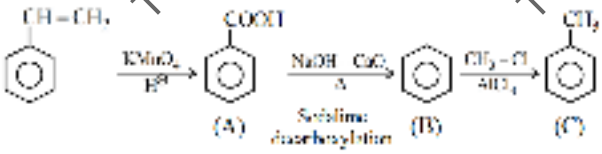


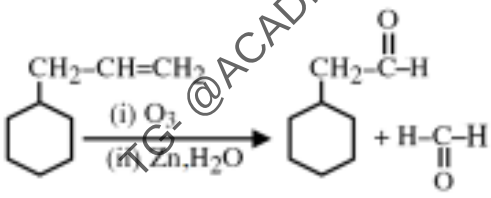
163) Reactivity towards electrophilic aromatic substitution reaction (ESR) α density in ring.



(2) c > a > d > b

164) Hydro boration oxidation

165) 

166) 

167)

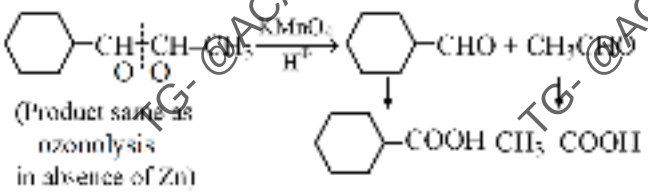
Na/Liquid NH_3 gives trans-alkene with nonterminal alkynes.

169)

Rate of Elimination ∝ Stability of alkene.

171)

NCERT 12th Page No. # 317

173) 

174)

Aniline do not show FCR.

175)

A = Benzene B = Friedel craft reaction

176) Silver acetylides are obtained by passing 1-alkyne in the ammonical solution of silver nitrate (tollen's reagent)

179)

TG- @ACADMIX

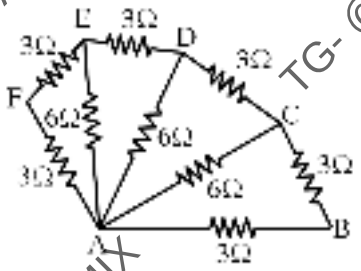
Fact Based

TG- @ACADMIX

PHYSICS

1)

The equivalent resistance between the points A and B of the network shown in figure is :

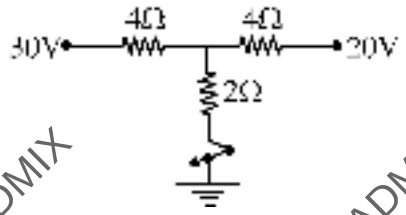


(1) 3Ω

(2) 6Ω

(3) 2Ω

(4) 1Ω



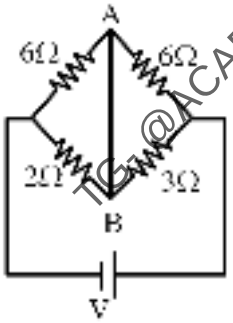
2) Find out current passing through switch when it is closed ?

(1) $\frac{25}{4}$A

(2) $\frac{25}{2}$A

(3) 25A

(4) Zero



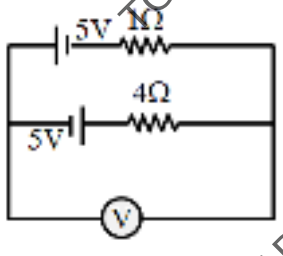
3) Find out direction of current in wire AB.

(1) A to B

(2) B to A

(3) Zero

(4) Depends on EMF of Battery



4) Find out Reading of Ideal Voltmeter.

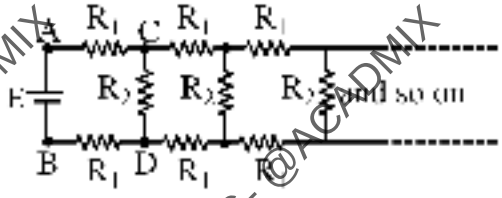
(1) 5V

(2) 3V

(3) Zero

(4) 1V

5) Find the equlvelent resistance between A and B of the network extending off to the inifinity shown in the figure for $R_1 = 1\Omega$, $R_2 = 4\Omega$:



(1) 4Ω

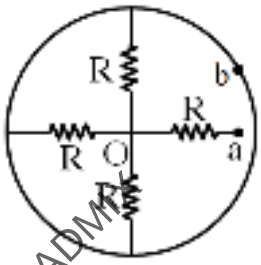
(2) 1Ω

(3) 2Ω

(4) 8Ω

6)

The equivalent resistance between points a and b of a network shown in the figure is given by :-



(1) $\frac{3}{4}R$

(2) $\frac{4}{3}R$

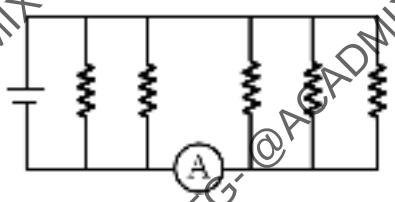
(3) $\frac{5}{4}R$

(4) $\frac{4}{5}R$

7) The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be :

(1) 300%
(2) 200%
(3) 100%
(4) 50%

8) Five identical resistors, each of value 1100Ω are connected to a 220V battery as shown. The



reading of the ideal ammeter A is :

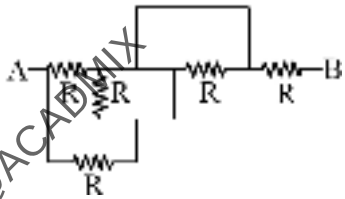
(1) 1/5 A
(2) 2/5 A
(3) 3/5 A
(4) 4/5 A

9) A voltmeter has resistance of 2000Ω and it can measure upto 2V. If we want to increase its range to 10V than the required resistance in series will be :

(1) 2000Ω
(2) 9000Ω
(3) 6000Ω
(4) 8000Ω

10) A(100 W, 200 V) bulb is connected to a 160 volts supply. The power consumption would be :

(1) 100W
(2) 125W
(3) 64W
(4) 80W



11) What is the equivalent resistance between A and B?

(1) $\frac{3R}{2}$
(2) 2R
(3) R
(4) $\frac{R}{2}$

12) There are 8.4×10^{22} free electrons per cm^3 in copper. The current in the wire is 0.21 A ($e = 1.6 \times 10^{-19}$ C). Then the drift velocity of electrons in a copper wire of 1 mm^2 cross section, will be :

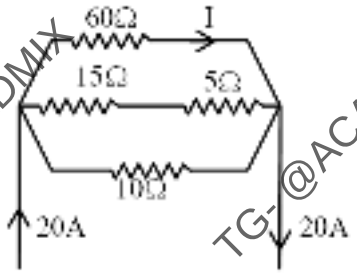
(1) 2.12×10^{-5} m/s

(2) 0.78×10^{-5} m/s

(3) 1.56×10^{-5} m/s

(4) None of these

13) Calculate current (I) in 60Ω :



(1) 1A

(2) 6A

(3) $\left(\frac{10}{3}\right)$ A

(4) 2A

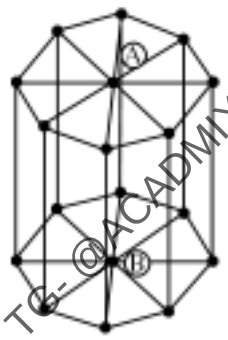
14) Two wires of equal diameters of resistivities ρ_1 and ρ_2 and lengths ℓ_1 and ℓ_2 respectively, are joined in series. The equivalent resistivity of the combination is :

(1) $\frac{\rho_1\ell_1 + \rho_2\ell_2}{\ell_1 + \ell_2}$

(2) $\frac{\rho_1\ell_2 + \rho_2\ell_1}{\ell_1 - \ell_2}$

(3) $\frac{\rho_1\ell_2 + \rho_2\ell_1}{\ell_1 + \ell_2}$

(4) $\frac{\rho_1\ell_1 - \rho_2\ell_2}{\ell_1 - \ell_2}$



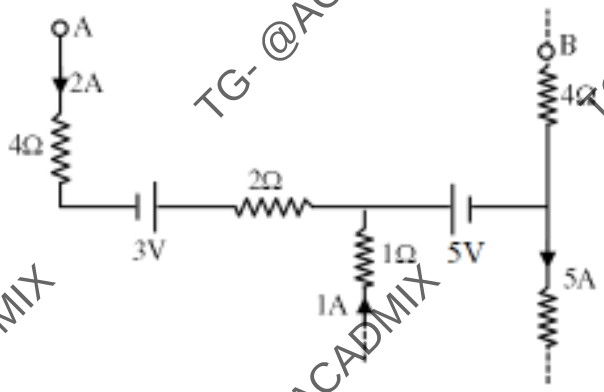
15) Find R_{AB} = ?, If resistance of each side is R :

(1) $\frac{3R}{8}$

(2) $\frac{2R}{5}$

(3) $\frac{R}{2}$

(4) 8R



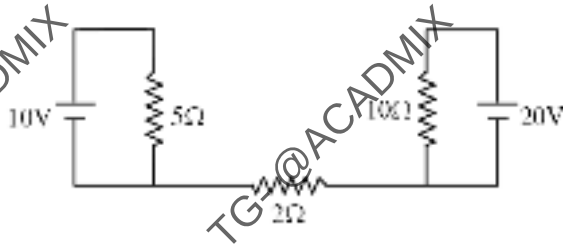
16) In given network find $V_A - V_B$:-

(1) 6V

(2) –6V

(3) 5V

(4) –10V



17) Find the current through 2Ω :

(1) Zero

(2) 1 amp

(3) 2 amp

(4) 4 amp

18) When four identical bulbs of 100 W, 150 V rating are connected in parallel to a 150 V supply the power drawn by them will be :

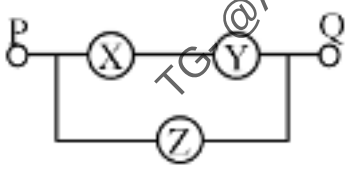
(1) 25 W

(2) 400 W

(3) 100 W

(4) 50 W

19) Three voltmeters X, Y and Z of resistance R, 2R and 3R respectively are connected as shown in figure. A potential difference is applied across P and Q, then reading of voltmeters are V_X, V_Y and



V_Z then :

(1) $V_X = V_Y = V_Z$

(2) $V_X \neq V_Y \neq V_Z$

(3) $V_X \neq V_Y = V_Z$

(4) $V_X \neq V_Y \neq V_Z$

20) A uniform wire of resistance 50Ω is cut into 5 equal parts. These parts are now connected in parallel. The equivalent resistance of combination is :

(1) 2Ω

(2) 10Ω

(3) 250Ω

(4) 6250Ω

21) Two identical metal spheres A and B have charges of 12 nC and –6 nC respectively. A neutral identical metal sphere C is touched to A, then to B and finally removed. What will be charged on sphere C finally.

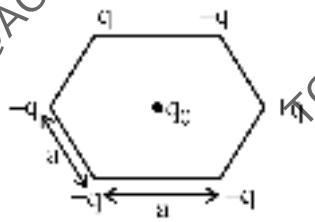
(1) 12 nC

(2) 6 nC

(3) 3 nC

(4) 0

22) Find net electric force feel by charge q_0 placed at centre of regular hexagon.



(1) $k\frac{2qq_0}{a^2}$

(2) $k\frac{qq_0}{a^2}$

(3) zero

(4) $\frac{3}{2}k\frac{qq_0}{a^2}$

23) The number of electrons in 2C of charge is :-

(1) 5×10^{29}

(2) 125×10^{17}

(3) 1.6×10^{19}

(4) 9×10^{11}

24) A point charge q is placed at x = 0. The electric field vector at x = L is $\vec{E}$. When another charge 4q is placed at x = 2L, what is the new electric field vector at x = L ?

(1) $4\vec{E}$

(2) $5\vec{E}$

(3) $-3\vec{E}$

(4) $-2\vec{E}$

25) For charges 2 μC and 3 μC, if force between them for some separation in air is F, then force between them in a medium of permittivity ∈ will be :-

(1) $\frac{\in_0}{\in}F$

(2) $\frac{\in}{\in_0}F$

(3) $\in\in_0 F$

(4) $\frac{F}{\in_0 \in}$

26) Five balls numbered (1) to (5) are checked pair wise. Pair (3, 2), (2, 4) and (4, 3) show electrostatic attractions while pairs (2, 1) and (4, 5) show repulsion. Ball (1) and ball (3) :-

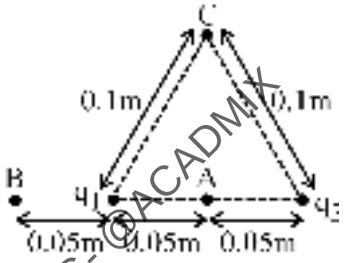
(1) Ball (1) → may be +ve.
Ball (3) → must be neutral.

(2) Ball (1) → must be (+ve).
Ball (3) → may be neutral.

(3) Ball (1) → must be (–ve).
Ball (3) → may be (+ve).

(4) Ball (1) → must be (–ve).
Ball (3) → must be (–ve).

27) Two point charges $q_1 = +10^{-8}$ C and $q_2 = -10^{-8}$ C, respectively are placed 0.1 m apart then, ratio of magnitudes of electric fields at A and C is :-



(1) 4 : 1

(2) 1 : 4

(3) 8 : 1

(4) 1 : 8

28) How does the superposition principle apply to the electric potential of charges ?

(1) Potential is scalar and simply add up.

(2) Potential cancel out

(3) Potential depend on the direction.

(4) All of these

29) If the net electric flux through a closed surface is zero can we conclude that the electric field at every point on the surface is zero ?

(1) Yes

(2) No

(3) Only if the charge inside is zero.

(4) It depends on the symmetry of the charge distribution.

30) If two bodies are rubbed and one of them acquires q_1 charge and another acquired q_2 charge, then ratio $q_1 : q_2$ is :-

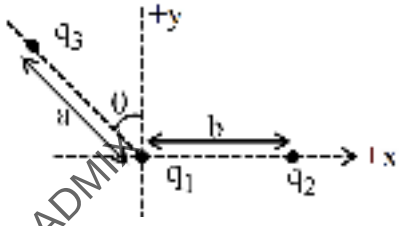
(1) 1 : 2

(2) 2 : 1

(3) –1 : 1

(4) 1 : 4

31) Three charges are placed as shown find components of force on $-q_1$ are :-



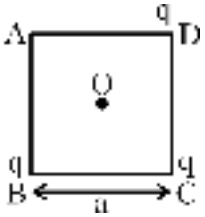
(1) $k\left(\frac{q_1q_2}{b^2}+\frac{q_1q_3\sin\theta}{a^2}\right)\hat{i}, \frac{kq_1q_3\cos\theta}{a^2}(-\hat{j})$

(2) $\left(k\frac{q_1q_2}{b^2}\right)\hat{i}, \frac{kq_1q_3}{b^2}(-\hat{j})$

(3) $\frac{kq_1q_3}{a^2}\hat{i}, \frac{kq_2q_1}{b^2}(-\hat{j})$

(4) $\frac{kq_1q_3}{b^2}\hat{i}, \frac{kq_1q_3}{a^2}\cos\theta(-\hat{j})+\frac{kq_1q_2}{b^2}$

32) Equal charges q are placed at the three corners of square ABCD of side "a". Find the ratio of potential at A to centre O is ?



(1) $\frac{2\sqrt{2}+1}{3\sqrt{2}}$

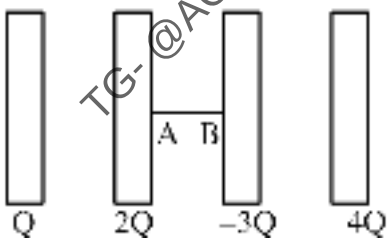
(2) $\frac{2+\sqrt{2}}{6}$

(3) $\frac{2\sqrt{2}+1}{6}$

(4) $\frac{2\sqrt{2}+1}{3}$

33) Four large metal plates are shown in fig. The middle two are then connected through a wire find

the charge that will flow through the wire.



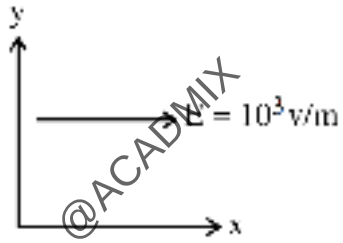
(1) 5Q from A to B

(2) $\frac{5Q}{2}$ from A to B

(3) 5Q from B to A

(4) No charge will flow

34) Find the workdone by an external agent in slowly shifting a charge q = 1 μC in the electric field $\vec{E} = 10^3 \hat{i} \frac{V}{m}$ from the point P(1, 2) m to Q(3, 4) m :



(1) 2×10^{-3} J

(2) 4×10^{-3} J

(3) -2×10^{-3} J

(4) -4×10^{-3} J

35) The electric potential in a region is given by $V = 3x^2 + 4x$. When an electron is placed at the origin it will experience

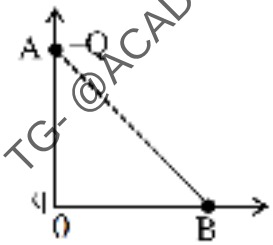
(1) Force along positive x

(2) Zero

(3) Force along negative x

(4) ∞

36) The work required to move point charge – Q from the point A(0, a) to another point B(a, 0) along the straight path AB



(1) $\frac{kqQ}{a}$

(2) $\frac{kqQ}{a^2}$

(3) Zero

(4) $-\frac{qQ}{4\pi\in_0\sqrt{2}a}$

37) Electric potential $V = -5x + 3y + \sqrt{15}z$, then find magnitude of the electric field.

(1) $3\sqrt{2}$
(2) $4\sqrt{2}$
(3) $5\sqrt{2}$
(4) 7

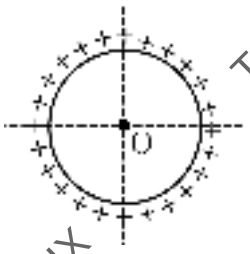
38) There is no net charge at any point ____A____ the conductor and any excess charge must reside at the ____B____.There A and B refer to :

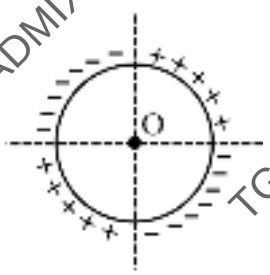
(1) inside, outer surface
(2) surface, centre
(3) outer surface, inside
(4) centre, inside

39) If xJ of work must be done to move electric charge equal to 4C from a place where potential is –10V to another place where potential is 5V, find the value of x.

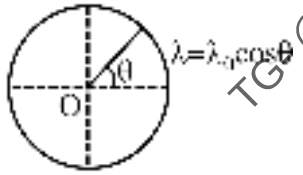
(1) 100 J
(2) 50 J
(3) 60 J
(4) 30 J

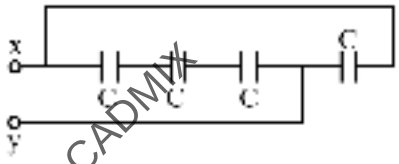
40) In which of the following figure electric field at point O is non-zero ?

(1) 

(2) 

(3) 

(4) 

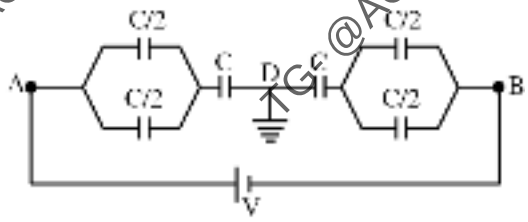
41) Find equivalent capacitance between x and y is - 

(1) $\frac{C}{4}$

(2) $\frac{3C}{4}$

(3) $\frac{C}{3}$

(4) $\frac{4C}{3}$

42) In the circuit as shown, find the potentials at A and B : 

(1) +V, Zero

(2) +V, –V

(3) $+\frac{V}{2}, -\frac{V}{2}$

(4) $+\frac{V}{3}, -\frac{V}{3}$

43) Let k and x are dielectric constant and dielectric strength of dielectric material respectively. A suitable material for using as a dielectric in a capacitor has :-

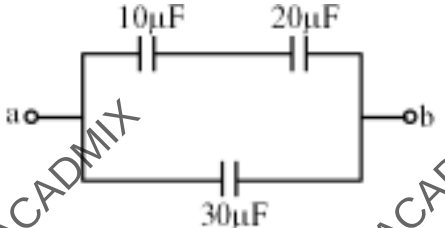
(1) Low 'k', High 'x'

(2) Low 'k', Low 'x'

(3) Low 'x', High 'k'

(4) High 'k', High 'x'

44) Potential difference between a and b is 50 volt. Find total charge stored in the circuit :



(1) $\frac{5500}{3}\mu C$

(2) 750 μC

(3) 3000 μC

(4) 1280 μC

45) A sphere of radius 1 cm has a potential 8000 V, then energy density near its surface will be :

(1) 64×10^5 J/m^3

(2) 8×10^3 J/m^3

(3) 32 J/m^3

(4) 2.83 J/m^3

CHEMISTRY

1)

Which of the following statement is correct ?

(1) $(\Delta_{sol}H^o)_{CuSO_4} = (\Delta_{lattice}H^o)_{CuSO_4} + (\Delta_{Hydration}H^o)_{CuSO_4}$

(2) $(\Delta_{sol}H^o)_{CuSO_4} = (\Delta_{lattice}H^o)_{CuSO_4} - (\Delta_{Hydration}H^o)_{CuSO_4}$

(3) $(\Delta_{sol}H^o)_{CuSO_4} = -(\Delta_{lattice}H^o)_{CuSO_4} + (\Delta_{Hydration}H^o)_{CuSO_4}$

(4) All of the above

2) Which the following reaction is correct representation of enthalpy of neutralization reaction ?

(1) $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(\ell)}$;
$\Delta H_{neut.} = -57.1$ kJ eq^{-1}

(2) $H^+_{(aq)} + OH^-_{(aq)} \rightarrow H_2O_{(\ell)}$;
$\Delta H_{neut} = -57.1$ kJ eq^{-1}

(3) $HCl_{(aq)} + Li(OH)_{(aq)} \rightarrow LiCl_{(aq)} + H_2O_{(\ell)}$;
$\Delta H_{neut.} = -13.7$ k Cal g eq^{-1}

(4) All of the above

3) If temperature of a reaction is increased from 10°C to 100°C then how many times rate of reaction will become :-

(1) 2^2

(2) 2^4

(3) 2^9

(4) 2^6

4) Slope of which plot will give the value of activation energy :-

(1) K v/s T

(2) $\frac{1}{K}$ v/s T

(3) lnK v/s T

(4) $\frac{\ln K \text{ v/s}}{\frac{1}{T}}$

5) The vapour pressure of pure benzene and toluene are 160 and 60 torr respectively. The mole fraction of toluene in vapour phase in contact with equimolar solution of benzene and toluene is :-

(1) 0.5

(2) 0.6

(3) 0.27

(4) 0.73

6) The color of phenophthalein indicator in acid solution is

(1) Pink

(2) Yellow

(3) Colourless

(4) Orange

7) 1L of 1M HCl and 1L of 1M H_2SO_4 are completely neutralized with NaOH separately. The heat liberated in two cases are x and y calories respectively. Then which of the following relation is correct ?

(1) x = y

(2) x > y

(3) x << y

(4) y = 2x

8) **Assertion :** Negatively charged As_2S_3 colloidal sol is obtained by double decomposition method using As_2O_3 and H_2S.

$As_2O_3 + 3H_2S \rightarrow As_2S_3 + 3H_2O$

Reason : Adsorption of As^{3+} ions take place on As_2S_3 when As_2O_3 solution is saturated with H_2S.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) Both **Assertion** and **Reason** are false.

9) In one molal solution that contains 0.5 mole of a solute, there is :

(1) 500 g of solvent

(2) 100 mL of solvent

(3) 1000 g of solvent

(4) 500 mL of solvent

10) In water saturated air the mole fraction of water vapour is 0.02. If the total pressure of the saturated air is 1.2 atm, the partial pressure of dry air is :

(1) 1.18 atm
(2) 1.76 atm
(3) 1.176 atm
(4) 0.98 atm

11) In an aqueous solution of NaCl mole fraction of solvent is 0.9 then relative lowering in vapour pressure will be –

(1) 0.2 atm
(2) 0.1
(3) 0.2
(4) 0.05

12) What will be the molarity of 49% w/w H_2SO_4 solution having density = 1.2 g/mL.

(1) 3 M
(2) 6 M
(3) 1 M
(4) 12 M

13) Which of the following aqueous solution has highest osmotic pressure at 25°C

(1) 0.1 M $AlCl_3$
(2) 0.01 M $Al_2(SO_4)_3$
(3) 0.5 M Glucose
(4) 0.20 M $CaCl_2$

14) A solution of two volatile liquids form maximum boiling azeotrope at a particular composition, then select the incorrect statement :-

(1) This solution shows negative deviation from Raoult's law
(2) HNO_3 and H_2O may form this type of solution.
(3) Vapour pressure of solution at azeotrope composition is maximum
(4) At this composition, components cannot be seperated by fractional distillation.

15) What will be the degree of dissociation of 0.1 M $Mg(NO_3)_2$ solution if Van't Hoff factor is 2.74?

(1) 75%
(2) 87%
(3) 100%
(4) 92%

TG- @ACADMIX

16) Which of the following solvent has maximum value of cryoscopic constant?

(1) Water (H_2O)

(2) Cyclohexane (C_6H_{12})

(3) Carbon disulphide (CS_2)

(4) Carbon tetrachloride (CCl_4)

17) Bromoform has a normal freezing point of 7.9°C and it's K_f = 14.4°C/molal. A solution of 1.3 g of an unknown in 100 g of bromoform freezes at 5.3°C. What is the molecular weight of the unknown :

(1) 7.2

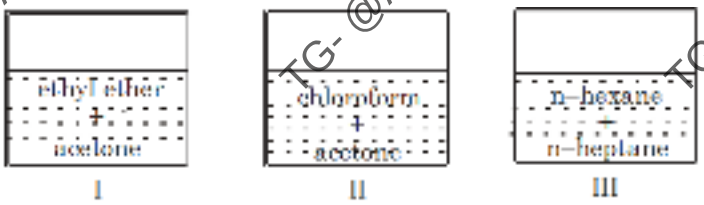
(2) 72

(3) 144

(4) 14.4

18)

State T(true)/F(false) for following statements regarding given liquid solutions :



I II III

(a) Final volume of solutions if equal volume of compounds are mixed, is in the order:
$V_I = V_{II} < V_{III}$
(b) ΔS_{mix} for solution I and solution II is positive but for solution III it is zero.
(c) Enthalpy change of mixing is negative for solution II and solution III but positive for solution I.

(1) TFF

(2) FFF

(3) TTF

(4) FTT

19) Ratio of solubilities of gases N_2 & O_2 in water (as mole of gas in large volume of water) from air at 25º & 1 atm will be, if air is 20 % by mole of O_2 & 80% by mole of N_2
Given : K_H (N_2) = 2×10^4 atm
K_H (O_2) = 10^4 atm

(1) 8 : 1

(2) 1 : 8

(3) 2 : 1

(4) 1 : 2

20) The electrolytic decomposition of dilute sulphuric acid with platinum electrode, cathodic reaction is:-

(1) Reduction of H_2O / H^+

(2) Oxidation of SO_4^{2-}

(3) Reduction of SO_4^{2-}

(4) Oxidation of H_2O

21) In an experiment, 0.04 F was passed through 400 mL of a 1M solution of NaCl. What would be the pH of the solution after the electrolysis :-

(1) 8

(2) 10

(3) 13

(4) 6

22) The following facts are available;
$2A^- + B_2 \rightarrow 2B^- + A_2$;
$2C^- + B_2 \rightarrow$ No reaction;
$2D^- + A_2 \rightarrow 2A^- + D_2$
Which of the following order is correct :-

(1) $E^o_{C^-|C_2} > E^o_{B^-|B_2} > E^o_{A^-|A_2} > E^o_{D^-|D_2}$

(2) $E^o_{C^-|C_2} < E^o_{B^-|B_2} < E^o_{A^-|A_2} < E^o_{D^-|D_2}$

(3) $E^o_{C^-|C_2} < E^o_{B^-|B_2} > E^o_{A^-|A_2} > E^o_{D^-|D_2}$

(4) $E^o_{C^-|C_2} > E^o_{B^-|B_2} < E^o_{A^-|A_2} < E^o_{D^-|D_2}$

23) The reaction mechanism for the reaction P → R is as follows : $P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Q$ (fast) ; $2Q + P \xrightarrow{K_3} R$ (slow), the rate law for the main reaction (P → R) is :

(1) $k_1[P]\,[Q]$

(2) $k_1k_2[P]$

(3) $\frac{k_1k_3[P]^2}{k_2}$

(4) $k_1k_2[P]^2$

24) $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ is a first order gaseous reaction with $K = 2.3 \times 10^{-5}$ sec^{-1} at 320°C. The % of SO_2Cl_2 decomposed on heating for 90 minute is : ($10^{0.054} = 1.13$)

(1) 11.5

(2) 1.15

(3) 0.118

(4) 18.11

25) A reaction takes place in three steps with individual rate constant and activation energy,

	Rate constant	Activation energy
Step 1	k_1	$E_{a_1} = 180\,kJ/mol$
Step 2	k_2	$E_{a_2} = 80\,kJ/mol$
Step 3	k_3	$E_{a_3} = 50\,kJ/mol$

overall rate constant, $k = \left(\frac{k_1 k_2}{k_3}\right)^{2/3}$

overall activation energy of the reaction will be :

(1) 140 kJ/mol

(2) 150 kJ/mol

(3) 130 kJ/mol

(4) 120 kJ/mol

26) The initial rate of reaction A + 5B + 6C → 3L + 3M has been determined by measuring the rate of disappearance of A under the following conditions :-

Experiment No.	$[A]_0$ (M)	$[B]_0$ (M)	$[C]_0$ (M)	Initial rate (M min^{-1})
1.	0.02	0.02	0.02	2.08×10^{-3}
2.	0.01	0.02	0.02	1.04×10^{-3}
3.	0.02	0.04	0.02	4.16×10^{-3}
4.	0.02	0.02	0.04	8.32×10^{-3}

Determine the rate law ?

(1) Rate = k[A] [B] $[C]^2$

(2) Rate = k[A] $[C]^2$

(3) Rate = k[A] [B] [C]

(4) Rate = k[A] [B] $[C]^{1/2}$

27) Two different species A and B show first order disintegration having the half lives 0.231 and 0.3465 second, respectively. The ratio of initial concentration of A and B is 3 : 2. The value of C_A/C_B after three half lives of A will be :-

(1) 3/4

(2) 2/3

(3) 4/5

(4) 1/7

28) Calculate value of emf of cell :-

Pt;H_2(0.1 atm)|HCl(pH=5)||HCl(pH=3)|H_2(g)(1 atm);Pt

$\left[\text{Given } \frac{2.303RT}{F} = 0.06\right]$

(1) 0.010 V

(2) 0.20 V

(3) 0.09 V

(4) 0.45 V

29) Find SRP of M^{+2}/M electrode from following information :-
$2H^+$ (aq) + M(s) ⇌ M^{2+}(aq) + H_2(g) ;
ΔG^o = + 96.5 KJ at 298 K

(1) 0.5 V

(2) –0.5 V

(3) 1 V

(4) –1 V

30) Based on the following information arrange four metals A, B, C and D in order of decreasing ability to act as reducing agents :
(I) Only A, B and C react with 1 M HCl to give H_2(g)
(II) When C is added to solutions of the other metal ions, metallic B and D are formed
(III) Metal C does not reduce A^{n+}

(1) C > A > B > D

(2) C > A > D > B

(3) A > C > D > B

(4) A > C > B > D

31) For saturated solution of Ag_2CrO_4 at infinite dilution $(\lambda^{\infty}_m)_{Ag^+}$ = 127 ohm^{-1} cm^2 mol^{-1} and $(\lambda^{\infty}_m)_{CrO_4^{-2}}$ = 273 ohm^{-1} cm^2 mol^{-1} respectively and specific conductance of solution is 2 × 10^{-2} ohm^{-1} cm^{-1} ; calculate solubility product of Ag_2CrO_4 :-

(1) 0.05

(2) 1.5

(3) 0.037

(4) 2.2 × 10^{-4}

32) Quantity of charge required to reduce 12.3 gm of Nitrobenzene to aniline is (molar mass of nitrobenzene = 123):-

(1) 57900 C

(2) 28950 C

(3) 19300 C

(4) 9650 C

33) $PbO_2 \rightarrow PbO \quad \Delta G_{298K} < 0$
$SnO_2 \rightarrow SnO \quad \Delta G_{298K} > 0$
Most probable oxidation state of Pb and Sn will be :

(1) Pb^{4+}, Sn^{4+}

(2) Pb^{4+}, Sn^{2+}

(3) Pb^{2+}, Sn^{2+}

(4) Pb^{2+}, Sn^{4+}

34) Electrode potential will be less for hydrogen electrode at pH.

(1) 2

(2) 3

(3) 4

(4) 5

35) **Statement I :-** During electrolysis of aqueous sodium acetate solution, the molar ratio of gases formed at cathode and anode is 1 : 3.
Statement II :- Acetate ion discharges at anode and H^+ ion, at cathode.

(1) Both **Statement I** and **Statement II** are true.

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are false.

36) **Statement I :-** In the Daniel cell, if concentrations of Cu^{+2} and Zn^{+2} ions are doubled, the E.M.F. of cell does not change.
Statement II :- Active mass of solids and pure liquids does not effect the E.M.F. of cell.

(1) Both **Statement I** and **Statement II** are true but **Statement II** is NOT the correct explanation of **Statement I**

(2) **Statement I** is true but **Statement II** is false.

(3) **Statement I** is false but **Statement II** is true.

(4) Both **Statement I** and **Statement II** are true and **Statement II** is the correct explanation of **Statement I**

37) What will happen if a aluminium spoon is used to stir a solution of copper nitrate ?

(1) The Spoon will get coated with copper

(2) Alloy of copper and aluminium is formed

(3) The solution becomes blue

(4) Spoon will disappear after some time

38) The equation that is incorrect is :-

(1) $(\Lambda_m^o)_{H_2O} = (\Lambda_m^o)_{HCl} + (\Lambda_m^o)_{NaOH} + (\Lambda_m^o)_{NaCl}$

(2) $(\Lambda_m^o)_{NaBr} - (\Lambda_m^o)_{NaCl} = (\Lambda_m^o)_{KBr} - (\Lambda_m^o)_{KCl}$

(3) $(\Lambda_m^o)_{KCl} - (\Lambda_m^o)_{NaCl} = (\Lambda_m^o)_{KBr} - (\Lambda_m^o)_{NaBr}$

(4) All are incorrect

39) Match the following

1	Neutralization Titration	P	formation of coordination compound
2	Precipitation titration	Q	oxidation reduction reaction
3	Redox titration	R	formation of salt and water
4	Complexometric titration	S	formation of precipitate

(1) 1-R, 2-S, 3-Q, 4-P

(2) 1-Q, 2-R, 3-P, 4-S

(3) 1-R, 2-P, 3-S, 4-Q

(4) 1-S, 2-Q, 3-R, 4-P

40) Match the half-life in column-I with the order in column-II for the reaction A → product.

Column-I		**Column-II**	
(A)	$t_{1/2} \propto a$	(P)	First order
(B)	$t_{1/2} \propto \frac{1}{a}$	(Q)	IIIrd order
(C)	$t_{1/2} \propto \frac{1}{a^2}$	(R)	IInd order
(D)	$t_{1/2}$ = constant	(S)	Zero order

(1) A-P, B-R, C-Q, D-S

(2) A-S, B-R, C-Q, D-P

(3) A-S, B-R, C-P, D-Q

(4) A-S, B-Q, C-R, D-P

41) **Statement I :-** The temperature coefficient of a single-step reaction may be negative.
Statement II :- The rate of reaction having negative order with respect to a reactant decreases with the increase in concentration of that reactant.

(1) **Statement I** is incorrect but **Statement II** is correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) Both **Statement I** and **Statement II** are correct.

42) The rate constant is given by the equation $K = PAe^{-E_a/RT}$. Which factor should register a decrease for the reaction to proceed more rapidly.

(1) P

(2) A

(3) E_a

(4) T

43) Freshly prepared precipitate sometimes gets converted to colloidal solution by :-

(1) Coagulation

(2) Peptisation
(3) Electrolysis
(4) Diffusion

44) **Assertion :-** Aquatic species are more comfortable in warm water than in cold water.
Reason :- K_H values for both N_2 and O_2 increases with increase in temperature.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

45) Compare boiling point of given solutions.
(I) 10^{-4}M NaCl (II) 10^{-3}M NaCl
(III) 10^{-2}M NaCl (IV) 10^{-4}M Urea

(1) III > I > II > IV
(2) I > II > III > IV
(3) III > II > I > IV
(4) II > I > III > IV

BIOLOGY

1) The arrangement of flowers on the floral axis is termed as –

(1) Venation
(2) Phyllotaxy
(3) Inflorescence
(4) Placentation

2) In which of following seed coat is membranous and generally fused with the fruit wall :-

(1) Gram
(2) Castor
(3) Coconut
(4) Maize

3) Scutellum is found in seeds of :–

(1) Maize
(2) Gram
(3) Pea
(4) Bean

4) In seeds of cereals the outer covering of endosperm separates the embryo by which layer–

(1) Epicarp
(2) Fibrous layer
(3) Aleurone layer
(4) Testa

5) Given below are the two statements :
Statement-I : In monocot plants primary root is short lived.
Statement-II : In monocot like wheat fibrous root system is present.

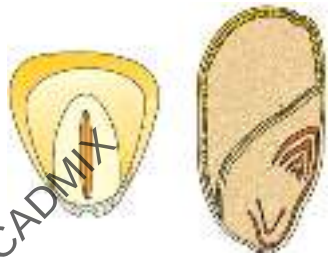
(1) Both statements are correct
(2) Only statement -I is correct
(3) Only statement -II is correct
(4) Both statements are incorrect

6) **Assertion (A) :** In majority of dicotyledonous plants, direct elongation of radicle leads to the formation of primary root.
Reason (R) : In monocotyledonous plants, fibrous root system is present.

(1) Both A and R are true and R is correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is not correct explanation of A.

7) Which of the following statement is correct ?

(1) In mango and coconut, fruits, developed from monocarpellary inferior ovaries.
(2) In coconut fruit, the epicarp is fibrous.
(3) In castor seeds, endosperm is a food storing structure
(4) Aleurone layer is present in seeds of gram.



8) Select the correct option with respect to given diagrams

(1) Seed coat is membranous
(2) Seed coat fused with fruit wall
(3) Plumule and radicle are not enclosed in sheaths.
(4) Both (1) and (2)

9) Find out incorrect match :-

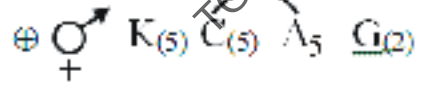
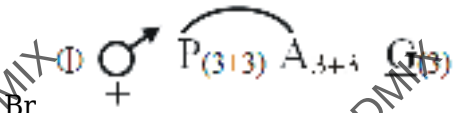
(1) Vexillary aestivation in petal - Bean
(2) Epitepalous stamens - Brinjal

(3) Apocarpous gynoecium - Lotus
(4) Basal Placentation - Marigold

10) Which of the following statement is not true ?

(1) Polyadelphous stamens are found in *Citrus*.
(2) Twisted aestivation is found in petals of lady finger flower.
(3) In cymose inflorescence the flowers are borne in acropetal order.
(4) False septum is found in ovary of mustard flower.

11) Match the column-I with column-II and select the correct option.

Column-I		Column-II	
(A)		I.	*Petunia*
(B)		II.	Lupin
(C)		III	Tulip
(D)		IV	Mustard

(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-IV, B-II, C-I, D-III

12) In __A__ type of phyllatoxy, a single leaf arises at each node, as in china rose, mustard and __B__ plants.

(1) [A] Alternate, [B] *Calotropis*
(2) [A] Opposite, [B] Guava
(3) [A] Alternate, [B] Sunflower
(4) [A] Opposite, [B] *Alstonia*

13) Fruit is developed without fertilisation is called :-

(1) Apospory
(2) Parthenogenesis
(3) Parthenocarpy
(4) Polyembryony

14) In which plant part two to four vascular bundles with radial and exarch xylem present :

(1) Monocot stem
(2) Dicot stem
(3) Monocot root
(4) Dicot root

15) On the basis of following terms find out the type of leaf :
Bulliform cells, mesophyll undifferentiated, Parallel venation.

(1) Isobilateral leaf
(2) Only dorsiventral leaf
(3) Both isobilateral and dorsiventral leaf
(4) Only dicotyledonous leaf

16) The cortex of dicotyledonous roots is consisted of :-

(1) Parenchyma
(2) Sclerenchyma
(3) Collenchyma
(4) (2) and (3) both

17) Stomatal apparatus include :-

(1) Stomatal aperature
(2) Guard cell
(3) Subsidiary cell
(4) All of these

18) In dicot root many cells of epiblema protude in the form of :-

(1) Cortex
(2) Multicellular root hair
(3) Unicellular root hair
(4) Hypodermis

19) In dicot stem the hypodermis is mainly consist of :

(1) Parenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Xylem

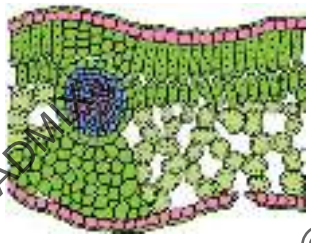
20) What type of vascular bundles are found in monocotyledons stem ?

(1) Open vascular bundles

(2) Conjoint closed vascular bundles

(3) Radial vascular bundles

(4) Conjoint open vascular bundles



21)

Which of following option is incorrect about given diagram :

(1) Loosely arranged mesophyll is known as spongy parenchyma.

(2) Xylem are present towards abaxial surface.

(3) Dorsiventral leaf.

(4) More stomata towards abaxial surface.



22)

Which of the following is not correct about diagram ?

(1) Hypodermis is sclerenchymatous

(2) Vascular bundles are conjoint closed

(3) Exarch protoxylem

(4) Pith is indistinct



23)

Identify the given diagram and select correct option about diagram.

(1) Conjoint open vascular bundle

(2) Present in stem of sun flower

(3) Unable to form secondary xylem and phloem tissue

(4) Present in monocot roots

24) Read carefully the statements about root and identify the incorrect statement :

(1) The anatomy of the monocot root is similar to the dicot root in many respect.

(2) Monocot roots generally show secondary growth.

(3) In dicot root thick walled parenchymatous cells are called pericycle.

(4) Casparian strips are present in endodermis.

25) Which of the following statement are not incorrect:-
(a) Epidermis is outermost layer always made up of multilayered parenchymatous cells
(b) Shoot hair are generally multicellular and branched structure
(c) Root hair are only present at the tip of root
(d) Subsidiary cell in stomata are modified endodermal cell

(1) a & b

(2) a, b, c

(3) b & d

(4) b, c, d

26) What is the name of the oldest viable seed excavated from Arctic Tundra ?

(1) *Strigms*

(2) *Lupinus arcticus*

(3) *Phoenix dectyfera*

(4) *Orobanche*

27) In some cereals such as rice and wheat, pollen grains lose viability within (a) of their release and in some members of Rosaceae, Leguminosae and Solanaceae, they maintain viability for (b).

(1) (a) 30 minutes, (b) months

(2) (a) 30 minutes, (b) years

(3) (a) 30 seconds, (b) months

(4) (a) months, (b) 30 minutes

28) In Autogamy :

(1) Plant should be dioecious

(2) Bisexual flowers requires synchrony in pollen release and stigma receptivity

(3) Anther and stigma should not lie close to each other

(4) Plant should be monoecious with unisexual flowers

29) The pollen tube releases both male gametes inside the cytoplasm of which degenerating cell of embryo sac ?

(1) Antipodal cell

(2) Synergid cell

(3) Egg cell

(4) Central cell

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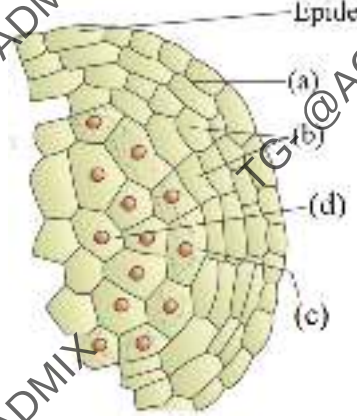
30) In *Zostera* pollination takes place through _____

(1) Wind

(2) Water

(3) Animals

(4) Insects



31)
Identify (a), (b), (c) and (d) in above diagram

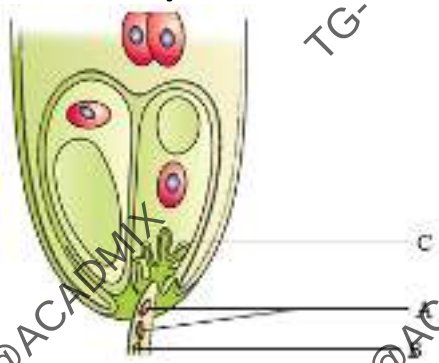
(1) (a) endothecium, (b) tapetum,
(c) middle layers, (d) microspore mother cells

(2) (a) microspore mother cells, (b) middle layers,
(c) tapetum, (d) endothecium

(3) (a) Endothecium, (b) middle layers,
(c) tapetum, (d) microspore mother cells

(4) (a) tapetum, (b) endothecium,
(c) microspore mother cells, (d) middle layers

32) Observe the given diagram carefully and select the correct option in which labelled part A, B and



C are correctly identified?

(1) A-Male gametes
B-Vegetative nucleus
C-Filiform apparatus

(2) A-Vegetative nucleus
B-Male gametes
C-Filiform apparatus

(3) A-Synergid
B-Central cell
C-Egg nucleus

A-Male gametes
(4) B-Central cell
C-Synergid

33) Examine the figure A-D given below and select the right option in which all the four figures A, B, C and D are identified correctly.

	A	B	C	D
				
1	Pollen grain tetrad	Mature embryo-sac	Ovule	Microspore
2	Pollen grain tetrad	Ovule	mature embryo sac	mature pollen grain
3	Megaspore tetrad	Ovule	mature embryo sac	mature pollen grain
4	Megaspore tetrad	Ovule	mature male gametophyte	mature pollen grain

(1) 1
(2) 2
(3) 3
(4) 4

34) Match the column-I with column-II and select correct option.

Column-I		Column-II	
A.	Funicle	I.	Mass of cells with abundant reserve food materials.
B.	Hilum	II.	Basal part of ovule.
C.	Integument	III.	One or two protective layers of ovule.
D.	Chalaza	IV.	Region where body of ovule fuses with funicle.
E.	Nucellus	V.	Stalk of ovule.

(1) A-I, B-II, C-III, D-IV, E-V
(2) A-V, B-IV, C-III, D-II, E-I
(3) A-IV, B-II, C-I, D-III, E-V
(4) A-I, B-III, C-V, D-II, E-IV

35) Functionally cross pollination but genetically self pollination is?

(1) Xenogamy
(2) Geitonogamy
(3) Autogamy
(4) Both (1) and (2)

36) Majority of insect pollinated flowers are :

(1) Large
(2) Colourful
(3) Fragrant and rich in nectar
(4) All of the above

37) The outermost layer of endosperms of maize grain is called

(1) Epidermis
(2) Tunica
(3) Aleurone
(4) Pericarp

38) In which type of pollination flowers have single ovule in ovary and numerous flowers packed into inflorescence ?

(1) Wind pollination
(2) Water pollination
(3) Bee pollination
(4) Bird pollination

39) Primary endosperm nucleus (PEN) is formed by the fusion of :-

(1) 2 polar nuclei + 1 synergid cell nucleus
(2) 1 polar nuclei + 1 antipodal cell nucleus + 1 synergid cell nucleus
(3) 2 polar nuclei + 1 male gamete nucleus
(4) 2 antipodal cell nuclei + 1 male gamete nucleus

40) Filiform apparatus are seen in :-

(1) Egg cell
(2) Antipodal cells
(3) Synergid cells
(4) Endothecium cells

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41)
This type of placentation is related to which of the following plant -

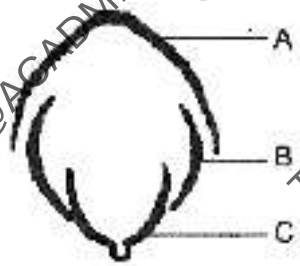
(1) Pea
(2) Mustard
(3) Dianthus
(4) Chinarose



42) The following diagram represents the position of the ovary in.

(1) Guava
(2) Cucumber
(3) Plum
(4) 1 and 2 both

43) Name the petals A, B and C in vexillary aestivation shown in the below figure.



(1) A-Standards, B-Wing, C-Perianth
(2) A-Standard, B-Keel, C-Wing
(3) A-Wing, B-Keel, C-Wing
(4) A-Standard, B-Wing, C-Keel

44) **Assertion:** In Cymose inflorescence, Growth of main axis is limited.
Reason: The main axis terminates in a flower.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

45) Read following two statements(I and II) and choose correct option –

Statement – I : Leaves develops at the node and bears a bud in its axil.
Statement – II : Leaves originate from shoot apical meristems and are arranged in basipetal order.

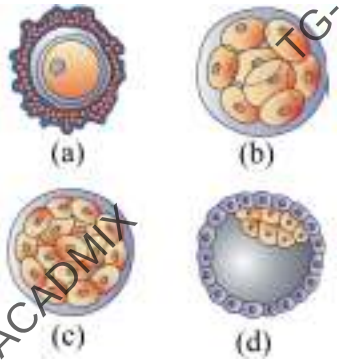
(1) Statement I and statement II both are correct
(2) Statement I and statement II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

46) Which statement is incorrect regarding female external genitalia ?

(1) Mons pubis is a cushion of fatty tissue covered by skin and pubic hairs
(2) Labia minora are paired fold of tissue under labia majora
(3) Clitoris lies at the upper junction of labia minora below the urethral opening
(4) Clitoris is a tiny finger like structure.

47)

The figure below show four stage (a, b, c, d) of human development. Select the option given correct identification together with number of cells present in it –



		Developmental stage	Number of cells
(1)	(a)	Zygote	2-4
(2)	(b)	Blastula	8-16
(3)	(c)	Morulla	16-32
(4)	(d)	Blastocyst	16-32

(1) 1
(2) 2
(3) 3
(4) 4

48) **Assertion :** The testes of human are situated outside the body in an extra-abdominal pouch called scrotum. **Reason :** Scrotum helps in maintaining the low temperature of testes by 2-2.5°C less than the normal body temperature, which is necessary for spermatogenesis.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

49) **Assertion :** Absence of menstruation is not a confirmatory indicator of pregnancy. **Reason :** Absence of menstruation can happen due to poor health or stress.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

50) The first movements of the foetus usually observed during the :

(1) First trimester of pregnancy
(2) Fourth month of pregnancy
(3) Fifth week of pregnancy
(4) Fifth month of pregnancy

51) How many days, pregnancy last in dogs, elephants and cats ?

(1) 30 days, 290 days, 60 days
(2) 30 days, 640 days, 60 days
(3) 60 days, 640 days, 65 days
(4) 65 days, 270 days, 30 days

52) Secondary follicle in ovary has-

(1) Primary oocyte
(2) Secondary oocyte
(3) Ootid
(4) Antrum

53) In oogenesis process, division of primordial germ cells and formation of oogonia occurs at the time of :-

(1) Before puberty
(2) Embryo stage
(3) After puberty
(4) In adult stage

54) In human beings longest phase of spermatogenesis is :-

(1) Resting phase
(2) Multiplication phase

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(3) Growth phase

(4) Maturation phase

55) Menstrual cycle is controlled by :-

(a) Estrogen and progesterone of ovary

(b) FSH of anterior pituitary gland

(c) LH of adenohypophysis

(d) Oxytocin of neurohypophysis

(1) Only a and b are correct.

(2) Only b and d are correct.

(3) Only a and c are correct.

(4) a, b and c are correct.

56) Which of the following is released by spermatozoa that acts on the corona radiata cells during fertilisation ?

(1) Relaxin

(2) Gonadotropin

(3) Hyaluronidase

(4) Trypsinogen

57) Level of estrogen and progesterone are minimum at:-

(1) Follicular phase

(2) Ovulation

(3) Secretory phase

(4) Bleeding phase

58)

Choose the incorrect for luteal phase :-

(1) Transformation of graafian follicle in corpus luteum

(2) Large amount of progesterone

(3) Maintenance of the endometrium by progesterone

(4) Heavy menstruation

59) Which hormone maintain corpus luteum during pregnancy?

(1) LH

(2) FSH

(3) hCG

(4) Both (1) & (2)

60) In the human female, menstruation can be deferred by the administration of :-

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(1) FSH only
(2) LH only
(3) Combination of FSH and LH
(4) Combination of estrogen and progesterone

61) Common duct formed by union of vas deferens and duct from seminal vesicle is

(1) urethra
(2) tunica-vasculosa
(3) ejaculatory duct
(4) spermatic duct

62) The mitotic division starts as the zygote moves through the _____ of the oviduct called cleavage towards the uterus.

(1) Ampulla
(2) Isthmus
(3) Fimbriae
(4) Infundibulum

63) Several mammary ducts join to form a wider A which is connected to B through which milk is sucked-out. Identify the A & B.

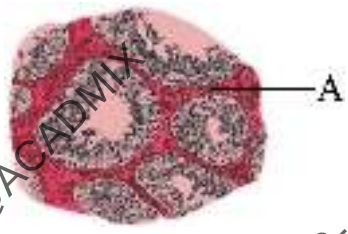
(1) A-Mammary tubule, B-Mammary ampulla
(2) A-Mammary Ampulla, B-Lactiferous duct
(3) A-Lactiferous duct, B-Mammary ampulla
(4) A-Mammary Ampulla, B-Nipple

64) How many meiosis required to form 160 zygotes?

(1) 220
(2) 320
(3) 160
(4) 200

65)

The given diagram is a cross section of seminiferous tubules. What is the function of structure A?



(1) Formation of sperms
(2) Nourishment of sperms
(3) Secretion of testosterone
(4) All of the above

66)

Which structure does not carry sperms ?

(1) Vas deferens
(2) Rete testis
(3) Epididymis
(4) Prostate gland

67) Read the following statements (A-D) and choose the correct option.
(A) Hymen is often torn during the first coitus.
(B) The clitoris is a tiny finger like structure which lies at the lower junction of two labia minora.
(C) Myometrium exhibits strong contraction during delivery of baby.
(D) Several mammary ducts join to form a wider mammary lobe.

(1) Only (A) and (C) are correct.
(2) (A), (B) and (C) are correct.
(3) (B), (C) and (D) are correct.
(4) (B) and (C) are incorrect.

68) **Statement-I :-** Labia majora extend down from mons pubis & surrounds vaginal & urethra opening.
Statement-II :- Clitoris is finger like projection at upper junction of labia minora above urethral opening.

(1) Statement I & II both are correct
(2) Statement I & II both are incorrect
(3) Statement I is correct but II is incorrect
(4) Statement I is incorrect but II is correct

69) First haploid cell during spermatogenesis is :-

(1) Spermatogonia
(2) Primary spermatocyte
(3) Secondary spermatocyte
(4) Spermatids

70) Semen is :-

(1) Seminal plasma along with sperm
(2) Secretion of seminal vesicle along with sperm
(3) Secretion of epididymis along with sperm

(4) Secretion of androgen along with sperm

71) Primary oocyte complete its first meiotic division in which structure ?

(1) Primary follicle
(2) Corpus luteum
(3) Secondary follicle
(4) Tertiary follicle

72) Theca layer is organised into an inner theca interna and outer theca externa in :-

(1) Primary follicle
(2) Secondary follicle
(3) Tertiary follicle
(4) Corpus luteum

73) Which hormone shows significant increase at the age of puberty -

(1) GnRH
(2) Growth Hormone
(3) Melanin
(4) All of the above

74) The second meiotic division during oogenesis completes:

(1) Before birth
(2) Just after birth
(3) At the time of puberty
(4) During fertilization

75) Which of the following is a copper releasing IUD ?

(1) Lippes loop
(2) Multiload 375
(3) Progestasert
(4) LNG-20

76) The medical termination of pregnancy (Amendment) act was enacted in :-

(1) 1951
(2) 1971
(3) 2017
(4) 2000

77) Test tube baby programme also known :-

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(1) Artificial insemination
(2) In vitro fertilisation
(3) In vivo fertilisation
(4) Gamete intra fallopian transfer

78) Find the odd one out about Saheli

(1) Contains steroidal preparation
(2) Once a week pill
(3) Few side effects
(4) High contraceptive value

79) __A__ programme create awareness among people about various reproductive related aspects and provides facilites for build up a reproductive healthy society.

(1) Family planning programme
(2) Reproductive and child health care (RCH)
(3) Contraceptive methods
(4) Lactational amenorrhea

80) Which of the following a natural/traditional contraceptive method ?

(1) Condoms
(2) Coitus interruptus
(3) Vaults
(4) Diaphragms

81) Emergency contraceptives are effective if used within :-

(1) 72 hrs of coitus
(2) 72 hrs of ovulation
(3) 72 hrs of menstruation
(4) 72 hrs of implantation

82)

Given below some Assisted Reproductive Technologies (ART). In which find out Invitro Fertilisation :-

(a) ZIFT (b) IUT (c) ICSI
(d) AI (e) GIFT (f) IUI

(1) a, b, c
(2) b, c, d
(3) d, e, f
(4) b, e, f

83) In which of the technique fertilisation occur inside the body.

(1) ZIFT
(2) IUT
(3) ICSI
(4) A.I.

84) Embryo with up to 8 blastomeres formed due to in vitro fertilization in transferred into -

(1) Cervix
(2) Uterus
(3) Fimbriae
(4) Fallopian tube

85) Choose the incorrect match pair :-

(1)	Multiload 375	Suppress sperm motility
(2)	Progestasert	Make uterus unsuitable for implantation.
(3)	Lippes loop	Make cervix hostile to the sperm
(4)	CuT	Suppress sperm motility

(1) 1
(2) 2
(3) 3
(4) 4

86) Find out the incorrect statement :

(1) Government of india legalized MTP in 1971.
(2) A majority of the MTPs are performed legally by unqualified quacks.
(3) In India, often the female is blamed for the couple being childless.
(4) In IVF, fertilisations occurs outside the body.

87) ***Assertion :*** Copper releasing IUD prevent fertilization.
Reason : Copper ions inhibit motility of sperms.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

88) Which of the following statement is incorrect about cervical caps & diaphragms?

(1) Only to be used by females only.

(2) Prevents user from contracting STDS

(3) Applied on cervix part

(4) A type of barrier method for contraception.

89) **Statement-I :** India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal.
Statement-II : Oral administration of small doses of either progestogens or progestrogen-estrogen combination is another contraceptive method used by the female.

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both Statement I and Statement II are incorrect.

90) Which of the following method of birth control can cause excessive menstrual bleedings pain and inflammation of uterine wall as side effects ?

(1) Condom

(2) Cervical cap

(3) Oral contraceptive

(4) Intra uterine device (IUDs)

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	1	2	1	2	1	3	4	3	1	3	4	1	1	1	1	2	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	2	3	1	1	3	1	2	3	1	3	1	3	1	3	4	1	3	4
Q.	41	42	43	44	45															
A.	4	3	4	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	3	4	3	3	4	3	1	3	3	2	4	3	2	4	2	2	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	3	1	1	1	1	3	1	4	4	1	4	1	1	1	4	1	1	2
Q.	86	87	88	89	90															
A.	1	3	2	3	3															

BIOLOGY

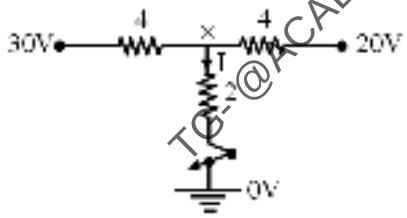
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	4	1	3	1	4	3	4	2	3	4	3	3	4	1	1	4	3	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	3	3	2	3	2	1	2	2	2	3	1	2	2	2	4	3	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	1	3	3	3	4	4	4	3	1	2	4	4	3	4	4	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	4	3	4	1	1	3	1	4	3	1	4	2	3	2	1	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	4	4	3	2	1	2	1	4										

TG- @ACADMIX

SOLUTIONS

PHYSICS

1) Simple series parallel combination start from 'AFE' side and combined.



2)

Apply KCL,

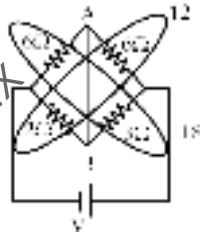
$$\frac{X-30}{4}+\frac{X-20}{4}+\frac{X-0}{2}=0$$

$\Rightarrow 4x = 50$

$x = 25/2$ V

$$I=\frac{\frac{25}{2}-0}{2}$$

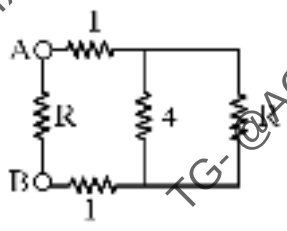
$$I=\frac{25}{4}A$$



3)

Current will flow from A to B.

4) T.P.D. $= \dfrac{5\times 4-5\times 1}{1+4}=3V$



5)

$$R=1+1+\frac{4R}{4+R}$$

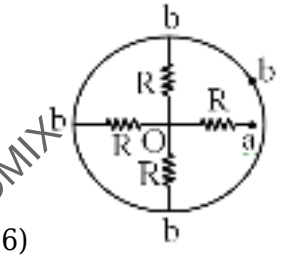
$$R=\frac{2(4+R)+4R}{4+R}=\frac{8+6R}{6+R}$$

$4R + R^2 = 8 + 6R$

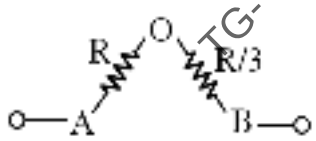
$R^2 - 4R + 2R - 8 = 0$

$R(R-4) + 2(R-4) = 0$

R = 4Ω
R = – 2Ω (Not possible)



6)
Re-Draw



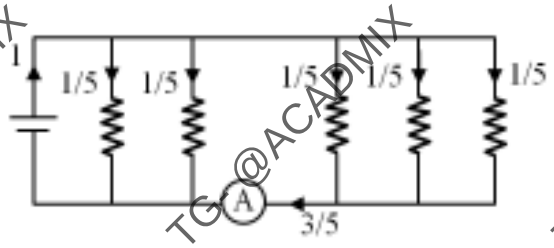
$R_{AB} = \frac{4R}{3}$

7) $R \propto \ell^2$

$\frac{R}{R^1} = \frac{\ell^2}{(2\ell)^2} = \frac{1}{4}$

$R^1 = 4R.$

% change $= \left(\frac{4R - R}{R}\right) 100\% = 300\%$



8)

$R_{eq} = \frac{1100}{5} = 220$

$I = \frac{220}{2200} = 1A$

9) $I_g = \frac{2}{2000} = 10^{-3}A$

$V = Ig (G + H)$

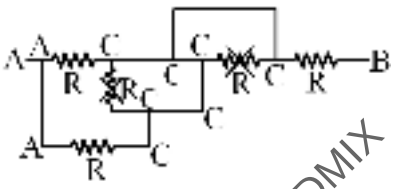
$10 = 10^{-3} (2000 + H)$

$H = 10^4 - 2000 = 8000\ \Omega$

10) $P_c = \frac{V_A^2}{R} = \frac{V_A^2}{V_A^2} P_R$

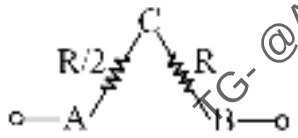
$= \left(\frac{160}{200}\right)^2 100$

$= \frac{16}{25} \times 100 = 64$ w



11)

Re-Draw



$R_{AB} = \frac{3R}{2}$

12) $I = neAV_d$

$0.21 = 8.4 \times 10^{22} \times 10^6 \times 1.6 \times 10^{-19} \times 10^{-6} V_d$

$V_d = \frac{0.21}{8.4 \times 1.6 \times 10^3}$ m/s

$= 1.56 \times 10^{-5}$ m/s

13) $I_1 : I_2 : I_3 = \frac{1}{60} : \frac{1}{20} : \frac{1}{10}$

$= 1 : 3 : 6$

$I_1 = \left(\frac{1}{10}\right) 20 = 2A$

14)

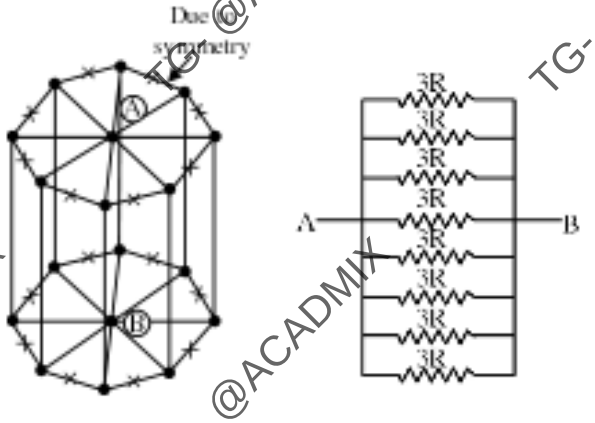


$R_{eq} = R_1 + R_2$

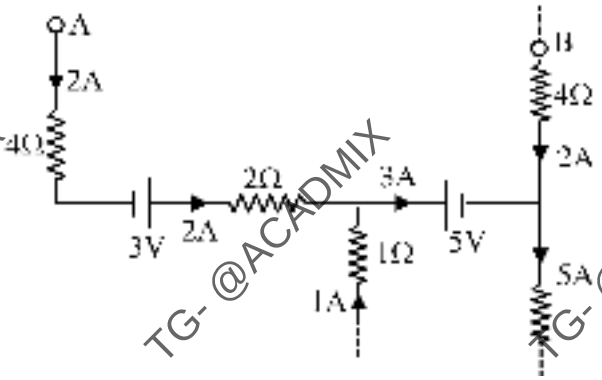
$\frac{\rho_{eq}(\ell_1 + \ell_2)}{A} = \frac{\rho_1 \ell_1}{A} + \frac{\rho_2 \ell_2}{A}$

$\rho_{eq} = \frac{\rho_1 \ell_1 + \rho_2 \ell_2}{\ell_1 + \ell_2}$

15)



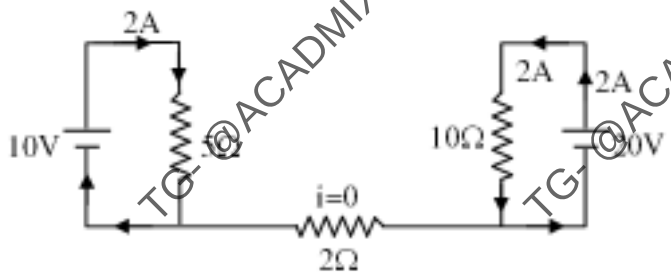
$R_{AB} = \frac{3R}{8}$



16)

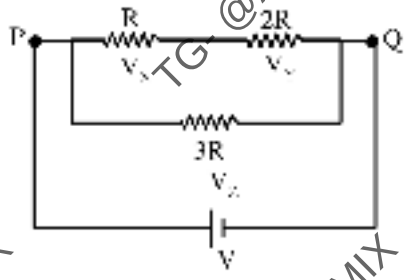
$V_A - 8 + 3 - 4 - 5 + 8 = V_B$

$V_A - V_B = 6V$



17)

18) $P_C = \frac{(150)^2}{R/4} = \frac{4(150)^2}{(150)^2/100} = 400\ W$



19)

$V_X = \frac{V}{3}$

$V_Y = \frac{2V}{3}$

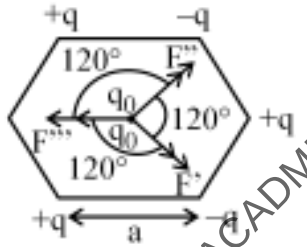
$V_Z = V$

20) $R_{eq} = \frac{R}{5} = \frac{50/5}{5} = 2\Omega$

21) $q_A = 12$ nC, $q_B = -6$ nC, $q_C = 0$

$q_A^1 = q_C^1 = \left(\frac{12nC + 0}{2}\right) = 6$ nC

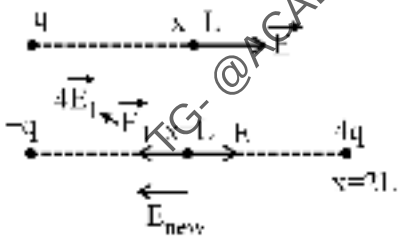
$$q_C^{11} = q_B^1 = \frac{6nC - 6nC}{2} = 0\ C$$



22)
F' = F" = F" = equal in magnitude
$F_{net} = 0$

23) q = ne

$$n = \frac{2}{1.6 \times 10^{-19}} = 1.25 \times 10^{19}$$



24)

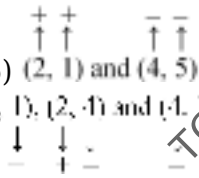
25) Force in AIR $F = \frac{1}{4\pi\epsilon_0}\frac{2\mu 3\mu}{r^2}$

force in medium $F_m = \frac{1}{4\pi\epsilon_0}\frac{2\mu 3\mu}{r^2}$

$$= \frac{\epsilon_0}{\epsilon_0}\left(\frac{1}{4\pi\epsilon}\frac{2\mu 3\mu}{r^2}\right)$$

$$= \frac{\epsilon_0}{\epsilon}\left(\frac{1}{4\pi\epsilon_0}\frac{2\mu 3\mu}{r^2}\right)$$

$$= \frac{\epsilon_0}{\epsilon}F$$



26) (2, 1) and (4, 5) ⇒ Repulsion
(2, 1), (2, 4) and (4, 3) ⇒ Attraction
Ball (3) must be neutral

27) For (A)

$$E_1 = 9 \times 10^9 \times \frac{10^{-8}}{(0.05)^2} = 3.6 \times 10^4 = E_2$$

$E_A = E_1 + E_2 = 7.2 \times 10^4$ N/C
For (C)

$E_C = 2E_1 \cos\left(\frac{120}{2}\right)$

$= 2 \times 9 \times 10^9 \times \frac{10^{-8}}{(0.1)^2} \times \frac{1}{2} = 9 \times 10^3$

$\frac{E_A}{E_C} = \frac{7.2 \times 10}{0.9 \times 10} = \frac{8}{1}$

28) Electric potential is a scalar quantity and net potential at any point is algebric sum of individual potential.

29) Net flux through closed surface is depend on inside charge only, while electric field on closed surface depend on both inside and outside charge.

30) By frictional method of charging, charge aquire by bodies are equal in magnitude and opposite in nature.

31) $\vec{F_x} = (F_{12} + F_{13}\sin\theta)\hat{i}$

$= k\left(\frac{q_1 q_2}{b^2} + \frac{q_1 q_3}{a^2}\sin\theta\right)\hat{i}$

$\Rightarrow F_y = F_{13}\cos\theta(-\hat{j})$

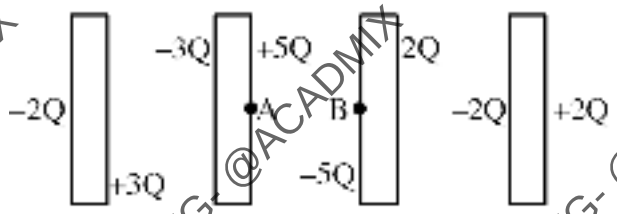
$= -k\frac{q_1 q_3}{a^2}\cos\theta\hat{j}$

32) $V_A = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{a} + \frac{q}{\sqrt{2}a} + \frac{q}{a}\right) = \frac{q}{4\pi\epsilon_0}\frac{1}{a}\left(2 + \frac{1}{\sqrt{2}}\right)$

$V^0 = \frac{1}{4\pi\epsilon_0}\left(\frac{3q}{a/\sqrt{2}}\right) = \frac{1}{4\pi\epsilon_0}\frac{3\sqrt{2}}{a}$

$\frac{V_A}{V_0} = \frac{\left(2\sqrt{2}+1\right)}{6}$

33)



34) Workdone against the electric field. $W = -q\int \vec{E}.\vec{dl}$

$= -q\int \vec{E}\hat{i}.\left(dx\hat{i} + dy\hat{i}\right)$

$= -qE(x_2 - x_1)$

$= -10^{-6} \times 10^3 \times (3 - 1)$

$= -2 \times 10^{-3} J$

35) $\vec{E}_x = -\frac{dv}{dx}\hat{i} = -(6x + 4)\hat{i}$

$\left(\vec{E}_x\right)_{origin} = -4\hat{i}$

Force on e $\vec{F} = q\vec{E}$

$=(-e)\ (-4\hat{i})$

$= 4e\hat{i}$

36) $W = q_0(V_B - V_A)$

$V_B = V_A = k\frac{q}{a}$

37) $E_x = -\frac{dV}{dx} = (-5) = 5$

$E_y = -\frac{dV}{dy} = -3$

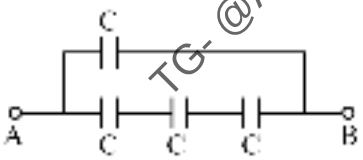
$E_z = -\frac{dV}{dz} = -\sqrt{15}$

$E = \sqrt{(5)^2 + (-3)^2 + (-\sqrt{15})^2}$

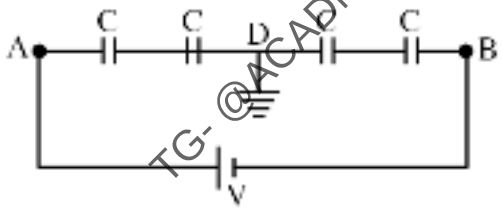
$= 7$

39) $W_{A \to B} = q_0(V_B - V_A)$

$= 4\,(5 - (-10)) = 60J$



41) $C_{eq} = \frac{C}{3} + C = \frac{4C}{3}$



42)

$V_A - V_D = \frac{V}{2} \Rightarrow V_A = \frac{V}{2}$

$V_D - V_B = \frac{V}{2} \Rightarrow V_B = -\frac{V}{2}$

44) $C_{eq} = \frac{110}{3}\mu F$

$C = \frac{Q}{V} \Rightarrow Q = C.V.$

$= \frac{110}{3}\mu F \times 50$

$= \frac{5500}{3}\mu C$

45) $U = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \left(\frac{V}{r}\right)^2$

$= 2.83\ J/m^3$

CHEMISTRY

51)

In acidic → Colourless
In Basic → Pink

52)

Number of g eq of HCl neutralized = 1 × 1 × 1 = 1
Number of g eq of H_2SO_4 neutralized = 1 × 2 × 1 = 2

2x = y

54) $m = \frac{\text{Moles of solute}}{\text{Weight of solvent (g)}} \times 1000$

$1 = \frac{0.5}{\text{Weight of solvent (g)}} \times 1000$

Weight of solvent (g) = 500 g

55)

$X_{H_2O} = 0.02$

$X_{gas} = 0.98$

$P_{total} = 1.2$ atm

partial pressure of dry-air = P_T × mole fraction of dry-air

Partial pressure of dry-air = 1.2 atm × 0.98 = **1.176 atm.**

56) $\frac{P^0 - P_S}{P^0} = i\, x_{solute}$

= 2 × 0.1 = 0.2

57)

% w/w = 49
Let weight of solution = 100 g
weight of H_2SO_4 = 49 g

vol. of solution = $\frac{\text{mass}}{d} = \frac{100}{1.2} = 83.33\text{ml}$

Molarity = $\frac{49}{98} \times \frac{1000}{83.33} = 6\text{M}$

58)

$\pi \propto ic$

59) At maximum boiling point composition, vapour pressure of solution will be minimum.

60) For $Mg(NO_3)_2$ $i = 1 + 2\alpha = 2.74$
$2\alpha = 1.74$
$\alpha = 0.87$
$\alpha = 87\%$

61) NCERT page # 53 (XII)

62)

$\Delta T_f = 2.6 = k_f \times \frac{n}{w(kg)}$

$2.6 = 14.4 \times \frac{1.3}{M.w \times 0.1}$
M.w. = 72

63)

I. Ethyl ether + acetone : + ve deviation from Raoult's law
II. Chloroform + acetone : –ve deviation from Raoult's law
III. n–Hexane + h–heptane: ideal solution

64)

in air, $P_{N_2} = 0.8$ atm, $P_{O_2} = 0.2$ atm
$P = K_H X$ (Henry's law)
$0.8 = 2 \times 10^4 X_{N_2}$ .....(1)
$0.2 = 10^4 X_{O_2}$ .....(2)
Ratio of solubility of N_2 & O_2 is 2 : 1

65) On dilution, H_2O in solution increases so reduction of H_2O occurs at cathode.

66)

At cathode

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

No. of Equ. of OH^- = No. of Faraday charge

$OH^- \Rightarrow 4\times10^{-2}$ equivalent

$\Rightarrow 4\times10^{-2}$ mol

$$\text{Conc. } [OH^-] = \frac{4 \times 10^{-2}}{\left(\frac{400}{1000}\right)} = 10^{-1}\,\text{mol}\,L^{-1}$$

$pOH = -\log(10^{-1}) \Rightarrow pOH = 1$

$pH = 13$

67)

Reaction -1 suggest Reduction of B_2 & oxidation of A^- is spontaneous,

Reaction -2 suggest Reduction of B_2 & oxidation of C^- is non-spontaneous.

Reaction -3 suggest Reduction of A_2 & oxidation of D^- is spontaneous.

68)

From slow step $r = k_3(Q)^2 (P)$ ...(i)

from reversible step $k_c = \frac{k_1}{k_2} = \frac{(Q)^2}{(P)}$ ...(ii)

from (i) & (ii) $r = k_3 . \frac{k_1}{k_2}(P)^2$

69)

$2.3 \times 10^{-5}\ s^{-1} \times 90 \times 60s = 2.303 \log\left(\frac{A_0}{A_t}\right)$

$\log\left(\frac{A_0}{A_t}\right) = 0.054$ $\Rightarrow [A_t = 0.885\ A^0]$

70) $Ae^{-E_a/RT} = \left[\frac{Ae^{-E_{a_1}/RT} Ae^{-E_{a_2}/RT}}{Ae^{+E_{a_3}/RT}}\right]^{2/3}$

$= \left[Ae^{(-E_{a_1}-E_{a_2}+E_{a_3})/RT}\right]^{2/3}$

$E_a = \frac{2}{3}[E_{a_1} + E_{a_2} - E_{a_3}]$

$= \frac{2}{3}[180 + 80 - 50] = 140\,\text{kJ/mol}$

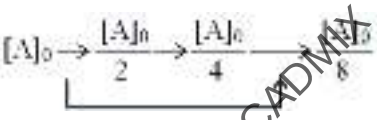
71)

[A]	[B]	[C]	rate
0.02	0.02	0.02	2.08×10^{-3}

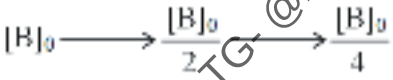
0.01	0.02	0.02	1.04×10^{-3}
0.02	0.04	0.02	4.16×10^{-3}
0.02	0.02	0.04	8.32×10^{-3}

rate $\propto [A]^1 [B]^1 [C]^2$

rate $= K [A] [B] [C]^2$

72) 

time – $3 \times 0.231 = 0.693$ sec



time – $2 \times 0.3465 = 0.693$ sec

$$\frac{C_A}{C_B} \rightarrow \frac{[A]_0/8}{[B]_0/4} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$$

73) NCERT Pg # 69-70, para-8.3

74)

$E^0 = 0 - \left(E^0_{M^{+2}/M}\right)$

$\Delta G^0 = -nF\left(-E^0_{M^{+2}/M}\right)$

$96.5 \times 1000 = +2 \times 96500 \times E^0_{M^{+2}/M}$

75)

A, B, C > H_2 > D

C > B, D

A > C

on combining

A > C > B > H_2 > D

76) $\Lambda^\infty_M (Ag_2CrO_4) = 2\Lambda^\infty_M (Ag^+) + \Lambda^\infty_M \left(CrO_4^{-2}\right)$

$= 2 \times 127 + 273$

$= 254 + 273 = 527$ ohm^{-1} cm^2 mol^{-1}

$\Lambda^\infty_M = \frac{k \times 1000}{S}$

$s = \frac{2 \times 10^{-2} \times 1000}{527}$

$Ksp = 4s^3$

77) $Q = n \times n_f \times 96500 = \frac{12.3}{123} \times 6 \times 96500$

$= 57900$ C

78)

$Pb^{+4} \rightarrow Pb^{+2}$ (Spontaneous)
$Sn^{+4} \rightarrow Sn^{+2}$ (Non-spontaneous)

BIOLOGY

91) NCERT-XI, Pg. # 61

92) NCERT-XI, Pg. # 66, 67

93)

NCERT, Pg. # 67

94) NCERT XI, Pg. # 67

95) XI-NCERT Page No. # 58

96)

XI-NCERT Page No. # 57, 58

97)

NCERT XI Pg : 65, 66, 67

98) NCERT XI Pg : 67

99)

XI-NCERT Page No. # 63, 64, 65

100)

XI-NCERT Page No. # 61, 62, 63, 64, 65

101) XI-NCERT Page No. # 67, 68

102) XI-NCERT Page No. # 61

103) NCERT-XI, Pg. # 65

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104) NCERT-XI, Pg. # 74

105) NCERT-XI, Pg. # 76-77

106) NCERT-XI, Pg. # 73, 74

107) NCERT-XI, Pg # 72

109) NCERT-XI, Pg # 75

110) NCERT-XI, Pg # 73

111) NCERT XII Pg # 76

112) NCERT XII Pg # 75, 76

113) NCERT-XI Pg#73

114) NCERT XI, Pg. # 73-74

115) NCERT XI, Page No. 72

116)

NCERT-XII Pg. # 22

117)

NCERT-XII Pg. # 7

118) NCERTXII Pg. # 11

119) NCERT XII Page No. 16

120) NCERT XII Pg. # 13

121)

NCERT-XII Pg. # 6

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122)

NCERT XII Pg. # 16

123) NCERT XII Pg. # 7, 9, 10

124)

NCERT XII Pg. # 9

126) NCERT XII, Pg. No. # 14

127) NCERT XII Pg. # 21

128) NCERT XII, Pg. No. # 13

129) NCERT Pg. # 18

130) NCERT, Pg. # 11

131) NCERT XI, Pg. No. # 65

132) NCERT-XI Pg. # 62 (d) in figure 5.13

133) NCERT-XI, Pg # 64

134) NCERT (XIth) Pg. # 61, 62

135) NCERT XI, Pg. # 59

136) NCERT Pg. # 46

137)

NCERT-XII Pg. # 52(E) & 57(H)

138) NCERT(XII) Pg# 43 Para:3.1

139) NCERT(XII) Pg# 50 Para:3.4

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140) NCERT(XII) Pg#54/59(H) Para: 3.6

141) NCERT(XII) Pg# 54 (E), 58 (H)

142) NCERT (XIIth) Pg. # 52 (E) 57(H) , Fig. (3.11)

143) NCERT-XII Page#48

144) Module Page#24

145) NCERT, Pg # 49,50
In general oxytocin do not have role in regulation of menstrual cycle.

146) NCERT, Pg # 51,52

147) NCERT Pg. # 50,51

148)

NCERT XII Pg # 55, para 1, Pg#51(E)

149)

Application based - Allen module.

150)

NCERT-XII, Pg. # 60

151)

NCERT XII Page # 27

152)

NCERT Pg. # 36

153) NCERT - XII, Pg-47 [Ist Para]

154) NCERT - XII, Pg-49 [Fig-3.8]

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155)

NCERT (XII) Pg # 44/77(H) Fig. : 3.2

156)

NCERT-XII # Page No. 43

157) NCERT Pg. # 30-31

158) NCERT Pg. # 30

159) NCERT-XII, Page # Eng. 47, Hindi 51

160) NCERT Page # Eng. 48, Hindi 52

161) NCERT Page # Eng. 48, Hindi 53

162) NCERT Page # Eng. 48, Hindi 53

163) NCERT Page - 47

164) Second meiotic division occur when sperm comes in contact with ovum in perivitelline space during fertilisation.

165) NCERT-XII, Pg. # 44

166) NCERT Pg. # 46

167) NCERT Pg. # 48

168) NCERT, Pg. # 45
Saheli is non-steroidal preparation.

169) NCERT Pg # 42

170) NCERT Pg. # 44

171) NCERT, Pg. # 45 (E)

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172) NCERT, Pg. # 48 (E)

173) NCERT Pg. # 48

174) NCERT Pg. # 48

175) NCERT XII Pg. # 44

176) NCERT XII, Pg. # 46 & 48

177) NCERT-XII, Pg. # 44

178) Module-8, Pg. # 104

179) NCERT Pg # 57

180)

NCERT Pg No # 60

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CHEMISTRY

1) Select the incorrect statement about graphite ?

(1) The bond order of C–C bond = 1.33
(2) Graphite is a electricity conductor
(3) All C atoms are sp^2 hybridised
(4) It is harder than diamond

2) Which compound is not a monobasic acid ?

(1) H_3PO_2
(2) H_3BO_3
(3) HNO_3
(4) H_3PO_3

3) BCl_3 and $AlCl_3$ exist in :-

(1) monomer and dimer respectively
(2) dimer and monomer respectively
(3) monomer and monomer respectively
(4) dimer and dimer respectively

4)

$A + H_2O \rightarrow B + HCl$
$B + 3H_2O \rightarrow C + 3HCl$

Compound (A), (B) and (C) will be respectively:
(1) PCl_5, $POCl_3$, H_3PO_3
(2) PCl_5, $POCl_3$, H_3PO_4
(3) $SOCl_2$, $POCl_3$, H_3PO_3
(4) PCl_3, $POCl_3$, H_3PO_4

5) Which is correct statement :-

(1) Compounds of $T\square^{+1}$ are less ionic than $T\square^{+3}$ but have more stability
(2) Compunds of $T\square^{+1}$ are more ionic than $T\square^{+3}$ but have less stability
(3) Compounds of $T\square^{+1}$ are more ionic than $T\square^{+3}$ and have more stability
(4) Compounds of $T\square^{+1}$ are less ionic than $T\square^{+3}$ and have less stability

6) Select incorrect match :-

(1) HOCl < HOBr < HOI - Acidic strength increasing order.

(2) $NH_3 < PH_3 < AsH_3 < SbH_3$ - Order of PK_b value

(3) HF < HCl < HBr < HI - Acidic character increasing order.

(4) $Ga_2O_3 < In_2O_3 < Tl_2O_3$ - Oxidizing power increasing order.

7) Which of the following element exist in liquid state during summer ?

(1) B

(2) Al

(3) Ga

(4) In

8) In which of the following reaction XeF_6 act as fluoride acceptor ?

(1) $XeF_6 + AsF_6 \rightarrow$

(2) $XeF_6 + SbF_5 \rightarrow$

(3) $XeF_6 + CsF \rightarrow$

(4) $XeF_6 + PF_5 \rightarrow$

9) The correct order of solubility in water for He, Ne, Ar, Kr, Xe is :

(1) He > Ne > Ar > Kr > Xe

(2) Ne > Ar > Kr > He > Xe

(3) Xe > Kr > Ar > Ne > He

(4) Ar > Ne > He > Kr > Xe

10) **Statement (I) :** Heavier members of carbon family do not form pπ-pπ bonds.
Statement (II) : Atomic orbitals of heavier members of carbon family are too large and diffuse to not have effective pπ-pπ overlapping.

(1) Both statement I and II are True

(2) Statement I is false but statement II is true.

(3) Statement I is True but statement II is False.

(4) Both statement I and II are False.

11) F_2 is stronger oxidising agent, than Cl_2 due to
(a) Low enthalpy of dissociation of F – F bond.
(b) High hydration enthalpy of F^-.
(c) High electron gain enthalpy of Cl.

(1) a and b

(2) a, b and c

(3) only a

(4) only b

12) Which of the following is most stable with respect to disproportionation for +3 oxidation state ?

(1) P

(2) As

(3) Sb

(4) Bi

13)

$Na_2S_2O_3.5H_2O$ is used in photography to :-

(1) Reduce AgBr to metallic Ag

(2) Remove reduced Ag

(3) Remove undecomposed AgBr as a soluble complex

(4) Convert metallic Ag to silver salt

14) Which one of the following complexes is not expected to exhibit isomerism ?

(1) $[Ni(NH_3)_4(H_2O)_2]^{2+}$

(2) $[Pt(NH_3)_2Cl_2]$

(3) $[Ni(H_2O)_2Cl_2]$

(4) $[Ni(en)_3]^{2+}$

15) Both geometrical and optical isomerism is shown by :-

(1) $[Co(en)_2Cl_2]^{+}$

(2) $[Co(NH_3)_5Cl]^{+2}$

(3) $[Co(NH_3)_4Cl_2]^{+}$

(4) $[Cr(OX)_3]^{-3}$

16) Which of the following complex represent diamagnetic and inner orbital complex :-

(1) $[Cr(H_2O)_6]^{3+}$

(2) $[Co(CN)_6]^{4-}$

(3) $[Ni(H_2O)_6]^{2+}$

(4) $[Fe(CN)_6]^{4-}$

17) Wavelength of absorption is minimum in the visible region for the complex ion

(1) $[Cr\ (H_2O)_6]^{3+}$

(2) $[Cr(NH_3)_6]^{3+}$

(3) $[Cr(NO_2)_6]^{3-}$

(4) $[Cr(en)_3]^{+3}$

18) Which is low spin complex :-

(1) $[Fe(CN)_6]^{3-}$

(2) $[Co(NO_2)_6]^{3-}$

(3) $[Mn(CN)_6]^{3-}$

(4) All of these

19) Which of the following order of stability of complex ion is incorrect ?

(1) $[Fe(C_2O_4)_3]^{3-} > [Fe(H_2O)_6]^{3+}$

(2) $[Fe(edta)]^{-} > [Fe(en)_3]^{3+}$

(3) $[Ni(en)_2]^{2+} > [Ni(DMG)_2]$

(4) $[Fe(CN)_6]^{3-} > [Fe(CN)_6]^{4-}$

20) EAN of central metal atom in $Co_2(CO)_8$ will be :-

(1) 35

(2) 36

(3) 37

(4) 33

21) The IUPAC name for
$[Pt(NH_3)_3(Br)(NO_2)(Cl)]Cl$ is :-

(1) Triamminechlorobromonitroplatinum (IV) chloride

(2) Triamminebromidochloridonitrito-N platinum(IV) chloride

(3) Triamminechlorobromoplatinum (IV) chloride

(4) Triamminechloronitrobromoplatiunum (IV) chloride

22) **Assertion (A) :** The metal carbon bond in metal carbonyls possesses both σ and π character
Reason (R) : The ligand to metal bond is a π bond and metal to ligand bond is a σ bond.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

23) The strongest C-O bond is present in-

(1) $[Cr(CO)_6]^{+}$

(2) $[Fe(CO)_5]$

(3) $[V(CO)_6]^{-}$

(4) all have equal C-O bond strength

24)

Correct option for the following columns after matching is

	Column-I		Column-II
	Formula		Maximum Moles of AgCl can be precipitated with excess $AgNO_3$
(I)	$PdCl_2.4NH_3$	(A)	4
(II)	$PtCl_4.6H_2O$	(B)	2
(III)	$CoCl_3.4NH_3$	(C)	0
(IV)	$PtCl_4.2HCl$	(D)	1

(1) I-B, II-A, III-D, IV-C

(2) I-D, II-A, III-B, IV-C

(3) I-B, II-D, III-A, IV-C

(4) I-B, II-D, III-C, IV-A

25) Which of the following compound does not give test of Cr^{+3} ion in aq. solution :-

(1) $K_2SO_4.Cr_2(SO_4)_3 24H_2O$

(2) $Cr_2(SO_4)_3$

(3) $[Cr(CN)_6]^{-3}$

(4) $CrCl_3$

26)

Arrange the following in decreasing order of ionic size

(1) $Ce^{3+} > Lu^{3+} > Gd^{3+} > Eu^{3+}$

(2) $Ce^{3+} > Eu^{3+} > Gd^{3+} > Lu^{3+}$

(3) $Eu^{3+} > Ce^{3+} > Gd^{3+} > Lu^{3+}$

(4) $Gd^{3+} > Eu^{3+} > Ce^{3+} > Lu^{3+}$

27) Which of the following is not a consequence of the Lanthanoid contraction ?

(1) 5d series elements have a higher IE_1 than 3d or 4d series

(2) Zr and Hf have a comparable size

(3) Zr and Hf occurs together in the earth crust in their minerals

(4) Fe and Co have a comparable size

28) Which of the following oxides of Cr is amphoteric:

(1) CrO_2

(2) Cr_2O_3

(3) CrO_5

(4) CrO_3

29) Which of the following pair can show +2 oxidation state normally ?

(1) $_{58}Ce$ and $_{65}Tb$
(2) $_{63}Eu$ and $_{70}Yb$
(3) $_{58}Ce$ and $_{63}Eu$
(4) $_{58}Ce$ and $_{70}Yb$

30) Which of the following chlorides produce chromyl chloride vapour when heated with conc. H_2SO_4 and $K_2Cr_2O_7$ (Means chromyl chloride test).

(1) NaCl
(2) AgCl
(3) Hg_2Cl_2
(4) $PbCl_4$

31) In the 3d series the enthalpy of atomization of zinc is the lowest (126 kJ mol^{-1}) due to -

(1) It have stronger interatomic interaction
(2) It have high melting point
(3) It have high degree of ionization
(4) In the formation of metallic bonds, no electrones from 3d orbitals are involved in case of zinc.

32) The product of oxidation of I^- with MnO_4^- in weak alkaline medium is -

(1) IO_3^-
(2) I_2
(3) IO^-
(4) IO_4^-

33) The incorrect statement(s) about Cr^{2+} and Mn^{3+} is (are)
[Atomic numbers of Cr = 24 and Mn = 25]

(1) Cr^{2+} is a reducing agent
(2) Mn^{3+} is an oxidizing agent
(3) Both Cr^{2+} and Mn^{3+} exhibit d^4 electronic configuration
(4) When Cr^{2+} is used as a reducing agent, the chromium ion attains d^5 electronic configuration

34) Highest fluoride and oxide of Mn are respectively:-

(1) MnF_7, Mn_2O_7
(2) MnF_5, Mn_2O_7
(3) MnF_4, Mn_2O_7
(4) MnF_6, Mn_2O_7

35) Which of the following 3d series metal has positive value of $E°(M^{2+}/M)$?

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(1) Zn
(2) Cu
(3) Mn
(4) Cr

36) Mark the compound which is soluble in hot water but insoluble in cold water.

(1) Lead chloride
(2) Mercurous chloride
(3) Stronsium sulphate
(4) Silver chloride

37) Which of the following sulphides has the maximum solubility product ?

(1) HgS
(2) PbS
(3) CuS
(4) MnS

38) The presence of magnesium is confirmed in the qualitative, analysis by the formation of a white crystalline precipitate of :

(1) $Mg(HCO_3)_2$
(2) $MgNH_4PO_4$
(3) $MgNH_4(HCO_3)_3$
(4) $MgCO_3$

39) In acidic solution of sodium nitroprusside when H_2S gas is passed then -

(1) Violet colour will form
(2) Violet ppt will form
(3) H_2S will form 'S'
(4) Colour will not be changed

40)

	Column-I		**Column-II**
(p)	$ZnCl_2 + H_2S$ (basic)	(a)	Blood red colour
(q)	$CuSO_4 + NH_4OH$ (excess)	(b)	White ppt form.
(r)	$FeCl_3$ + KSCN sol^n	(c)	deep blue solution
(s)	$Pb(NO_3)_2$ + KI	(d)	Yellow ppt form

(1) p-c, q-a, r-d, s-b
(2) p-b, q-c, r-a, s-d
(3) p-d, q-b, r-a, s-c
(4) p-b, q-a, r-c, s-d

41) Nessler reagent is used to detect.

(1) NH_4^+
(2) PO_4^{-3}
(3) MnO_4^-
(4) SO_4^{-2}

42) **Assertion :** Be, Mg salts do not give flame test
Reason : Be, Mg have high ionisation energy.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion
(3) Assertion is True but the Reason is False
(4) Both Assertion & Reason are False

43) Which of the following is not a blue coloured compound.

(1) $Fe_4[Fe(CN)_6]_3$
(2) $CuSO_4.5H_2O$
(3) $[Cu(NH_3)_4]SO_4$
(4) $[Ni(DMG)_2]$

44) Which of the following anions is not easily removed from aqueous solution by precipitation ?

(1) Cl^-
(2) SO_4^{2-}
(3) NO_3^-
(4) CO_3^{2-}

45) The brown ring test for NO_3^- is due to formation of the complex ion with formula

(1) $[Fe(H_2O)_6]^{2+}$
(2) $[FeNO(CN)_5]^{2-}$
(3) $[Fe(H_2O)_5NO]^{2+}$
(4) $[Fe(H_2O)(NO)_5]^{2+}$

BIOLOGY

1) Choose incorrect statement –

(1) Experimental verification of the chromosomal theory of inheritance was done by T.H. Morgan and his colleagues.

(2) Morgan worked with the tiny fruit flies, *Drosophila melanogaster*.

(3) A single mating could produce a large number of progeny in fruit flies.

(4) *Drosophila melanogaster* has few types of hereditary variations that can be seen with low power microscope.

2) Read the following statement & select correct option with respect to Human skin colour :
Statement-I : Human have the genotype with all the dominant allele will have lightest skin colour.
Statement-II : Human have the genotype with 3 dominant allele and 3 recessive allele will have an intermediate skin colour.

(1) Both **statement-I** and **statement-II** are incorrect.

(2) **Statement-I** is correct but **statement-II** is incorrect.

(3) **Statement-II** is correct but **statement-I** is incorrect.

(4) Both **statement-I** and **statement-II** are correct.

3) Which of the following statements are correct regarding multiple allele (ABO blood group) :
(A) Gene I has three alleles I^A, I^B and i.
(B) The allele I^A and I^B produce same type of sugar.
(C) In ABO blood group there are six different phenotypes are possible.
(D) In human, each person possesses any two of the three I gene alleles.
Choose the correct answer from the option given below :

(1) A and D only

(2) A, B and C only

(3) A, B, C and D

(4) B and C only

4) Match the following column-I with column-II.

	Column-I		**Column-II**
(a)	Heterozygous	(I)	Appear only in Homozygous condition
(b)	Dominants	(II)	Dissimilar allele
(c)	Gene	(III)	Appear in F_1 and F_2
(d)	Recessive	(IV)	Functional unit of inheritance

Choose the correct option :

(1) a-II, b-III, c-I, d-IV

(2) a-II, b-III, c-IV, d-I

(3) a-III, b-IV, c-II, d-I

(4) a-II, b-I, c-IV, d-III

5) **Assertion (A)** - In Mendel's experiment the alleles do not show any blending and that both the traits are recovered as such in the F_2 generation. **Reason (R)** - Factors of a pair segregate

from each other such that a gamete receives only one of the two factors.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

6) Read the following statements :-
(A) DNA is double stranded & both strands are complementary.
(B) DNA resist changes by evolving repair process.
(C) RNA used to act as genetic material as well as catalyst.
(D) RNA being a catalyst was reactive & hence stable.
Choose the correct option :

(1) A,B & D statements are correct

(2) B,C & D statements are correct

(3) C,D & A statements are correct

(4) A,B & C statements are correct

7) Read the following statements :-
(A) A gene is defined as the functional unit of inheritance.
(B) The DNA sequence coding for tRNA or r-RNA molecule also define a gene.
(C) The exons are said to be those sequences that appear in mature or processed RNA
(D) Inheritance of a character is not affected by promoter and regulatory sequences of structural gene.
Select the set of correct statements in given option :

(1) A, B and C

(2) B, C and D

(3) A, C and D

(4) A, B and D

8) Correctly match the column I and II.

	Column-I		**Column-II**
(a)	RNA	(i)	Double stranded, complementary
(b)	Promoter	(ii)	Polynucleotide phosphorylase
(c)	DNA	(iii)	First genetic material
(d)	Severo Ochoa enzyme	(iv)	Binding site for RNA polymerase.

(1) a–iii, b–iv, c–i, d–ii

(2) a–i, b–iii, c–ii, d–iv

(3) a–iv, b–iii, c–i d–ii

(4) a–iii, b–ii, c–iv d–i

9) **Assertion** : In HGP for sequencing DNA from a cell is isolated and converted into random fragment of relatively smaller sizes.
Reason : DNA is a long polymer and there are technical limitations in sequencing long pieces of DNA.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

10) **Statement – I :** The backbone of a polynucleotide chain is formed due to sugar and phosphates.
Statement – II : In RNA, every nucleotide residue has an additional –OH group present at 2'–position in the deoxyribose sugar.

(1) Statement I is correct and statement II is incorrect
(2) Statement I is incorrect and statement II is correct
(3) Statement I and II both are correct
(4) Statement I and II both are incorrect

11) Ribozyme are :

(1) RNA act as catalyst
(2) DNA act as catalyst
(3) Protein act as catalyst
(4) 1 & 2 Both

12) Protein digesting enzymes ___A____ and RNA digesting enzyne ___B___ did not affect transformation but digestion with ___C___ did inhibit transformation.
Select the correct option for A,B and C

	A	B	C
(1)	Proteases	RNases	DNases
(2)	Endonucleases	RNases	DNases
(3)	Exonucleases	RNases	DNases
(4)	RNases	Proteases	DNases

(1) 1
(2) 2
(3) 3
(4) 4

13) Which strain of the bacteria in Griffith's experiment are responsible for pneumonia ?

(1) *Streptococcus pneumoniae* - S strain
(2) *Streptococcus bulgaricus* - S strain
(3) *Streptococcus pneumoniae* - R strain

(4) *Streptococcus pneumoniae* - Both S & R strain

14) Match the following :

(A)	hn-RNA	(i)	RNA Polymerase I
(B)	t-RNA	(ii)	RNA Polymerase II
(C)	m-RNA	(iii)	RNA Polymerase III
(D)	5s-rRNA		

(1) A-(i), B-(ii), C-(iii), D-(ii)

(2) A-(ii), B-(ii), C-(iii), D-(ii)

(3) A-(ii), B-(iii), C-(ii), D-(iii)

(4) A-(i), B-(iii), C-(iii), D-(ii)

15) What will be the sequence of m-RNA produced from the following coding strand :-
5'ATTGCCAATTGCTAT3'

(1) 5' AUUGCCAAUUGCUAU3'

(2) 5'UAACGGUUAACGUAU3'

(3) 5'TAACGTTAACGATA3'

(4) 5' TATCGTTAACCGTTA3'

16) Select the incorrect statement regarding the salient features of the double helix structure of DNA :

(1) It is made of two polynucleotide chains

(2) The two chains have anti-parallel polarity

(3) Guanine is bonded with thymine by three H-bonds

(4) The two chains are coiled in a right handed fashion.

17) Which of the following statement is correct regarding the process of replications in prokaryotes.

(1) The DNA - dependent DNA polymerase catalyse polymerisation only in one direction, that is 5' → 3'

(2) On template with polarity 5' → 3' the replication is continuous

(3) On template with polarity 3' → 5' the replication is discontinuous.

(4) The DNA dependent DNA polymerase can initiate the process of replication.

18) Read the following statements :
(a) AUA codes for methionine (met) and it also act as initiator codon.
(b) Regulation of lac operon by repressor is referred to as positive regulation.
(c) The Human genome project was completed in 2003.
(d) Inheritance of character is also affected by the promotor and regulatory sequences of structural gene.
Select the correct option :

(1) a, b

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(2) c, d
(3) a, c
(4) a, d

19) During maturation process, In tailing adenylate residues are added at 3'-end of hnRNA ?

(1) In accordance to initiation-factor (σ)
(2) In a template dependent manner.
(3) In accordance to termination-factor (ρ)
(4) In a template independent manner.

20) DNA polymorphism play very important role in :

(1) DNA finger printing
(2) Evolution
(3) Speciation
(4) All of the above

21)

What is the effect of low glucose and high lactose on the lac operon.

(1) It remains off
(2) It turns on
(3) It switches between on and off states rapidly
(4) It become partially activated

22) In eukaryotes, the regulation could be exerted at -
a. Transcriptional level
b. RNA processing level
c. Transport of m-RNA
d. Translation level.
Which of the above option is correct.

(1) (a) and (b) only
(2) (a), (b) and (c) only
(3) (a), (b), (c) and (d)
(4) (b), (c) and (d) only

23) The pitch of the helix is 4.4 nm and there are roughly 10 bp in each turn. The distance between a bp in a helix is approximately.

(1) 0.44 mm
(2) 0.34 nm
(3) 0.34 mm
(4) 0.44 nm

24) Which DNA is more stable -

(1) G - 40%, A - 10%

(2) G - 30%, A - 20%

(3) G - 25%, A - 25%

(4) G - 20%, A - 30%

25) Select the odd one with reference to DNA :

(1) Able to undergo mutations

(2) Able to express itself directly

(3) Chemically stable

(4) Act as genetic material

26) Select the correct statement regarding post transcriptional changes in eukaryotic cells :-

(1) Capping is addition of Guanine at 5' end of RNA.

(2) Ribozymes are involved in process of splicing.

(3) mRNA includes both exons and introns.

(4) During tailing 200-300 adenine residues are added at 3' end of hnRNA, complementary to template.

27) The two allele of a gene pair are located on __A__ of __B__.
Fill in the blanks with correct option :

	A	B
(1)	Homologous site	Non Homologous chromosome
(2)	Non Homologous site	Homologous chromosome
(3)	Homologous site	Homologous chromosome
(4)	Non Homologous site	Non Homologous chromosme

(1) 1

(2) 2

(3) 3

(4) 4

28) Select the incorrect option w.r.t. organisms and their set of chromosomes.

	Organism	Male	Female
(1)	Human	2A + XY	2A + XX
(2)	Birds	2A + ZW	2A + ZZ
(3)	Grosshopper	2A + XO	2A + XX
(4)	*Drosophila*	2A + XY	2A + XX

(1) 1

(2) 2
(3) 3
(4) 4

29) Sahiwal cow is result of :-

(1) Natural selection of wild cow
(2) Artificial selection of wild cow
(3) Somatic variation
(4) Test cross

30) Which disorder is inherited through allosomes ?

(1) Phenylketonuria
(2) Haemophilia
(3) Thalassemia
(4) Sickle cell anemia

31) Select the incorrect match -

(1) Test Cross – Tt × tt
(2) Out Cross – Tt × TT
(3) Monohybrid cross – Tt × Tt
(4) Dihybrid test cross – TTRR × TtRr

32) Identify wrong statement :-

(1) During meiosis, every one pair segregates independently of another pair is true for chromosomes.
(2) Experimental verification of chromosomal theory of inheritance was done by Morgan and his colleagues.
(3) Honey bee show haploid-diploid mechanism of sex determination.
(4) In most of birds, male is heterogametic and female is homogametic.

33) In polygenic inheritance, the effect of each dominant allele on phenotype is :-

(1) Dominant
(2) Recessive
(3) Additive
(4) Subtractive

34) Phenylketonuria manifests itself through phenotypic expression characterised by _____.

(1) Increase in hair and skin pigmentation
(2) Mental retardation and a reduction in hair and skin pigmentation.
(3) Increase in mental capacity

(4) No effect

35) If a white flowered snapdragon plant is crossed with pink flowered snapdragon plant, the produced off spring will be :-

(1) All pink flowered

(2) All white flowered

(3) Pink as well as white flowered

(4) Red, pink as well as white flowered

36) Match list-I and list-II

	List-I (Character)		**List-II (Contrasting traits)**
(A)	Seed shape	(I)	Violet / white
(B)	Flower colour	(II)	Yellow / green
(C)	Seed colour	(III)	Tall / dwarf
(D)	Stem height	(IV)	Round / wrinkled

Choose the correct answer from the options given below :-

(1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(2) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(3) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(4) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

37) If you want to find out the correct genotype of a dominant plant (BB/Bb). Which of the following genotype will you cross with it ?

(1) BB

(2) Bb

(3) bb

(4) BB/Bb

38) Which of the following can be explained on the basis of independent assortment ?
(A) In dihybrid cross 9 : 3 : 3 : 1 ratio is observed in F_2 generation.
(B) Allele do not show any blending.
(C) In a hybrid, segregation by one pair of characters is independent of the other pair of character.
Choose the correct answer from the option given below :-

(1) Both A and C

(2) Only B and C

(3) Only A

(4) Only B

39) Which one of the following cannot be explained on the basis of Mendel's law of dominance ?

(A) Expression of both the characters in F_2 generation.
(B) Allele do not show any blending.
(C) 3 : 1 proportion obtained at F_2 generation.
Choose the correct answer from the options given below :-

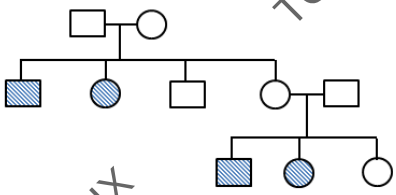
(1) A and B
(2) B and C
(3) A and C
(4) Only B

40) In F_2 generation of dihybrid cross, F_2 ratio is deviated from 9 : 3 : 3 : 1 this is because :-

(1) two genes are independent
(2) two genes are closely located on same chromosome
(3) 2 genes are distantly located on same chromosome
(4) two genes segregate independently

41)

Given pedigree show the inheritance of :



(1) Autosomal Recessive
(2) Autosomal dominant
(3) Sex linked Recessive
(4) Sex linked dominant

42) Which of the following statement is incorrect ?

(1) Colour blindness is due to a defect in the red or green cone of the eye.
(2) Haemophilia is a dominant disease.
(3) The genes that lead to red-green colour blindness are on the X chromosome.
(4) A daughter will not normally be colour blind unless her mother is atleast a carrier and her father is colour blind.

43) How many of them are correctly matched ?

(i)	Roquefort cheese	Bacteria
(ii)	Wine	Saccharomyces cerevisiae
(iii)	Citric acid	Aspergillus niger
(iv)	Biogas	Trichoderma sp.

(1) Only ii

(2) ii, iii
(3) i, iv
(4) i, ii, iii, iv

44) Which of the following statement is incorrect ?

(1) Biogas is a mixture of gases.
(2) Biogas predominantely contain methane.
(3) Methanogens bacteria are main producer of biogas.
(4) Biogas technology in India was developed by NBRI and CDRI.

45) Which of the following is not a benefit that plant have due to mycorrhizal association ?

(1) absorbs phosphorus from soil and passes it to the plant.
(2) resistance to root-borne pathogens.
(3) tolerance to salinity and drought
(4) helps in pollination

46) Which of the following tools of recombinant DNA technology is incorrectly paired with its use?

(1) Restriction enzyme – Production of blunt ends
(2) DNA ligase – Creates sticky ends of restriction fragments
(3) DNA polymerase – used in a PCR to amplify section of DNA
(4) Reverse transcriptase – Production of cDNA from RNA

47) Which enzyme is required for the ligation of vector DNA molecules with desired DNA molecules in recombinant DNA technology ?

(1) DNA polymerase
(2) DNA ligase
(3) Alkaline phosphatase
(4) Reverse transcriptase

48) Cutting of gel for removal of DNA is called :

(1) Elution
(2) Upstream processing
(3) Downstream processing
(4) Transformation

49) Read the following statements and select the **incorrect** statement(s).
a. Cellulase is used to degrade the cell wall of plant.
b. Chitinase is used to degrade the cell wall of fungus.
c. Lysozyme is used to degrade the cell wall of animal.
d. Purified DNA ultimately precipitates out after the addition of chilled ethanol.

(1) Only a and b

(2) Only c
(3) Only c and d
(4) Only b, c and d

50) What happens when DNA fragment is inserted in to plasmid PBR^{322} using enzyme Pvu-I or Pst-I ?
(a) Inactivation of amp^R gene
(b) Inactivation of tet^R gene
(c) Unable to grow in presence of ampicilin
(d) Unable to grow in presence of tetracycline
(e) Unable to replicate

(1) b & d are correct
(2) a and c are correct
(3) a, c & e are correct
(4) b, d & e are correct

51) **Assertion :** Biotechnology work to improve the quality of human life especially in the field of food production and health.
Reason : Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

52) Which of the following statement is **not correct** ?

(1) Genetically engineered insulin is produced in *E-Coli.*
(2) In man insulin is synthesised as a proinsulin.
(3) The proinsulin has an extra peptide called B-peptide.
(4) The functional insulin has A and B chains linked together by disulphide bonds.

53)

Recombinant DNA technology is used in formation of :-

(1) Curd
(2) Wine
(3) Bread
(4) DNA vaccine

54) Construction of first recombinant DNA was done by using native plasmid of :-

(1) *Salmonella typhimurium*
(2) *Escherichia coli*
(3) *Bacillus thuringiensis*

(4) Yeast

55) *Bacillus thuringiensis* has been used to kill certain insects such as
(a) Tobacco budworm
(b) Army worm
(c) Beetles
(d) Flies
(e) Mosquitoes

(1) b, c and d
(2) a, b and e
(3) a, c, d and e
(4) a, b, c, d and e

56)

Insect tolerant gene from *Bacillus thuringiensis* is introduced using Ti plasmid of

(1) *Escherchia coli*
(2) *Agrobacterium tumifaciens*
(3) *Haemophilus influenzae*
(4) *Arabidopsis thaliana*

57) Recombinant DNA technology uses :-
(A) Restriction endonucleases
(B) DNA ligase
(C) Cloning vector
(D) Electrophoresis

(1) A, B only
(2) B, C only
(3) C, D only
(4) A, B, C, D all

58) Match column-I with column-II, and choose the correct combination from the option given below :

	Column-I		**Column-II**
(a)	Molecular scissor	(i)	Large scale production of products
(b)	Plasmid	(ii)	Recombinant protein
(c)	Bioreactors	(iii)	pBR 322
(d)	Heterologous host	(iv)	Eco RI

(1) a-iii, b-iv, c-ii, d-i
(2) a-iv, b-iii, c-i, d-ii
(3) a-i, b-ii, c-iii, d-iv
(4) a-ii, b-i, c-iv, d-iii

59) Identify the plasmid among the following :-

(1) Hind II
(2) pBR322
(3) λ phage
(4) Hae III

60) Which of the following is used as a vector for plant cells ?

(1) Biolistic
(2) Microinjection
(3) Retrovirus
(4) Ti Plasmid

61) Plasmids are :-

(1) Semi-autonomously replicating DNA
(2) Semi-autonomously replicating chromosomal DNA
(3) Non-autonomously replicating extra-chromosomal DNA
(4) Autonomously replicating extra-chromosomal DNA

62) Ligation of foreign DNA at which of the following site will result in loss of tetracycline resistance of pBR322 :-

(1) Pst I
(2) Pvu I
(3) EcoR I
(4) BamH I

63) ....A....and....B....are the features that are required to facilitate cloning into a vector plasmid.

	A	B
(1)	Origin of replication	Cloning sites
(2)	Selectable marker	C-peptide
(3)	RNA interference	Desired gene
(4)	RNA polymerase	Primase

(1) 1
(2) 2
(3) 3

(4) 4

64) Which one of the following palindromic base sequences in DNA can be easily cut by EcoRI ?

(1) 5' _____ CACGTA _____ 3'
3' _____ CTCAGT _____ 5'

(2) 5' _____ CGTTCG _____ 3'
3' _____ ATGGTA _____ 5'

(3) 5' _____ GATATG _____ 3'
3' _____ CTACTA _____ 5'

(4) 5' _____ GAATTC _____ 3'
3' _____ CTTAAG _____ 5'

65) Choose the **incorrect** match :–

(1) Ori-responsible for controlling the copy number of linked DNA
(2) Rop–Codes for the protein involved in replication of plasmid.
(3) *Hind* III –Restriction site on pBR322
(4) *Sal* I – Selectable marker on plasmid of *Salmonella typhimurium*.

66) Given below are the two statements-
Statement-I : The separated DNA fragments are visualized only after staining it with eithidium bromide.
Statement-II : DNA is loaded on anode.
Choose the correct statement-

(1) Both statements are correct
(2) Only statement-I is correct
(3) Only statement-II is correct
(4) Only statement-I is incorrect

67)

Match List I with List II :

List I		**List II**	
A.	Denaturation	I	DNA primer
B.	Extension	II	Heat
C.	Taq polymerase	III	Deoxy Nucleotides
D.	Annealing	IV	*Thermus aquaticus*

Choose the correct answer from the options given below :
(1) A–(II), B–(III), C–(I), D–(IV)
(2) A–(II), B–(III), C–(IV), D–(I)
(3) A–(III), B–(II), C–(IV), D–(I)

(4) A–(III), B–(II), C–(I), D–(IV)

68) Match the following enzymes with their functions -

(a)	Restriction endonuclease	(i)	Joins the DNA fragments
(b)	Exonuclease	(ii)	Extends primer on genomic template
(c)	DNA ligase	(iii)	Cuts DNA at specific position
(d)	Taq polymerase	(iv)	Removes nucleotides from the ends of DNA

Select the correct option from the following :-

(1) a-iii, b-i, c-iv, d-ii

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-ii, c-i, d-iii

(4) a-ii, b-iv, c-i, d-iii

69) **Assertion :-** *Agrobacterium tumefaciens* is natural genetic engineer of plant.
Reason :- *Agrobacterium* is rarely used in plant genetic engineering.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

70) In gel-electrophoresis, all the DNA fragment move towards the base of the gel with variable velocities because :-

(1) DNA is positively charged

(2) DNA is double stranded

(3) DNA fragments have different molecular weights

(4) Both 2 and 3

71) The experimental manipulation of DNA of different species and producing recombinant DNA is known as :-

(1) Electrophoresis

(2) Recombinant DNA technology

(3) Transformation

(4) Somatic hybridization

72) What is true for Bt-toxin:-

(1) Bt-toxin protein exist as active toxin in bacterium

(2) Bt-toxin is converted into an active toxin due to acidic pH in gut of insect

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(3) Bt-toxin is converted into an active toxin due to alkaline pH in gut of insects

(4) Bt-toxin is formed by Biotechnology

73) Fill up the blanks.

I. Using *Agrobacterium* vectors, ...A.... specific genes are introduced into the host plant.

II. RNAi takes place in all eukaryotic organisms as a method of .....B.....

III. The genes .....C.... and .....D..... control cotton bollworms.

IV. .....E..... has been introduced in Bt corn to protect the corn from corn borer.

A to E in above statements can be:-

(1) A-fungal, B-pest resistance,
C-cry II Ab, D-cry II Ab, E-cry I Ac

(2) A-nematode, B- Cellular defence
C-cry I Ac, D-cry II Ab, E-cry I Ab

(3) A-insects, B-cellular defense,
C-cry II Ab, D-cry I Ac, E-cry I Ab

(4) A-virus, B-protection against viral infection,
C-cry I Ac, D-cry II Ab, E-cry II Ac

74) Read the following statement and choose correct option :-

I. Recombinant DNA technology is used to improve crop plants by increasing their productivity, by making them more nutritious and by developing disease resistance

II. Bt cotton is resistant to bollworm infestation

III. *Bacillus thuringiensis* is not harmed by self Cry protein because of its occurrence as protoxin (inactive)

IV. Protoxin Cry protein is changed into active Cry protein in the stomach of insects due to acidic pH in stomach

(1) All are correct

(2) I and IV are correct

(3) Only IV is false

(4) All are false

75) Indian government organisation which make decision regarding validity of GM research and safety of introducing GMO for public services is:-

(1) IARI

(2) GEAC

(3) CDFD

(4) Eli lily

76) Toxin protein crystals of Bacillus thuringiensis cannot kill the bacteria. Why?

(1) The toxin cannot pass through bacterial cell wall

(2) It is produced as protoxin which gets activated to toxin in the insect gut

(3) The toxin gets activated to protoxin in the insect gut

(4) The toxin cannot pass through the bacterial cell membrane

77) Read the following statements :-
(a) Reduced reliance on chemical pesticides
(b) Helped to increase post-harvest losses
(c) Increased efficiency of mineral usage by plants
(d) Enhanced nutritional value of food
How many statements are incorrect for application of GM crop

(1) One
(2) Two
(3) Three
(4) Four

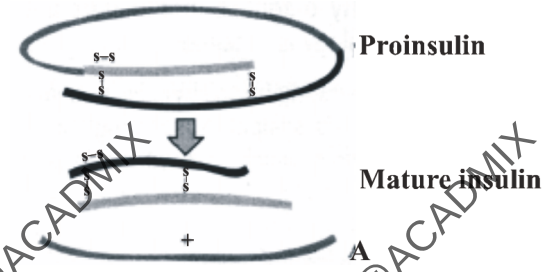
78) Given below are two statements:
Statement-I : Bacterial infection by *Bacillus thuringiensis* can kill caterpillar larvae.
Statement-II : The bacterial disease will kill the caterpillars, but leave other insects unharmed.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement-I** and **Statement-II** are incorrect.
(2) **Statement-I** is correct but **Statement-II** is incorrect.
(3) **Statement-I** is incorrect but **Statement-II** is correct.
(4) Both **Statement-I** and **Statement-II** are correct.

79) Important objective of biotechnology in agriculture is to :

(1) increase nitrogen content in plant.
(2) decrease the seed number.
(3) increase the plant weight.
(4) produce pest resistant varieties of plant.

80) Given figure represents the maturation of pro-insulin into insulin. Identify the product A.



(1) Polypeptide chain A
(2) Polypeptide chain B
(3) Polypeptide chain C
(4) Polypeptide chain D

81)
Probe is a :

(1) A single stranded DNA
(2) A single stranded DNA or RNA
(3) A single stranded RNA
(4) A single stranded protein

82) **Assertion :** Golden rice is a vitamin-A enriched rice.
Reason : Meristem culture is used to obtain virus free plants.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

83) Match List-I with List-II :

	List-I		**List-II**
(A)	Molecular Diagnosis	(I)	Polio vaccine
(B)	Transgenic mice	(II)	PCR
(C)	Transgenic rice	(III)	Vitamin-A enriched
(D)	Nematode resistance tobacco	(IV)	RNAi

(1) A-II, B-I, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-I, C-III, D-II

84) Which of the following is not true regarding the use of transgenic animals ?

(1) In study of normal physiology and development
(2) Understanding of development of diseases
(3) α-1 antitrypsin can be synthesised using them
(4) Toxicity testing in such animals allow us to obtain results in more time

85) Read the given statements and select the correct option :
Statement-I : Plants, bacteria, Fungi and animals whose genes have been altered by manipulation are called genetically modified organism (GMO).
Statement-II : Genetic modification of organism can have unpredictable results when such organisms are introduced into the ecosystem.

(1) Both statement-I and statement-II are correct
(2) Statement-I is correct but statement-II is incorrect
(3) Statement-I is incorrect but statement-II is correct
(4) Both statement-I and statement-II are incorrect

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86) A transgenic food crop which may help in solving the problem of acquired nightblindness in developing countries is :-

(1) Bt-rice
(2) Bt-Brinjal
(3) Golden rice
(4) Both (1) and (3)

87) The illegal and unlawful development of biomaterials without payment to the inhabitants of their origin is called :

(1) Biopatent
(2) Biotechnology
(3) Bio-war
(4) Biopiracy

88) Which statement is/are correct about GEAC?
(a) It makes decision regarding validity of GM research.
(b) Most of the industrialized nations are rich financially but poor in biodiversity and traditional knowledge.
(c) Transgenic animals are designed to increase our understanding of how genes contributed to development of disease
(d) Ensures safety of introducing GM organism for public services

(1) Statement (a) and (b)
(2) Statement (b) and (c)
(3) Statement (c) and (d)
(4) Statement (a) and (d)

89) **Statement-A :** At present about 30 recombinant therapeutics have been approved for human use the world over.
Statement-B : 27 documented varieties of Basmati rice are grown in India.

(1) Both A and B are true
(2) Only A is true
(3) Only B is true
(4) Both A and B are false

90) **Statement I** :- Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattles and pigs.
Statement II :- Adenosine deaminase enzyme is crucial for the immune system to function.

(1) Statement I is correct and statement II is incorrect
(2) Statement I is incorrect and statement II is correct
(3) Statement I and statement II both are correct
(4) Statement I and statement II both are incorrect.

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PHYSICS

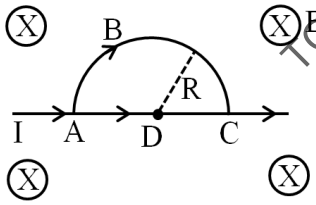
1) **Statement-I :** Ampere circuital law is valid for current changing with respect to time.
Statement-II : Newton's third law is valid for parallel long wire carrying constant current.

(1) Statement-I and II both are correct
(2) Statement-I and II both are incorrect
(3) Statement-I is correct and II is incorrect
(4) Statement-I is incorrect and II is correct

2) Which of the following is unit of magnetic dipole moment?

(1) Tesla
(2) Ampere
(3) Weber
(4) Bohr magneton

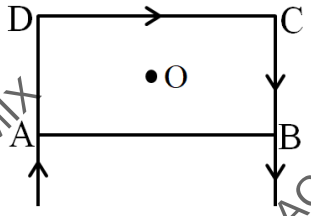
3) Current gets distributed in the loop as shown. The loop is kept uniform magnetic field 'B'. The net



force acting on the loop is?

(1) Zero
(2) I(2R)B
(3) IRB
(4) Cannot be calculated

4) Current I enters at A in a square loop of uniform resistance and leaves at B. The ratio of magnetic



field at center of square due to segment AB to that due to DC is-

(1) $\frac{3}{1}$
(2) $\frac{4}{1}$
(3) $\frac{3}{2}$
(4) $\frac{5}{2}$

5) A solenoid has 2000 turns. Its area of cross section is 1.6×10^{-4} m^2 and carries current of 4A. Find

out its magnetic dipole moment.

(1) 1.28 A-m^2

(2) 12.8 A-m^2

(3) 0.128 A-m^2

(4) 1.28 × 10^{-2} A-m^2

6)

Match the column (Where symbol has usual meaning)

	Column-I		**Column-II**
(P)	Diamagnetic	(x)	$\chi >> 1$
(Q)	Paramagnetic	(y)	$0 < \chi < 1$
(R)	Ferromagnetic	(z)	$-1 < \chi < 0$

(1) (P–y), (Q–z), (R–x)

(2) (P–x), (Q–y), (R–z)

(3) (P–z), (Q–y), (R–x)

(4) (P–z), (Q–x), (R–y)

7) **Statement-I :** Diamagnetism exist in all material.
Statement-II : The phenomenon of perfect diamagnetism in super conductor is called "mesiner" effect.

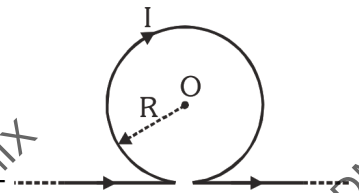
(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are correct.

8) Magnetic field at point O will be :-



(1) $\frac{\mu_0 I}{2R}\otimes$

(2) $\frac{\mu_0 I}{2R}\odot$

(3) $\frac{\mu_0 I}{2R}\left(1-\frac{1}{\pi}\right)\otimes$

(4) $\frac{\mu_0 I}{2R}\left(1+\frac{1}{\pi}\right)\odot$

9) A circular loop of radius R carries current I. Another circular loop of radius (r < < R) carries a current i is placed at the center of larger loop. The planes of the two circles are at right angle to

each other. Find the magnetic torque acting on the smaller loop.

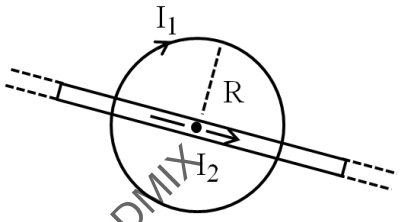
(1) Zero

(2) $\frac{\mu_0 \pi i I r^2}{4R}$

(3) $\frac{\mu_0 \pi i I r^2}{2R}$

(4) $\frac{\mu_0 \pi i I r^2}{R}$

10) Current flows through a circular ring as shown. An infinite current carrying wire is kept perpendicular to the plane of ring and passing through its center. Find the magnetic interaction



force per unit length between the ring and the wire.

(1) $\frac{\mu_0 I_1 I_2}{2\pi R}$

(2) $\frac{\mu_0 I_1 I_2}{\pi R}$

(3) $\frac{2\mu_0 I_1 I_2}{\pi R}$

(4) 0

11) A thin circular disk of radius R is uniformly charged with density ($\sigma > 0$) per unit area. The disk rotates about its axis with a uniform angular speed ω. The magnetic moment of the disk is-

(1) $2\pi R^4 \sigma\omega$

(2) $\pi R^4 \sigma\omega$

(3) $\frac{\pi R^4 \sigma\omega}{2}$

(4) $\frac{\pi R^4 \sigma\omega}{4}$

12) A proton, deutron and an α-particle are accelerated by same potential, enter in uniform magnetic field perpendicularly. Ratio of radii of circular path respectively :-

(1) $1 : \sqrt{2} : \sqrt{2}$

(2) 2 : 2 : 1

(3) 1 : 2 : 1

(4) 1 : 1 : 1

13) A flexible steel wire of length L has magnetic moment M. It is then bent into a semi circular, arc the new magnetic moment will be-

(1) M

(2) $\frac{2M}{\pi}$

(3) $\frac{M}{L}$

(4) ML

14) Two protons move parallel to each other with an equal velocity v = 300 km/sec. Find the ratio of forces of magnetic and electric interaction of the protons.

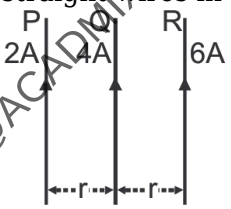
(1) 10^{-10}

(2) 10^{-6}

(3) 10^{8}

(4) 10^{-4}

15) P, Q and R long parallel straight wires in air, carrying currents as shown. The direction of



resultant force on R is :-

(1) Towards left

(2) Towards right

(3) The same as that of current in Q

(4) Perpendicular to plane of paper

16) A charge is moving in magnetic field $\vec{B} = 2\hat{i} + 2\hat{j} + 2\hat{k}$. If the acceleration of the particle at an instant is $(X\hat{i} + 2\hat{j} - 6\hat{k})$ m/sec^2, then value of X is-

(1) 4

(2) 2

(3) 3

(4) 1

17) A charge 'q' moves in a region where electric field and magnetic field both exist, then force on it :-

(1) $q(\vec{v} \times \vec{B})$

(2) $q\vec{E} + q(\vec{v} \times \vec{B})$

(3) $q\vec{E} + q(\vec{B} \times \vec{v})$

(4) $q\vec{B} + q(\vec{E} \times \vec{v})$

18) In a straight wire of length 0.5 m placed in magnetic field $\vec{B} = 2\hat{i} + 4\hat{j}$ T and current of 1 ampere is flowing along positive X-axis. Magnitude and direction of force acting on wire will be-

(1) $\sqrt{18}$N along positive X-axis direction

(2) $\sqrt{20}$N along positive X-axis direction

(3) 2N along positive Z-axis direction

(4) 4N along positive Z-axis direction

19) A long solenoid of radius 1 mm has 100 turns per mm. If 1A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is

(1) 12.56×10^{-2} T

(2) 12.56×10^{-4} T

(3) 6.28×10^{-4} T

(4) 6.28×10^{-2} T

20) A circular loop of radius r carries current i. How should a long straight wire carrying a current 3i be placed in the plane of circle so that magnetic field at the center of loop becomes zero

(1) At distance $\frac{3r}{\pi}$ from the center of loop

(2) At distance $\frac{2r}{\pi}$ from the center of loop

(3) At distance $\frac{2r}{3\pi}$ from the center of loop

(4) Not possible

21) A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B. The work done by field when particle completes one full circle is :-

(1) Zero

(2) BQ(2πR)

(3) $\left(\frac{mv^2}{R}\right)(2\pi R)$

(4) BQ(v) (2πR)

22) The current and voltage function in an AC circuit is :-

$I = 100 \sin\left(100t - \frac{\pi}{4}\right)$ mA

$V = 100 \sin\left(100t + \frac{\pi}{4}\right)$ volt

The power dissipated in the circuit is :-

(1) 0

(2) 2.5 W

(3) 5 W

(4) 10 W

23) Which of the following is false for the power factor of an AC circuit.

(1) Power factor is the ratio of real power to apparent power

(2) Power factor is the ratio of resistance to impedance

(3) Unit of power factor is watt

(4) Power factor is zero for purely inductive circuit.

24) A bulb of resistance 10 Ω connected to an inductor of inductance L, is in series with a source 100V, 50 Hz. If the phase angle between the voltage and current is $\frac{\pi}{4}$ radian. The value of L will be :-

(1) 0.0318 H

(2) 0.318 H

(3) 0.0787 H

(4) 0.787 H

25) An RLC series resonance circuit has a resonance frequency of $\frac{5000}{2\pi}$Hz and impedance at resonance is 56Ω. If quality factor of circuit is 25, than value of the inductance will be :-

(1) 0.42 H

(2) 0.35 H

(3) 0.56 H

(4) 0.28 H

26) In series LCR circuit resonance frequency is f_r, then select correct statement/statements.
Statement-I :- If frequency $f < f_r$, then current leads voltage in circuit.
Statement-II :- If frequency $f > f_r$, then current lags voltage in circuit.

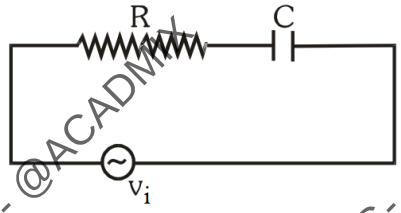
(1) **Statement I** is correct but **Statement II** is incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) Both **Statement I** and **Statement II** are incorrect.

27) A 50 Hz a.c. source of 20 volts is connected across R and C as shown in figure below. The



voltage across R is 12 volts. The voltage across C is :

(1) 8 V

(2) 16V

(3) 10 V

(4) Not possible to determine unless values of R and C are given

28) In an AC circuit with resistance R = 2Ω and alternating voltage $V = V_0 \sin\omega t$, where $V_0 = 2V$ and

$\omega = \frac{\pi}{2}$ rad/s, the power delivered at t = 0.5 sec. is :-

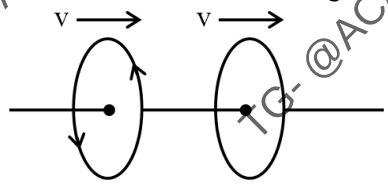
(1) 0 W

(2) 1 W

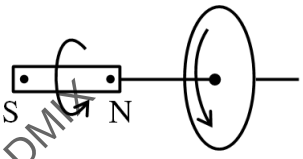
(3) 2 W

(4) 3 W

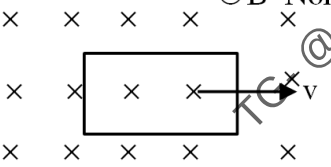
29) In which of the following EMF will not induced.



(a)



(b)



(c)

(1) Only (b)

(2) Only (a)

(3) a, b

(4) a, b, c

30) A conducting cycle wheel there are four spokes (rod) of length ℓ is rotating with angular velocity ω in uniform magnetic field B. The induced EMF between axel (entre of wheel) and rim is 10 V. Find emf between axel and rim if number of spokes increase to 8 (eight).

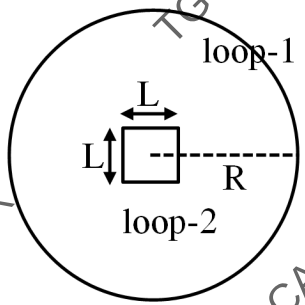
(1) 5 V

(2) 10 V

(3) 15 V

(4) 20 V

31) Find the mutual inductance between two loops 1 and 2 both loops are placed in same plane &



co-axial as shown in fig. and (L << R)

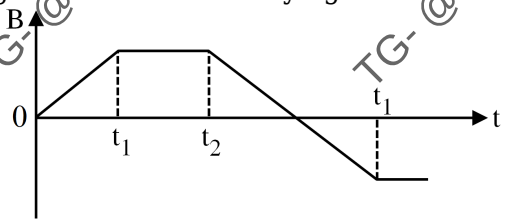
(1) $\frac{\mu_0 L^2}{2R}$

(2) $\frac{\mu_0 L}{2R^2}$

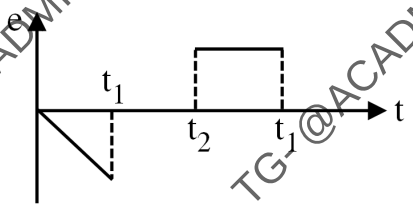
(3) $\frac{\mu_0 L}{2R}$

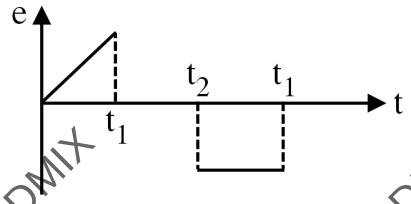
(4) $\frac{\mu_0 L^2}{2R^2}$

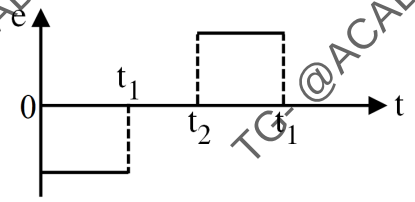
32) A conducting wire loop is placed transverse magnetic field which varying with time as shown in

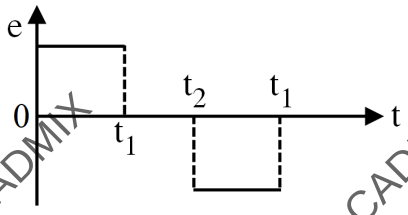


figure, then induced emf v/s time graph for loop is-

(1)


(2)


(3)


(4)


33) A rectangular loop of dimension 33 cm × 11 cm is changed to circle in time 0.6 sec. The magnetic field present is 0.3 T. The induced emf is-

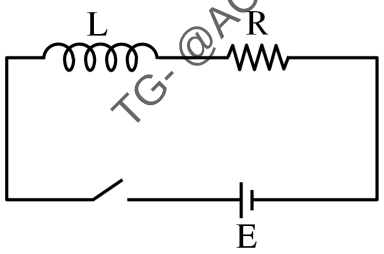
(1) –12.65 mV

(2) +12.65 mV

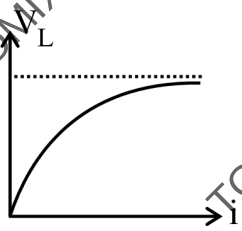
(3) –6.33 mV

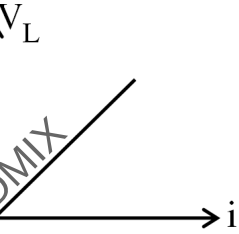
(4) +6.33 mV

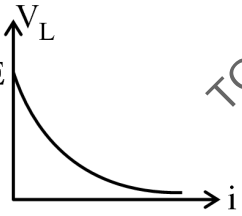
34) In figure switch is closed at t = 0, If V_L is the voltage induced across inductor and i is

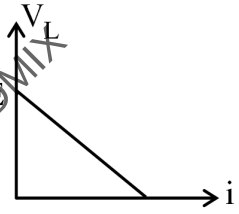


instantaneous current, the correct variation of V_L versus i is given by-

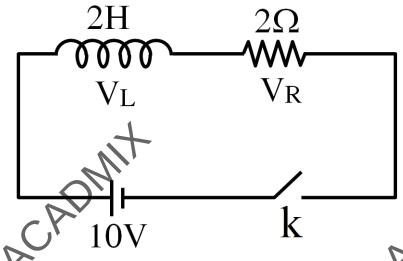
(1)


(2)


(3)


(4)


35) In the circuit shown in figure switch 'k' is closed at time t = 0. Match the following two columns.



Column-I		**Column-II**	
(a)	V_L at t = 0	(p)	Zero
(b)	V_R at t = 0	(q)	10 V
(c)	V_L at t = 1s	(r)	10/e V
(d)	V_R at t = 1s	(s)	$\left(1-\frac{1}{e}\right)10V$

(1) (a)→(q), (b)→(r), (c)→(s), (d)→(p)

(2) (a)→(q), (b)→(p), (c)→(r), (d)→(s)

(3) (a)→(p), (b)→(s), (c)→(q), (d)→(r)

(4) (a)→(p), (b)→(r), (c)→(q), (d)→(s)

36) The equivalent inductance of two inductors is 2.4 mH when connected in parallel and 10 mH when connected in series. The difference between two inductance is-

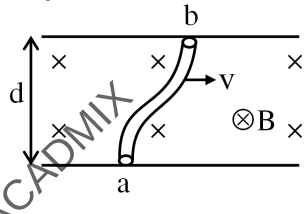
(1) 4 mH

(2) 3mH

(3) 2mH

(4) 16mH

37) A wire ab moves with velocity $v = v_0\hat{i}$ in uniform magnetic field $\vec{B} = -B_0\hat{K}$. Find emf induced between end a and b of wire.



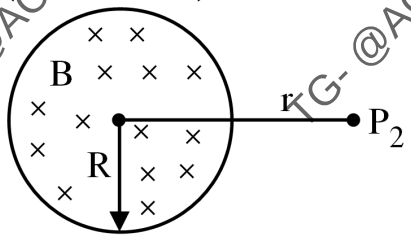
(1) Zero

(2) vBd end a is +ve.

(3) $\frac{vBd}{2}$, end a is +ive

(4) vBd, end b is +ive.

38) In given diagram, the magnetic field changes with time according to the given expression. B = $(2t^3 - 4t^2 + 0.8)$T and r = 2R = 5 cm



Magnitude of force on electron located at point P_2 at time t = 2 sec. will be-

(1) 8×10^{-21} N

(2) 8×10^{-19} N

(3) 5×10^{-2} N

(4) 4×10^{-20} N

39) The frequency of radio wave of wave length 30 m is

(1) 2 MHz

(2) 5 MHz

(3) 10 MHz

(4) 15 MHz

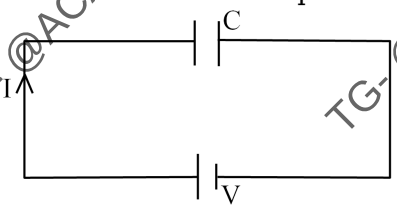
40) A parallel plate capacitor consists two circular plates of radius R = 0.1m. They are separated by a distance d = 0.5mm. If electric field between the plates change as $\frac{dE}{dt} = 5 \times 10^{13}\ Vm^{-1}s^{-1}$. Find displacement current between plates.

(1) 13.9 A
(2) 13.9×10^{-2} A
(3) 13.9×10^{-3} A
(4) 13.9×10^{-4} A

41) In electromagnetic wave the electric and magnetic field are not related as.

(1) They are perpendicular to each other
(2) They have phase difference of 90°
(3) They have no phase difference
(4) They have equal energy density

42) At a particular instant the current in the circuit given below is I. The displacement current between the plates of the capacitor shown below is -



(1) zero
(2) I
(3) $\frac{I}{2}$
(4) $\frac{I}{4}$

43) Light wave travelling along y direction. If the corresponding $\vec{E}$ vector at any time is along x axis, the direction of $\vec{B}$ vector at that time is along.

(1) y axis
(2) x axis
(3) z axis
(4) –z axis

44) Electromagnetic wave is produced due to-

(1) uniform moving charge
(2) static charge
(3) Accelerated charge
(4) All of these

45) A parallel plate capacitor is charged to 60 μC, due to a radioactive source, the plate loses charge

at the rate of 1.8×10^{-8} C/s. The magnitude of displacement current is-

(1) 3.6×10^{-8} A

(2) 1.8×10^{-8} A

(3) 4.1×10^{-11} A

(4) 5.7×10^{-12} A

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ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	1	2	3	1	3	3	3	1	1	4	3	3	1	4	3	4	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	1	3	2	4	2	2	1	4	1	4	3	2	1	4	2	4	2
Q.	41	42	43	44	45															
A.	1	1	4	3	3															

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	1	2	4	4	1	1	1	1	1	1	1	3	1	3	1	2	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	4	1	2	3	3	2	2	2	4	4	3	2	3	1	3	1	4	2
Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
A.	1	2	2	4	4	2	2	1	2	2	1	3	4	1	4	2	4	2	2	4
Q.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125
A.	4	4	1	4	4	2	2	2	3	3	2	3	2	3	2	2	1	4	4	3
Q.	126	127	128	129	130	131	132	133	134	135										
A.	2	2	1	4	1	3	4	4	1	3										

PHYSICS

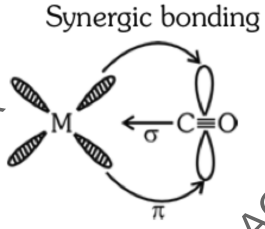
Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
A.	4	4	2	1	1	3	4	3	3	4	4	1	2	2	1	1	2	3	1	1
Q.	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
A.	1	1	3	1	4	3	2	2	3	2	1	3	1	4	2	3	4	1	3	1
Q.	176	177	178	179	180															
A.	2	2	4	3	2															

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6) In oxy acid down side in group basic strength increases.
HOCl < HOBr < HOI (Basic strength)

12)

Bi^{3+} is more stable due to inert pair effect



22)

23)

Page-254, NCERT-XII, Part-I

24) $[Pd(NH_3)_4]Cl_2$
$[Pt(H_2O)_6]Cl_4$
$[Co(NH_3)_4Cl_2]Cl$
$H_2[PtCl_6]$

25)

$[Cr(CN)_6]^{-3}$
complex part

26)

Size decreases along a period $Ce^{3+} > Eu^{3+} > Gd^{3+} > Lu^{3+}$

34)

MnF_4 (maximum fluoride)
Mn_2O_7 (maximum oxide)

35)

Cu has positive of (M^{+2}/M) E° in the 3d series due to high enthalpy of atomization.

39) No free S^{2-} ion available in acidic medium of H_2S.

41)

Nessler reagent is used to detect NH_4^+ ion.

BIOLOGY

46) NCERT XII, Pg. # 67

47) NCERT XII Pg#69

48) XII NCERT Pg # 61

49) NCERT XII Pg#56,57

50) NCERT XII Pg. No.#59

51) NCERT XII Pg # 80,81,88

52) NCERT XII Pg#92,93

53) NCERT XII Pg. No. - 104, 108, 112 (E)
114, 119, 122 (H)

54) NCERT XII Page-No. 119

55) NCERT, Pg. # 97

56) NCERT-XII, Pg # 115

57) NCERT XII Page-No. 101

58) NCERT-XII, Page No. # 100

59) NCERT-XII, Page No. # 111, 110

60) NCERT Pg. # 92

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61) NCERT Pg. # 81

62) NCERT Pg. # 90

63) NCERT-XII, Pg # 96,101,102,93(E) / 102,106,112,113(H)

64) NCERT, Pg # 95

65) NCERT, Pg # 104

66) Module

67) NCERT Pg. No. 115

68) NCERT Pg # 81, 82

69) NCERT Pg. No. # 92

70) NCERT XII Pg. # 104

71) NCERT Pg # 95

72) NCERT XII Pg. # 65

73) NCERT XII Page No. # 86, 87

74) NCERT XII Pg#88

75) NCERT Pg. 73

76) NCERT Pg. # 74

77) NCERT, Pg. # 66, 67, 70

78) NCERT, Pg. # 69

79) NCERT, Pg. # 69 and 74

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80) NCERT-XII, Pg. # 60

81) NCERT-XII, Pg. # 55

82) NCERT Pg. # 58

83) NCERT Pg. # 71

84) NCERT Pg. # 59

85) NCERT Pg. # 67

86) NCERT Pg#73

87) NCERT-XII, Pg. # 74, 73

88) NCERT, Pg. # 151, 152, 153, 155

89) NCERT, Pg. # 155

90) NCERT, Pg. # 158

91) NCERT-XII Pg. # 198-203

92) NCERT-XII, Pg. # 195,198,199

93) NCERT XIIth Pg # 198

94)

NCERT XII Pg # 171

95) NCERT (XII) Pg. # 199

96) NCERT XII Pg # 207

97)

NCERT XII Pg # 210, 211

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98) NCERT-XII, Pg # 167

99) NCERT (XIIth) Pg. # 164

100)
NCERT XII, Pg. No. 157

101)
NCERT XII, Pg. No. 170

102)
NCERT XII, Pg.#171, 172, 173

103)
NCERT Pg. No. 174; 172; 173

104) NCERT Page # 169

105) NCERT XII Pg.# 170

106) NCERT Pg # 170

107) NCERT, Pg. # 169

108) NCERT Pg # 169

109) NCERT (XII) Pg. # 167

110)
NCERT Pg. No. 169

111)
NCERT-XII, Pg. # 168

112) NCERT-XII, Pg. # 172, 173

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113) NCERT-XII Pg. # 165, 166, 167, 173

114) NCERT XII Pg # 170

115)

NCERT pg.no.168

116)

NCERT pg.no.165, 166

117) NCERT XII Pg. # 179

118) NCERT-XII Pg # 170, 180

119)

NCERT XII Pg # 180

120) NCERT Pg. # 184

121)

NCERT Pg. No. 179

122) NCERT Pg. # 179

123) NCERT-XII, Pg. # 157

124) NCERT-XII, Pg. # 179

125)

NCERT Pg. No. 182

126)

NCERT XII, Pg. No. 183

127) NCERT-XII, Pg. # 178, 179

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128) NCERT-XII, Pg # 180,182, 184

129) NCERT XII Pg.# 183, 184

130) NCERT-XII, Pg. # 179

131) NCERT XII, Pg.# 179

132)

NCERT-XII, Pg # 185

133) NCERT Pg # 184

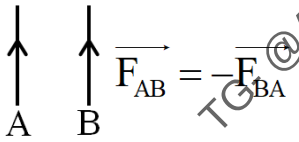
134) NCERT Pg # 185

135) NCERT Pg # 181, 182

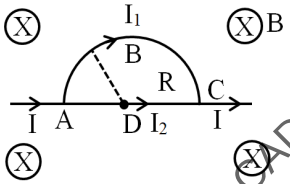
PHYSICS

136) Ampere circuital law is valid for constant current not for current varying with respect to time.

For parallel current carrying wire.



$\overrightarrow{F_{AB}} = -\overrightarrow{F_{BA}}$



138)

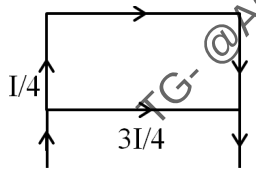
$F_{ABC} = I_1 (2R)B$

$F_{ADC} = I_2 (2R)B$

Since dir of both force same

$\vec{F}_{ABC} + \vec{F}_{ADC} = (I_1 + I_2)\ 2RB$

$= I(2R)B$



139)

$$\frac{I_1}{I_2} = \frac{3}{1}$$

$$\frac{B_1}{B_2} = \frac{3}{1}$$

140) M = NIA

$M = 2 \times 10^3 \times 4 \times 1.6 \times 10^{-4}$

$M = 1.28$ A-m^2

141)

For Diamagnetic

$-1 < \chi < 0$

For paramagnetic

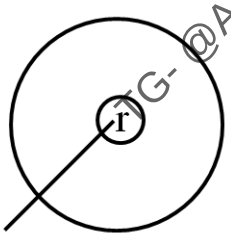
$0 < \square < 1$

For ferromagnetic

$\chi >> 1$

143) $B_0 = \frac{\mu_0 I}{2R} \otimes + \frac{\mu_0 I}{2\pi R} \odot$

$= \frac{\mu_0 I}{2R}\left(1 - \frac{1}{\pi}\right) \otimes$



144)

$\tau = MB \sin\theta$

$\tau = (i)(\pi r^2)\frac{\mu_0 I}{2R}\sin 90^\circ$

$\tau = \frac{\mu_0 \pi i(I) r^2}{2R}$

145) As magnetic field due to the ring would be opposite to direction of current so it would not exert any force.

146) $\frac{M}{L} = \frac{q}{2m}$

$M = \frac{q}{2m}(I\omega)$

$$M = \frac{(\sigma)\left(\pi R^2\right)}{2m}\left(\frac{mR^2}{2}\omega\right)$$

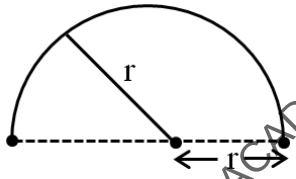
$$M = \frac{\sigma\pi R^4\omega}{4}$$

147) $r = \frac{\sqrt{2mqV_{acs}}}{qB} \propto \sqrt{\frac{m}{q}}$

$r_p : r_d : r_a$

$\sqrt{\frac{m}{e}} : \sqrt{\frac{2m}{e}} : \sqrt{\frac{4m}{2e}}$

$1 : \sqrt{2} : \sqrt{2}$



148)

$M = m \times L$

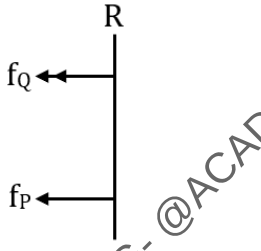
$M^1 = m\,(2r)$ $(\pi r = \ell)$

$M^1 = 2m\left(\frac{\ell}{\pi}\right)$ $r = \frac{\ell}{\pi}$

$M^1 = \frac{2M}{\pi}$

149) $\frac{F_m}{F_c} = \frac{v^2}{c^2}$

$\frac{F_m}{F_c} = \frac{(3\times10^5)^2}{(3\times10^8)^2} = 10^{-6}$



150) $f_{net} \Rightarrow$ (towards left)

like currents always attracts to each other

151) $\vec{F} \downarrow \vec{B}$

$\vec{a} \downarrow \vec{B}$

$\vec{a}.\vec{B} = 0$

$(X\hat{i} + 2\hat{j} - 6\hat{k}) \cdot (2\hat{i} + 2\hat{j} + 2\hat{k}) = 0$

$2X + 4 - 12 = 0$

X = 4

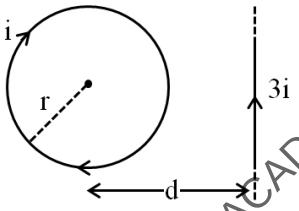
153) $\vec{F} = I(\vec{\ell} \times \vec{B})$

$\vec{F} = 1\left[\frac{1}{2}\hat{i} \times (2\hat{i} + 4\hat{j})\right]$

$\vec{F} = 2\hat{k}$

154) $B = \mu_0 ni = \mu_0 \frac{N}{\ell} i$

$\therefore B = 4\pi \times 10^{-7} \times \frac{100}{10^{-3}} \times 1 = 12.56 \times 10^{-2}\,T$



155)

if $B_{net} = 0$

$B_1 = B_2$

$\frac{\mu_0 i}{2r} = \frac{\mu_0 (3i)}{2\pi(d)}$

$d = \frac{3r}{\pi}$

156) Since $\vec{F} \perp \vec{B}$

$\vec{F} \perp \vec{v}$

so $W_{field} = 0$

157) $P = \frac{V_0 I_0}{2} \cos\phi = \frac{100 \times 100}{2} \times \cos\frac{\pi}{2} = 0$

158) Power factor = $\cos\phi = \frac{R}{Z} = \frac{P_{avg}}{P_{virtual}}$

159) $\tan\phi = \frac{X_L}{R}$

$\tan\frac{\pi}{4} = \frac{2\pi f \cdot L}{R} = 1$

$2\pi f.L = R$

$L = \frac{R}{2\pi f} = \frac{10}{2 \times \frac{22}{7} \times 50}$

$L = \frac{7}{220} = 0.0318\,H$

160) $Q = \frac{2\pi f_0 L}{R}$ At resonance Z = R = 56

$$L = \frac{QR}{2\pi f_0}$$

$$L = \frac{25 \times 56}{2\pi \times \frac{5000}{2\pi}} = \frac{25 \times 56}{5000} = 0.28\,H$$

161) By theory

162) Use $V^2 = V_R^2 + V_C^2$

163) Power $P = \frac{V^2}{R} = \frac{V_0^2}{R}\sin^2\omega t$

$$P = \frac{(2)^2}{2}\sin^2\left(\frac{\pi}{2} \times \frac{1}{2}\right) = 2\sin^2\left(\frac{\pi}{4}\right)$$

$$P = 2 \times \left(\frac{1}{\sqrt{2}}\right)^2 = 1\,W$$

164) No flux change in Coil/loop

165) emf independent of No. of spokes.

166) $M = \frac{\phi_2}{I_1} = \frac{B_1 A_2}{I_1}$

$$= \frac{\mu_0 I}{2R} \cdot \frac{L^2}{I}$$

$$= \frac{\mu_0 L^2}{2R}$$

167) $e = -\frac{d\phi}{dt} = -A\frac{dB}{dt}$

168) $\phi = -B\frac{\Delta A}{\Delta t} = -\frac{B}{\Delta t}[A_{Cir} - A_{Rectangle}]$

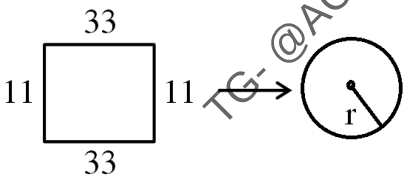
$$= \frac{-0.3}{0.6}\left[\pi r^2 - a \times b\right]$$

$$= -\frac{1}{2}\left[\frac{22}{7} \times 14 \times 14 - 11 \times 33\right] \times 10^{-4}$$

$$= -\frac{11}{2}[2 \times 28 - 33] = -\frac{11}{2}[56 - 33] \times 10^{-4}$$

$$= -\frac{11}{2} \times 23 \times 10^{-4} = -126.5 \times 10^{-4}$$

$$= -12.65mV$$



$88 = 2 \times \frac{22}{7} \times r$

r = 14 cm

169) $V_L = E\, e^{-Rt/L} \Rightarrow e^{-Rt/L} = \frac{V_L}{E}$

$I = I_o (1 - e^{-Rt/L}) \Rightarrow I = I_o \left(1 - \frac{V_L}{E}\right)$

170) $V_L + V_R = E$

$\Rightarrow V_L = -V_R + E$

$V_L = -RI + E$

171) In series $L_1 + L_2 = 10$ ..(1)

In parallel $\frac{L_1 L_2}{L_1 + L_2} = 2.4$

$L_1L_2 = 24$

$L_1 = \frac{24}{L_2}$

Put in eq. (1) $\frac{24}{L_2} + L_2 = 10$

$L_2^2 - 10L_2 + 24 = 0$

$(L_2 - 4)(L_2 - 6) = 6$

$L_1 = 4mH, L_1 = 6 mH$

$L_2 = \frac{24}{4}$ $L_2 = \frac{24}{6}$

$L_2 = 6mH,\ L_2 = 4\ mH$

Difference

$L_1 - L_2 = 6 - 4 = 2$ mH

172) Right hand palm rule.

$e = \left(\vec{v} \times \vec{B}\right) . \vec{\ell}$

e = vBd

= end b is +ive.

173) $E_{out} = \frac{R^2}{2r}\frac{dB}{dt} = \frac{R^2}{2(2R)}[6t^2 - 8t]$

at t = 2 $E_{out} = \frac{R^2}{4R}[6 \times 4 - 8 \times 2] = \frac{5 \times 10^{-2}}{2 \times 4}(8)$

$E_{out} = 5 \times 10^{-2} \frac{N}{C} = E_0$

Force on electron $F = qE_P = 1.6 \times 10^{-19} \times 5 \times 10^{-2}$

$F = 8 \times 10^{-21}$ N

174) $\upsilon = \frac{c}{\lambda} = \frac{3 \times 10^8}{30}$

$= 10^7$ Hz

= 10 MHz

175) $I_d = \epsilon_0 A \frac{dE}{dt}$

$= 8.85 \times 10^{-12} \times 3.14 \times (0.1)^2 \times 5 \times 10^{13}$

= 13.9 A

176) Theory

177) $I_d = I_c$

= I

178) $\hat{C} = \hat{E} \times \hat{B}$

$\hat{j} = \hat{i} \times$ ....

B along -z axis

179) Theory

180) $I_d = I_c$

$= 1.8 \times 10^{-8}$ C/s

$= 1.8 \times 10^{-8}$ A

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English

(1016CMD303032240051)

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024 - 2025)

Test Pattern: NEET(UG)

02-04-2025

PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAZV, MAAZ, MAAW, SPARK PRO

Test Booklet Code

L6

This Booklet contains **24** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case**.
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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SUBJECT : BIOLOGY

Topic : SYLLABUS-6

1. **Assertion :-** The interaction of cattle egret birds and cattle is an example of commensalism.
Reason:- It is an association between members of two different species in which one is benefitted while other is unaffected.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

2. Given below are two statements.
Statement - I :- Majority of the parasites harm the host, they may reduce the survival, growth and reproduction of the host and reduce its population density.
Statement - II :- Parasites help in maintaining species diversity in a community.
In the light of above statements, choose the most appropriate answer from the option given below :-

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

3. Select the correct option with respect to parasitism ?

(1) Copepods on marine fishes.

(2) Sea anemone and Clown fish

(3) Abingdon tortoise and goats

(4) Fig tree and Wasps

4. Identify the incorrect statement-

(1) Darwin spoke of the struggle for existence and survival of the fittest in nature.

(2) Interspecific competition is a potent force in organic evolution.

(3) Totally unrelated species could also compete for the same resources.

(4) In shallow south American lakes, visiting flaminogoes and resident fish compete for phytoplankton

5. Identify the incorrect statement :-

(1) Predators act as 'conduits' for energy transfer across trophic levels.

(2) In the absence of predators, prey species could achieve very high population density.

(3) Predators also help in maintaining species diversity in a community.

(4) Predators in nature are not prudent.

6. Choose the incorrect option from the given below :

(1) The Mediterranean orchid *Ophyrs* employs "sexual deceit" to get pollination done by a species of bee.

(2) In mycorrhizal association, photosynthesis is done by higher plants

(3) Orchid flower and wasp species are example of mutualism.

(4) Lichen is a relationship between algae and fungi

7. Population interactions represented by (+), (−) and (−), (0) refers to respectively :-

(1) Parasitism and Competition

(2) Predation and Amensalism

(3) Amensalism and Parasitism

(4) Amensalism and Predation

8. Match the column-I with column-II and give correct answer :

Column-I (Name of Interaction)		Column-II (Examples)	
(A)	Mutualism	(a)	Lice on humans
(B)	Commensalism	(b)	Fig tree and wasp
(C)	Parasitism	(c)	Abingdon tortoise and goats
(D)	Competition	(d)	Clown fish and sea anemone

(1) A-d, B-a, C-b, D-c (2) A-d, B-c, C-a, D-b

(3) A-b, B-d, C-a, D-c (4) A-b, B-d, C-c, D-a

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9. Match List - I with List - II :-

List - I		List - II	
(a)	Natality	(i)	The number of deaths in the population during a given period.
(b)	Mortality	(ii)	The number of births in the population during a given period.
(c)	Immigration	(iii)	The number of individuals of the same species that have come into the habitat from else where during the time period under consideration.
(d)	Emigration	(iv)	The number of individuals of the population who left the habitat and gone else where during the time period under consideration.

Choose the correct answers from the option given below :-

(1) (a) - (ii); (b) - (i); (c) - (iii); (d) - (iv)

(2) (a) - (i); (b) - (ii); (c) - (iii); (d) - (iv)

(3) (a) - (iv); (b) - (iii); (c) - (i); (d) - (ii)

(4) (a) - (iv); (b) - (iii); (c) - (ii); (d) - (i)

10. If in a pond there are 60 lotus plant last year and through reproduction 15 new plants are added. What would be the birth rate during the year?

(1) 0.25 offspring/lotus/year

(2) 4 offspring/lotus/year

(3) 0.4 offspring/lotus/year

(4) Zero offspring/lotus/year

11. Match the column correctly for this below equation $N_t = N_0e^{rt}$

	A		B
(i)	N_t	(a)	The base of natural logarithms
(ii)	N_0	(b)	Population density after time t
(iii)	r	(c)	Intrinsic rate of natural increase
(iv)	e	(d)	Population density at time zero

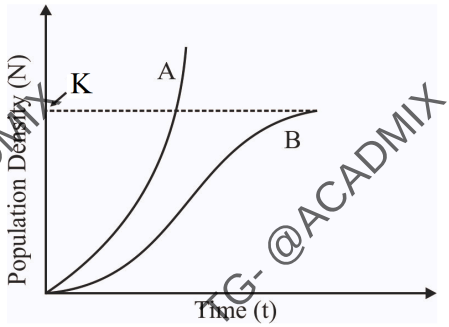
(1) (i) - b, (ii) - d, (iii) - c (iv) - a

(2) (i) - a, (ii) - c, (iii) - b (iv) - d

(3) (i) - d, (ii) - a, (iii) - b (iv) - c

(4) (i) - c, (ii) - b, (iii) - d (iv) - a

12. Study the population growth curves (A and B) in given graph and select the "incorrect option."



(1) Curve (A) show exponential growth represent by equation $\frac{dN}{dt} = rN$

(2) Curve (B) show logistic growth represent by equation $\frac{dN}{dt} = rN\frac{(K-N)}{K}$

(3) Exponential growth curve is considered as more realistic than the logistic growth curve.

(4) Curve (A) also represented by $N_t = N_0e^{rt}$

13. Organism breed many times in their life and produces a small number of large sized offspring :

(1) Pacific salmon fish, Bamboo

(2) Oyster, Pelagic fishes

(3) Pacific salmon fish, Pelagic fishes

(4) Birds, Mammals

14. Which of the following best explains why there are usually not more than five trophic levels in a food chain ?

(1) There are not enough organisms to fill more than five levels

(2) There is too much competition among the organisms at the lower levels to support more animals at higher levels.

(3) The statement is not true, there can be unlimited trophic levels

(4) Energy is lost at each trophic level

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15. Which of following is a function of ecosystem?

(1) Productivity (2) Decomposition

(3) Energy flow (4) All of these

16. Given below are two statements: One is labelled as assertion (A) and the other is labelled as Reason (R).

Assertion (A) : There is unidirectional flow of energy in ecosystem.

Reason (R) : All organisms are dependent for their food on producers, either directly or indirectly.

In the light of the above statements, choose the correct answer from the options given below.

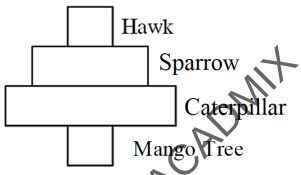
(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

17. The diagram shows a pyramid of number in a food chain -



What type of organism are caterpillars ?

(1) Primary producer

(2) Primary consumer

(3) Quaternary consumer

(4) Secondary consumer

18. Select the correct order of different step occuring during decomposition.

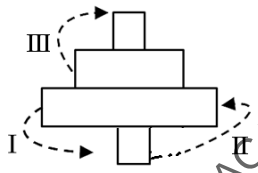
(1) Leaching → Catabolism → Fragmentation → Humification

(2) Fragmentation → Catabolism → Leaching → Humification

(3) Catabolism → Fragmentation → Leaching → Humification

(4) Catabolism, Fragmentation, Leaching, occurs simultaneously

19. Given diagram shows the number pyramid of terrestrial ecosystem, maximum amount of energy transfer will occur in which step?



(1) I (2) II

(3) III (4) Both III and II

20. Below two statement are given

Statement - I : Vertical distribution of different species occupying different level is called stratification.

Statement - II : Productivity is expressed in term of $gm^{-2} yr^{-1}$.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

21. Match the column I with column II.

Column – I		Column – II	
(a)	Fragmentation	(i)	Formation of humus
(b)	Leaching	(ii)	Breakdown of detritus into smaller particles
(c)	Humification	(iii)	Water soluble inorganic nutrients go down into the soil and get precipitated as unavailable salts

(1) a – i, b – ii, c – iii (2) a – iii, b – ii, c – i

(3) a – ii, b – i, c – iii (4) a – ii, b – iii, c – i

22. The primary producers of the deep-sea hydrothermal vent ecosystem are :

(1) Blue-green algae

(2) Coral reefs

(3) Red algae

(4) Chemosynthetic bacteria

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23. Primary productivity depend on -

(1) A variety of environmental factor

(2) Availability of nutrients

(3) Photosynthetic capacity of plants

(4) All of these

24. Which one is correct representation of Net Primary Productivity ?

(1) GPP − R = NPP (2) NCP − R = NPP

(3) Both (1) and (2) (4) GPP + R = NPP

25. Which of the following factor is responsible for slow rate of decomposition :-

(1) Nitrogen

(2) Water soluble substance

(3) Low temperature

(4) Aerobic condition

26. Choose the incorrect statement :-

(1) All animals depend on plant directly or indirectly for their food needs.

(2) Primary consumer will be herbivores.

(3) The consumer that feed on herbivores are carnivores, or secondary consumers

(4) Those animals that depend on the herbivores for food are called as secondary carnivores.

27. Match list-I with list-II :-

	List-I		List-II
(A)	Tertiary consumer	(I)	Plant
(B)	Secodnary consumer	(II)	Lion
(C)	Secondary producer	(III)	Wolf
(D)	Primary producer	(IV)	Zooplankton

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-II, B-I, C-III, D-IV

28. Colonisation of tropical Pacific Islands by humans is said to have led to the extinction of :-

(1) Less than 1000 species of native birds.

(2) More than 2000 species of native birds.

(3) 359 species of vertebrates.

(4) 87 species of vascular plants.

29. Analysis of records shows that extinctions across taxa are not random, then which one of the following groups are more vulnerable to extinction :-

(1) Vertebrates (2) Invertebrates

(3) Amphibians (4) Vascular plants

30. When a species becomes extinct, the plant and animal species associated with it in an obligatory way ?

(1) Will proliferate

(2) Also become extinct

(3) Will show no change

(4) Will select another host

31. The extinction of passenger pigeon was due to

(1) Increased number of predatory birds

(2) Over-exploitation

(3) Non- availability of the food

(4) Disease in pigeon

32. What is the correct order (increasing diversity) of plants in world ?

A. Fungi

B. Angiosperms

C. Algae

D. Mosses

(1) D ⟶ C ⟶ B ⟶ A

(2) B ⟶ C ⟶ D ⟶ A

(3) D ⟶ C ⟶ A ⟶ B

(4) D ⟶ A ⟶ C ⟶ B

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33. Considering that an overwhelmingly large proportion of the species are waiting to be discovered in tropics, biologist make a statistical comparison of the temperate-tropical species richness of an exhaustively by studied group of :

(1) Amphibians (2) Insects

(3) Molluscs (4) Fern and allies

34. One of the *ex-situ* conservation methods for endangered species is :-

(1) Wildlife Sanctuaries

(2) Biosphere Reserves

(3) Cryopreservation

(4) National Parks

35. When there are situations where an animal or plant is endangered or threatened and needs urgent measures to save it from extinction, which is the desirable approach ?

(1) *In-situ* conservation

(2) *Ex-situ* conservation

(3) *In-vivo* conservation

(4) *In-vitro* conservation

36. Which one of the following is not an example of hot spot in India ?

(1) Western Ghat (2) Indo-Burma

(3) Eastern Ghat (4) Himalaya

37. According to Robert May the global species diversity is about :

(1) 50 million (2) 25 million

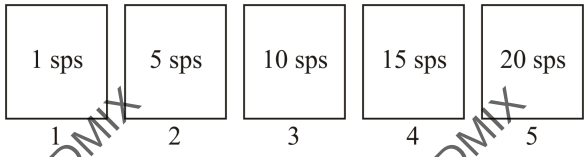
(3) 15 million (4) 7 million

38. In rivet popper hypothesis, Paul Ehrlich compared the rivets in an air plane to :

(1) Genus (2) Community

(3) Species (4) Ecosystem

39. Given below are 5 plotes :



Which one of the following plot has least year to year variation in productivity ?

(1) 4 (2) 3 (3) 2 (4) 5

40. Species-Area relationship is described by the following equation :

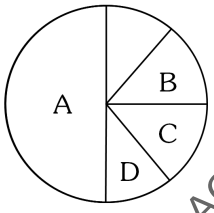
$\log S = \log C + Z \log A$

Where Z is :

(1) Area (2) Slope of the line

(3) Species richness (4) Species diversity

41. Given pie-chart shows number of species of major taxa of vertebrates.



Find out the A, B, C and D :

(1) A-Fishes, B-Mammals, C-Birds, D-Reptiles

(2) A-Mammals, B-Birds, C-Reptiles, D-Amphibians

(3) A-Birds, B-Fishes, C-Amphibians, D-Mammals

(4) A-Fishes, B-Birds, C-Reptiles, D-Amphibians

42. The variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges known as :

(1) Species diversity

(2) Ecological diversity

(3) Genetic diversity

(4) Molecular diversity

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43. __A__ long term ecosystem experiment using outdoor plot and found that plots with more species showed less year to year variation in total biomass. Identify the A ?

(1) Tilman's (2) Paul Ehrlich's

(3) Reiter's (4) Odum's

44. How many mega diversity countries are in the world ?

(1) 15 (2) 13 (3) 12 (4) 5

45. Which one of the following introduced illegaly in our rivers for aquaculture purposes ?

(1) Nile perch

(2) Cichlid fishes

(3) African catfish *Clarias gariepinus*

(4) Salmon fish

46. Which statement is true :-

(1) All pathogens are parasite

(2) All parasites are pathogen

(3) All parasites are viruses

(4) None

47.



A



B

Choose the correct option about given above :-

	A	B
(1)	Hallucinogen	Depressant
(2)	Morphine	Datura
(3)	Heroin	Bhang
(4)	Datura	Morphine

48. Read the following statements (A - D)

(A) When people are healthy they are more efficient at work

(B) Health for a long time, was considered as a state of body and mind where there was a balance of certain humors

(C) William Harvey thought that persons with 'blackbile' belonged to hot personality and would have fevers

(D) Pathogens have to adapt to life within the environment of the host

How many of the above statement/statements is/are incorrect?

(1) 3 (2) 4 (3) 1 (4) 2

49. Who disproved the "good humor hypothesis" of health using thermometer to demonstrate normal body temperature in person with blackbile ?

(1) Hippocrates

(2) Indian Ayurveda system of medicine

(3) William Harvey

(4) Both (1) and (2)

50. Incubation period for AIDS is usually :-

(1) 5-10 months (2) 5-10 years

(3) 2-12 months (4) 2-12 weeks

51. Which of the following statements is incorrect?

(1) Health, for a long time was considered as a state of body and mind.

(2) When people are healthy, they are more efficient at work.

(3) Balanced diet, personal hygiene and regular exercise are very important to maintain good health.

(4) Health simply means 'absence of disease' or physical fitness.

52. Which one is a wrong statement about passive immunity ?

(1) Induced by vaccination

(2) Immune response is short lived

(3) Immediate effect

(4) No immunological memory

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53. Given below are two statements : One is labelled as Assertion-A and the other is labelled as Reason-R.
Assertion (A) : Interferons work like cytokine barriers.
Reason (R) : Interferons protectes non-infected cells from further bacterial infection.
(1) Both A and R are true but R is NOT the correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is correct explanation of A.

54. What measure, a doctor would take to reduce the allergic symptoms of patient.
(1) Inject vaccine (2) Inject adrenaline
(3) Inject IgE (4) Inject histamine

55. T-lymphocytes are produced in :-
(1) Bone marrow (2) Spleen
(3) Thymus (4) Pancreas

56. Antigen binding site is present on which terminal of peptide chain in antibodies :
(1) N-terminal
(2) C-terminal
(3) Either N or C terminal
(4) In between N and C terminal

57. **Statement-I :-** Active immunity is very fast and takes time to give its full effective response.
Statement-II :- When ready made antibodies are directly give to protect the body against foreign agents, it is called passive immunity.
(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

58. Bluish discoloration of lips and nails due to infection of Alveoli may be caused due to :-
(1) *Orthomyxovirus*
(2) *Haemophilus influenzae*
(3) *Rhino* virus
(4) *Corynebacterium diptheriae*

59. Which of the following set of diseases is caused by bacteria.
(1) Typhoid and Malaria
(2) Herpes and Dengue
(3) Tetanus and common cold
(4) Plague and pneumonia

60. Read the following statements carefully
(A) Loss of appetite
(B) Sustained high fever.
(C) Parasite in large intestine
(D) In severe cases intestinal perforation is seen.
Select the correct statements for typhoid :-
(1) One (2) Two
(3) Three (4) Four

61. 'A healthy person acquires the infection by using towels, clothes or comb of the infected individual and the infection occurs easily in moist and warm conditions' Above statement is applicable to -
(1) Viral infections
(2) Bacterial infections
(3) Fungal infections
(4) Helminthic infections

62. Malignant malaria which is most serious one and can even be fatal is caused by :-
(1) *Plasmodium malariae*
(2) *Plasmodium vivax*
(3) *Plasmodium falciparum*
(4) *Plasmodium ovale*

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63. Active immunity may be gained by?

(1) Introducing weakened antigen or inactive antigen in our body.

(2) Detoxified toxins introduced in our body.

(3) Injection of serum immunoglobulins artificially in infected person.

(4) Passive transfer of mother IgG to baby.

64. Many diseases are spread by insect vectors. One such major vector is mosquito which spreads all of the following diseases, except :-

(1) Chikungunya

(2) Filariasis

(3) Ascariasis

(4) Malaria

65. Which of the following is the symptom of ascariasis?

(1) Internal bleeding and muscular pain

(2) Fever and anemia

(3) Blockage of intestinal passage

(4) All of these

66. Match List-I with List-II :

	List-I		List-II
(A)	Pneumonia	(I)	Fungus
(B)	Malaria	(II)	Nematode
(C)	Ringworm	(III)	Protozoa
(D)	Elephantiasis	(IV)	Bacteria

Choose the correct answer from the options given below :-

(1) A-IV, B-III, C-I, D-II

(2) A-II, B-I, C-III, D-IV

(3) A-IV, B-II, C-III, D-I

(4) A-III, B-II, C-I, D-IV

67. **Assertion (A) :** Graft from any source - an Animal, primate, any human being cannot be made randomly.
Reason (R) : Tissue matching, blood group matching are essential before undertaking any graft / transplant.

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is not the correct explanation of A.

68. Given below are two statements -
Statement 1 : Charas is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound.
Statement 2 : Tobacco has been used by human beings for more than 400 years.
Choose the correct answer from the following –

(1) Both Statement 1 and 2 are false

(2) Statement 1 is true and Statement 2 is false.

(3) Statement 1 is false and Statement 2 is true.

(4) Both Statement 1 and 2 are true.

69. In which part, parasite (Sporozoites) reproduces asexually ?

(1) Salivary gland of human

(2) Gut of mosquito

(3) Liver cells of human

(4) Gut of human

70. Which of the following options are not incorrect regarding Heroine ?

(1) It is also called as morphine

(2) It is a stimulant

(3) Excessive dosage of heroine causes hallucinations

(4) Heroine taken by snorting and injection

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71. Alfred Wallace, a naturalist worked on which island ?

(1) Srilanka (2) Galapagos

(3) South America (4) Malay Archepelago

72. According to Panspermia theory, life has :

(1) Come out of dead and decaying matter

(2) Originated from non-living organic molecules

(3) Come from another planet

(4) Created by God

73. Industrialization during 1920's in England led to :-

(1) Increase in population of white coloured biston moths.

(2) Increase in population of birds feeding on biston moths.

(3) Increase in population of dark coloured biston moths.

(4) Decrease in population of birds feeding on biston moths

74. Founder effect can be seen in a population because of

(1) No change in allele frequency

(2) Genetic drift

(3) Gene flow

(4) Natural Selection

75. The chronological order of human evolution from first human like being-hominid to recent is:-

(1) *Homo habilis* ⟶ *Homo erectus* ⟶ *Neanderthal* ⟶ *Homo sapiens sapiens*

(2) *Australopithecus* ⟶ *Homo erectus* ⟶ *Homo habilis* ⟶ *Homo sapiens*

(3) *Homo erectus* ⟶ *Homo habilis* ⟶ *Neanderthal* ⟶ *Homo sapiens*

(4) *Australopithecus* ⟶ *Homo erectus* ⟶ *Homo sapiens* ⟶ *Ramapithecus*

76. How many of the following statements are correct?

(a) Homology indicate common ancestry and based on divergent evolution.

(b) Different structures evolving for same function-convergent evolution

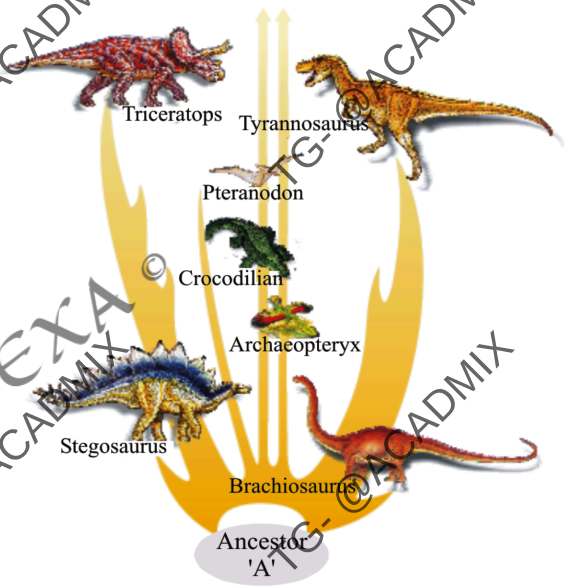
(c) Same structure developed along different direction-divergent evolution

(d) Analogy indicate common ancestory and based on convergent evolution

(1) Three (2) Four

(3) Two (4) One

77. Identify the ancestor-A in the family tree of dinosaur in given figure :-



(1) Synapsids

(2) Thecodonts

(3) Therapsids

(4) Pelycosaurus

78. Which of the following were formed in Stanley-Miller's experiment ?

(1) Fat, Carbohydrate, Pigments

(2) Carbohydrate, Nucleic acid, Protein

(3) Aspartic acid, Alanine, Glycine

(4) Butyric acid, Fatty acid, Nucleic acid

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79. In the given figure which one of the following represent the correct example of convergent evolution ?

(1) Mouse  Tasmanian tiger cat 

(2) Bob cat  Tasmanian wolf 

(3) Lemur  Spotted cuscus 

(4) Anteater  Marsupial Mole 

80. Which statement is **correct** for industrial melanism?

(1) It is a directional natural selection where median population is selected.

(2) It is a directional natural selection where either of the two extremities are selected at particular time.

(3) It is a stabilising natural selection where median population is selected.

(4) It is a stabilising natural selection where one of the extremities is selected.

81. **Statement-I :** Wings of bat and flippers of whale are analogous organs.
Statement-II : Analogy is based on divergent evolution.

(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

82. Which of the following options show correct sequence of evolution of plant forms :-

(1) Tracheophyte ⟶ Ferns ⟶ Bryophytes ⟶ Seed ferns

(2) Cycads ⟶ Gnetales ⟶ Ginkgos ⟶ Psilophyton

(3) Rhynia ⟶ Psilophyton ⟶ Progymnosperms ⟶ Seed ferns ⟶ Dicotyledons

(4) Lycopods ⟶ Conifers ⟶ Sphenopsids ⟶ Monocotyledons

83. In hardy-weinberg principle, frequency of homozygous receive (aa) is represented by :-

(1) p^2 (2) q^2 (3) 2pq (4) pq

84. Which of the following were evolved into the first amphibians ?

(1) Sauropsids (2) Jaw less fishes
(3) Coelacanths (4) Salamanders

85. Embryological support for evolution was proposed by :

(1) Alfred wallace (2) Ernst Heckel
(3) Malthus (4) Karl Ernst Von Baer

86. Direct ancester of Rhynia-type plants :

(1) Chlorophytes (2) Tracheophytes
(3) Psilophyton (4) Seed ferns

87. The following could be considered as an example of anthropogenic action :

(1) Antibiotic resistant organisms
(2) Darwin's finches
(3) Dinosaurs
(4) Tasmanian wolf

88. Which of the following is not a factor to affect Hardy-Weinberg equilibrium ?

(1) Gene flow (2) Genetic drift
(3) Mutation (4) Sexual selection

89. In which condition the gene ratio remains constant for any species?

(1) Natural selection (2) Random mating
(3) Mutation (4) Gene flow

90. Which of the following is the correct sequence of evolutionary history of mammals?

(i) Pelycosaurs (ii) Therapsids
(iii) Synapsids (iv) Early reptiles

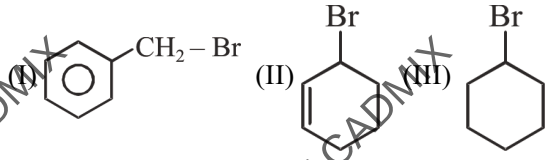
(1) iv, iii, ii, i (2) iv, ii, iii, i
(3) iv, iii, i, ii (4) iv, i, iii, ii

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SUBJECT : CHEMISTRY

Topic : SYLLABUS-6

91. Compare the rate of S_N1 reaction



(1) I > II > III (2) II > III > I

(3) II > I > III (4) III > I > II

92. Which of the following reaction is correctly matched?

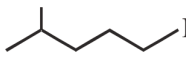
(1) $CH_3-\underset{CH_3}{\underset{|}{CH}}-O-CH_3 \xrightarrow{conc.HI} S_N1$

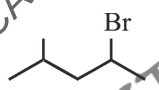
(2) $CH_3-\underset{CH_3}{\overset{CH_3}{\overset{|}{\underset{|}{C}}}}-O-CH_2-CH_3 \xrightarrow{conc.HI} S_N2$

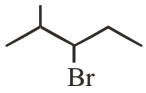
(3) $C_6H_5-CH_2-O-CH_3 \xrightarrow{conc.HI} S_N1$

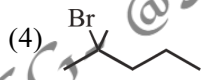
(4) $CH_3\text{-}CH_2\text{-}O\text{-}CH_2\text{-}CH{=}CH_2 \xrightarrow{conc.HI} S_N2$

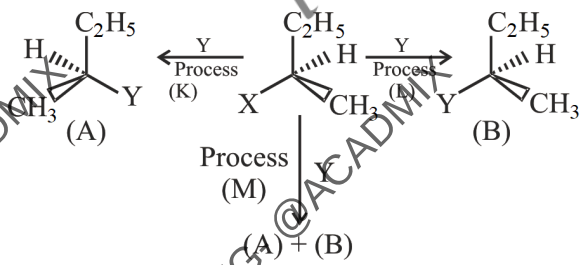
93. Which compound undergoes SN^2 reaction at fastest rate :-

(1) 

(2) 

(3) 

(4) 

94. 

Process (K), (L) and (M) are called respectively.

(1) Inversion, Retention, Racemisation

(2) Retention, Inversion, Racemisation

(3) Racemisation, Inversion, Retention

(4) Inversion, Racemisation, Retention

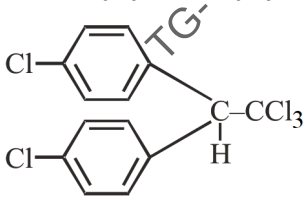
95. In the following reaction

$C_6H_5-CH_2Br \xrightarrow[2.\ H_3O^+]{1.\ Mg.Ether} X,$

The product 'X' is :-

(1) $C_6H_5CH_2OH$ (2) $C_6H_5CH_3$

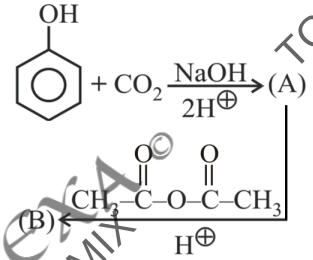
(3) $C_6H_5CH_2CH_2C_6H_5$ (4) $C_6H_5CH_2OCH_2C_6H_5$

96. 

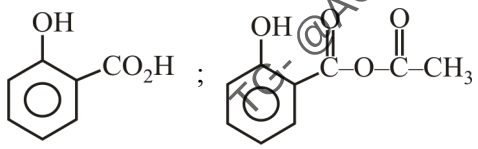
The formula suggests given compounds is :-

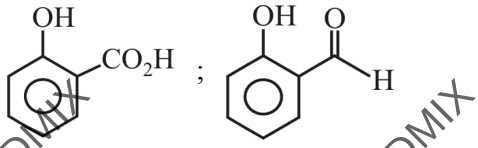
(1) BHC (2) DNA

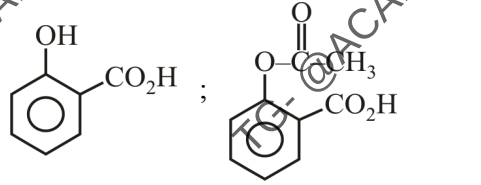
(3) DDT (4) RNA

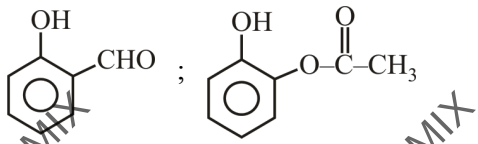
97. 

Product (A) and (B) respectively are :-

(1) 

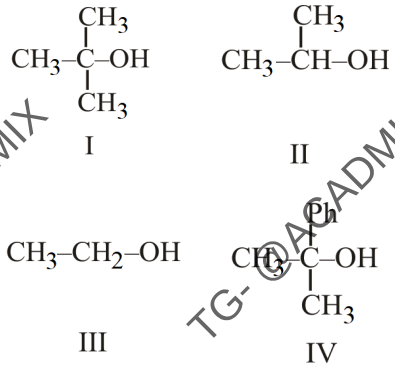
(2) 

(3) 

(4) 

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98. Arrange the following in correct order of their reactivity while reacting with $HCl/ZnCl_2$:-



I II III IV

(1) IV > I > II > III (2) I > II > III > IV

(3) IV > III > II > I (4) III > II > I > IV

99. Williamson ether synthesis follows :-

(1) S_N1 reaction (2) S_N2 reaction

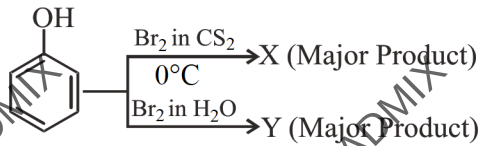
(3) E_1 reaction (4) E_2 reaction

100. Which of the following alcohol is not oxidised by PCC :-

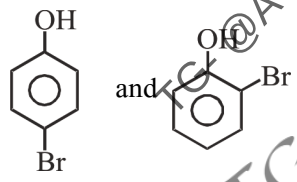
(1) n-butyl alcohol (2) sec-butyl alcohol

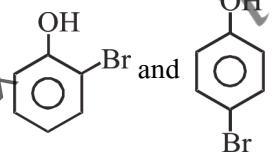
(3) iso-butyl alcohol (4) tert-butyl alcohol

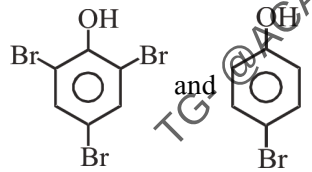
101.

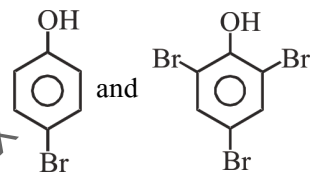


What is X and Y in above reaction respectively

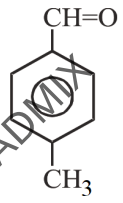
(1) 

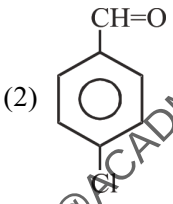
(2) 

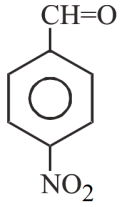
(3) 

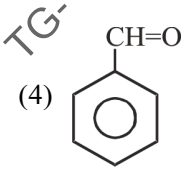
(4) 

102. Which is most reactive towards nucleophilic addition reaction :-

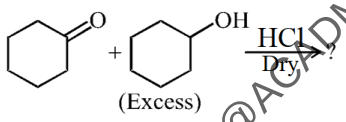
(1) 

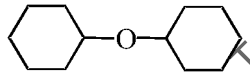
(2) 

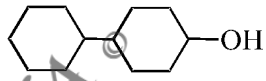
(3) 

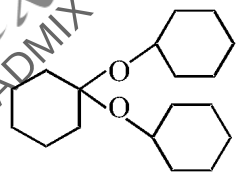
(4) 

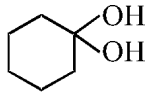
103. What is the major product of the following reaction:-



(1) 

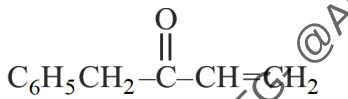
(2) 

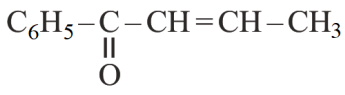
(3) 

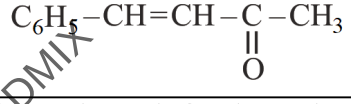
(4) 

104. The Cross aldol condensation product of benzaldehyde and acetone is -

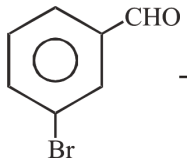
(1) $C_6H_5CH = C(CH_3)_2$

(2) 

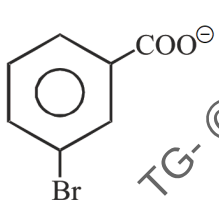
(3) 

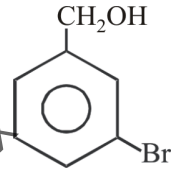
(4) 

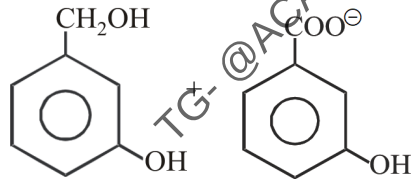
TG- @ACADMIX
TG- @STUDYNEXA
Watermarkly

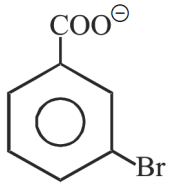
105. 

Products will be

(1) 

(2) 

(3) 

(4) 

106. Which of the following will produce silver mirror with Tollen's reagent (ammonical $AgNO_3$)

(1) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$ (2) HCOOH

(3) Glucose (4) All of these

107. What is compound Z ?

$CH_3-CH_2-CH_2-Br \xrightarrow{NaCN} X \xrightarrow[\Delta]{H_3O^{\oplus}} Y \xrightarrow[H^{\oplus}]{CH_3-CH_2-OH} Z$

(1) $CH_3-CH=CH\overset{\overset{O}{\|}}{C}OH$

(2) $CH_3-CH_2-CH_2CH=NOCH_2CH_3$

(3) $CH_3-CH_2-CH_2CH(OCH_2CH_3)_2$

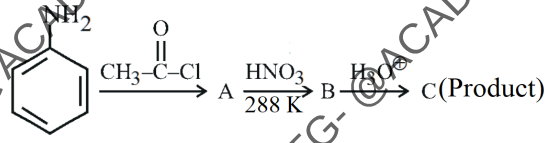
(4) $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-OCH_2CH_3$

108. Which of the following amine can be prepared by Gabriel Phthalimide synthesis ?

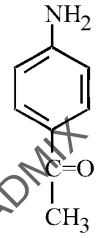
(1) Dimethyl Amine (2) Aniline

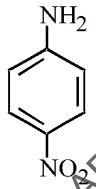
(3) Ethyl amine (4) Trimethyl amine

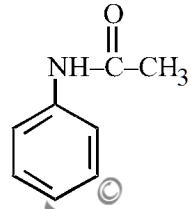
109. Suppose following reaction

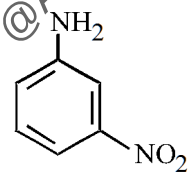


The major product 'C' will be :-

(1) 

(2) 

(3) 

(4) 

110. In the chemical reaction,

$CH_3CH_2NH_2 + CHCl_3 + 3KOH \longrightarrow$

The product is :-

(1) C_2H_5NC (2) $CH_3CH_2CH_2-NC$

(3) C_2H_5CN (4) $CH_3CH_2CONH_2$

111. Given below are two statements :

Statement I : Primary aliphatic amines react with nitrous acid to form aliphatic salt which being unstable, lebrate N_2 gas and alcohol.

Statement II : Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

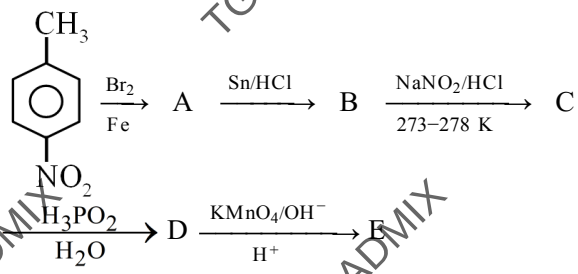
(1) Both **Statement-I** and **Statement-II** are incorrect.

(2) **Statement-I** is correct but **Statement-II** is incorrect.

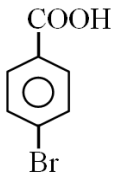
(3) **Statement-I** is incorrect but **Statement-II** is correct.

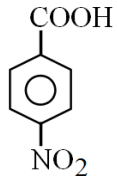
(4) Both **Statement-I** and **Statement-II** are correct.

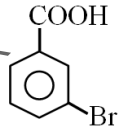
TG- @ACADMIX
TG-@STUDYNOTA ©
Watermarkly

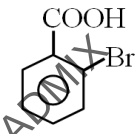
112. 

E is :-

(1) 

(2) 

(3) 

(4) 

113. Maltose is formed by :-

(1) α-D-glucose and α-D-glucose

(2) α-D-glucose and β-D-glucose

(3) α-D-glucose and β-D-fructose

(4) α-D-glucose and β-D-galactose

114. Match the list-I with list - II

	List -I		List - II
(a)	1° structure of protein	(i)	The shape in which a polypeptide chain can exist.
(b)	2° structure of protein	(ii)	Overall folding of polypeptide chains
(c)	3° structure of protein	(iii)	The spatial arrangement of two or more polypeptide chains with respect to each other.
(d)	4° structure of protein	(iv)	Specific sequence of amino acids

Select the correct option :-

(1) (a) – i, b – ii, c – iii, d – iv

(2) (a) – iv, b – ii, c – i, d – iii

(3) (a) – iv, b – i, c – ii, d – iii

(4) (a) – ii, b – iii, c – iv, d – i

115. In the Lassaigne's test for nitrogen in an organic compound, the prussian blue color is obtained due to formation of :-

(1) $Na_4[Fe(CN)_6]$

(2) $Fe_4[Fe(CN)_6]_3$

(3) $Fe_2[Fe(CN)_6]$

(4) $Fe_3[Fe(CN)_6]_4$

116. Which of the following C–X bond enthalpies order is correct :

(1) $CH_3-F > CH_3-Cl > CH_3-Br > CH_3-I$

(2) $CH_3Cl > CH_3Br > CH_3F > CH_3I$

(3) $CH_3I > CH_3Br > CH_3Cl > CH_3F$

(4) $CH_3I > CH_3Cl > CH_3F > CH_3Br$

117. [Major]A $\xleftarrow{AgCN}$ CH_3-CH_2-Br $\xrightarrow{KCN}$ B[Major]

A and B will be :

(1) CH_3CH_2CN, CH_3CH_2NC

(2) CH_3CH_2NC, CH_3CH_2CN

(3) CH_3CH_2CN, CH_3CH_2CN

(4) CH_3CH_2NC, CH_3CH_2NC

118. $RCOOH \xrightarrow[H^{\oplus}]{R'OH} A \xrightarrow[\text{Catalyst}]{H_2} B + C$

Identify B and C :

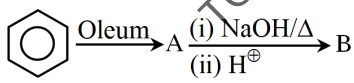
(1) RCHO + R'OH

(2) RCH_2OH + R'OH

(3) ROH + R'OH

(4) RCOOH + R'OH

TG- @ACADMIX TG- @STUDYNEXA Watermarkly

119. 

B will be :

(1) Ph–OH (2) Ph–CHO

(3) Ph–H (4) Ph–COOH

120. When chlorine is treated with carboxylic acid in the presence of red phosphorus then α-chloro acid will be formed, this reaction is known as :

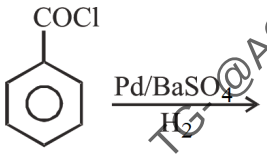
(1) Hunsdicker reaction

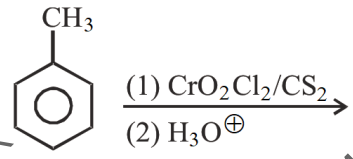
(2) Hell-Volhard - Zelinsky reaction

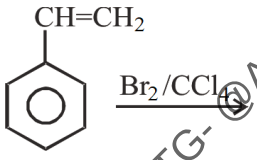
(3) Friedel - crafts reaction

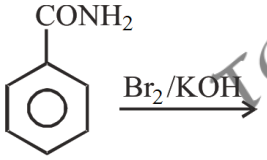
(4) Rosenmund reduction

121. Which of the following is "Etard reaction" :

(1) 

(2) 

(3) 

(4) 

122. Select incorrect statement :-

(1) 2-Chloro butane give more saytzeff product on reaction with alc. KOH

(2) 2-Fluro butane give more Hoffman product on reaction with alc KOH

(3) 2-Chlorobutane give more Hoffman product on reaction with t-BuOK

(4) Alc KOH preferebly give substitution reaction with 2-Bromobutane.

123. Which of the following vitamin can be stored in our body :

(1) A

(2) C

(3) B_{12}

(4) 1 and 3 both

124. $C_6H_5-CH_2-Cl \xrightarrow{NH_3} \xrightarrow{2CH_3Cl}$ Major product :

(1) $C_6H_5-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{N}}-CH_3$

(2) $C_6H_5-CH_2-NH-CH_3$

(3) $C_6H_5-CH_2-NH-C_6H_5$

(4) $C_6H_5-CH_2-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{N}}-CH_2C_6H_5$

125. Arrange followings in their boiling point order :

$n-C_4H_9NH_2$ (I) $(C_2H_5)_2NH$ (II) $C_2H_5N(CH_3)_2$ (III)

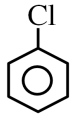
(1) I > II > III

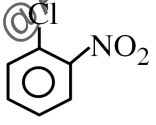
(2) I < II < III

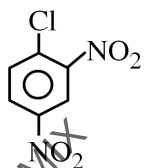
(3) I < III < II

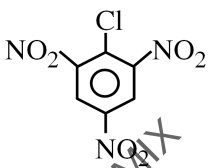
(4) I > III > II

126. Which of the following is more reactive towards nucleophile?

(1) 

(2) 

(3) 

(4) 

TG- @ACADMIX
TG- @STUDYNEXA
Watermarkly

127. In which of the following reaction product is not correctly matched?

(1) $RCH_2OH \xrightarrow[573K]{Cu} RCHO$

(2) $R-\underset{\underset{OH}{|}}{CH}-R' \xrightarrow[573K]{Cu} R-\underset{\underset{O}{||}}{C}-R'$

(3) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH \xrightarrow[573K]{Cu} CH_3-\underset{\underset{O}{||}}{C}-CH_3+CO_2+H_2O$

(4) All are correctly matched

128. Williamson ether continuous process is not suitable to get :-

(1) CH_3OCH_3

(2) $C_2H_5OC_2H_5$

(3) $CH_3-O-C_2H_5$

(4) None of these

129. Carboxylic acids are reduced to primary alcohol by :

(1) $LiAlH_4$ (2) B_2H_6

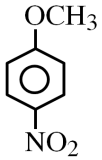
(3) $NaBH_4$ (4) 1 & 2 Both

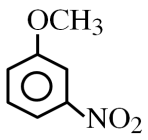
130. Which of the following gives sec. alcohol when reacted with grignard reagent, followed by hydrolysis?

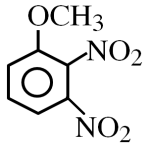
(1) HCHO (2) CH_3CHO

(3) $CH_3-\underset{\underset{O}{||}}{C}-CH_3$ (4) $CH_3-\underset{\underset{O}{||}}{C}-Ph$

131. When anisole reacts with a mixture of conc. H_2SO_4 & Conc. HNO_3, then major product will be-

(1) 

(2) 

(3) 

(4) 2 & 3 Both

132. Which of the followings will give haloform test?

(1) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(2) $CH_3+CH=O$

(3) $CH_3-\underset{\underset{O}{||}}{C}-CH_3$

(4) All of the above

133. In which of the following reaction, product is not correct?

(1) $\underset{\underset{CH_2OH}{|}}{\overset{\overset{CHO}{|}}{(CHOH)_4}} \xrightarrow{NH_2OH} \underset{\underset{CH_2OH}{|}}{\overset{\overset{CH=N-OH}{|}}{(CH-OH)_4}}$

(2) $\underset{\underset{CH_2OH}{|}}{\overset{\overset{CHO}{|}}{(CHOH)_4}} \xrightarrow{Br_2,water} \underset{\underset{COOH}{|}}{\overset{\overset{COOH}{|}}{(CHOH)_4}}$

(3) $\underset{\underset{CH_2OH}{|}}{\overset{\overset{CHO}{|}}{(CHOH)_4}} \xrightarrow{HI/\Delta} CH_3-(CH_2)_4-CH_3$

(4) $\underset{\underset{CH_2OH}{|}}{\overset{\overset{CHO}{|}}{(CHOH)_4}} \xrightarrow{HNO_3} \underset{\underset{COOH}{|}}{\overset{\overset{COOH}{|}}{(CHOH)_4}}$

134. Which of the following is non reducing ?

(1) Sucrose (2) Fructose

(3) Maltose (4) Lactose

135. During denaturation of protein which of the following structure is not destroyed

(1) Primary

(2) Secondary

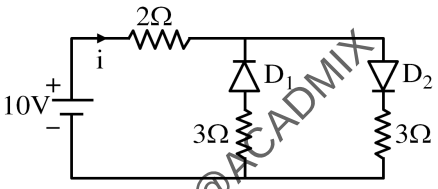
(3) Tertiary

(4) All of the above

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Topic : SYLLABUS-6

136. Find current in 2Ω resistance in the given circuit.



(1) 2.85 A. (2) 6.66 A.

(3) 2 A. (4) 5 A.

137. What value of a series resistance is required to control the current through an LED to 20 mA with a forward voltage drop 2V when connected to 8V supply ?

(1) 600 Ω (2) 300 Ω

(3) 400 Ω (4) 800 Ω

138. **Assertion (A) :** A p-n junction diode is kept in reverse bias while using it as a photodiode.

Reason (R) : Current in forward bias is more than current in reverse bias.

(1) Both A & R are true and R is correct explanation of A.

(2) Both A & R are true but R is not correct explanation of A.

(3) A is correct but R is incorrect.

(4) Both A & R are incorrect.

139. Identify the wrong statement

(1) In conductors the valence band and conduction band are overlaped.

(2) The resistivity of semiconductor is lower than metals.

(3) Substance with energy gap of the order of 10 eV are insulators.

(4) The resistivity of a semiconductor is lower than insulator.

140. Boolean expression $Y = (A + B) \cdot (\bar{A} \cdot \bar{B})$ equivalent to :-

(1) 1 (2) 0

(3) A + B (4) A · B

141. In terms of energy bands at room temperature, semiconductor have :

(1) Full conduction band

(2) Full valence band

(3) Large energy gap

(4) Moderate energy gap

142. According to energy band theory, a solid can be classified as a semiconductor if the energy gap between the valence band and conduction band is :-

(1) $3\text{ eV} < \Delta E_g < 6\text{ eV}$

(2) $\Delta E_g > 3\text{ eV}$

(3) $\Delta E_g < 3\text{ eV}$

(4) $\Delta E_g = 0$

143. Which of the following is true in the depletion region of a P-N junction ?

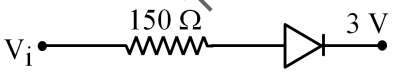
(1) It is rich in free charge carriers.

(2) It is void of any charge.

(3) It has immobile ions creating an electric field.

(4) It has an equal number of electrons and holes.

144. In the given circuit diode is ideal. When V_i increase from 2V to 6V then change in the current is :-

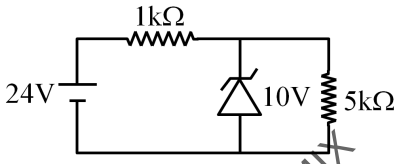


(1) 0 (2) 20 mA

(3) 40 mA (4) $\frac{80}{3}$mA

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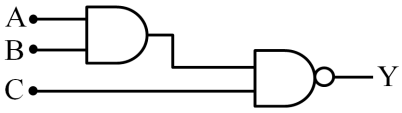
145. The power consumed in zener diode will be-



(1) 60 mW (2) 120 mW

(3) 160 mW (4) 240 mW

146. When all the three inputs are first high and then low, then output Y respectively will be



(1) 1, 0 (2) 1, 1

(3) 0, 0 (4) 0, 1

147. The logic behind NOR gate is that it gives :-

(a) High output when one input is low and other is high.

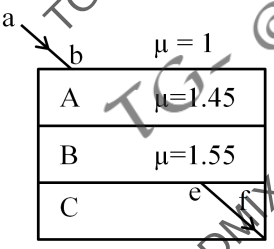
(b) Low output when one input is low and other is high.

(c) High output when both inputs are low.

(d) High output when both inputs are high.

(1) a, c (2) a, d (3) b, c (4) b, d

148. The refractive index of slab C if ray ef is parallel to ray ab is :-



(1) 1.5 (2) 1.45 (3) 1.55 (4) 1

149. A fish at a depth of 12 cm in water $\left(\mu = \frac{4}{3}\right)$ is viewed by an observer on the bank of a lake. To what height the image of the fish is raised :-

(1) 9 cm (2) 12 cm

(3) 3.8 cm (4) 3 cm

150. An object 2 cm tall is placed at 8 cm in front of a mirror. In order to produce an upright image of 6 cm height one needs a :-

(1) Convex mirror of radius of curvature 24 cm.

(2) Concave mirror of radius of curvature 24 cm.

(3) Plane mirror of height 12 cm

(4) Concave mirror of radius of curvature 4 cm

151. One of the refracting surfaces of a prism of angle 30° is silvered. A ray of light incident at angle of 60° retraces its path. The refractive index of the material of prism is :-

(1) $\sqrt{2}$ (2) $\sqrt{3}$

(3) 3/2 (4) 2

152. At what distance from a convex lens of focal length 30 cm, an object should be placed so that the size of the real image be $1/2^{th}$ of the object :-

(1) 30 cm (2) 60 cm

(3) 15 cm (4) 90 cm

153. A small air bubble is at the centre of a glass sphere of radius 5 cm. When it is viewed from outside what will be the position of image (Refractive index of glass is 1.5) :-

(1) At the same place as that of object

(2) 10/3 cm from the surface near the eye

(3) 7.5 cm from the surface of sphere near the eye

(4) At infinity

154. A lens of power +2 dioptre is placed in contact with a lens of power −1 dioptre. The combination will behave like :-

(1) A convergent lens of focal length 50 cm

(2) A divergent lens of focal length 100 cm

(3) A convergent lens of focal length 100 cm

(4) A convergent lens of focal length 200 cm

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155. In a compound microscope, the intermediate image is :-

(1) virtual, erect and magnified

(2) real, erect and magnified

(3) real, inverted and magnified

(4) virtual, erect and reduced

156. The magnifying power of an astronomical telescope can be increased, if we :-

(1) Increase the focal length of the objective

(2) Increase the focal length of the eye piece

(3) Decrease the focal length of the objective

(4) Decrease the focal length of the objective and at the same time increase the focal length of the eye piece.

157. The image of a real object formed by a plane mirror is :

(1) Erect, real and of equal size

(2) Erect, virtual and of equal size

(3) Inverted, real and of equal size

(4) Inverted, virtual and of equal size

158. Find the deviation caused by a prism having refracting angle 4° and refractive index 3/2.

(1) 4° (2) 2° (3) 8° (4) 6°

159. The refractive indices of violet and red light are 1.54 and 1.52 respectively, if the angle of the prism is 10°, the angular dispersion is

(1) 0.02° (2) 0.2°

(3) 3.06° (4) 30.6°

160. For constructive interference to take place between two monochromatic light waves of wavelength λ, the path difference should be -

(1) $(2n-1)\frac{\lambda}{4}$ (2) $(2n-1)\frac{\lambda}{2}$

(3) $n\lambda$ (4) $(2n+1)\frac{\lambda}{2}$

161. In two separate set-ups of the Young's double slit experiment, fringes of equal width are observed when lights of wavelengths in the ratio 1 : 2 are used. If the ratio of the slit separation in the two cases is 2 : 1, the ratio of the distances between the plane of the slits and the screen in the two set-ups is -

(1) 4 : 1 (2) 1 : 1 (3) 1 : 4 (4) 2 : 1

162. For which colour the fringe width is minimum -

(1) Violet (2) Red

(3) Green (4) Yellow

163. Two sources of light of wavelengths 2500 Å and 3500 Å are used in YDSE simultaneously. Which orders of bright fringes of two wavelength patterns coincide -

(1) 3rd order of 1st source and 5th order of the 2nd

(2) 7th order of 1st and 5th order of 2nd

(3) 5th order of 1st and 3rd order of 2nd

(4) 5th order of 1st and 7th order of 2nd

164. A screen is placed 2m away from the single narrow slit. Calculate the slit width if the first minimum lies 5 mm on either side of the central maximum. Incident plane waves have a wavelength of 5000 Å -

(1) 2×10^{-4} m (2) 2×10^{-3} cm

(3) 2×10^{-5} m (4) none

165. The condition for obtaining a maxima in the diffraction pattern due to single slit is -

(1) $a \sin\theta = n\lambda$ (2) $a \sin\theta = (2n+1)\frac{\lambda}{2}$

(3) $a \sin\theta = (2n-1)\lambda$ (4) $a \sin\theta = \frac{n\lambda}{2}$

166. In case of diffraction pattern of a single slit under polychromatic illumination first minima of wavelength λ_1 is found to coincide with the second maxima of wavelength λ_2, then

(1) $3\lambda_1 = 2\lambda_2$ (2) $2.5\lambda_1 = \lambda_2$

(3) $\lambda_1 = 2.5\lambda_2$ (4) $5\lambda_1 = \lambda_2$

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167. Which of the following phenomena exhibits particle nature of light -

(1) Interference (2) Diffraction

(3) Polarisation (4) Photoelectric effect

168. Two polarisers are parallel and then one of them is rotated through 60°. The percentage of incident unpolarised light transmitted is -

(1) 12.5 (2) 25.0 (3) 37.5 (4) 50

169. A stopping potential of 0.82 V is required to stop the photocurrent in a photo cell by light of wavelength 4000 Å. For light of wavelength 3000 Å, required stopping potential will be (approx.) :-

(1) 2.28 eV (2) 0.82 eV

(3) 4.13 eV (4) 1.85 eV

170. At stopping potential if wavelength of light is kept fixed but the intensity is increased two times, then photo current :-

(1) becomes two times

(2) becomes more than two times

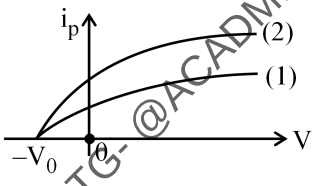
(3) becomes half

(4) still remains zero

171. Photocurrent (i_p) versus voltage (V) graph for two different experiments of photo electric effect are shown, in which radiation beams of same intensity are falling. On the basis of graph consider following results :-

(A) $\upsilon_1 < \upsilon_2$ (B) $\upsilon_1 > \upsilon_2$

(C) $\phi_1 < \phi_2$ (D) $\phi_1 > \phi_2$

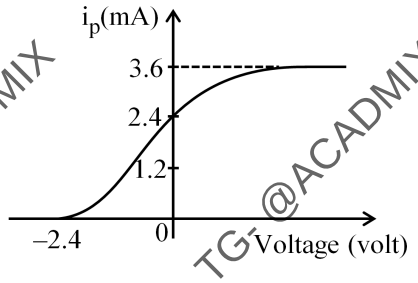


Which of the above is/are correct ? (υ is frequency and ϕ is work function)

(1) A and C (2) B and D

(3) A and D (4) B and C

172. Consider photocurrent (i_p) versus voltage (V) graph as shown below :-



Match the physical quantities described in column-I to its numerical value given in column-II :

	Column-I		Column-II
(A)	KE of photoelectrons (eV)	(p)	1.2
(B)	Saturation current (mA)	(q)	2.4
(C)	Stopping potential (V)	(r)	3.6

(1) A-p, q, B-r, C-q

(2) A-q, B-q, C-q

(3) A-p, B-q, C-r

(4) A-r, B-p, C-q

173. Find ratio of de broglie wavelength of electron and proton if they enter in uniform magnetic field so that the radii of their path are equal. Assume velocity of electron and proton are perpendicular to magnetic field.

(1) $\frac{1}{1}$ (2) $\frac{1}{4}$

(3) $\frac{2}{1}$ (4) $\frac{1}{2}$

174. Mass and energy of a proton are 1.67×10^{-27} kg and 2.0 eV respectively. Its de-Broglie wavelength is :-

(1) 2×10^{-10} m

(2) 2×10^{-11} m

(3) 1×10^{-10} m

(4) 1×10^{-11} m

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175. **Statement-1 :-** The density of nuclear matter is independent of the size of the nucleus.

Statement-2 :- The mass density of the atom does not follow above rule.

(1) Only statement-1 is correct.

(2) Only statement-2 is correct.

(3) Both statements are correct.

(4) None is correct.

176. (A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass no.

(D) Density of the nucleus is proportional to cube root of the mass no.

(E) Density of the nucleus is independent of the mass no.

Choose the correct from the above statements :-

(1) A and D only (2) A and E only

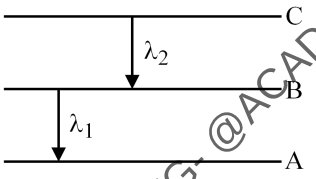
(3) B and E only (4) A and C only

177. The mass of $_2He^4$ nucleus is 0.0305(u) lesser than the sum of the masses of all its nucleons. The binding energy per nucleon of $_2He^4$ nucleus is nearly :-

(1) 5.1 MeV (2) 6.1 MeV

(3) 7.1 MeV (4) 8.1 MeV

178. As per given figure A, B and C are the first, second and third excited energy level of H-atom respectively. If the ratio of the two wavelengths (i.e. $\frac{\lambda_1}{\lambda_2}$) is $\frac{7}{4n}$, then the value of n will be :-



(1) 5 (2) 4 (3) 3 (4) 2

179. The decay of a proton to neutron is :-

(1) Not possible as proton mass is less than the neutron mass.

(2) Possible only inside the nucleus.

(3) Not possible but neutron to proton conversion is possible.

(4) Always possible as it is associated only with β^- decay.

180. Hydrogen is the simplest atom of nature. If all Bohr postulates are followed

m = mass of electron

n = orbit number

choose correct option :

(a) Magnetic moment of orbiting electron in n^{th} orbit is $\frac{mv}{nh}$.

(b) Ratio of angular momentum to linear momentum of electron directly proportional to n^2.

(c) Minimum wavelength of emitted photon is $\frac{1}{R}$.

(1) Only a (2) a, b

(3) b, c (4) a, b, c

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ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	1	4	4	3	2	3	1	1	1	3	4	4	4	2	2	4	2	4	4	4	4	1	3	4	2	2	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	3	2	3	2	3	4	3	4	2	4	3	1	3	3	1	3	3	3	2	4	1	2	2	1	1	3	2	4	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	1	3	4	1	3	3	3	4	4	3	3	2	1	1	2	3	3	2	2	3	2	3	2	2	1	4	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	1	1	2	3	3	1	2	4	4	3	3	4	1	4	4	3	2	1	2	4	1	3	2	1	2	2	1	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	1	1	4	3	3	4	2	1	4	2	1	1	3	2	2	2	2	4	3	3	2	2	4	3	4	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	1	3	3	1	2	2	2	3	1	1	2	1	2	3	4	1	4	4	2	1	1	2	3	2	3	1	2	3

HINT – SHEET

1. **Ans (1)**
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2. **Ans (2)**
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3. **Ans (1)**
NCERT-XII, Pg. # 200

4. **Ans (4)**
NCERT Page No. 198

5. **Ans (4)**
NCERT Pg. # 197

6. **Ans (3)**
NCERT, Pg. # 201, 202

7. **Ans (2)**
NCERT, Pg. # 197

8. **Ans (3)**
NCERT, Pg. # 199,200,201,202

9. **Ans (1)**
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10. **Ans (1)**
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11. **Ans (1)**
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12. **Ans (3)**
MODULE Pg. No. # 194

13. **Ans (4)**
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14. **Ans (4)**
NCERT-XII, Pg. # 211

15. **Ans (4)**
NCERT XII Pg#206

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16. **Ans (2)**
NCERT XII, Page No. 209

17. **Ans (2)**
NCERT XII Pg No. # 212, 213

18. **Ans (4)**
NCERT XII Pg. # 207
Catabolism, Fragmentation, Leaching, Occurs Simultaneously

19. **Ans (2)**
NCERT XII Pg. # 212, 213

20. **Ans (4)**
NCERT-XII, Pg. No. # 206, 207

21. **Ans (4)**
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22. **Ans (4)**
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23. **Ans (4)**
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24. **Ans (1)**
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25. **Ans (3)**
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26. **Ans (4)**
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27. **Ans (2)**
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28. **Ans (2)**
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29. **Ans (3)**
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30. **Ans (2)**
NCERT Pg. # 223

31. **Ans (2)**
NCERT-XII, Pg # 223

32. **Ans (3)**
NCERT-XII, Pg. # 218

33. **Ans (2)**
NCERT, Pg # 217

34. **Ans (3)**
NCERT-XII, Pg # 225

Solution/Explanation:

In given options cryopreservation is a type of ex-site conversion,

Hence (3) option is correct.

35. **Ans (2)**
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36. **Ans (3)**
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37. **Ans (4)**
NCERT, Pg # 217

38. **Ans (3)**
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39. **Ans (4)**
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40. **Ans (2)**
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41. **Ans (4)**
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42. **Ans (3)**
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43. **Ans (1)**
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44. **Ans (3)**
NCERT, Pg # 219

45. **Ans (3)**
NCERT Pg. # 223

46. **Ans (1)**
NCERT XII Pg. # 146

47. **Ans (3)**
NCERT (E) Pg. # 159

48. **Ans (3)**
NCERT XII, Pg. # 145

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49. **Ans (3)**
NCERT XII Pg. # 145

50. **Ans (2)**
NCERT Pg. # 170

51. **Ans (4)**
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52. **Ans (1)**
Module No. # 11, Pg. No. # 7

53. **Ans (2)**
NCERT-XII, Pg. # 135

57. **Ans (3)**
NCERT Pg. # 136(E),149(H)

59. **Ans (4)**
NCERT-XII, Hint Pg. # 147

60. **Ans (3)**
NCERT Page No. # 146

62. **Ans (3)**
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66. **Ans (1)**
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67. **Ans (3)**
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68. **Ans (3)**
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69. **Ans (3)**
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70. **Ans (4)**
NCERT Pg#142

71. **Ans (4)**
NCERT : PAGE 129

72. **Ans (3)**
NCERT Pg. No. 127

73. **Ans (3)**
NCERT-XII Pg#116

74. **Ans (2)**
NCERT : PAGE 137

75. **Ans (1)**
NCERT (XII), Pg # 140, 141

76. **Ans (1)**
NCERT (XII), Pg # 130, 131

77. **Ans (2)**
NCERT XII Pg. # 114

78. **Ans (3)**
NCERT (XII) Page no. 112
Allen module

79. **Ans (3)**
NCERT (XII) Pg. # 118

80. **Ans (2)**
NCERT : PAGE 131 & 132 (FIG 7.4)

81. **Ans (2)**
NCERT Page No. 131/143

82. **Ans (3)**
NCERT XII Pg # 138

83. **Ans (2)**
NCERT, Pg. # 121

84. **Ans (3)**
NCERT P.No. 123
First amphibian evolved from coelacanth (Lobefins)

85. **Ans (2)**
NCERT- P.No.113
Embryological support for evolution Proposed by Ernst Heckel.

86. **Ans (2)**
NCERT-XII, Pg. # 138 (Fig. 7.9)

87. **Ans (1)**
NCERT-XII, Pg. # 132

88. **Ans (4)**
NCERT-XII, Pg. # 137, Para-7.7

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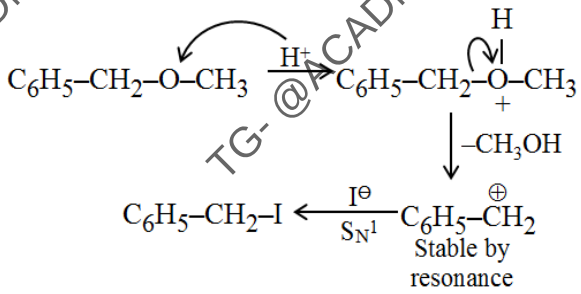
89. **Ans (2)**

Module-11 Page No. #107

90. **Ans (3)**

NCERT Pg.# 123

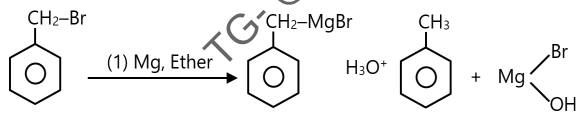
92. **Ans (3)**



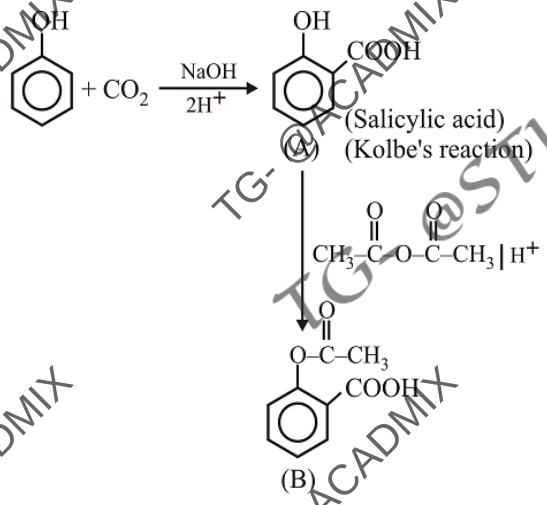
94. **Ans (1)**

NCERT (XIIth) Part II, Pg. # 299

95. **Ans (2)**

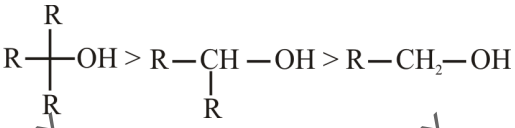


97. **Ans (3)**



98. **Ans (1)**

Rate towards lucas reagent $\propto$ [Stability of $C^{\oplus}$]



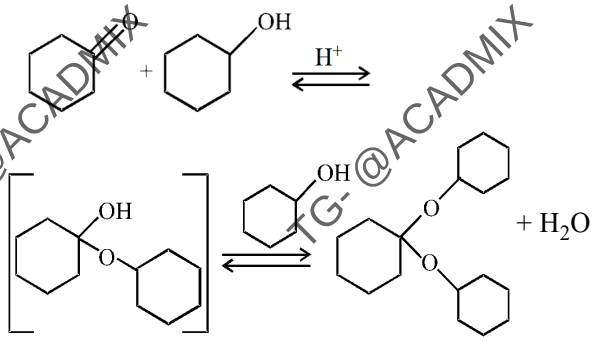
99. **Ans (2)**

Fact

100. **Ans (4)**

3 degree alcohol does not oxidise by PCC.

103. **Ans (3)**



(NCERT XII/part II, pg. 359)

105. **Ans (1)**

The given reaction is Cannizzaro reaction in which the aldehydes which does not have α undergoes both oxidation and reduction.

This reaction is also called as disproportionation reaction.

106. **Ans (4)**

Because sucrose is non reducing.

107. **Ans (4)**

$X = CH_3-CH_2-CH_2-C \equiv N$

$Y = CH_3-CH_2-CH_2-CO_2H$

$Z = CH_3-CH_2-CH_2-\underset{\underset{C}{\|}}{O}-OEt$

$Y \longrightarrow Z$ esterification reaction

108. **Ans (3)**

1° amine in which NH_2 not attached to benzene can be prepared by Gabriel Phthdimide synthesis

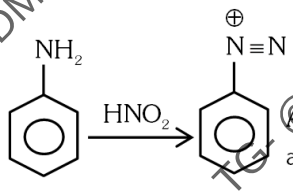
$\therefore$ Ethyl Amine ($CH_3CH_2NH_2$) can be prepared

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111. Ans (2)

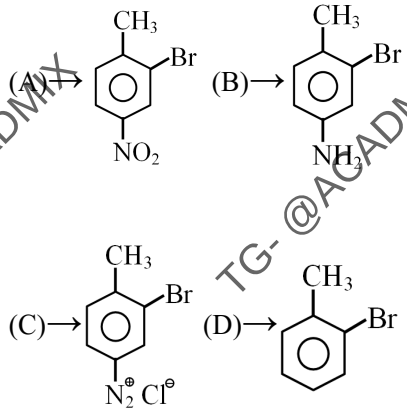
$$R-NH_2 \xrightarrow{HNO_2} R-N_2^{\oplus}$$

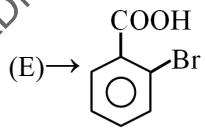
Alkyl diazonium ion
(unstable)



Alkyl diazonium ion stable at low temperature (0–5°C)

112. Ans (4)





115. Ans (2)

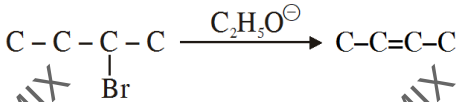
$$6CN^{\ominus} + Fe^{+2} \longrightarrow [Fe(CN)_6]^{-4}$$

$$3[Fe(CN)_6]^{-4} + 4Fe^{+3} \longrightarrow Fe_4[Fe(CN)_6]_3$$

120. Ans (2)

Name reaction

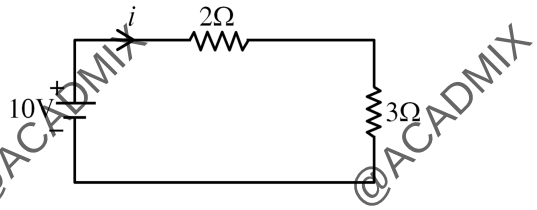
122. Ans (4)



(Elimination)
E_2

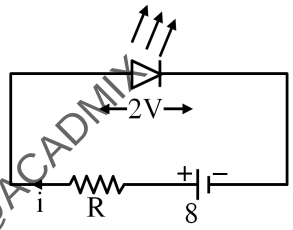
136. Ans (3)

As shown D_1 is in R.B. so circuit below



$$\Rightarrow \quad i = \frac{10}{2} = 2\ \text{Amp}$$

137. Ans (2)



$$\therefore \quad i = \frac{8-2}{R}$$

$$\therefore \quad R = \frac{6}{20\text{mA}} = \frac{6000}{20}$$

$$= 300\ \Omega$$

138. Ans (2)

Photodiode is used in reverse bias so, A is correct

Current in F.B. > current in F.B. so R is correct.

but photodiode is used to detect light intensity or photo conductivity which can only be observed while using it in reverse bias, so R is not correct explanation of A.

139. Ans (2)

The resistivity of semiconductor is higher than metals but lower than insulator.

140. Ans (2)

$$Y = (A+B)\cdot(\bar{A}\cdot\bar{B})$$

$$= A\cdot\bar{A}\cdot\bar{B} + B\cdot\bar{A}\cdot\bar{B}$$

$$= 0 + 0 = 0$$

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141. Ans (4)

In conductor's - No energy gap (≈ 0 eV)

In insulators - Large energy gap (> 3 eV)

In semiconductors - Moderate energy gap (≈ 1 eV)

142. Ans (3)

For semiconductor $\Delta E_g \leq 3$ eV

143. Ans (3)

The depletion layer or region contains immobile positive ions on the n-side and immobile negative ions on the p-side, creating an electric field.

144. Ans (2)

when $V_i = 2V$ then $I = 0$

$V_i = 6V$ then $I = \frac{V}{R} = \frac{6-3}{150\Omega} = 20$ mA

145. Ans (2)

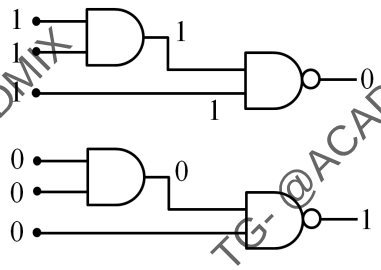
$I_L = \frac{V_Z}{R_L} = \frac{10}{5k\Omega} = 2mA$

$I = \frac{V - V_Z}{R_S} = \frac{24-10}{1k\Omega} = 14mA$

$I_Z = I - I_L = 14 - 2 = 12mA$

$P_Z = V_Z I_Z = 10 \times (12 \times 10^{-3}) = 120mW$

146. Ans (4)



147. Ans (3)

For NOR gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

148. Ans (4)

Angle with normal is same so R.I. will also be same.

149. Ans (4)

Shift = $d - \frac{d}{\mu}$

150. Ans (2)

$m = \frac{h_i}{h_0} = \frac{f}{f-u} \Rightarrow f = -12$ cm

$\Rightarrow R = -24$ cm

158. Ans (2)

$S = (\mu - 1)A$

$= \left(\frac{3}{2} - 1\right) \times 4$

$= 2°$

159. Ans (2)

$\mu_v = 1.54 \ \mu_r = 1.52 \quad A = 10°$

Angular dispersion $(\delta) = (\mu_v - \mu_r).A$

$= 0.02 \times 10 = 0.2°$

161. Ans (1)

$\beta = \frac{\lambda D}{d}$

$\frac{D_1}{D_2} = \frac{d_1}{d_2} \times \frac{\lambda_2}{\lambda_1}$

169. Ans (4)

In case-1

$0.82 = \frac{12400}{4000} - \phi \Rightarrow \phi = 3.1 - 0.82 = 2.28$ eV

$\therefore$ In case-2

$eV_s = \frac{12400}{3000} - 2.28 = 4.13 - 2.28 = 1.85$ eV

170. Ans (4)

For a given wavelength, stopping potential is independent of intensity.

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171. Ans (2)

Intensity $I = \frac{xh\nu}{A}$ and for same I $\Rightarrow x \propto \frac{1}{\nu}$

from the graphs, it is clear that $(i_p)_1 < (i_p)_2 \Rightarrow x_1 < x_2$

$\Rightarrow \nu_1 > \nu_2$

but here V_0 is same, therefore from photo electric equation

$eV_0 = h\nu - \phi \Rightarrow \nu_1 > \nu_2$ means $\phi_1 > \phi_2$

173. Ans (1)

$\lambda = \frac{h}{qBr} \Rightarrow \lambda \propto \frac{1}{q}$

$\frac{\lambda_e}{\lambda_P} = \frac{e}{e} = \frac{1}{1}$

174. Ans (2)

$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$

$\lambda_p = \frac{0.286}{\sqrt{V}} = \frac{0.286}{\sqrt{2}} = \frac{0.286}{1.4}$

$= 2 \times 10^{-11}$ m

175. Ans (3)

NCERT, Pg. # 320

See NCERT points to ponder page 320

176. Ans (2)

(A) $v \propto A$

(B) $\rho \longrightarrow$ constant

177. Ans (3)

$_2He^4 \longrightarrow \Delta m = 0.0305(u)$

B.E. = 0.0305 × 931 MeV

B.E. = 28.39 MeV

$\frac{B.E.}{A} = \frac{28.39}{4} = 7.09$ MeV

High temperature and high pressure are the necessary condition for fusion process.

178. Ans (1)

$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

A to B $\frac{1}{\lambda_1} = RZ^2\left(\frac{1}{(2)^2} - \frac{1}{(3)^2}\right)$

C to B $\frac{1}{\lambda_2} = RZ^2\left(\frac{1}{(3)^2} - \frac{1}{(4)^2}\right)$

n = 5

179. Ans (2)

$P \longrightarrow n + \beta^+ + \nu$ (neutrino)

180. Ans (3)

(a) $\frac{q}{2m} = \frac{M}{L} \Rightarrow M = \frac{e}{2m}\frac{nh}{2\pi}$

(b) $\frac{L}{P} = \frac{mvr}{mv} = r$ & $r \propto n^2$

(c) $\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

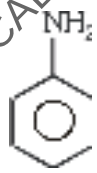
$n_1 = 1 \quad n_2 = \infty$

$\lambda = \frac{1}{R}$

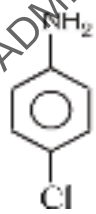
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CHEMISTRY

1) Arrange these compounds in decreasing order of basic strength

(i) 

(ii) 

(iii) 

(iv) 

(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (iii) > (i) > (iv)

(3) (iv) > (i) > (iii) > (ii)

(4) (iv) > (i) > (ii) > (iii)

2) Which one is an elimination reaction –

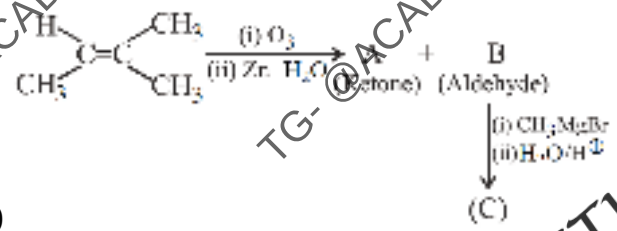
(1) $CH_3CH_3 + Cl_2 \xrightarrow{h\nu} CH_3CH_2Cl + HCl$

(2) $CH_3Cl + KOH(aq) \rightarrow CH_3OH + KCl$

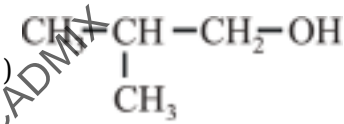
(3) $CH_2 = CH_2 + Br_2 \xrightarrow{CCl_4} CH_2BrCH_2Br$

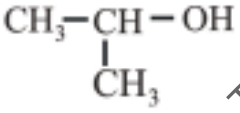
(4) $C_2H_5Br + KOH(alc.) \rightarrow C_2H_4 + KBr$

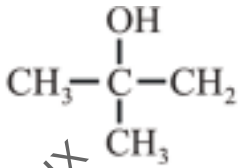
3)

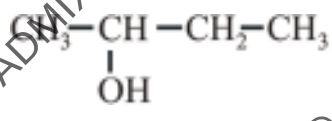


What is C ?

(1) 

(2) 

(3) 

(4) 

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$$2CH_3CH_2COONa \xrightarrow[\text{(Aqueous solution)}]{\text{Electrolysis}}$$

4) Incorrect statement about following reaction is:

(1) Carbon free radical is involved

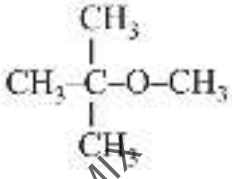
(2) CO_2 gas is released at cathode

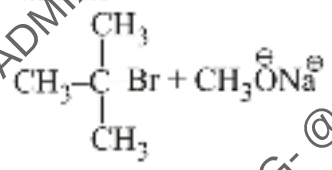
(3) pH of solution is increases

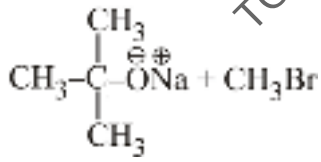
(4) Butane is formed as major product

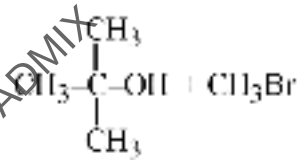
5)

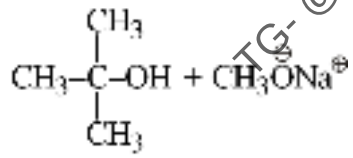
To make this ether, which of the following reactions will be most suitable:



(1) 

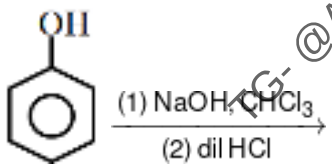
(2) 

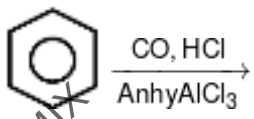
(3) 

(4) 

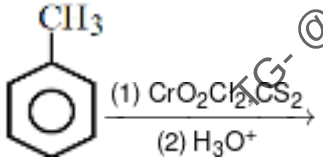
6)

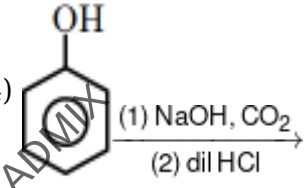
Product of following reaction will not contain –CHO group?

(1) 

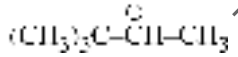
(2) 

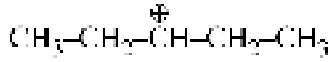
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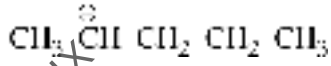
(3) 

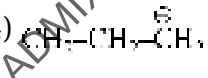
(4) 

7) Which of the following carbocation is most stable :-

(1) 

(2) 

(3) 

(4) 

8) Which of the following will give yellow precipitate with I_2 + NaOH ?

(1) Propanamide

(2) Butan-2-ol

(3) Acetophenone

(4) Both (2) and (3)

9) ***Assertion :-*** Alkyl iodide can be prepared by treating alkyl chloride/bromide with NaI in acetone.
Reason :- NaCl/NaBr are soluble in acetone while NaI is not.

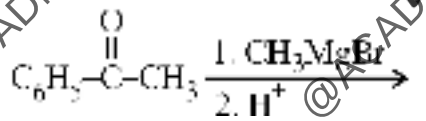
(1) Both **Assertion** & **Reason** are True & the **Reason** is a correct explanation of the **Assertion**.

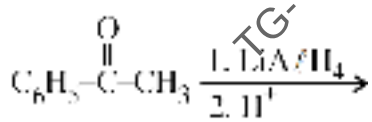
(2) Both **Assertion** & **Reason** are True but **Reason** is not a correct explanation of the **Assertion**.

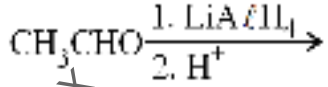
(3) **Assertion** is True but the **Reason** is False.

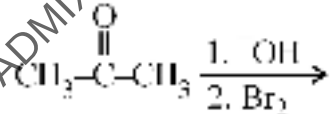
(4) Both **Assertion** & **Reason** are False.

10) Which one of the following reactions would produce secondary alcohol-

(1) 

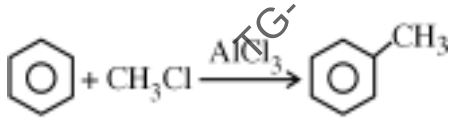
(2) 

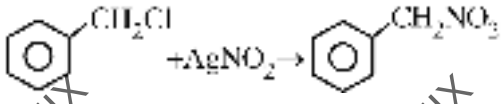
(3) 

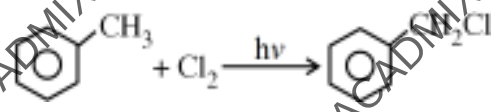
(4) 

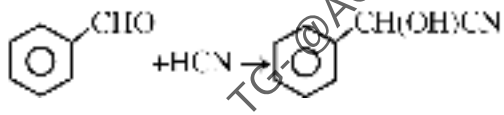
TG- @ACADMIX
TG- @STUDYNEXA ©

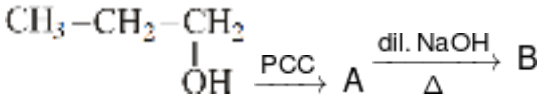
11) Which one of the following is a free radical substitution :-

(1) 

(2) 

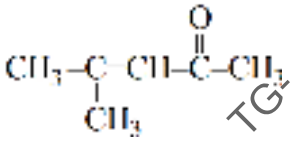
(3) 

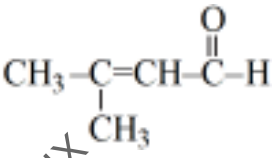
(4) 

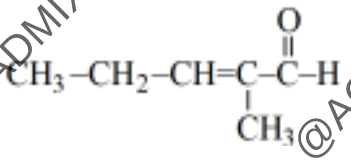
12) 

Find the product B :

(1) $CH_3–CH_2–CH=CH–CH_2–CHO$

(2) 

(3) 

(4) 

13) $CH_3–CH=O$ and $C_6H_5–CH=O$ can be distinguished by :-

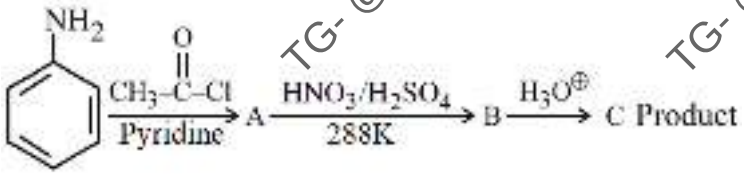
(1) Tollen reagent

(2) Fehling reagent

(3) Grignard reagent

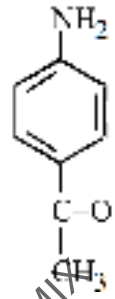
(4) $LiAlH_4$

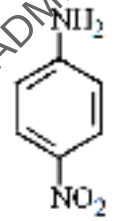
14) Suppose following reaction

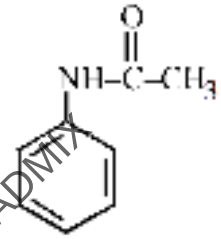


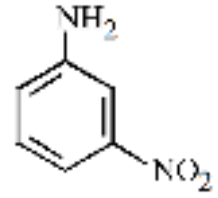
The major product 'C' will be :-

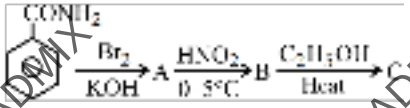
TG- @ACADMIX
Tg* @STUDYNEXA ©

(1) 

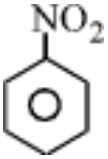
(2) 

(3) 

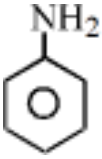
(4) 

15) 

What is C ?

(1) 

(2) 

(3) 

(4) 

16) Correct order of boiling point of given aqueous solution is -

(A) 10% $\frac{w}{W}$ glucose

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TG- @STUDYNEXA ©

(B) 10% $\frac{w}{W}$ urea

(C) 10% $\frac{w}{W}$ Na_2SO_4

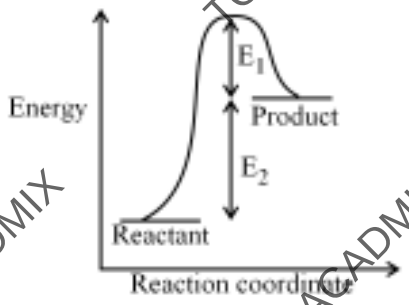
(D) 10% $\frac{w}{W}$ sucrose

(1) C < B < A < D

(2) D < A < B < C

(3) D < A < C < B

(4) A < D < B < C



17) Consider the given figure and choose the correct answer -

(1) Activation energy of forward reaction is E_1 and product is more stable than reactant

(2) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

(3) Activation energy of backward reaction is E_1 and product is more stable than reactant.

(4) Activation energy of backward reaction is $E_1 + E_2$ and product is less stable than reactant

18) Species which does not undergo disproportionation reaction is -

(1) ClO_2^-

(2) ClO_4^-

(3) ClO^-

(4) ClO_3^-

19) Density of 2M NaOH (aq.) solution is 1.25 g/ml, then molality (m) of solution is -

(1) 2 m

(2) 2.25 m

(3) 1.7 m

(4) 3 m

20) Molar solubility of zirconium phosphate with molecular formula $(Zr^{+4})_3(PO_4^{3-})_4$ is given by relation -

(1) $\left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$

(2) $\left(\frac{K_{sp}}{5348}\right)^{\frac{1}{6}}$

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TG- @STUDYNEXA ©

(3) $\left(\frac{K_{sp}}{8435}\right)^{\frac{1}{7}}$

(4) $\left(\frac{K_{sp}}{9612}\right)^{\frac{1}{3}}$

21) 200 mL of 2M NaOH solution is added to 400 mL of 0.5 M NaOH solution, then molarity (M) of resulting solution is -

(1) 2.5

(2) 1.5

(3) 1

(4) 2

22) Consider the following cases of standard enthalpy of reaction

$C_2H_6(g) + \frac{7}{2} O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$;
$\Delta H^o_1 = -z$ kJ mol^{-1}

$C(graphite) + O_2(g) \rightarrow CO_2(g)$;
$\Delta H^o_2 = -x$ kJ mol^{-1}.

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$;
$\Delta H^o_3 = -y$ kJ mol^{-1}.

On the basis of above data, calculate $\Delta_f H^o$ of $C_2H_6(g)$ (in kJ mol^{-1}) :-

(1) z – 2x – 3y

(2) z – x – y

(3) 2x + 3y – z

(4) 2x + 3y + z

23) Given below are two statements

Statement-I :- for a given shell, total number of allowed orbitals is given by n.

Statement-II :- For any subshell, maximum number of electrons which can be filled is given by 2ℓ + 1.

In the light of above statements, choose the correct answer from the options given below :

(1) **Statement-I** is true but **Statement-II** is false

(2) **Statement-I** is false but **Statement-II** is true

(3) Both **Statement-I** and **Statement-II** are true

(4) Both **Statement-I** and **Statement-II** are false

24) $Fe^{+3} \xrightarrow{0.3\ V} Fe^{+2} \xrightarrow{-0.1\ V} Fe$

In the above diagram, the standard electrode potential are given in volts, then value of $E^o_{Fe^{+3}/Fe}$ is -

(1) 0.3 V

(2) 1.3 V

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(3) –0.066 V

(4) –0.3 V

25) Match list-I with list-II

List-I		List-II	
A.	Isothermal process for an ideal gas	I.	$w = \Delta U$
B.	Adiabatic process for an ideal gas	II.	$\Delta S = nR\ln\frac{P_1}{P_2}$
C.	Spontaneous process at all temperature	III.	$\Delta S = +ve$ $\Delta G = 0$
D.	Melting of ice at 1 atm and 0°C	IV.	$\Delta H = -ve, \Delta S = +ve$

Choose the correct answer from the options given below

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-II, B-III, C-IV, D-I

(4) A-II, B-IV, C-III, D-I

26) 10^{-3} M aqueous solution of ethylamine has pH = 9 then its ionisation constant is -

(1) 10^{-15}

(2) 10^{-7}

(3) 10^{-10}

(4) 10^{-5}

27) Which of the following is a lyophobic sol?

(1) Gold Sol

(2) Gum Sol

(3) Gelatin Sol

(4) Starch Sol

28) 2×10^{-3} moles of CO_2 is left after removing 10^{20} molecules from $x \times 10^{-3}$ moles of CO_2 sample, then x is –

(1) $\frac{13}{6}$

(2) 6

(3) $\frac{1}{6}$

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(4) $\frac{19}{6}$

29) Given at 25°C

$H_{(aq.)} + OH_{(aq.)} \rightarrow H_2O(\ell)$;

$\Delta_{neut.} H = -57$ kJ mol^{-1}

$HA(aq.) + NaOH_{(aq.)} \rightarrow NaA_{(aq.)} + H_2O(\ell)$

$\Delta_{neut.} H = -53$ kJ mol^{-1}

$HB(aq.) + NaOH\ (aq.) \rightarrow NaB(aq.) + H_2O(\ell)$

$\Delta_{neut.} H = -51$ kJ mol^{-1}

Correct statement is -

(1) $\Delta H_{ionisation}$ (HA) = 53 kJ mol^{-1}

(2) $\Delta H_{ionisation}$ (HB) = 51 kJ mol^{-1}

(3) pK_a (HA) < pK_a (HB)

(4) pK_a (HA) < pK_a (HB)

30) How much electricity in terms of Faraday is required to produce 20 g of Ca from molten $CaCl_2$ -

(1) 2F

(2) 1F

(3) 0.5 F

(4) 4F

31) Two elements X and Y having electronic configuration $1s^2\ 2s^2$ and $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ respectively, then correct property of expected compound formed by combination of X and Y.

(1) It is ionic in nature

(2) It is covalent in nature

(3) It is neutral

(4) Can't predicted

32) Which of the following species isostructural, planner and non polar :-

(1) ClF_3, BrF_3, XeO_3

(2) $ClO_4^-, SO_4^{2-}, PO_4^{3-}$

(3) $I_3^{\oplus}, XeF_3^{\oplus}, S_3^{2-}$

(4) $CO_2, HgCl_2, XeF_2$

33) The correct order of bond energy is :-

(1) $F_2 < Cl_2 < Br_2 < I_2$

(2) $I_2 < Cl_2 < Br_2 < F_2$

(3) $I_2 < F_2 < Br_2 < Cl_2$

(4) $Cl_2 < Br_2 < F_2 < I_2$

TG- @ACADMIX
TG- @STUDYNEXA

34) Which ligand is used for removing excess of copper and iron ?

(1) EDTA and DMG

(2) Cyanoamine and Naphthol

(3) D-pencillamine and desferrioxime B

(4) Ph_3P and NH_3

35) Match the column-I with column-II

	Column-I (Property)		**Column-II (Metal)**
i	An element which can show +8 oxidation state	a	Mn
ii	3d series element that can show upto +7 oxidation state	b	Cr
iii	3d series element with highest melting point	c	Os
		d	Fe

(1) i–c, ii–a, iii–b

(2) i–a, ii–b, iii–c

(3) i–a, ii–d, iii–d

(4) i–c, ii–b, iii–a

36) Match the column

	Column-I		**Column-II**
I.	$[Co(en)_3]^{3+}$	a.	Optical isomerism
II.	$[CoCl_2(en)_2]^+$	b.	Geometrical isomerism
III.	$[Co(NH_3)_5(SO_4)]Br.$	c.	Ionisation isomerism
IV.	$[Cr(NH_3)_6][Co(CN)_6]$	d.	co-ordination isomerism

(1) I – a; II – b; III – d; IV – c

(2) I – a; II – b; III – c; IV – d

(3) I – a; II – c; III – b; IV – d

(4) I – a; II – b; III – d; IV – c

37) Which element has least ΔH_{eg} in group 16th.

(1) Se

(2) O

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TG- @STUDYNEXA ©

(3) Te

(4) Po

38) The acid having O–O bond is :-

(1) $H_2S_2O_3$

(2) $H_2S_2O_6$

(3) $H_2S_2O_8$

(4) $H_2S_2O_7$

39)

Electronegativity scale of Pauling is based upon :-

(1) Bond energy

(2) Bond length

(3) Atomic radius

(4) Covalent radius

40)

Among the following least and most polar bonds are respectively :-
(a) C-I (b) N-O (c) C-F (d) P-F

(1) d and c

(2) a and d

(3) b and d

(4) b and c

41)

Reagent used for identifying nickel ion is :-

(1) Dimethyl glyoxime

(2) Potassium Ferrocyanide

(3) EDTA

(4) Phenolphthalein

42)

Zero group basic radical is :-

(1) NH_4^+

(2) Fe^{+3}

(3) Co^{+2}

(4) Ca^{+2}

TG- @ACADMIX

43) Which carbide of lanthanoid is not form when lanthanoid react with carbon at high temperature ?

(1) Ln_4C_3

(2) Ln_2C_3

(3) LnC_2

(4) Ln_3C

44)

Hybridisation of $[PtCl_4]^{-2}$ is :-

(1) sp^3

(2) dsp^2

(3) sp^3d^2

(4) d^2sp^3

45) Which of the following electronic arrangement is/are possible for inner orbital complex?

(I) $t_{2g}^3e_g^2$ (II) $t_{2g}^6e_g^1$ (III) $t_{2g}^3e_g^0$ (IV) $t_{2g}^4e_g^2$

Select the correct code:

(1) I, IV

(2) II, III

(3) III only

(4) III, IV

BIOLOGY

1) When the incisions of the lamina reach upto the mid rib breaking it into number of leaflets are called.

(1) Petiole

(2) Phyllotaxy

(3) Simple leaf

(4) Compound leaf

2) $\oplus \female K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$ is the floral formula of :-

(1) Soyabean

(2) Pea

(3) Mustard

(4) Gram

3) Match the following column :

	Column-I		Column-II
(A)	Edible mesocarp	(i)	Coconut
(B)	Endospermous seed	(ii)	Mango
(C)	Fibrous mesocarp	(iii)	Bean
(D)	Non-endospermous seed	(iv)	Castor
(E)	Ovules	(v)	Future fruit
(F)	Ovary	(vi)	Future seed

	A	B	C	D	E	F
(1)	vi	ii	iii	iv	v	i
(2)	iv	v	iv	iii	ii	i
(3)	ii	iv	i	iii	vi	v
(4)	v	vi	iii	i	iv	ii

(1) 1
(2) 2
(3) 3
(4) 4

4) Subsidiary cells are the specialised cell in the

(1) Vicinity of guard cell
(2) Vicinity of root hair
(3) Vicinity of trichomes
(4) Vicinity of phloem

5) When xylem and phloem within vascular bundle are arranged in alternate manner along the different radii, the arrangement is called?

(1) Conjoint closed
(2) Radial
(3) Conjoint open
(4) 1 & 3 both

6) Which of the following statements is not true ?

(1) Guard cells are dumble-shaped in grasses.
(2) Outer wall of guard cells is thin and inner wall is thick.
(3) Guard cells possess chloroplast.
(4) Xylem, guard cells and the surrounding subsidiary cells are together called egg apparatus.

7) What is common among *Spirogyra*, *Laminaria* and *Polysiphonia*?

(1) All have immotile gametes.
(2) All have glycogen as stored food.
(3) All have chlorophyll a and chlorophyll b
(4) All are autotrophic organisms.

8) Which of the following basis used by Carolus Linnaeus for two kingdom system?

(1) Cell structure
(2) Presence or absence of cell wall
(3) Phylogenetic relationship
(4) Stored food

9) Pteridophytes do not have :-

(1) Vascular tissues
(2) Stem
(3) Leaf
(4) Flower

10) Which of the following shows isogamy ?

(1) *Pinus*
(2) *Cycas*
(3) *Spirogyra*
(4) *Cedrus*

11) Which of the class of kingdom-fungi belongs to aquatic members and obligate parasites like *Albugo*?

(1) Phycomycetes
(2) Asomycetes
(3) Basidiomycetes
(4) Deuteromycetes

12) **Statement-I :** Dominant stage of pteridophytes consists of protonema and leafy stage.
Statement-II : Most pteridophytes produce two kind of spores-microspores and megaspores.

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.
(3) **Statement I** is incorrect but **Statement II** is correct.
(4) Both **Statement I** and **Statement II** are correct.

13) Select the correct statement :

(1) Biological names are separately underlined when they printed.
(2) Artifical system of classification is based on all observable characters

(3) Numerical taxonomy gives equal weightage to each character

(4) Always a genus have only one specific epithet

14) Enzyme that catalyse removal of group from substrate by mechanism other than hydrolysis and leaving double bonds are :

(1) Transferases

(2) Lyases

(3) Oxidoreductases

(4) Isomerases

15) Which of the following statement is incorrect :-

(1) Auxin inhibit flowering in pineapple

(2) Gibberellin promote bolting in beet.

(3) Zeatin is a naturally occurring cytokinin

(4) ABA stimulates the closing of stomata

16) The current availability of CO_2 level is limiting to the :

(1) C_4 plants

(2) C_3 plants

(3) C_4 & C_3 plants both

(4) C_2 plants

17) The primary acceptor of CO_2 in maize plant is :-

(1) MA

(2) PEP

(3) OAA

(4) PGAL

18) The TCA cycle start with the condensation of acetyl CoA with _____ .

(1) PEP

(2) PGAL

(3) Oxaloacetic acid (OAA)

(4) Starch

19) Breakdown of glucose in pyruvic acid is called :-

(1) C-3 Cycle

(2) Kreb's cycle

(3) Glycolysis

(4) ETS

20) **Assertion :** Glycerol would enter in the respiratory pathway after being converted to starch.
Reason : Glycogen is the most favoured substrate for respiration in plants.

(1) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(2) Both assertion and reason are correct and reason is correct explanation of assertion.

(3) Assertion and reason both are incorrect.

(4) Assertion is incorrect but reason is correct.

21) (A) They tolerate high temperature. (B) They lack photorespiration.
(C) They have great productivity of biomass.
The above characteristics are shown by.

(1) C_3 plants

(2) C_4 plants

(3) Dicots only

(4) Monocots only

22) Biological control methods adopted in agricultural pest control are based on ability of _A_ to regulate _B_ population ?

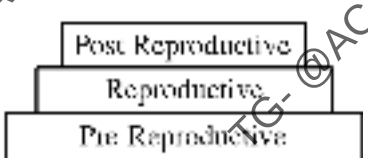
(1) A - Parasite; B - Predator

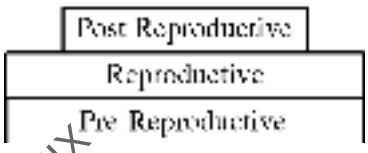
(2) A - Prey; B - Predator

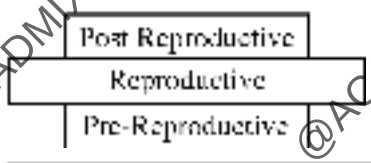
(3) A - Predator; B - Parasite

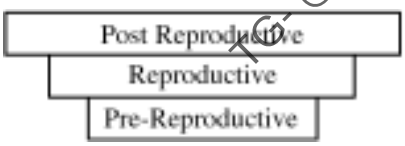
(4) A - Predator; B-Prey

23) Which among the given age pyramids of human population represents stable population -

(1)


(2)


(3)


(4)


24) Choose the correct statement -

(1) Morphology and anatomy of endoparasites are greatly simplified.

(2) Parasites act as conduits of energy transfer across trophic levels.

(3) Brood parasitism is a true parasitism.

(4) Parasitic mode of life ensures catch, kill and eat by parasites.

25) Bacterial and fungal enzymes degrade detritus into simpler inorganic substances, this process is called -

(1) Fragmentation
(2) Leaching
(3) Catabolism
(4) Humification

26) Select the correct statement -

(1) Small standing crop of phytoplankton can support large standing crop of zooplankton.
(2) In grassland ecosystem pyramid of energy is always upright.
(3) In grassland ecosystem pyramid of biomass is upright
(4) All of the above

27) India has more than 50000 different strains of rice and 1000 varieties of mango, this can be explained by -

(1) Genetic diversity
(2) Species diversity
(3) Ecological diversity
(4) Community diversity

28) The IUCN Red list (2004) documents the extinction of how many species in last 500 years ?

(1) 15,500 species
(2) 784 species
(3) 359 species
(4) 338 species

29) Match column-I with column-II.

	Species		**% of species facing threat of extinction**
a.	Mammal species	i.	12%
b.	Bird species	ii.	31%
c.	Gymnosperm species	iii.	23%
d.	Amphibian species	iv.	32%

(1) a-iii, b-i, c-ii, d-iv
(2) a-iii, b-i, c-iv, d-ii
(3) a-ii, b-iv, c-iii, d-i
(4) a-iii, b-iv, c-i, d-ii

30) Assuming that fur colour of an animal is dark, range of colour shade and white. A cross is made between a male (AABBCC) with dark fur colour and female (aabbcc) with white fur colour. What would be the fur colour of F_1 generation ?

(1) All intermediate colour
(2) All dark colour
(3) All white colour
(4) Range of colour shade

31) When modified allele produces a non functional enzyme then the phenotype or trait will only be dependent on the functioning of :

(1) Modified allele
(2) Unmodified allele
(3) Both modified allele and unmodified allele
(4) Not dependent on any type of allele

32) It was found that in few cases the F_1 had a phenotype that did not resemble either of two parents and was in between the two. This is a case of -

(1) Dominance
(2) Incomplete dominance
(3) Co-dominance
(4) Pleiotropism

33) Which of the following statements are **correct** about Down's syndrome ? **(a)** The cause of this genetic disorder is the presence of an additional copy of the chromosome no. 18. (Trisomy of 18)
(b) This disorder was first described by Langdon Down.
(c) The affected individual is short statured with large round head.
(d) Furrowed tongue and partially open month.

(1) All a, b, c, d are correct
(2) only a and b are correct
(3) only b and c are correct
(4) only b and d are correct

34) Match list-I with list-II.

	List-I (Type of cross)		**List-II (Phenotypic ratio)**
(a)	Monohybrid cross	(i)	1 : 1 : 1 : 1
(b)	Dihybrid cross	(ii)	1 : 1
(c)	Incomplete Dominance	(iii)	1 : 2 : 1
(d)	Monohybrid test cross	(iv)	9 : 3 : 3 : 1
(e)	Dihybrid test cross	(v)	3 : 1

(1) a-v, b-iv, c-iii, d-ii, e-i
(2) a-iv, b-v, c-iii, d-ii, e-i
(3) a-v, b-iii, c-iv, d-ii, e-i
(4) a-i, b-iv, c-iii, d-ii, e-v

35) Baculoviruses are employed as :-

(1) Biogas production
(2) Biocontrol agents
(3) Sewage treatment agents
(4) Biofertilizer

36) Match column I with column II and select the correct option using the codes given below :

	Column I		**Column II**
(A)	*Saccharomyces cerevisiae*	(i)	Production of product for removal of clot
(B)	*Monascus purpureus*	(ii)	Production of immunosuppressive agents
(C)	*Trichoderma polysporum*	(iii)	Production of blood cholesterol lowering agent
(D)	*Streptococcus*	(iv)	Commercial production of ethanol

(1) A-iv, B-iii, C-ii, D-i
(2) A-i, B-ii, C-iii, D-iv
(3) A-iii, B-iv, C-ii, D-i
(4) A-ii, B-i, C-iii, D-iv

37) Nitrogen base of RNA are-

(1) TCGU
(2) AGCT
(3) AGCU
(4) CTAU

38) In the structure of DNA at each step of ascent, the strand turns-

(1) 20º
(2) 36º
(3) 40º
(4) 75º

39) Purines are-

(1) Single ring structures

(2) Double ring structures

(3) Linear structures

(4) Highly acidic compounds

40) Chargaff's equivalency rule is valid for-

(1) dsRNA and ssDNA

(2) dsDNA

(3) ssRNA and ssDNA

(4) ssRNA and dsDNA

41) **Assertion :** The repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation. **Reason :** The bulk DNA forms a minor peak.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

42) In prokaryote type of RNA polymerase is -

(1) 2

(2) 3

(3) 5

(4) 1

43) The body of ovule fuses with the funicle in the region called ___A__ .

(1) A-Placenta

(2) A-Micropyle

(3) A-Leaf base

(4) A-Hilum

44) Which of the following is a correct statement ?

(1) Embryo develops at chalazal end of the anther.

(2) The residual, persistant nucellus is known as embryosac.

(3) The seed of *Lupinus arcticus* germinated and flowered after an estimated record of 10,000 years of dormancy.

(4) In some cereals such as rice and wheat, seed lose viability with in 10 sec of their release.

45) **Statement-I :** Majority of plants use water for pollination.
Statement-II : Wind pollination is quite rare in grasses.

(1) Statement I is correct but statement II is incorrect.

(2) Statement I is incorrect but statement II is correct.

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(3) Both statement I and statement II are correct.
(4) Both statement I and statement II are incorrect.

46) Which of the following structure play a role in motility in bacteria?

(1) Pili
(2) Fimbriae
(3) Flagella
(4) Plasmid

47) How many cycles of nuclear division and cell division are completed respectively during meiosis ?

(1) One, One
(2) One, Two
(3) Two, Two
(4) Two, One

48) During different stages of cell division, cell shows structured chromosomes in place of _____.

(1) Cell wall
(2) Ribosome
(3) Nucleus
(4) Plasma membrane

49) Which of the following characters is/are common to both mitochondria and chloroplast?
(A) Semi-autonomous cell organelles
(B) Presence of DNA
(C) Role in photosynthesis
(D) Prokaryotic type of ribosome.

(1) A and C only
(2) A and B only
(3) A, B and D
(4) A, B and C

50) During maturation of insulin in human :-

(1) C peptide is removed
(2) A and B peptides are removed
(3) C peptide is added
(4) Only A peptide is removed

51) Why do non-recombinant transfarments bacteria grow on a medium containing both ampicillin and tetra cyclin ?

(1) Because they don't contain any plasmid

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(2) Because they carry both antibiotic resistance gene intact.
(3) Because they have "Ori site".
(4) Because they do not have "Ori site".

52) Select the group of aromatic amino acids :

(1) Tyrosine, Phenylalanine, tryptophan
(2) Glutamic acid, Phenylalanine, tryptophan
(3) Valine, Glutamic acid, glycine
(4) Lysin, Valine, glycine

53) **Statement-I** : Only primary structure of protein is necessary for the many biological activities of proteins.
Statement-II : Adult human haemoglobin is an example of quaternary structure of protein.

(1) **Statement I** and **Statement II** both true
(2) **Statement I** and **Statement II** both false
(3) **Statement I** is false but **Statement II** is true
(4) **Statement I** is true but **Statement II** is false

54) Simple cuboidal epithelium is found in -

(1) wall of blood vessels
(2) PCT
(3) Stomach
(4) Bowmans capsule

55) First true coelom is found in which phylum ?

(1) Annelids
(2) Arthropods
(3) Mollusca
(4) Aschelminthes

56) Animals of which phylum shows metagenesis ?

(1) Porifera
(2) Annelida
(3) Coelenterata
(4) Enhinodermeta

57) In which of the following vertebrates, digestive tract has additional chambers the crop and gizzard ?

(1) *Periplenata*
(2) *Columba*

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(3) *Chelone*

(4) *Camelus*

58) Read the following statements carefully and choose the correct statements ?

(A) All living members of the class cyclostomata are ectoparasites on some fishes.

(B) Closed type of circulation is present in cyclostomata.

(C) In chondrichthyes notochord persists throughout the life.

(D) Chondrichthyes have circular and sucking mouth without jaw.

(1) A and D

(2) B, C and D

(3) Only A

(4) A, B and C

59) Excretory product of Tadpole larva of Frog.

(1) Ammonia

(2) Urea

(3) Uric acid

(4) (1) and (2) both

60) The fluid present in the space between the cells can be described as -

(1) Blood – Lymph

(2) Lymph – Protein content

(3) Blood – (RBCs + Platelets + Larger proteins)

(4) Lymph + Plasma Protein

61) Which of the following situation is responsible for the closure of cuspid valve (Bicuspid and tricuspid) ?

(1) Increasing the atrial pressure than ventricular pressure.

(2) Decreasing of atrial pressure than vena cava

(3) Increasing the ventricular pressure than Atrial Pressure.

(4) Increasing the ventricular Pressure than Aortic Pressure

62) Match the Column - I and Column - II and choose the correct option -

	Column - I		**Column - II**
(a)	AB^+ Blood	(i)	Universal Donor
(b)	Basophils	(ii)	Most abundant Blood cell
(c)	Erythrocytes	(iii)	Least in Blood
(d)	O^- ve blood	(iv)	Universal receipient

(1) a-(i); b-(ii); c-(iii); d-(iv)

(2) a-(iv); b-(ii); c-(iii); d-(i)
(3) a-(iv); b-(iii); c-(ii); d-(i)
(4) a-(iv); b-(i); c-(ii); d-(iii)

63) How many given parts of respiratory tract are supported by incomplete cartilaginous rings ?
Trachea, Tertiary bronchi, Terminal bronchioles, Secondary bronchi, Alveoli, initial bronchioles.

(1) 3
(2) 4
(3) 5
(4) 2

64) The volume of air that will remain in the lungs after normal expiration is known as -

(1) EC
(2) FRC
(3) TLC
(4) ERV

65) Identified the correct and incorrect statements and select the correct option.
(a) Emphysema is a chronic disorder leading to fibrosis and causing serious lungs damage.
(b) Carbonic anhydrase enzyme also present in plasma.
(c) The binding of O_2 with Hb is primarily related to PO_2.
(d) Nearly 3 ml O_2 and 7ml CO_2 transported as in dissolved state per 100 ml blood.

(1) a and b is correct and c and d is incorrect.
(2) a and d is correct while b and c is incorrect.
(3) b and c is correct while a and d is incorrect.
(4) a, b and d is correct c is incorrect.

66) Glomerular filtrate formed by the glomerulus can be described as -

(1) Blood - formed elements
(2) Plasma + WBC
(3) Blood - (formed elements + Plasma Protein)
(4) Plasma + Platelets

67) Excess secretion of growth hormone in adults especially in middle age can result in severe disfigurement (especially of the face) called:

(1) Acromegaly
(2) Cretinism
(3) Dwarfism
(4) Addison’s disease

68) The condition associated with diminished ability of the kidney to conserve water leading to water loss and dehydration is known as:

(1) Diabetes Insipidus
(2) Renal failure
(3) Cretinism
(4) Addison's disease

69) Grave's disease is associated with abnormal secretion of hormones of :

(1) Thyroid Gland
(2) Parathyroid Gland
(3) Pineal Gland
(4) Pancreas

70) Match the following columns and choose the correct option :

Hormone		**Function**	
A.	Adrenaline	(i)	BMR Regulation
B.	Thyroxine	(ii)	Emergency response
C.	Melatonin	(iii)	Immunosuppressive
D.	Cortisol	(iv)	Sleep-wake cycle

Options:
(1) A-ii, B-i, C-iv, D-iii
(2) A-i, B-ii, C-iii, D-iv
(3) A-iv, B-iii, C-ii, D-i
(4) A-iii, B-ii, C-iv, D-i

71) Given below are two statements :
Statement-I : When blood pressure is increased, ANF is secreted which causes dilation of the blood vessels.
Statement-II : Progesterone supports pregnancy.

(1) Both Statement-I and Statement-II are incorrect
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I and Statement-II are correct

72) Which of the following is the structural and functional unit of neural system ?

(1) Nephron
(2) Sarcolemma
(3) Neuron
(4) Synapse

73) Which part of the brain controls respiration and cardiovascular reflexes ?

(1) Hypothalamus
(2) Cerebrum
(3) Medulla oblongata
(4) Cerebellum

74) **Assertion :** Electrical synapses are faster than chemical synapses.
Reason : In electrical synapse, impulse transmission occurs through neurotransmitter.

(1) Assertion and Reason both false.
(2) Assertion is true but reason is false.
(3) Assertion and Reason both true.
(4) Assertion is false but reason is true.

75) Which part of the muscle stores Ca^{+2} ions ?

(1) Sarcolemma
(2) Myofibrils
(3) Z-line
(4) Sarcoplasmic reticulum

76) Which of the following is the disorder related to skeletal system ?

(1) Myasthenia gravis
(2) Osteoporosis
(3) Muscular dystrophy
(4) Tetany

77) Which gene is responsible to stop further division in a normal cell?

(1) Proto oncogene
(2) Tumor suppressor gene
(3) Gene related with programme cell death
(4) Oncogene

78) Identify the incorrect statement about common cold?

(1) It usually last for 3-7 days
(2) It spreads through air
(3) It is caused by Rhino virus
(4) It affects alveoli & accumulate fluid in lungs

79) In which of the following types of natural selection more individuals acquire peripheral character value at both ends of the distribution curve?

(1) Stabilizing selection
(2) Disruptive selection

(3) Directional selection

(4) All of the above

80) Which of the following man was with a brain size of 1400 cc, used hide to protect their body and burried their dead?

(1) Homo habilis

(2) Homo erectus

(3) Cromagnon man

(4) Neanderthal man

81) Select the correct answer about the figure given below. 

(1) Cu-T

(2) Cu-7

(3) Implants

(4) Lippes loop

82) Which of the following set of sexually transmitted diseases is completely curable if detected early and treated properly?

(1) Geninal warts and Genital herpes

(2) Genital herpes and Hepatitis-B

(3) Gonorrhoea and Syphilles

(4) Hepatitis-B and HIV injection

83) Match the column-I with column-II and select the correct answer about them.

	Column-I		**Column-II**
(a)	Fimbriae	(i)	A cushion of fatty tissue covered by skin and pubic hairs
(b)	Bulbourethral gland	(ii)	a tiny finger like structure which lies at the upper junction of both the labia minora
(c)	Clitoris	(iii)	Help in the collection of ovum after ovulation

(d)	Mons pubis	(iv)	Help in the lubrication of penis during coitus

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iii), b-(iv), c-(i), d-(ii)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(iii), b-(iv), c-(ii), d-(i)

84) Which of the following scientists proposed that the first form of life could have come from pre-existing non-living organic molecules like RNA, protein etc?

(1) Lamarck and S.L. Miller

(2) Darwin and Hugo-De varies

(3) Darwin and A.R. Wallace

(4) Oparin and Haldane

85) When the same structures developed along different directions due to adaptation to different needs. This is called-

(1) Analogy

(2) Convergent evolution

(3) Panspermia

(4) Divergent evolution

86) The excessive use of drugs or antibiotics in microbes has only resulted in the selection of drug or antibiotic resistant varieties in a much lesser time scale. This is an example of evolution by :-

(1) Industrial melanism

(2) Adaptive radiation

(3) Anthropogenic action

(4) Convergent evolution

87) In the male reproductive system which of the following cells provide nutrition to germ cells and secrete androgen bonding protein also?

(1) Leydig cells

(2) Immunologically competent cells

(3) Sertoli cells

(4) Follicular cells

88) Match the column-I with column-II and select the correct option about them.

	Column-I		**Column-II**
(a)	Lactational Amenorrhea	(i)	Male withdraws his penis from vagina just before ejaculation
(b)	Coitus interruptus	(ii)	Suppress the sperm motility

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(c)	Diaphragms	(iii)	Absence of menstrual cycle during the period of intense lactation
(d)	Cu-T	(iv)	Cover the cervix during coitus

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iii), b-(i), c-(iv), d-(ii)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(ii), b-(iv), c-(i), d-(iii)

89) Which of the following hormones are produced by placenta in a women only during pregnancy ?

(1) Progesterone, Estrogen, Relexin, FSH

(2) Progesterone, HPL, Relexin, LH

(3) HCG, HPL, Estrogen, FSH

(4) HCG, HPL, Relexin

90) According to Hardy-Weinberg principle the allelic frequencies in a sexually reproducing and randomly mating population are-

(1) Stable but not constant from generation to generation

(2) Not stable but is constant from generation to generation

(3) Stable and is constant from generation to generation

(4) Represented by $2p^2 + pq + q^2 = 0$

PHYSICS

1) A charged particle q is shot towards another charged particle Q which is fixed, with a speed u. It approaches Q upto a closest distance r and then returns. If q were given a speed u/4, the closest distance of approach would be :



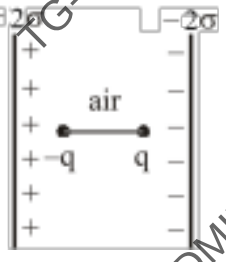
(1) 4r

(2) 16r

(3) r/2

(4) r/4

2) An electric dipole of dipole moment P is kept perpendicular to the plane of two large charged conducting plates as shown. The potential energy of the dipole is given as -



(1) zero

(2) $\frac{-2\sigma P}{\epsilon_0}$

(3) $\frac{-\sigma P}{\epsilon_0}$

(4) $\frac{\sigma P}{\epsilon_0}$

3) The capacity and the energy stored in a charged parallel plate condenser with air between its plates are respectively C_0 and W_0. If the air is replaced by glass (dielectric constant = 4) between the plates, the capacity of the plates and the energy stored in it will respectively be :-

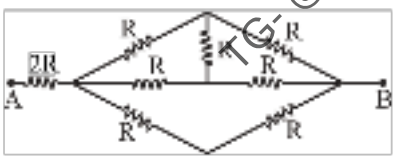
(1) $4C_0, 4W_0$

(2) $4C_0, \frac{W_0}{4}$

(3) $\frac{C_0}{4}, 4W_0$

(4) $\frac{C_0}{4}, \frac{W_0}{4}$

4) In the network of resistances as shown in figure, the effective resistance between points A and B is :-



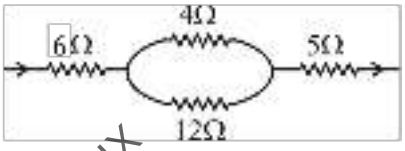
(1) 8 R

(2) 5R

(3) (5/3) R

(4) (8/3) R

5) If the power dissipated in the 12 Ω resistance is 12 W, then the power dissipated in 6 Ω will be:-



(1) 96 W

(2) 3 W

(3) 8 W

(4) 48 W

6) Consider the following statements –
(A) Biot savart's law gives magnetic field due to a current element which is a vector quantity.
(B) A point charge moving with uniform velocity produces magnetic field only.

(1) Both A & B are correct.

(2) A is correct, B is wrong

(3) A is wrong, B is correct

(4) Both are wrong

7) A bar magnet having a magnetic moment of 10×10^4 JT^{-1} is free to rotate in a horizontal plane. A horizontal magnetic field B = 5×10^{-4} T exist in the space. The work done in taking the magnet slowly from a direction parallel to the field to a direction 60° from the field is

(1) 0.5 J
(2) 25 J
(3) 8 J
(4) 10 J

8) In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.5×10^{10} Hz and amplitude 1260 V/m. The amplitude of oscillating magnetic field will be :

(1) 2.1×10^{-6} Wb/m^2
(2) 2.1×10^{-7} Wb/m^2
(3) 4.2×10^{-6} Wb/m^2
(4) 4.2×10^{-7} Wb/m^2

9) A convex lens of focal length 20 cm, a concave lens of focal length 20 cm and a concave lens of focal length 30 cm are placed in contact. The power of the combination in dioptre is :

(1) +3.33
(2) –3.33
(3) +6.67
(4) –6.67

10) Two beams of light having Intensities I and 9I interfere to produce a fringe pattern on a screen. The phase difference between the beam is "π" at point A and "2π" at point B. Then find out the difference between the resultant intensities at A and B.

(1) 4I
(2) 8I
(3) 12I
(4) 9I

11) Find the polarization angle of a light ray going from air in a medium in which the speed of light is 2×10^8 m/s :-

(1) $\tan^{-1}\left(\frac{2}{3}\right)$
(2) $\tan^{-1}\left(\frac{6}{5}\right)$
(3) $\tan^{-1}\left(\frac{5}{6}\right)$

(4) $\tan^{-1}\left(\frac{3}{2}\right)$

12) ${}_{100}X^{244} \xrightarrow{\alpha} \xrightarrow{\alpha} \xrightarrow{\alpha} \xrightarrow{\alpha} {}_{Q}Y^{P}$, then P & Q will be :-

(1) 232, 96
(2) 230, 98
(3) 228, 92
(4) 232, 98

13) Let T_2 and T_1 be the energy of an electron in the first and third excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is :

(1) 4 : 1
(2) 4 : 9
(3) 9 : 4
(4) 1 : 4

14) When a certain metal surface is illuminated with light of frequency ν, the stopping potential for photo electric current is V_0. When the same surface is illuminated by light of frequency ν/3, the stopping potential is $\frac{V_0}{6}$. The threshold frequency for photoelectric emission is :-

(1) $\frac{\nu}{6}$
(2) $\frac{\nu}{3}$
(3) $\frac{\nu}{5}$
(4) $\frac{2\nu}{5}$

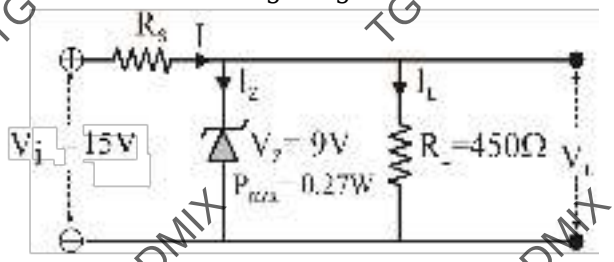
15) The innermost orbit of the hydrogen atom has a diameter of 1.06 Å. What is the diameter of the fifth orbit ?

(1) 2.65 Å
(2) 10.6 Å
(3) 26.5 Å
(4) 106 Å

16) How many photons of wavelength λ = 3300 nm must strike a totally reflecting screen per second at normal incidence so as to exert a force of 1μN ?

(1) 5×10^{21} photon/sec
(2) 2.5×10^{21} photon/sec
(3) 10×10^{21} photon/sec
(4) 5×10^{19} photon/sec

17) A 9V stabilised voltage supply is required to run a car stereo system from the car's 15V battery. A zener diode with V_z = 9V and P_{max} = 0.27 W is to be used as the voltage regulator. The load



resistance is 450Ω. The value of series resistor is :

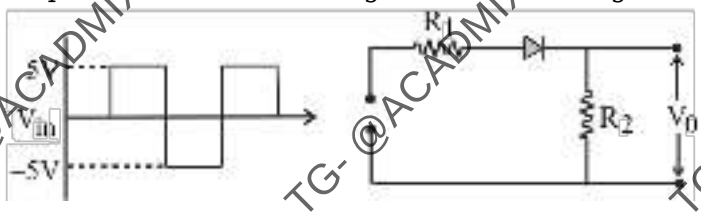
(1) 120Ω

(2) 150Ω

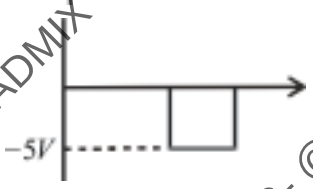
(3) 100Ω

(4) 90Ω

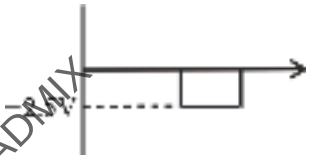
18) A waveform shown when applied to the following circuit will produce which of the following output waveform. Assuming ideal diode configuration and $R_1 = R_2$:-



(1) 

(2) 

(3) 

(4) 

19) A gas is filled in a vessel of volume 20 m^3 at pressure 30 kPa. If it is connected with an evacuated container of volume 10 m^3 then find change in pressure of gas (At constant temperature) :-

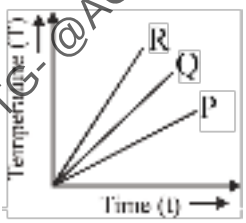
(1) $\frac{20}{3}$kPa

(2) 20 kPa

(3) $\frac{10}{3}$kPa

(4) 10 kPa

20) Which of the substances P, Q or R of same mass has the highest specific heat if heat supply rate

is same? The temperature v/s time graph is shown 

(1) P
(2) Q
(3) R
(4) All have equal specific heat

21) The equation of a plane progressive wave is

$$y = 0.01 \sin 10\pi\left[t - \frac{x}{10}\right]$$

When it is reflected at a rarer medium, its amplitude becomes 90% of its pervious value. The equation of reflected wave is :

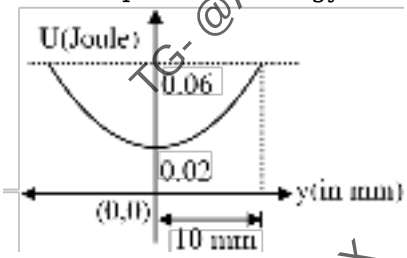
(1) $y = 0.009 \sin 10\pi\left[t - \frac{x}{10}\right]$
$y = 0.9 \sin 10\pi$
(2) $\left[t + \frac{x}{10}\right]$
(3) $y = -0.009 \sin 10\pi\left[t - \frac{x}{10}\right]$
(4) $y = 0.009 \sin 10\pi\left[t + \frac{x}{10}\right]$

22) Two sound sources produce progressive waves given by :-

$y_1 = 5 \sin (250\ \pi t)$
and $y_2 = 3 \sin (260\ \pi t)$

A person situated nearby will hear :-

(1) five beats per second of maximum & minimum intensity ratio 4 : 3
(2) five beats per second of maximum & minimum intensity ratio 16 : 1
(3) ten beats per second of maximum & minimum intensity ratio 7 : 2
(4) ten beats per second of maximum & minimum intensity ratio 4 : 3

23) The variation of potential energy of harmonic oscillator is as shown in figure. The spring constant is 

(1) 1×10^2 N/m
(2) 150N/m

(3) 8×10^2 N/m

(4) 3×10^2 N/m

24) Two rain drops reach the earth with different terminal velocities having ratio 16 : 9. Then, the ratio of their volume is :-

(1) 27 : 64

(2) 9 : 16

(3) 16 : 9

(4) 64 : 27

25) From an inclined plane a solid sphere, a disc, a ring and a spherical shell of same radius are rolled without slipping from the same height simultaneously. The decreasing order of their velocity when reaching at the base will be :-

(1) Ring, shell, disc, sphere

(2) Shell, sphere, disc, ring

(3) Sphere, disc, shell, ring

(4) Ring, sphere, disc, shell

26) Two point masses A & B of masses m and 2m are attached at the ends of the rod and both are revolving together about their centre of mass. Then the ratio of linear speeds of A and B will be



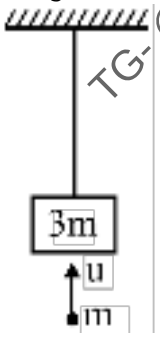
(1) 1 : 1

(2) 4 : 1

(3) 2 : 1

(4) 1 : 2

27) A bullet of mass m moving vertically upwards instantaneously with a velocity 'u' hits the hanging block of mass '3m' and gets embedded in it, as shown in the figure. The height through which the block rises after the collision (assume sufficient space above block) is :



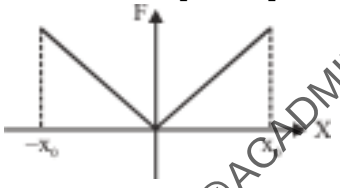
(1) $u^2/2g$

(2) $u^2/32g$

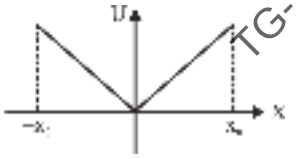
(3) $u^2/8g$

(4) $u^2/4g$

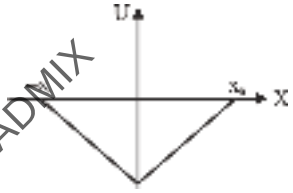
28) Figure shows a plot of the conservative force F in a unidimensional field. The plot representing the function corresponding to the potential energy (U) in the field is



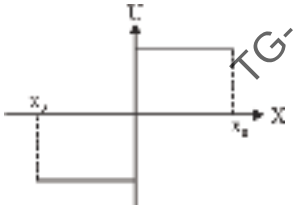
(1)



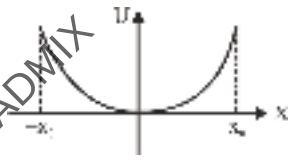
(2)



(3)



(4)



29) Given $F = \frac{a}{x}e^{-bt}$ where F is force, x is distance and t is time. The dimensions of $\left(\frac{b}{a}\right)$ is :-

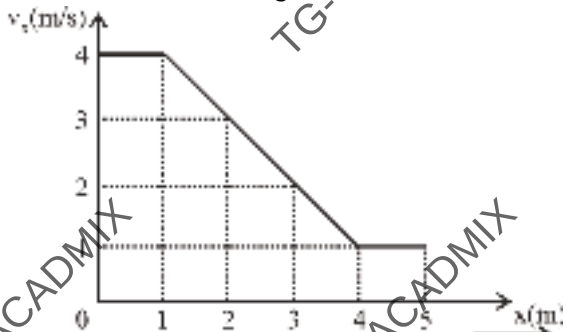
(1) $M^{-1}L^{-1}T$

(2) $M^{-1}L^{-2}T$

(3) $ML^{-1}T^{-1}$

(4) $M^{-1}L^{1}T^{1}$

30) A body moves in a straight line for which graph of v_x vs x is shown in the figure. Find the magnitude of acceleration of the body (in m/s^2) at x = 2m.

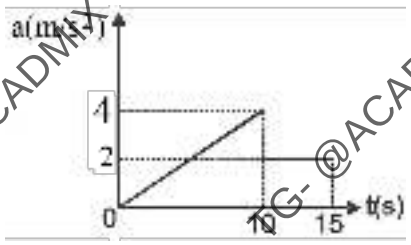


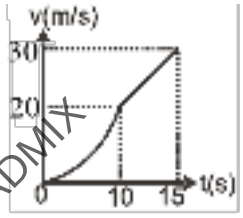
(1) 2

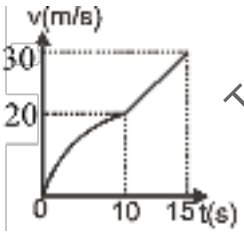
(2) 0

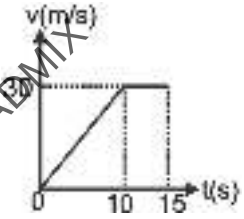
(3) 4

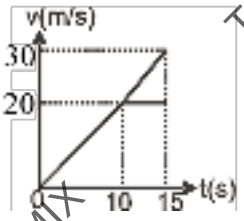
(4) 3

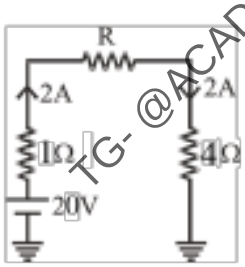
31) A particle starts from rest and undergoes an acceleration which varies with time as shown in figure. The velocity-time graph from figure will have shape. 

(1) 

(2) 

(3) 

(4) 

32) The value of R in the circuit should be :- 

(1) 1 Ω

(2) 1.5 Ω

(3) 2 Ω

(4) 5 Ω

33) In a circuit a coil of resistance 4Ω, then magnetic flux changes from 4.0 Wb to 16.0 Wb in 0.2 sec. The induced charge in the coil during this time is :–

(1) 5.0 C

(2) 3.0 C

(3) 1.0 C

(4) 0.8 C

34) Refractive index of equiconvex lens is 1.7 & its radius of curvature is 21 cm. Its focal length will be :-

(1) 15 cm

(2) 40 cm

(3) 2 cm

(4) 1 cm

35) If the critical angle for total internal reflection from a medium to vacuum is 37°, the velocity of light in the medium is

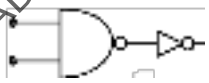
(1) 3×10^8 m/s

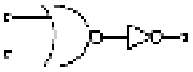
(2) 1.8×10^8 m/s

(3) 6×10^8 m/s

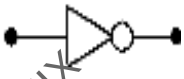
(4) $\sqrt{3} \times 10^8$ m/s

36) Which represents AND gate?

(1) 

(2) 

(3) 

(4) 

37) If the length of a cylinder on heating increases by 4%, the area of its base will increase by:

(1) 5%

(2) 8%

(3) 6%

(4) 4%

38) A black body has wavelength λ_m corresponding to maximum radiant energy at temperature 2000K. Its corresponding wavelength at temperature 6000K will be.

(1) $\frac{3}{2}\lambda_m$

(2) $\frac{2}{3}\lambda_m$

(3) $\frac{\lambda_m}{3}$

(4) $\frac{9}{10}\lambda_m$

39) A wire has strain of 10^{-3} when a stress of 1×10^9 N/m^2 is applied to it. Young's modulus of material of wire is

(1) 10^{11} N/m^2

(2) 10^{12} N/m^2

(3) 10^8 N/m^2

(4) 10^6 N/m^2

40) The height at which the weight of a body becomes 1/16th of its weight on the surface of earth (radius of earth is R)

(1) h = 3R

(2) h = R

(3) $h = \frac{R}{2}$

(4) h = 2R

41) A ring of radius 1 m and mass 20 kg is rotating about its diameter with angular velocity of 40 rad/s. Its kinetic energy is

(1) 10 J

(2) 1000 J

(3) 2000 J

(4) 8000 J

42) A string breaks if its tension exceeds 10 newton. A stone of mass 100 gm tied to this string of length 50 cm is rotated in a horizontal circle. The maximum angular velocity of rotation can be

(1) 10 rad/s

(2) $10\sqrt{2}$ rad/s

(3) 100 rad/s

(4) 200 rad/s

43) Distance moved by a screw in 2 rotation is 1 mm and total number of circular divisions is 100. The leastcount of the screwguage is :-

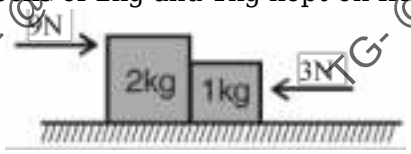
(1) 0.005 mm

(2) 0.001 cm

(3) 0.0001 cm

(4) 0.001 mm

44) Two forces of 9N and 3N are acting on the two blocks of 2kg and 1kg kept on frictionless floor.



What is the force exerted on 2kg block by 1kg block ?

(1) 1N
(2) 2N
(3) 4N
(4) 5N

45) A particle projected from origin follows the trajectory $y = x - \frac{x^2}{9}$, where x-axis is along horizontal direction and y-axis is along vertical direction. Find the horizontal range of projectile :-

(1) 27 m
(2) 9 m
(3) $3\sqrt{3}$ m
(4) $9\sqrt{3}$ m

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ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	4	2	2	2	4	3	4	3	2	3	4	2	2	4	2	2	2	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	4	3	2	2	1	1	4	2	2	4	3	3	1	2	2	3	1	2
Q.	41	42	43	44	45															
A.	1	1	1	2	2															

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	3	1	2	4	4	2	4	3	1	1	3	2	1	2	2	3	3	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	2	1	3	4	1	2	1	1	2	2	4	1	2	1	3	3	2	2
Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
A.	3	4	4	3	4	3	3	3	3	1	2	1	3	2	1	3	2	4	1	3
Q.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125
A.	3	3	2	2	3	3	1	1	1	1	4	3	3	2	4	2	2	4	2	4
Q.	126	127	128	129	130	131	132	133	134	135										
A.	1	3	4	4	4	3	3	2	4	3										

PHYSICS

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
A.	2	2	2	4	1	2	2	3	2	3	4	3	4	3	3	2	1	3	4	1
Q.	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
A.	4	2	3	4	3	3	2	4	2	4	1	4	2	1	2	1	2	3	2	1
Q.	176	177	178	179	180															
A.	4	2	1	4	2															

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SOLUTIONS

CHEMISTRY

1)

Basic strength $\propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I}$

2) $CH_3 - CH_2 - Br + KOH\,(alc) \rightarrow CH_2 = CH_2 + KBr$

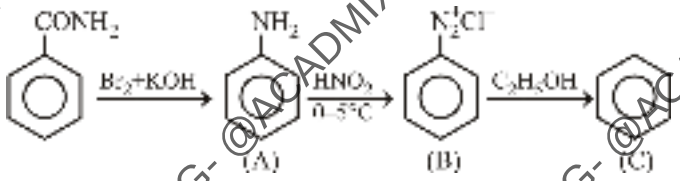
Hybridization of carbon atom is changed from sp^3 to sp^2, therefore the above reaction is an elimination reaction.

5)

E2 will be favoured in option 1 whereas S_{N^2} will be favoured in option 2.

10) $C_6H_5 - \overset{O}{\overset{\|}{C}} - CH_3 \xrightarrow[(ii)\ H_3O^{\oplus}]{(i)\ LiAlH_4}$ $C_6H_5-\underset{\text{2° alcohol}}{\overset{OH}{\overset{|}{CH}}}-CH_3$

15)



31) Due to polarisation as compound is $BeCl_2$

45)

Inner orbital complex means in presence of strong field ligand.

BIOLOGY

46) NCERT Pg. No. # 59

48) NCERT Pg. No. # 64, 65, 66

49) NCERT Pg. No. # 71

51) NCERT Pg. No. # 71

52) NCERT XI Page No. 26, 27, 28

53) NCERT XI, Page No. 10

54) NCERT XI, Page No. 30, 32

55) NCERT XI, Page No. 24

56) NCERT XI, Page No. 17

57) NCERT XI, Page No. 30, 32

58) NCERT XI, Page No. 4, 7, 23, 24

59) NCERT Pg # 117

60) NCERT Pg # 175-177

61) NCERT Pg # 150

62) NCERT Pg # 146

63) NCERT Pg # 158

64) NCERT Pg # 156

65) NCERT Pg # 162

66) NCERT Pg # 146

67) NCERT Page No. 197

68) NCERT Page No. 192

69) NCERT Page No. 200

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70) NCERT Page No. 207

71) NCERT Page No. 212

72) NCERT Page No. 217

73) NCERT Page No. 221

74) NCERT Page No. 222

75) NCERT-XII, Pg#69

76) NCERT-XII, Pg#61

77) NCERT-XII, Pg#60

78) NCERT-XII, Pg#76

79) NCERT-XII, Pg#55,56,57,59

80) NCERT XII Pg # 157

81) NCERT XII Pg # 152, 153

82) NCERT Pg # 81

83) NCERT Pg # 81

84) NCERT Pg # 81

85) NCERT Pg # 81

86) NCERT Pg # 105

87) NCERTG Pg # 94

88) NCERT Pg # 9

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89) NCERT Pg # 8, 19, 20, 22

90) NCERT Pg # 12, 13

91) NCERT-XI, Page No.: 91

92) NCERT-XI, Page No.: 125

93) NCERT-XI, Page No.: 101

94) NCERT-XI, Page No.: 97, 98

95) NCERT-XII Pg#182(E), 199(H)

96) NCERT-XII Pg#170(E), 219(H)

97) NCERT-XII Pg#106(E), 106(H)

98) NCERT-XII Pg#111,112 (E), 111,112(H)

99) NCERT-XII Pg#209

100) NCERT-XII Pg#39

101) NCERT-XII Pg#41

102) NCERT-XII Pg#49

103) NCERT-XII Pg#47

104) NCERT-XII Pg # 82

105) NCERT Pg. No. 203

106) NCERT Page No. # 199/200

107) NCERT Pg. No. 194/195

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108) NCERT Pg. No. 184

109) NCERT Pg. No. 187

110) NCERT Pg. No. 189/190

111) NCERT Pg. No. # 208

112) NCERT Page No. #241

113) NCERT Page No. #242

114) NCERT Page No. #244

115) NCERT Page No. #242, 243, 244

116) NCERT Page No. #247

117) NCERT Pg. No. # 232

118) NCERT Pg. No. # 236

119) NCERT Pg. No. # 234

120) NCERT Pg. No. # 219

121) NCERT Pg. No. # 227

122)

NCERT Page No.: 141(E), 154(H)

123) NCERT Page No.: 131(E)

124) NCERT Page No.: 121

125) NCERT Page No.: 125

126) NCERT Page No.: 44

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127) NCERT Page No.: 47, Para.: 3.4

128) NCERT Page No.: 28-30

129) NCERT Page No.: 111

130) NCERT Page No.: 114

131) NCERT Page No.: 116

132) NCERT Page No.: 27

133) NCERT Page No.: 44

134) NCERT Page No.: 37

135) NCERT Page No.: 120, 121

PHYSICS

136)

Applying law of energy conservation

T.KE = T. PE

$$\frac{1}{2}mU^2 = \frac{KQq}{r} \quad \ldots(i)$$

$$\frac{1}{2}m\left(\frac{U}{4}\right)^2 = \frac{KQq}{r'} \quad \ldots(ii)$$

(i) ÷ (ii)

$$16 = \frac{r'}{r} \Rightarrow r' = 16r$$

137) $U_{dipole} = -\vec{P}\cdot\vec{E} = -PE\cos\theta$

$$= -P\left(\frac{2\sigma}{\varepsilon_0}\right)\cos(0^\circ) = \frac{-2\sigma P}{\epsilon_0}$$

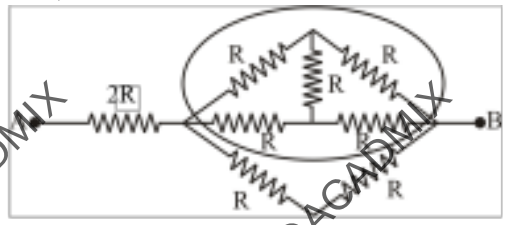
138) When a dielectric K is introduced in a parallel plate condenser its capacity becomes K times.

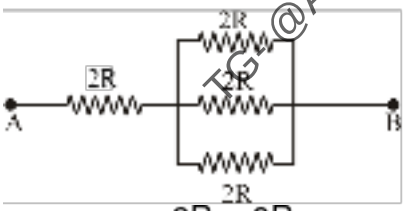
Hence $C' = 4C_0$.

Energy stored $W_0 = \frac{q^2}{2C_0}$

$$\therefore W' = \frac{q^2}{2C'} = \frac{q^2}{2 \times 4C_0} \Rightarrow W' = \frac{W_0}{4}$$

139) Balanced W.S.B.

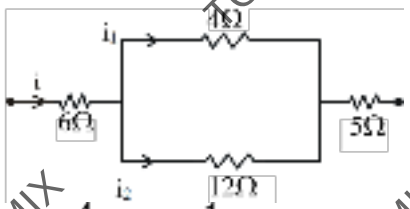




$$R_{AB} = 2R + \frac{2R}{3} = \frac{8R}{3}$$

140)

$P_{12} = 12W$



$$i_2 = \frac{4}{16} \times i = \frac{1}{4}i$$

$$\square\ P_{12} = i_2^2 \times 12$$

$$12 = \left(\frac{i}{4}\right)^2 \times 12$$

$i^2 = 16$

$i = 4A$

so $P_3 = i^2 \times 6 = (4)^2 \times 6 = 96$ W

142)

W = MB (1 – cos 60°) = 25 J

143) $\frac{E}{B} = C$

$$B = \frac{E}{C} = \frac{1260}{3 \times 10^8} = 4.2 \times 10^{-6}\ \text{Wb/m}^2$$

144)

For equivalent focal length,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{(+20)} + \frac{1}{(-20)} + \frac{1}{(-30)}$$

$$\Rightarrow \frac{1}{f} = -\frac{1}{30} \Rightarrow f = -30\text{cm}$$

Since, power i.e., P (in D)

$$= \frac{1}{f(\text{in m})} = -\frac{100}{30} = -\frac{10}{3}$$

= –3.33 D

145)

At point A,

$I_A = I + 9I + 2\sqrt{I}\sqrt{9I}\cos\pi$

$= 4I$

At point B,

$I_B = I + 9I + 2\sqrt{I}\sqrt{9I}\cos 2\pi$

$= 16I$

$\therefore I_B - I_A = 16I - 4I = 12I$

146) $\because \mu = \frac{c}{v} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2}$

From Brewstar's law

$\therefore i_P = \tan^{-1}(\mu) = \tan^{-1}\left(\frac{3}{2}\right)$

147) ${}_{100}X^{244} \xrightarrow{4\alpha} {}_{92}Y^{228}$

148) First excited state $\Rightarrow$ n = 2

$$T_2 = -13.6\frac{z^2}{n^2} = -\frac{13.6}{4}\text{eV}$$

Third excited state $\Rightarrow$ n = 4

$$T_1 = -13.6\frac{z^2}{n^2} = -\frac{13.6}{16}\text{eV}$$

$$T_1 : T_2 = \frac{1}{16} : \frac{1}{4} = 1 : 4$$

149) Using, $V^0 = \frac{h\nu}{e} - \frac{h\nu_0}{e}$ .....(1)

$$\frac{V_0}{6} = \frac{h\nu/3}{e} - \frac{h\nu_0}{e} \quad \text{.....(2)}$$

On solving, $\nu_0 = \frac{\nu}{5}$

150)

Diameter or radius $\propto n^2$

$$\frac{D}{1.06\ \overset{o}{A}} = \left(\frac{5}{1}\right)^2$$

$\Rightarrow D = 25 \times 1.06\ Å$

$= 26.50\ Å$

151) Momentum of the incident photon $p = \frac{h}{\lambda}$,

Momentum after reflection $= -\frac{h}{\lambda}$

Change in momentum $= \Delta p = \frac{2h}{\lambda}$

If n is the number of photons falling per second on the screen then force

$$F = \frac{\Delta p}{\Delta t} = \frac{2nh}{\lambda}$$

$$\Rightarrow n = \frac{F\lambda}{2h} = \frac{1 \times 3300 \times 10^{-9} \times 10^{-6}}{2 \times 6.6 \times 10^{-34}}$$

$= 2.5 \times 10^{21}$ photons s^{-1}

152) $I_Z = \frac{P_Z}{V_Z} = \frac{0.27W}{9V} = 0.03A$

$I_L = \frac{V_L}{R_L} = \frac{9V}{450\,\Omega} = 0.02A$

$I_S = I_Z + I_L = 0.05A$

$R_S = \frac{V_i - V_L}{I_S} = \frac{(15-9)V}{0.05\,A}$

$R_s = 120\Omega$

153) The P-N junction will conduct only when it is forward biased i.e. when + 5V is fed to it, so it will conduct for 2nd and 4th quarter part of signal shown and when it conducts potential drop 5 volt will be across both the resistors, so output voltage across R_2 is 2.5V.

$\therefore V_0 = + 2.5V$

154) $P_1V_1 = P_2V_2$

$30 \times 20 = P_2 \times 30$

Change in pressure $= 30 - 20 = 10kPa$

155)

For constant heat flow rate (Q)

$msdT = Qdt$

$$\frac{dT}{dt} \propto \frac{1}{s}$$

For high specific heat, slope $\frac{dT}{dt}$ is lower that is for P.

156) $A_r = \frac{90}{100} \times A_i = \frac{9}{10} \times 0.01 = 0.009$

No phase change at rarer medium.

So $y = 0.009 \sin 10\pi \left[t + \frac{x}{10}\right]$

157) $\Delta n = n_1 - n_2$

$= 130 - 125 = 5 b/s$

$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{5+3}{5-3}\right)^2 = \frac{64}{4} = \frac{16}{1}$

158) $\frac{1}{2}kA^2 = U_{max} - U_{min}$

$\Rightarrow \frac{1}{2}k\frac{100}{10^6} = 0.04$

$\Rightarrow k = 8 \times 10^2$ N/m

159) $\frac{v_1}{v_2} = \frac{16}{9} = \frac{r_1^2}{r_2^2}\frac{r_1}{r_2} = \frac{4}{3}$

$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \frac{64}{27}$

160) Velocity of the rolling body when it reach the bottom

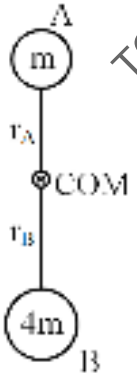
$v = \sqrt{\frac{2gh}{1 + \frac{K^2}{R^2}}}$

$v \propto \left(\sqrt{1 + \frac{K^2}{R^2}}\right)^{-1}$

$\left(\frac{K^2}{R^2}\right)_{ring} = 1, \left(\frac{K^2}{R^2}\right)_{disc} = \frac{1}{2}$;

$\left(\frac{K^2}{R^2}\right)_{sphere} = \frac{2}{5}, \left(\frac{K^2}{R^2}\right)_{shell} = \frac{2}{3}$

161)



$\frac{r_A}{r_B} = \frac{2m}{m} = 2$

$m_A r_A = m_B r_B \Rightarrow$

$\because \omega_A = \omega_B$

$\frac{v_A}{r_A} = \frac{v_B}{r_B}$

$\frac{v_A}{v_B} = \frac{r_A}{r_B} = 2$

162)

inelastic collision

mu = 4mV

$V = \frac{u}{4}$

COME $\frac{1}{2}4m\left(\frac{u}{4}\right)^2 = 4\,mgh$

$h = \frac{u^2}{32g}$

163) $f(x) = -\frac{dU}{dx}(x)$

or $U(x) = -\int F(x)\,dx$

Here F(x) = –kx, where k is a positive constant.

U(x) = $+k\int x dx = \frac{1}{2}kx^2$

i.e. parabola

164) $\frac{b}{a} = \frac{T^{-1}}{ML^2T^{-2}}$

$= M^{-1}L^{-2}T$

165) Here $\frac{dv}{dx} = -1$ and v = 3 at x = 2

so $a = v\frac{dv}{dx} = 3(-1) = -3$

 Magnitude of acceleration = 3 m/s^2

166) $a \propto t \Rightarrow V \propto t^2$

a = constant $\Rightarrow V \propto t$

167) Apply KVL from A to B

–20 + 2 × 1 + 2 × R + 2 × 4 = 0

R = 5 Ω

Hence option (4)

168) induced charge $q = \frac{\Delta\phi}{R} = \frac{16-4}{4} = \frac{12}{4}$
= 3.0 C

169) $\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$
$= (1.7 - 1) \times \left[\frac{2}{21}\right]$
f = 15 cm

170) $\mu = \frac{1}{\sin\theta_c}$
$\frac{c}{v} = \frac{1}{\sin 37^\circ}$
$v = \frac{c}{(5/3)} = \frac{9 \times 10^8}{5} = 1.8 \times 10^8 m/s$

171) NAND + NOT = AND

172) $\frac{\Delta A}{A} = \frac{2\Delta k}{L}$ = 8%

173) $\lambda_m \propto \frac{1}{T}$
$\frac{\lambda_{m_1}}{\lambda_{m_2}} = \frac{T_2}{T_1}$
$\frac{\lambda_m}{\lambda_{m_2}} = \frac{6000}{2000}$
$\lambda_{m_2} = \frac{\lambda_m}{3}$

174) $Y = \frac{\text{stress}}{\text{strain}} = \frac{10^9}{10^{-3}} = 10^{12} N/m^2$

175) $\frac{g}{16} = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$
$4 = 1 + \frac{h}{R}$ $\boxed{h = 3R}$

176)
R = 1 m ; m = 20 kg ; ω = 40 rad s^{-1}

K.E. = $\frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{mR^2}{2}\right)\omega^2$

$= \frac{1}{2} \times \left(\frac{20 \times 1}{2}\right)(40)^2 = 8000$ J

177) $T_{max} = m\omega_{max}^2 . r$

$10 = \frac{1}{10} \times \omega_{max^2} \times 50 \times 10^{-2}$

$\omega_{max} = 10\sqrt{2}$ rad/sec

178)

$\text{Pitch} = \frac{1\text{mm}}{2\text{Rotations}} = \frac{1\text{mm}}{2}$

$\text{LC} = \frac{\text{Pitch}}{100} = \frac{\frac{1}{2}\text{mm}}{100}$

= 0.005 mm

179) $a = \frac{9-3}{2+1} = 2\text{m/s}^2$



N – 3 = 1 × a

N = 3 + 2 = 5N

180) $y = x - \frac{x^2}{9} = x\left[1 - \frac{x}{9}\right]$

compare with

$y = x\tan\theta\left[1 - \frac{x}{R}\right]$

$\boxed{R = 9\text{m}}$